

FRM Part II Exam

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Questions with Answers - Risk Management and Investment
Management

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Reading 146: Factor Theory

Q.2375 Which of the following statements is (are) correct?

- I. The CAPM is based on the assumption that investors have homogeneous expectations
- II. The CAPM is based on the assumption of market equilibrium
- III. The CAPM is based on the assumption of diversified portfolio

- A. Statements I & II
- B. Only statement I
- C. All statements are correct
- D. Statements II & III

The correct answer is **C**.

The CAPM is based on the assumption that investors hold a combination of the market portfolio and risk-free assets. Furthermore, it assumes that the set of means, volatilities, and correlations are the same for all investors, i.e., the investors have homogeneous expectations. In addition, the market is at equilibrium, where demand is equal to supply. It also assumes investors do not hold concentrated portfolios. Rather, they hold well-diversified portfolios that eliminate specific risks.

Things to Remember

The Capital Asset Pricing Model (CAPM) is a financial model that is used to determine the expected return on an investment given its risk. It is based on several assumptions, including the assumption that investors have homogeneous expectations, that the market is in equilibrium, and that investors hold diversified portfolios. These assumptions are necessary to simplify the model and make it applicable to a wide range of scenarios. However, in reality, these assumptions may not always hold true. For example, investors may have different expectations based on their individual circumstances, markets may not always be in equilibrium due to various factors, and many investors may not hold diversified portfolios due to various reasons. Therefore, while the CAPM is a useful tool for estimating the expected return on an investment, it is important to understand its limitations and use it in conjunction with other financial models and tools.

Q.2376 Which of the following are true about multifactor models?

- A. Multifactor models take into consideration multiple risk factors
- B. Idiosyncratic risk is diversified
- C. Multifactor models contain multiple betas
- D. All of the above

The correct answer is **D**.

Multifactor models in finance are designed to evaluate multiple risk factors simultaneously. This is in contrast to the Capital Asset Pricing Model (CAPM), which only considers risk relative to the market portfolio. Multifactor models are more comprehensive and can model both linear and non-linear dependencies of risk factors. Each risk factor in a multifactor model is associated with a distinct beta, which measures the sensitivity of the expected excess asset returns to the expected excess market returns. Furthermore, both CAPM and multifactor models assume that idiosyncratic risks, which are unique to individual assets, are diversified away in a well-diversified portfolio. Therefore, all the statements in the options are correct.

Choice A is incorrect. While it is true that multifactor models consider multiple risk factors, this statement alone does not fully capture the characteristics and components of these models. Multifactor models also contain multiple betas and diversify idiosyncratic risk, which are not mentioned in this choice.

Choice B is incorrect. Although idiosyncratic risk can be diversified in multifactor models, this statement does not encompass all the features of these models. They also take into account multiple risk factors and contain several betas, aspects that are overlooked in this option.

Choice C is incorrect. The presence of multiple betas in a multifactor model is accurate but incomplete as a description of these types of financial tools. They also consider various risk factors and diversify idiosyncratic risks, elements that are omitted from this choice.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a single-factor model that considers only the market risk factor.
- Multifactor models can include factors such as size, value, momentum, and other specific risk factors in addition to market risk.
- Betas in multifactor models represent the sensitivity of an asset's returns to each specific risk factor included in the model.
- Idiosyncratic risk refers to the risk that is specific to an individual asset and can be diversified away in a well-diversified portfolio.
- Multifactor models are used to better capture the complexities of asset pricing and risk management compared to single-factor models like CAPM.

Q.2377 Country A, a developing economy, has recently witnessed government's renewed interest

and effort in reforming the financial sector. A recent report published by a non-governmental organization (NGO) indicates the presence of rampant insider trading. The report also points towards the lack of depth in the financial market which poses significant problems for investors and market participants. However, the presence of natural resources in the country makes it an attractive destination for foreign investors. A fund manager investing in country A utilizes the CAPM to compute the expected return. Which of the following options is (are) correct?

- I. Deviations from the CAPM will be observed due to insider trading
- II. Deviations from the CAPM will be observed due to the developing economy
- III. Deviations from the CAPM will be observed due to a lack of market depth
- IV. Deviations from the CAPM will be observed due to the country's underdeveloped corporate sector

- A. I only
- B. I and II
- C. I and III
- D. III and IV

The correct answer is **C**.

The Capital Asset Pricing Model (CAPM) is a theoretical model used to determine the expected return on an investment given its risk. The model assumes that markets are efficient, meaning that all relevant information is freely available to all market participants and that investors are rational and risk-averse. However, in the case of Country A, these assumptions are violated due to the presence of insider trading and a lack of market depth.

Insider trading refers to the practice of trading a public company's stock or other securities by individuals with access to nonpublic, material information about the company. In this scenario, information is not freely available to all investors, giving some an unfair advantage. This violates the CAPM's assumption of information symmetry and could lead to deviations from the model's predictions.

Similarly, a lack of market depth could also lead to deviations from the CAPM. Market depth refers to the market's ability to sustain relatively large market orders without impacting the price of the security. In a market with low depth, large trades could significantly move the price, violating the CAPM's assumption that investors are price takers. Therefore, both insider trading and a lack of market depth could cause deviations from the CAPM, making choice C the correct answer.

Choice A is incorrect. The statement I only does not correctly identify the factors that could cause deviations from the CAPM in this scenario. The CAPM assumes perfect markets, which means there are no transaction costs, taxes, or restrictions on borrowing. However, in a developing economy like Country A where insider trading is prevalent and market depth is lacking, these assumptions are violated. Therefore, relying solely on statement I would not fully capture the potential deviations from the CAPM.

Choice B is incorrect. While both statements I and II identify some of the factors that could cause deviations from the CAPM in this scenario, they do not cover all possible factors. For instance, they do not account for other market imperfections such as information asymmetry and investor irrationality which can also lead to deviations from the CAPM.

Choice D is incorrect. Statements III and IV alone do not correctly identify all of the factors that could cause deviations from the CAPM in this scenario. While these statements may highlight some potential issues with using CAPM in a developing economy like Country A (such as lack of liquidity or high transaction costs), they fail to consider other important aspects such as insider trading and lack of market depth which are also significant challenges for investors and can lead to deviations from expected returns calculated using CAPM.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model used to determine the expected return on an investment based on its risk.
 - Efficient markets assume all relevant information is freely available to all market participants and investors are rational and risk-averse.
 - Insider trading involves trading securities based on nonpublic, material information, violating information symmetry.
 - Market depth refers to the ability of a market to handle large orders without impacting security prices.
 - Market imperfections like information asymmetry and investor irrationality can also lead to deviations from the CAPM.
-

Q.2379 Country A is a developed economy with a vibrant financial sector. It witnessed a major financial crisis around 15 years ago. The economy recovered quite well and has consistently registered positive economic growth over the years. A recent survey conducted by an equity research firm indicated that the country's fund managers have built a well-diversified portfolio inclusive of all the sectors of the economy. The survey also highlighted the fact that the fund managers generally used historical data of the last 10 years and mean-variance utility to determine the correlations between the different assets. Asked why they preferred data from the last 10 years, most managers were of the view that recent data was the best estimator of the correlation between the different assets.

Which of the following statements is accurate regarding the case?

- A. The correlation between the assets will be underestimated.
- B. The correlation between the assets will be overestimated.

- C. The correlation between the assets is independent of the time period.
- D. The correlation between the assets is not paramount in this scenario

The correct answer is **A**.

During a financial crisis, the returns on most assets tend to move in the same direction, which means that the correlation between different assets increases. In this case, the fund managers in Country A are using data from the last 10 years to estimate the correlation between the assets. This time period does not include the financial crisis that occurred 15 years ago. Therefore, the correlation between the assets will be underestimated because the data from the crisis period, which would have shown a higher correlation, is not being considered. This underestimation could potentially lead to inaccurate risk assessments and suboptimal portfolio diversification strategies.

Choice B is incorrect. The correlation between the assets will not be overestimated. This is because the fund managers are using recent data, which does not include the financial crisis period. Therefore, they are likely to underestimate the correlation as they are missing out on significant market events that could have affected asset correlations.

Choice C is incorrect. The correlation between assets is not independent of the time period. Correlations can change over time due to various factors such as changes in market conditions, economic cycles, and regulatory changes among others.

Choice D is incorrect. As explained above, none of these options accurately reflect the situation described in this scenario.

Things to Remember

- Correlation between assets is a key concept in portfolio management as it helps in understanding how different assets move in relation to each other.
- Historical data is often used to estimate correlations, but it is important to consider the time period and any significant events that may have impacted correlations.
- Mean-variance utility is a common approach used in portfolio optimization to find the optimal balance between risk and return.
- Portfolio diversification is crucial to reduce risk by spreading investments across different assets and sectors.
- Market events, such as financial crises, can significantly impact asset correlations and should be considered when estimating correlations for portfolio management.

Q.2380 Country A is a developed economy with a vibrant financial sector. It witnessed a major financial crisis around 15 years ago. The economy recovered quite well and has consistently registered positive economic growth over the years. A recent survey conducted by an equity research firm indicated that the country's fund managers have built a well-diversified portfolio inclusive of all the sectors of the economy. The survey also highlighted the fact that the fund managers generally used historical data of the last 10 years and mean-variance utility to determine the correlations between the different assets. Asked why they preferred data from the last 10 years, most managers were of the view that recent data was the best estimator of the correlation between the different assets.

Suppose the country faces a major economic crisis, which of the following is the most appropriate statement?

- A. The fund managers would not be affected as their portfolios are well diversified.
- B. Portfolio diversification would most likely fail and lead to losses.
- C. The returns from the fund would most likely beat other similar funds.
- D. None of the above.

The correct answer is **B**.

Portfolio diversification would most likely fail and lead to losses. This is because during a financial crisis, the correlation between various assets tends to increase. This phenomenon was observed during the 2007/2008 financial crisis. When the correlation between assets increases, the returns on these assets tend to move in the same direction. This effectively diminishes the benefits of diversification. Diversification is a risk management strategy that mixes a wide variety of investments within a portfolio. The rationale behind this technique is that a portfolio constructed of different kinds of investments will, on average, yield higher returns and pose a lower risk than any individual investment found within the portfolio. However, in a financial crisis, this strategy may fail as the correlation between assets increases, causing them to move in the same direction and leading to potential losses.

Choice A is incorrect. While diversification is a key strategy in risk management, it does not guarantee immunity from losses during a major economic crisis. Even though the fund managers have constructed well-diversified portfolios, they are still exposed to systemic risk which affects all sectors of the economy and cannot be eliminated through diversification.

Choice C is incorrect. The assertion that the returns from the fund would most likely beat other similar funds during an economic crisis is not necessarily true. The performance of a fund during a crisis depends on various factors including its asset allocation, risk management strategies and market conditions among others. Therefore, it's not guaranteed that this particular fund would outperform others.

Choice D is incorrect. As explained above, both choices A and C are inaccurate statements in the context of an economic crisis hence option D which states 'None of the above' cannot be correct as option B correctly states that portfolio diversification could fail leading to losses in such scenarios.

Things to Remember

- **Systemic Risk:** Systemic risk is the risk of collapse of an entire financial system or entire market, as opposed to risk associated with any one individual entity, group or component of a system.
 - **Correlation:** Correlation is a statistical measure that describes the degree to which two securities move in relation to each other. Positive correlation means the securities move in the same direction, while negative correlation means they move in opposite directions.
 - **Mean-Variance Utility:** Mean-variance analysis is a technique used to determine the optimal allocation of assets in an investment portfolio. It considers the expected return and volatility of each asset to find the best combination that offers the highest return for a given level of risk.
 - **Historical Data:** Using historical data to estimate future correlations and returns is a common practice in financial analysis, but it comes with limitations as past performance may not always be indicative of future results.
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Q.2381 Country A is a developed economy with a vibrant financial sector. It witnessed a major financial crisis around 15 years ago. The economy recovered quite well and has consistently registered positive economic growth over the years. A recent survey conducted by an equity research firm indicated that the country's fund managers have built a well-diversified portfolio inclusive of all the sectors of the economy. The survey also highlighted the fact that the fund managers generally used historical data of the last 10 years and mean-variance utility to determine the correlations between the different assets. Asked why they preferred data from the last 10 years, most managers were of the view that recent data was the best estimator of the correlation between the different assets.

The best model that can be applied to determine the behavior of assets during a crisis is:

- A. The Capital Asset Pricing Model
- B. Fama-French model
- C. Gaussian copula model
- D. Pricing kernel model

The correct answer is **D**.

A pricing kernel model is the most appropriate for predicting asset behavior during a crisis. This is because during a crisis, many risky assets tend to perform poorly, which aligns with a multi-factor model specifically, a pricing kernel model where many assets are exposed to the same set of risk factors. These factors can include liquidity, slow economic growth rate, and uncertainty regarding government or monetary policy. In the context of Country A, the fund managers' use of historical data and mean-variance utility to determine correlations between different assets would be best served by a pricing kernel model. This model would allow them to account for multiple risk factors that could impact the performance of the assets in their portfolio during a crisis.

Choice A is incorrect. The Capital Asset Pricing Model (CAPM) is not the most suitable model in this scenario. CAPM assumes a linear relationship between the expected return and systematic risk of an asset, which may not hold true during a financial crisis. In times of crisis, correlations between assets can change dramatically and unpredictably, making it difficult for CAPM to accurately predict asset behavior.

Choice B is incorrect. This model is designed to better explain stock returns by incorporating multiple factors

Choice C is incorrect. This is a model that uses the correlation structure derived from the multivariate normal distribution to model the dependencies among different financial instruments or entities

Things to Remember

- **Financial Crisis:** A financial crisis is a situation where the value of financial institutions or assets drops rapidly. It can lead to a severe economic downturn and impact various sectors of the economy.
- **Mean-Variance Utility:** Mean-variance analysis is a method used to determine the optimal allocation of assets in a portfolio based on the trade-off between expected return and risk (variance). Fund managers often use this approach to construct diversified portfolios.
- **Historical Data:** Using historical data to estimate correlations between assets is a common practice in portfolio management. However, the choice of time period for historical data can impact the accuracy of the estimates.
- **Pricing Kernel Model:** A pricing kernel model is a multi-factor model that considers various risk factors affecting asset prices. It is particularly useful during crises when many assets are exposed to similar risk factors.

- **Systematic Risk:** Systematic risk refers to the risk that is inherent in the entire market or economy and cannot be diversified away. It is important to consider systematic risk when analyzing asset behavior during crises.
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Q.2382 Country A is a developed economy with a vibrant financial sector. It witnessed a major financial crisis around 15 years ago. The economy recovered quite well and has consistently registered positive economic growth over the years. A recent survey conducted by an equity research firm indicated that the country's fund managers have built a well-diversified portfolio inclusive of all the sectors of the economy. The survey also highlighted the fact that the fund managers generally used historical data of the last 10 years and mean-variance utility to determine the correlations between the different assets. Asked why they preferred data from the last 10 years, most managers were of the view that recent data was the best estimator of the correlation between the different assets.

The major shortcoming of using mean-variance utility to implement diversification is that:

- A. It assumes the correlation between assets classes is constant.
- B. It assumes the correlation between asset classes is not constant.
- C. It assumes the correlation between asset classes increases in a linear manner.
- D. It assumes the correlation between asset classes increases exponentially.

The correct answer is **A**.

The mean-variance utility model assumes that the correlation between asset classes remains constant. This assumption is a significant limitation of the model. In reality, the correlation between asset classes can change over time due to various factors such as changes in market conditions, economic factors, and investor behavior. For instance, during periods of market stress, correlations between asset classes can increase significantly as investors sell off risky assets and move towards safer investments. This phenomenon, known as 'flight to quality', can lead to a breakdown in the diversification benefits of a portfolio. Therefore, the assumption of constant correlation can lead to inaccurate risk assessments and sub-optimal portfolio allocations. It is important for fund managers to be aware of this limitation when using the mean-variance utility model for portfolio diversification.

Choice B is incorrect. The mean-variance utility does not assume that the correlation between asset classes is not constant. In fact, it assumes the opposite - that correlations are constant over time, which can be a significant drawback as it may not accurately reflect changes in market conditions.

Choice C is incorrect. The mean-variance utility does not assume that the correlation between asset classes increases in a linear manner. This assumption would imply a predictable and steady increase in correlation, which is often not the case in financial markets where correlations can fluctuate based on various factors.

Choice D is incorrect. The mean-variance utility does not assume that the correlation between asset classes increases exponentially. This would suggest an accelerating rate of increase in correlation over time, which again may not accurately reflect real-world market dynamics.

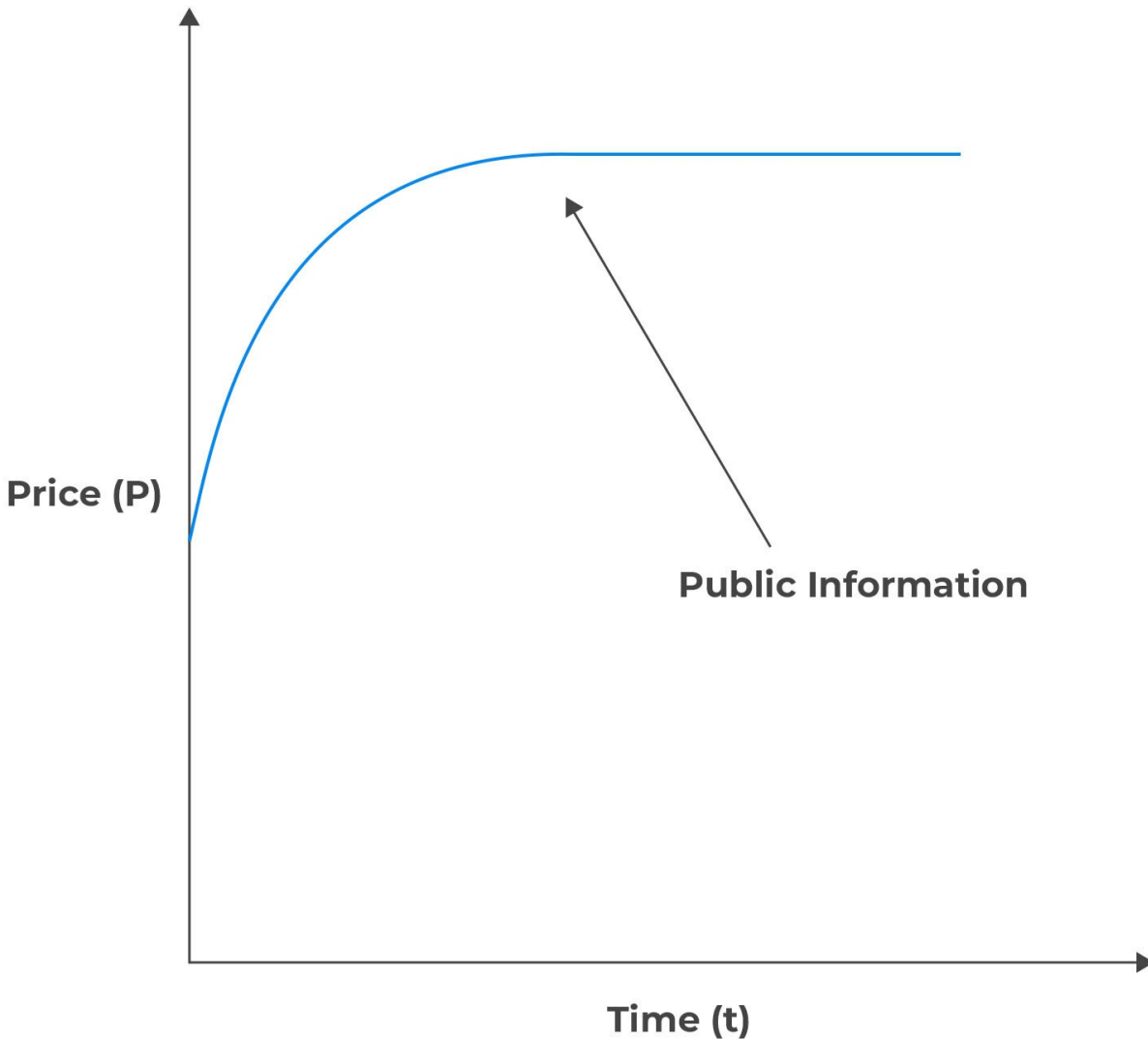
Things to Remember

- Mean-variance utility model is a key concept in modern portfolio theory (MPT) which helps investors to make decisions about asset allocation based on risk and return trade-offs.
- Correlation between asset classes is a crucial factor in portfolio diversification as it helps in reducing overall portfolio risk by including assets that are not perfectly correlated.
- Market conditions, economic factors, and investor behavior can all impact the correlation between asset classes, making it important for fund managers to regularly review and update their assumptions.
- Using historical data for estimating correlations can be useful, but it is important to consider the limitations of assuming constant correlations over time.
- Alternative methods like dynamic conditional correlation models or copula functions can be used to capture time-varying correlations more accurately in portfolio optimization.

Q.2384 Country A has a well-developed financial market. Information is freely available and is accessible by each market participant. A company involved in oil exploration recently discovered huge oil reserves. The stock price of the company after the disclosure of the information is presented below.



Change in Stock Price with Availability of Public Information



An investment management firm situated in country A promises a return in excess of the market return. Select the most appropriate statement:

- A. Returns will not exceed the market return.
- B. Active management may indeed result in excess return.
- C. Passive management may indeed result in excess return.
- D. None of the above.

The correct answer is **B**.

The stock market does not react to the new information regarding the discovery of huge oil reserves by the company. It takes time for the market to digest the new information. In such markets, active investment management will take advantage of this and thus beat the market hence resulting in excess returns.

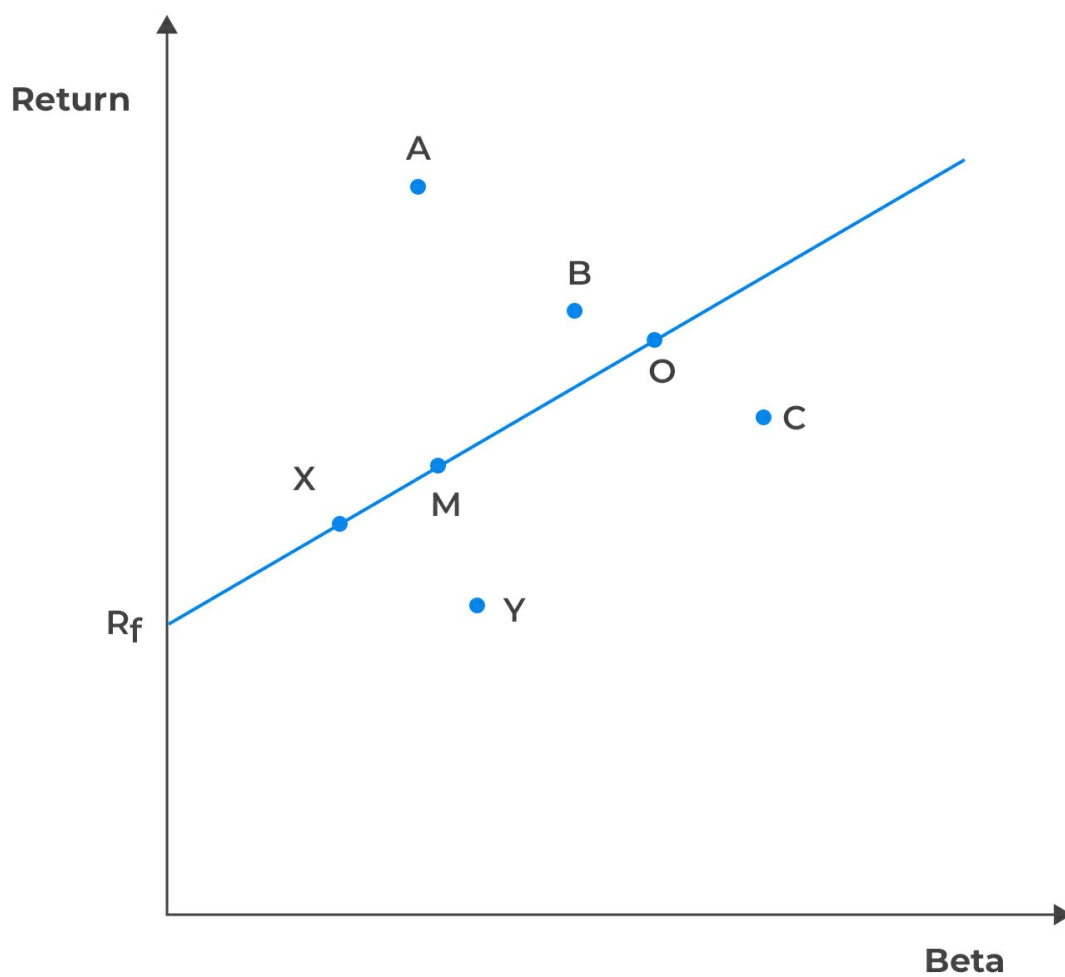
Things to Remember

- Efficient Market Hypothesis (EMH) states that all available information is already reflected in stock prices, making it impossible to consistently outperform the market through active management.
 - Active management involves making investment decisions in an attempt to outperform a specific benchmark or the overall market return.
 - Passive management, on the other hand, involves investing in a way that closely mirrors the performance of a specific market index, such as the S&P 500.
 - Market efficiency can be categorized into three forms: weak form efficiency, semi-strong form efficiency, and strong form efficiency, each representing different levels of information already reflected in stock prices.
 - Investment management firms often use various strategies and techniques to try to generate excess returns for their clients, such as fundamental analysis, technical analysis, and quantitative analysis.
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Q.2385 Consider the graph of expected returns presented below. The line represents the security market line derived from the Capital Asset Pricing Model (CAPM). A, B, C, M, O, X, and Y all represent different stocks.



The Security Market Line



According to CAPM, stock A is:

- A. Undervalued
- B. Overvalued
- C. Dependent on the financial market
- D. Appropriately valued

The correct answer is **A**.

Stock A lies above the SML. The return on the stock is more than its expected return as predicted by the CAPM. As such, the stock is undervalued and, therefore, a good buy.

Generally, assets that lie above SML are deemed undervalued, as they provide a higher expected return compared to the anticipated return for their risk level. This indicates that the asset might be underestimated by investors, creating a chance to purchase at a low price and sell at a high price. On the other hand, assets positioned below the SML are deemed as overvalued because they deliver a lower expected return than what is anticipated for their risk level. This suggests that the asset might be overestimated by investors, and it could be wise to steer clear of it or wait for its price to decrease.

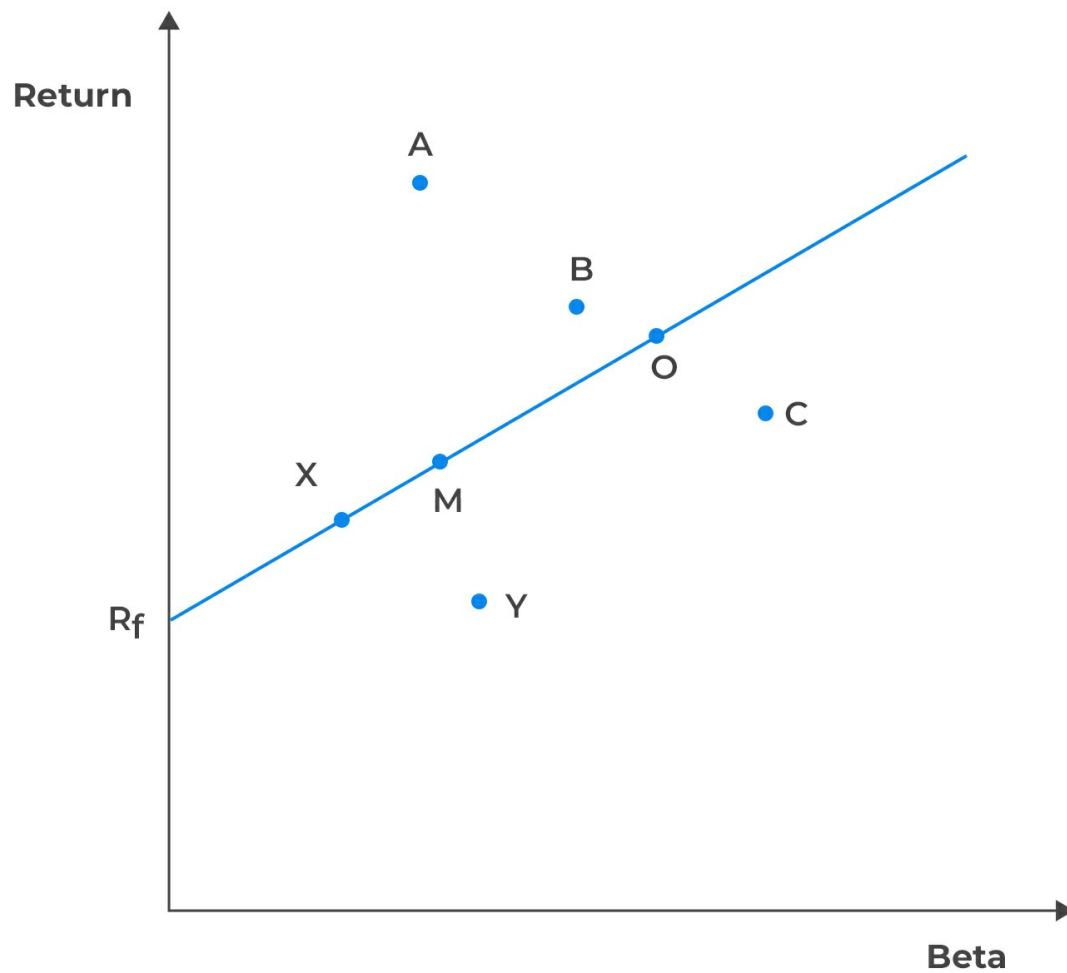
Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between systematic risk and expected return for assets, particularly stocks.
 - The Security Market Line (SML) is a graphical representation of the CAPM, showing the relationship between expected return and beta (systematic risk).
 - Stocks that lie above the SML are considered undervalued, offering a higher expected return for their risk level.
 - Stocks that lie below the SML are considered overvalued, providing a lower expected return for their risk level.
 - Investors can use the SML to evaluate whether a stock is undervalued, overvalued, or appropriately valued based on its position relative to the line.
-

Q.2386 Consider the graph presented below. The line represents the security market line derived from the Capital Asset Pricing Model (CAPM). A, B, C, M, O, X, and Y all represent different stocks.



The Security Market Line



According to CAPM, stock C is:

- A. Undervalued
- B. Overvalued
- C. Dependent on the financial market
- D. Appropriately valued

The correct answer is **B**.

Stock C lies below the SML, thus its return is less than that predicted by the CAPM. As such, the stock is overvalued and is therefore a good sell.

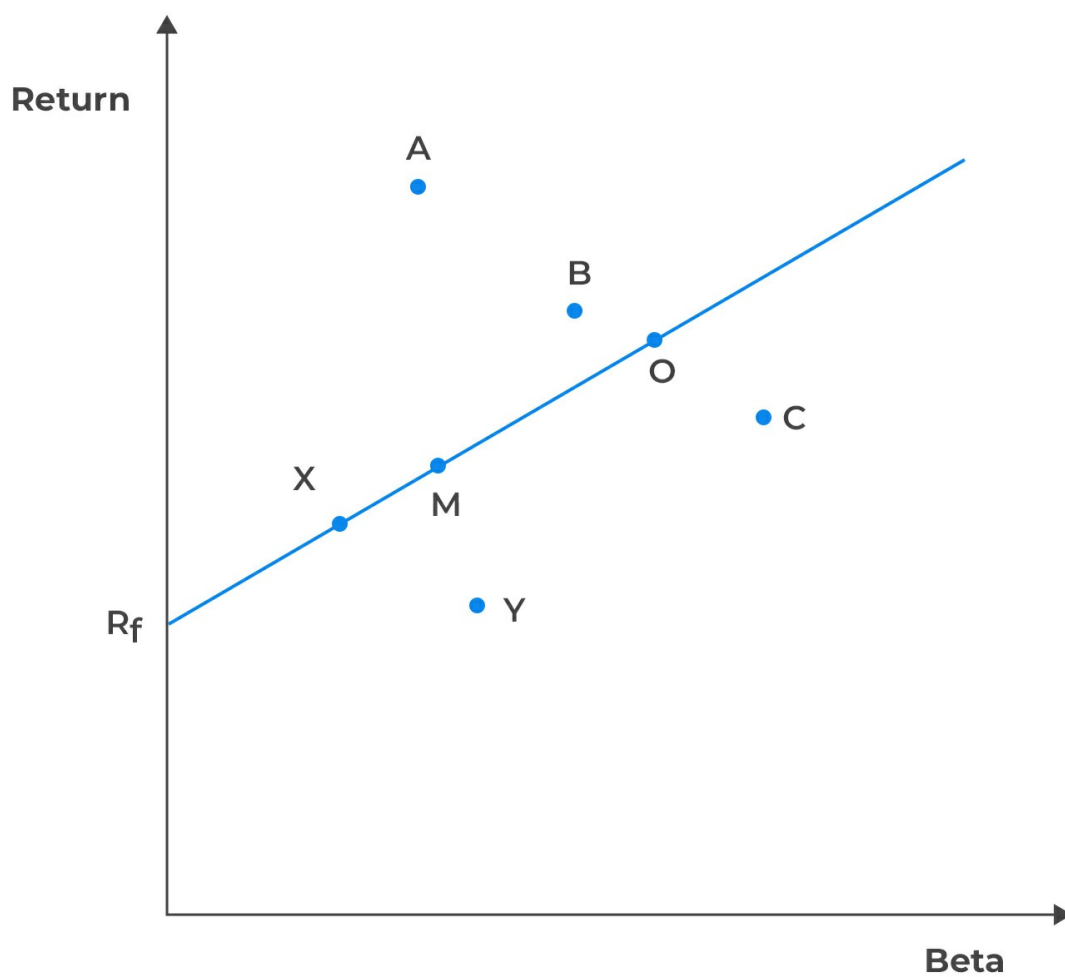
Things to Remember

- The Capital Asset Pricing Model (CAPM) is a widely-used method to determine the expected return on an investment based on its risk.
 - The Security Market Line (SML) is a graphical representation of the CAPM, showing the relationship between expected return and beta (systematic risk).
 - Stocks above the SML are considered undervalued, while stocks below the SML are considered overvalued according to CAPM.
 - Investors can use the SML to evaluate whether a stock is providing the expected return for its level of risk.
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Q.2387 Consider the graph presented below. The line represents the security market line derived from the Capital Asset Pricing Model (CAPM).



The Security Market Line



Stock X is subjected to a capital gain tax of 25%. What's the most likely effect of taxes on stock X?

- A. The observed return of the stock will be as predicted by CAPM.
- B. The observed return of the stock will be more than that predicted by CAPM.
- C. The observed return of the stock will be less than that predicted by CAPM.
- D. None of the above.

The correct answer is **C**.

CAPM is based on the assumption that assets are not subject to taxation. However, when tax is

levied on the assets the expected return is less than that predicted by CAPM. Hence, the observed return of stock X will be less than the CAPM predicted return.

Further Breakdown:

To hold Stock X, which is affected by taxes, individuals will require a higher return. For example, investors will require a 13.33% return on Stock X so that, after taxes, they will be getting the same 10% return as on a similar-beta stock.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between risk and expected return
 - CAPM is based on the assumption that assets are not subject to taxation
 - When taxes are levied on assets, the expected return is less than that predicted by CAPM
 - Investors may require a higher return on taxed assets to compensate for the tax impact
-

Q.2388 Which statement is not true regarding the Capital Market Line (CML)?

- A. The risk measure for the CML is standard deviation.
- B. The CML is the line from the risk-free rate through the market portfolio.
- C. The CML is the best attainable capital allocation line.
- D. The CML is also called the security market line.

The correct answer is **D**.

The statement that the Capital Market Line (CML) is also called the Security Market Line (SML) is incorrect. While both the CML and SML are fundamental concepts in modern portfolio theory and are used to illustrate the risk-return tradeoff in financial markets, they are not the same. The CML represents the efficient frontier of a market in the presence of a risk-free asset, using standard deviation of portfolio returns as a measure of risk. On the other hand, the SML is a line that shows the systematic, or market, risk versus return of the whole market at a certain time and shows all the risk-averse market's securities. The risk measure for the SML is beta, which measures the sensitivity of a security's returns to the overall market returns. Therefore, the CML and SML are distinct concepts with different risk measures and interpretations.

Choice A is incorrect. The risk measure for the CML is indeed standard deviation. This is because the CML represents portfolios that maximize expected return for a given level of risk, and this risk is measured as standard deviation.

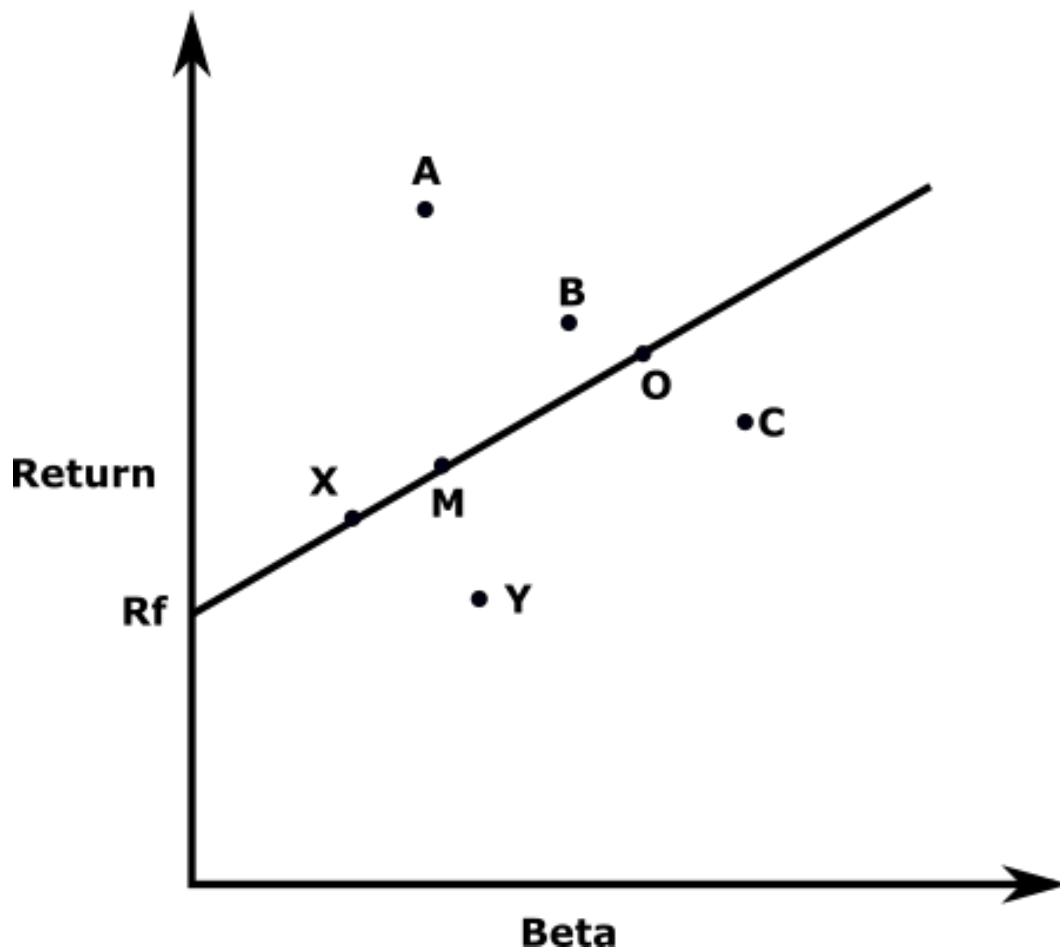
Choice B is incorrect. The statement that the CML is the line from the risk-free rate through the market portfolio accurately describes one of its key characteristics. It starts at the point representing a risk-free asset and extends up and to the right, passing through a point representing a risky market portfolio.

Choice C is incorrect. The statement that "The CML is the best attainable capital allocation line" correctly describes one of its properties. It shows all possible combinations of risky and risk-free assets, with each point on it offering an optimal combination given an investor's level of risk tolerance.

Things to Remember

- **Efficient Frontier:** The efficient frontier is a set of optimal portfolios that offer the highest expected return for a given level of risk or the lowest risk for a given level of expected return.
- **Risk-Free Rate:** The risk-free rate is the theoretical rate of return of an investment with zero risk of financial loss, typically represented by government bonds.
- **Market Portfolio:** The market portfolio is a theoretical portfolio that contains all risky assets in the market, weighted by their market values.
- **Beta:** Beta is a measure of the volatility, or systematic risk, of a security or a portfolio in comparison to the market as a whole.
- **Sharpe Ratio:** The Sharpe ratio is a measure of risk-adjusted return, calculated by dividing the excess return of an investment over the risk-free rate by the standard deviation of the investment's returns.

Q.2389 Consider the graph presented below. The line represents the security market line derived from the Capital Asset Pricing Model (CAPM).



Assume that the economy faces a crisis which leads to a recession. In such a scenario, the SML will most likely:

- A. decrease.
- B. increase.
- C. remain the same.
- D. It is impossible to determine the likely change in the SML with the given information.

The correct answer is **B**.

In the CAPM, the Security Market Line (SML) represents the relationship between expected return and systematic risk for individual securities, where the slope of the line represents the market risk premium. The CAPM assumes that investors can borrow and lend at the risk-free rate, which may vary over time. Similarly, the market risk premium varies over times. Therefore, any change in the position or slope of the SML on the graph can represent changes in the risk-free rate, the market risk premium, or expected returns for a given level of systematic risk.

In a recession, investors generally become more risk-averse and demand higher returns for any given level of risk. This means that they would require a higher expected return on a stock to compensate them for the additional risk they are taking on. As a result, the SML is likely to increase.

A is incorrect. A decrease in the SML would mean that expected returns are decreasing for a given level of risk, which is the opposite of what would happen in a recession.

C is incorrect. A recession would generally cause investors to demand higher returns for any given level of risk, which would cause a change in the position of the SML on the graph, rather than leaving it in the same position.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between risk and expected return
 - Security Market Line (SML) in CAPM represents the relationship between expected return and systematic risk
 - The slope of the SML represents the market risk premium
 - In a recession, investors generally become more risk-averse and demand higher returns for any given level of risk
 - Changes in the position of the SML on the graph represent changes in expected returns for a given level of systematic risk
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Q.3017 Which of the following reasons best explains why it is advisable for asset owners to hold the market portfolio as compared to individual stocks?

- A. The individual stocks are not only exposed to the market factor which carries the risk premium, but they also have idiosyncratic risk.
- B. Individual stocks represent a systemic risk, and it is not passive since all risky assets have risk premiums that are only determined by their exposure to the market portfolio.
- C. The market portfolio is not held by every investor, which has a strong implication that is not outright rejected in the data.
- D. It can not be determined

The correct answer is **A**.

This statement aligns with the principles of Modern Portfolio Theory (MPT), which suggests diversification to reduce unsystematic (idiosyncratic) risk. Idiosyncratic risks are unique to a particular company or industry and can be diversified away by holding a broad portfolio of assets, such as the market portfolio.

The market factor, or systematic risk, affects all stocks and cannot be diversified away. However, it is generally compensated by a risk premium according to the Capital Asset Pricing Model (CAPM). In contrast, idiosyncratic risks, which individual stocks carry, do not typically offer additional expected returns despite increasing the portfolio's overall risk. Holding a market portfolio helps in achieving broad diversification, thereby minimizing the impact of idiosyncratic risks while still capturing the market's overall returns.

B is incorrect. Individual stocks do not represent systemic risks; they represent idiosyncratic risk and a share of systematic risk.

C is incorrect. This option does not address why holding a market portfolio might be preferable to holding individual stocks. Instead, it seems to argue about the prevalence of market portfolio adoption among investors, which doesn't provide a direct benefit or rationale.

Things to Remember

- **Systematic Risk:** The risk that is inherent to the entire market or an entire market segment. It cannot be diversified away and is compensated by the market risk premium.
 - **Idiosyncratic Risk:** The risk that is specific to a particular asset or industry. It can be diversified away by holding a diversified portfolio.
 - **Modern Portfolio Theory (MPT):** A theory that suggests investors can construct an efficient frontier of optimal portfolios that offer the maximum expected return for a given level of risk.
 - **Capital Asset Pricing Model (CAPM):** A model that describes the relationship between risk and expected return and is used in the pricing of risky securities.
 - **Diversification:** Spreading investments across different assets to reduce risk.
-

Q.3018 Which of the following best describes assets with the high positive beta?

- A. High beta assets have lower volatility, or systemic risk, in comparison to the market and, therefore, tend to move slowly in response to market changes.
- B. High beta assets are assets that tend to go up when the market goes up and goes

down when the market goes down.

C. High beta assets are assets that have a tendency of going down when the market goes up and moves up as the market goes down.

D. High beta assets are assets whose prices are less volatile than the market portfolio.

The correct answer is **B**.

High beta assets are indeed assets that tend to go up when the market goes up and go down when the market goes down. The beta coefficient measures the volatility of an asset in relation to the overall market. A high positive beta, typically greater than 1, indicates that the asset is more volatile than the market. This means that if the market increases or decreases, the price of the high beta asset will increase or decrease to a greater extent. This characteristic makes high beta assets potentially more profitable, but also riskier, as they are more susceptible to market fluctuations.

Choice A is incorrect. High beta assets do not have lower volatility or systemic risk compared to the market. In fact, they are more volatile and tend to move more rapidly in response to market changes due to their high sensitivity.

Choice C is incorrect. This statement describes assets with a negative beta, not a high positive beta. Assets with a negative beta tend to move in the opposite direction of the market.

Choice D is incorrect. High beta assets are not less volatile than the market portfolio; rather, they are more volatile due to their higher sensitivity to market movements.

Things to Remember

- Beta coefficient measures the volatility of an asset in relation to the overall market.
- A high positive beta, typically greater than 1, indicates that the asset is more volatile than the market.
- High beta assets are more susceptible to market fluctuations due to their high sensitivity.
- Assets with a negative beta tend to move in the opposite direction of the market.
- High beta assets are not less volatile than the market portfolio; rather, they are more volatile due to their higher sensitivity to market movements.

Q.4586 Which of the following *least likely* represents an assumption of the CAPM?

- A. Investors have a long period investment horizon
- B. Investors have identical expectations
- C. Investors trade without transaction or taxation costs
- D. Investors can access all information at the same time

The correct answer is **A**.

CAPM assumes that investors have a short period investment horizon.

B is incorrect. CAPM assumes that investors have homogenous expectations. It assumes that investors hold the same mean-variance efficient (MVE) portfolio and is the market portfolio.

C is incorrect. CAPM assumes that investors trade without transaction costs and that returns are not affected by taxes.

D is incorrect. The CAPM assumes investors can access all information at the same time. According to the model, different information means no common efficient frontier line can be drawn.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between risk and expected return and is used in the pricing of risky securities.
- Assumptions of CAPM include rational investors, risk-averse behavior, a single-period investment horizon, perfect capital markets, homogeneous expectations, no taxes, and no transaction costs.
- The CAPM equation is: $E(R_i) = R_f + \beta_i(E(R_m) - R_f)$, where $E(R_i)$ is the expected return on asset i , R_f is the risk-free rate, β_i is the asset's beta, and $E(R_m)$ is the expected return on the market portfolio.
- Investors in CAPM are assumed to hold a diversified portfolio that includes all risky assets in the market in proportion to their market values.
- The CAPM helps in determining the required rate of return for an asset given its risk compared to the overall market.

Q.4589 Which of the following lessons from multifactor models are similar to the lessons from the CAPM?

- A. The average investor holds the market portfolio
- B. Exposure to factor risk must be rewarded
- C. Diversification of idiosyncratic risks works
- D. All of the above

The correct answer is **D**.

All the above lessons from multifactor models are similar to the lessons from the CAPM.

The average investor holds the market portfolio. Under both CAPM and multifactor models, the average investor holds the market.

Exposure to factor risk must be rewarded. Under CAPM assumptions, the market factor is priced in equilibrium. However, under multifactor models, risk premiums exist for each factor, assuming no arbitrage or equilibrium.

Diversification of idiosyncratic risks works. In CAPM, the market diversifies away the idiosyncratic risk. In multifactor models, the tradeable version of a factor diversifies the idiosyncratic risk.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between systematic risk and expected return for assets, particularly stocks.
- Multifactor models are extensions of the CAPM that incorporate multiple sources of risk or factors that can affect asset returns.
- Factor models help in understanding the drivers of asset returns and in pricing assets based on their exposure to these factors.
- Factors in multifactor models can include macroeconomic variables, industry-specific factors, or other sources of risk that can impact asset returns.
- Investors can use multifactor models to better understand the risk-return tradeoff of their portfolios and to make more informed investment decisions.

Q.4590 Multifactor models use the stochastic discount factor (SDF) to define bad times over multiple factors. Which of the following statements is *most likely* true about SDF?

- A. The SDF represents an index of bad times
- B. The CAPM is a special case of this SDF model
- C. The SDF is also called a pricing kernel
- D. All of the above

The correct answer is **D**.

SDF is denoted as m in the multifactor model. The SDF can be described as an index of bad times, which are indexed by many distinct factors and different states of nature. The single variable m is used to capture all the bad times, providing an extremely powerful notation to capture bad times with multiple variables. The CAPM is a special case of this model, where m is linear in the market return. That is,

$$m = a + b r_m$$

Where a and b are constants.

Q.4591 Which of the following statements is *most likely* true about factor risk exposure and rewards in CAPM?

- A. A high risk premium implies a low exposure to a factor
- B. A high risk premium implies a low expected return of an asset
- C. A high risk premium implies a higher exposure to the factor and a high expected return of the asset
- D. A high risk premium implies a high exposure to the factor and a low expected return of the asset

The correct answer is **C**.

According to the Capital Asset Pricing Model (CAPM), an asset's risk is measured in relation to its factor exposure. A factor with a high risk premium implies that there is a high exposure to that factor. This is because the risk premium is a reward for bearing the risk associated with that factor. Therefore, if the risk premium is high, it means that the risk associated with the factor is high, and hence the exposure to that factor is also high. Furthermore, a high risk premium also implies a high expected return of the asset. This is because the expected return of an asset is a

compensation for the risk borne by the investor. Therefore, if the risk premium is high, it means that the expected return of the asset is also high. This is in line with the risk-return tradeoff principle, which states that the potential return rises with an increase in risk. Therefore, choice C accurately describes the relationship between a high risk premium, factor exposure, and the expected return of an asset in the context of CAPM.

Choice A is incorrect. A high risk premium does not necessarily imply a low exposure to a factor. In fact, the opposite is often true: higher risk premiums are typically associated with higher factor exposures, as investors require additional compensation for taking on more risk.

Choice B is incorrect. A high risk premium does not imply a low expected return of an asset. Again, the opposite tends to be true: assets with higher risk premiums generally have higher expected returns because they carry greater risks and investors demand more compensation for bearing these risks.

Choice D is incorrect. While it's correct that a high risk premium implies a high exposure to the factor, it's not accurate that this leads to a low expected return of the asset. As mentioned earlier, assets with greater factor exposures and thus higher risks tend to have higher expected returns as investors seek additional rewards for bearing these increased risks.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between risk and expected return in the context of investing.
- Risk premium is the excess return above the risk-free rate that an investor expects to receive for bearing the risk of a particular investment.
- Factor exposure refers to the sensitivity of an asset's returns to changes in a particular factor, such as market returns or interest rates.
- The risk-return tradeoff principle states that higher expected returns are associated with higher levels of risk.
- Investors demand higher expected returns for assets with higher risk premiums to compensate for the increased risk.

Q.4592 In both CAPM and multifactor models, the risk is measured by factor betas. Which of the following is *most likely* true about factor betas and risk premiums?

- A. Low betas imply high risk premiums
- B. High betas imply low risk premiums

C. High betas imply high risk premiums

D. None of the above

The correct answer is **C**.

High betas imply high risk premiums. In the context of both the Capital Asset Pricing Model (CAPM) and multifactor models, beta is a measure of an asset's sensitivity to changes in the market. A high beta indicates that the asset's returns are highly sensitive to market changes. This means that when the market performs poorly, the asset is likely to perform poorly as well. This high sensitivity to market changes, or high beta, implies a high level of risk. Investors require compensation for bearing this risk, which is provided in the form of a risk premium. Therefore, assets with high betas are expected to offer high risk premiums to compensate investors for the increased risk they are taking on by investing in these assets.

Choice A is incorrect. Low betas do not imply high risk premiums. In the context of CAPM and multifactor models, a low beta indicates that the asset's returns are less sensitive to changes in the market. This implies a lower level of systematic risk and therefore, typically corresponds to a lower risk premium.

Choice B is incorrect. High betas do not imply low risk premiums. On the contrary, high betas indicate that an asset's returns are highly sensitive to changes in the market, implying a higher level of systematic risk. As such, investors would require a higher risk premium as compensation for bearing this increased level of risk.

Choice D is incorrect. The statement "None of the above" is inaccurate because Choice C correctly states that high betas imply high risk premiums which aligns with both CAPM and multifactor models' interpretation of factor betas and their relationship with risk premiums.

Things to Remember

- Factor betas measure an asset's sensitivity to changes in a particular factor or factors, such as the overall market or specific macroeconomic variables.
- In the CAPM model, the risk premium for an asset is determined by its beta in relation to the market portfolio.
- High factor betas indicate higher sensitivity to the factor and therefore higher risk premiums are required by investors.
- Low factor betas indicate lower sensitivity to the factor and therefore lower risk premiums are required by investors.
- Risk premiums are the additional return that investors demand for taking on the risk of investing in a particular asset.

Q.4594 Which of the following is the *most likely* reason as to why low beta assets are attractive to investors?

- A. Low beta assets outperform the market when the market is crashing
- B. Low beta assets attract higher risk premiums
- C. Low beta portfolios deliver higher volatility than expected and have shown significantly better results in practice
- D. None of the above

The correct answer is **A**.

Low beta assets are considered attractive to investors primarily because they tend to outperform the market during periods of market downturns or crashes. The beta of an asset is a measure of its sensitivity to market movements. A low beta indicates that the asset is less volatile than the market, meaning it tends to fluctuate less in price. This characteristic makes low beta assets a safer investment during periods of market instability. When the market is crashing, investors often seek refuge in low beta assets as they are less likely to experience severe price drops. Furthermore, low beta assets have a tendency to outperform high beta assets during these periods, providing investors with a degree of protection against potential losses.

Choice B is incorrect. Low beta assets do not attract higher risk premiums. In fact, the opposite is true. High beta assets are considered riskier and therefore attract higher risk premiums as compensation for the increased level of risk.

Choice C is incorrect. Low beta portfolios do not deliver higher volatility than expected; they are typically associated with lower volatility. The statement that low-beta portfolios have shown significantly better results in practice is also misleading, as their performance depends on various factors and cannot be generalized.

Choice D is incorrect. As explained above, low beta assets can provide a degree of protection against market downturns, making them an attractive investment option for certain investors.

Things to Remember

- Beta measures the sensitivity of an asset's returns to market returns.
- Low beta assets are less volatile and provide a degree of protection during market downturns.
- Investors often use low beta assets as a hedge against market risk.
- High beta assets are riskier and tend to attract higher risk premiums.
- Low beta assets may not always outperform high beta assets in all market conditions.

Q.4595 Consider Security K, which has a standard deviation of investment returns of 4%. If:

- the standard deviation of the market return is 5%;
- the correlation between K's return and that of the market is 0.75;
- the risk-free rate is 5%; and
- the expected return on the market is 10%.

Calculate the beta of Security K.

- A. 0.3
- B. 0.375
- C. 0.6
- D. 0.75

The correct answer is **C**.

$$\begin{aligned}\beta_K &= \frac{\text{Cov}(R_K, R_M)}{V_M} \\ &= \frac{\rho_{KM} \sigma_K \sigma_M}{\sigma_M^2} \\ &= \frac{0.75 \times 4\% \times 5\%}{5\%^2} \\ &= 0.600\end{aligned}$$

Reading 147: Factors

Q.2390 A recent survey indicated that the market is expecting a rate cut equivalent to 100 basis points.

Which of the following statements is most accurate?

- A. A rate cut of 105 basis points will create a shock.
- B. A rate cut of 95 basis points will create a shock.
- C. A rate cut of 100 basis points will create a shock.
- D. A rate cut of 150 basis points will create a shock.

The correct answer is **D**.

A rate cut of 150 basis points will create a shock. The market is expecting a rate cut equivalent to 100 basis points. If the Central Bank decides to cut the rate by 150 basis points, it will be significantly different from what the market is expecting. This difference between the expected and actual rate cut will act as a shock to the market. The term 'shock' in this context refers to an unexpected event that disrupts the market. It is important to note that the level of a factor often does not matter as much as a shock to the factor. In this case, the shock arises from the difference between the market's expectation and the actual rate cut. This shock can have various effects on the market, including increased volatility and changes in asset prices.

Choice A is incorrect. A rate cut of 105 basis points will not create a shock in the market as it is very close to the anticipated reduction of 100 basis points. The market has already factored in this change and hence, it would not lead to any significant changes or shocks.

Choice B is incorrect. Similarly, a rate cut of 95 basis points will also not create a shock as it falls within the range of expected reduction. Although it's slightly less than what was anticipated, the difference isn't substantial enough to cause any major disruptions in the market.

Choice C is incorrect. A rate cut of exactly 100 basis points won't create a shock either because this is precisely what the market has been anticipating. Therefore, such an eventuality has already been priced into current valuations and won't result in any unexpected outcomes.

Things to Remember

- **Market Expectations:** Market expectations play a crucial role in determining the impact of policy decisions on the market. Any deviation from these expectations can lead to shocks.
- **Monetary Policy:** Rate cuts are a common tool used by central banks to stimulate economic activity by reducing borrowing costs for businesses and consumers.

- Volatility: Shocks in the market can lead to increased volatility, which can impact asset prices and investor behavior.
 - Forward Guidance: Central banks often provide forward guidance to communicate their future policy intentions, which can influence market expectations.
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Q.2391 You are the given the following information:

Country	Economic growth	Inflation	Volatility
A	High	Low	Low
B	Low	High	High

Investors in country A must invest in:

- A. Government Bonds
- B. Equities
- C. Corporate Bonds
- D. Zero-Coupon bonds

The correct answer is **B**.

Equities, also known as stocks, represent ownership in a company. When an investor buys a company's stock, they are buying a piece of the company and, as such, are entitled to a portion of the company's profits. The performance of equities is closely tied to the performance of the overall economy. In a high-growth economy, companies are likely to see increased profits, which can lead to higher stock prices. Therefore, in a country with high economic growth, low inflation, and low volatility, equities are likely to provide a higher return than bonds. This is because the low inflation rate preserves the purchasing power of future profits, and the low volatility reduces the risk of large price swings. Therefore, in such an environment, equities are a more attractive investment option than bonds.

Choice A is incorrect. Government bonds are typically considered a safe investment during times of economic uncertainty or volatility. Given that Country A is experiencing high economic growth, low inflation, and low volatility, the returns from government bonds may not be as attractive compared to other investment options.

Choice C is incorrect. Corporate bonds can provide higher yields than government bonds but they also carry higher risk. In a stable and growing economy like Country A's, equities are likely to offer better returns due to the potential for capital appreciation.

Choice D is incorrect. Zero-coupon bonds do not pay interest until maturity and are typically purchased at a discount to their face value. They can be an attractive option in an environment with high inflation as they protect against price increases over time. However, given that Country A has low inflation, this would not be the most suitable investment option.

Q.2392 During a recession, which of the following categories of investments will most likely outperform all other classes of investments?

- A. Large-cap stocks

- B. High-yield bonds
- C. Small-cap stocks
- D. Government bonds

The correct answer is **D**.

During a recession, government bonds are likely to outperform other classes of investments. This is because during a recession, investors tend to move their investments to safer assets, and government bonds are considered one of the safest investments. This is due to the fact that they are backed by the full faith and credit of the government. Additionally, during a recession, central banks often lower interest rates to stimulate the economy. When interest rates fall, the price of existing bonds rises, which increases the return on bonds. According to the table “Means and Volatilities Conditional on Factor Realizations,” during a recession government bonds have a mean return of 12.3%. This is above large-cap stocks, which yield on average 5.6%, small-cap stocks, at 7.8%, and high-yield bonds, at 7.4%.

Choice A is incorrect. Large-cap stocks, while generally more stable than small-cap stocks, are still subject to the overall economic conditions. During a recession, consumer spending decreases which can lead to lower revenues and profits for these companies, negatively impacting their stock prices.

Choice B is incorrect. High-yield bonds are typically issued by companies with higher credit risk. In a recessionary environment where default risk increases due to deteriorating financial conditions of issuers, these bonds tend to underperform as investors seek safer investment options.

Choice C is incorrect. Small-cap stocks are usually more volatile and risky compared to large-cap stocks. During a recession, these companies may face greater challenges in terms of reduced demand for their products or services and limited access to capital markets which can lead to significant declines in their stock prices.

Things to Remember

- Government bonds are considered one of the safest investments due to being backed by the full faith and credit of the government.
- Central banks often lower interest rates during a recession to stimulate the economy, which can increase the return on bonds as their prices rise.
- Large-cap stocks are subject to overall economic conditions, and during a recession, consumer spending decreases, impacting their stock prices.
- High-yield bonds are typically issued by companies with higher credit risk, making them more vulnerable in a recessionary environment with increased default risk.

- Small-cap stocks are more volatile and risky compared to large-cap stocks, facing challenges like reduced demand and limited access to capital markets during a recession.
-

Q.2395 All the following are factors of the Fama-French Model, EXCEPT:

- A. Market factor.
- B. Difference in expected returns of high-beta stocks minus small-beta stocks.
- C. Difference in expected returns of a portfolio of high book-to-market stocks minus a portfolio of low book-to-market stocks.
- D. Difference in expected returns of small stocks minus big stocks.

The correct answer is **B**.

The Fama-French three-factor model does not include the difference in expected returns of high-beta stocks minus small-beta stocks as a factor. The model was developed by Eugene Fama and Kenneth French to better predict the returns of stocks and portfolios. It expands on the capital asset pricing model (CAPM) by adding two additional factors: size and value. The size factor is the difference in returns between small and large stocks, while the value factor is the difference in returns between high and low book-to-market stocks. Beta, a measure of a stock's volatility in relation to the market, is not a factor in the Fama-French model. While beta is used in the CAPM to measure market risk, Fama and French found that size and value factors provide a better explanation of stock returns.

Choice A is incorrect. The market factor, which is the excess return of the market portfolio over the risk-free rate, is indeed a part of the Fama-French three-factor model. This factor represents systematic or non-diversifiable risk.

Choice C is incorrect. The difference in expected returns between high book-to-market stocks and low book-to-market stocks, also known as value premium, is another factor in this model. High book-to-market stocks are considered undervalued or "value" stocks and tend to have higher returns than low book-to-market (growth) stocks.

Choice D is incorrect. The difference in expected returns between small-cap and large-cap companies, also known as size premium, forms the third factor of this model. Small companies tend to have higher risk and therefore higher expected returns compared to large companies.

Things to Remember

- The Fama-French three-factor model is an asset pricing model that adds two additional factors (size and value) to the traditional CAPM model.

- The size factor in the Fama-French model represents the difference in returns between small and large stocks.
 - The value factor in the Fama-French model represents the difference in returns between high and low book-to-market stocks.
 - Beta is not a factor in the Fama-French model, as it focuses on size and value factors to explain stock returns.
 - The market factor, representing the excess return of the market portfolio over the risk-free rate, is a key component of the Fama-French model.
-

Q.2396 All the following explains the reason behind the so-called value premium, EXCEPT:

- A. Value firms are less flexible and have more unproductive capital during bad times.
- B. Investors have psychological biases such as mental accounting and loss aversion.
- C. Investors underestimate the growth prospects of value stocks.
- D. Value stocks generate higher returns during bad economic times.

The correct answer is **D**.

The statement that 'Value stocks generate higher returns during bad economic times' is not a valid explanation for the value premium. While it is true that value stocks may perform better during economic downturns due to their inherent stability and lower valuation, this is not the primary reason behind the value premium. The value premium is primarily attributed to the higher risk associated with value stocks, which demands a higher return as compensation. This risk arises from factors such as the inflexibility of value firms and their possession of more unproductive capital during bad times, as well as the underestimation of their growth prospects by investors. Furthermore, investor psychology, including biases such as mental accounting and loss aversion, also plays a significant role in the existence of the value premium. Therefore, while value stocks may indeed generate higher returns during bad economic times, this is not the primary reason behind the value premium, making choice D the correct answer.

Choice A is incorrect. This statement accurately represents a reason behind the existence of the value premium. Value firms are indeed less flexible and have more unproductive capital during bad times, which can lead to higher risk and therefore higher expected returns as compensation for this risk.

Choice B is incorrect. This statement also accurately represents a reason behind the existence of the value premium. Behavioral finance theories suggest that investors' psychological biases

such as mental accounting and loss aversion can lead them to overvalue growth stocks and undervalue value stocks, thereby creating a value premium.

Choice C is incorrect. This statement correctly identifies another potential reason for the existence of the value premium. Investors may underestimate the growth prospects of value stocks, leading these stocks to be priced lower than their intrinsic values and thus generating higher returns when their true values are realized.

Things to Remember

- Value premium is the phenomenon where value stocks outperform growth stocks over the long term.
 - Value stocks are typically characterized by low price-to-book ratios, high dividend yields, and low price-to-earnings ratios.
 - Investor sentiment and market cycles can also impact the performance of value stocks.
 - Value investing is a strategy that involves buying undervalued stocks with the expectation that their prices will eventually reflect their true value.
 - The value premium is a key concept in the field of finance and is often studied in the context of asset pricing models.
-

Q.2397 A high book-to-market value ratio is indicative of a:

- A. small-cap stock
- B. large-cap stock
- C. value stock
- D. growth stock

The correct answer is **C**.

A value stock is characterized by a high book-to-market ratio. This is because the price of a value stock appears low relative to the company's financial performance, as measured by such fundamentals as the company's assets, revenue, dividends, earnings, and cash flows. The price is low in relation to their net worth, as measured by the accounting book value. Therefore, a high book-to-market ratio is indicative of a value stock.

Choice A is incorrect. Small-cap stocks are not necessarily associated with a high book-to-market ratio. The size of the company (small-cap or large-cap) does not directly influence this ratio, as it is more related to the company's financial performance and market perception.

Choice B is incorrect. Similar to small-cap stocks, large-cap stocks also do not have a direct correlation with the book-to-market ratio. This metric primarily reflects the financial performance and market valuation of a company, irrespective of its size.

Choice D is incorrect. Growth stocks typically have low book-to-market ratios because their market values tend to be high due to expected future earnings growth, which reduces this ratio.

Things to Remember

- Book-to-market ratio is calculated by dividing the book value of a company by its market value.
- Value stocks are often associated with characteristics such as low price-to-earnings ratios, high dividend yields, and low price-to-book ratios.
- Growth stocks, on the other hand, are expected to have high future earnings potential, leading to higher market valuations relative to book values.
- Market perception and investor sentiment play a significant role in determining a stock's classification as either a value or growth stock.
- Investors may use the book-to-market ratio as one of many metrics to assess the investment potential of a stock.

Q.2399 The Fama-French model with momentum strategy ($\beta_{i,WML}$) added as an investment factor is presented below:

$$E(r_i) = r_f + \beta_{i,MKT}E(r_m - r_f) + \beta_{i,SMB}E(SMB) + \beta_{i,HML}E(HML) + \beta_{i,WML}E(WML)$$

A fund manager employs this strategy to generate profits. He buys stocks with increasing prices while selling the stocks with decreasing prices. For stocks with increasing prices, $\beta_{i,WML}$, will be:

- A. Positive
- B. Negative
- C. Zero
- D. Equal to 1

The correct answer is **A**.

The $\beta_{i,WML}$, in this context, represents the sensitivity of the stock's returns to the momentum factor, which is the difference in returns between winning stocks (those with increasing prices) and losing stocks (those with decreasing prices). When a stock's price is increasing, it is considered a 'winning' stock. Therefore, the $\beta_{i,WML}$ for such a stock would be positive. This is because the stock's returns are moving in the same direction as the momentum factor. In other words, as the price of the winning stock increases, the return of the momentum factor (WML) also increases, indicating a positive relationship between the two. Hence, the $\beta_{i,WML}$ for stocks with increasing prices is positive.

Choice B is incorrect. A negative $\beta_{i,WML}$ would imply that the stock's price is moving in the opposite direction of the market trend, which contradicts our assumption that these stocks are on an upward price trajectory. Therefore, a negative $\beta_{i,WML}$ does not characterize stocks with increasing prices in a momentum strategy.

Choice C is incorrect. A zero $\beta_{i,WML}$ would suggest that there is no relationship between the stock's price movement and the market trend. This does not align with our scenario where we are specifically selecting stocks based on their positive correlation with market trends (upward price trajectory). Hence, a zero $\beta_{i,WML}$ cannot represent such stocks.

Choice D is incorrect. $\beta_{i,WML} = 1$ implies that the stock's returns move perfectly in sync with the market returns. However, this doesn't necessarily mean that these stocks are experiencing an upward trend in their prices as required by our momentum strategy. Therefore, having a beta equal to 1 does not accurately characterize these types of stocks.

Q.2403 A fund manager tracks the prices of his portfolio P. He regresses the market return against the stock return for a period of 5 years. There has been no financial crisis in the country for the last 10 years. The regression equation obtained is presented below:

$$E(r_i) = 5.23\% + 1.2(r_p)$$

Suppose the country now faces a major financial crisis. In such a scenario, the stock's beta will be:

- A. 1.2
- B. Higher than 1.2
- C. Lower than 1.2
- D. Equal to 1

The correct answer is **B**.

Beta is a measure of a stock's risk in relation to the market or, in other words, it represents the systematic risk of a stock. Systematic risk is the risk that affects all securities in the market, and it cannot be eliminated through diversification. It is influenced by factors such as inflation rates, exchange rates, political instability, war, and financial policies.

In the context of this question, the country is facing a major financial crisis. This event is a systematic risk as it affects all securities in the market. In such a scenario, the riskiness of the stock increases. Therefore, the beta of the stock, which measures this risk, would also increase. This is why the beta would be higher than 1.2 in the event of a financial crisis.

It's important to note that the beta of a stock is not constant. It changes to reflect the prevailing market conditions. Therefore, it's reasonable to expect that the beta would increase in a situation where the market risk has increased due to a financial crisis.

Choice A is incorrect. The beta of the stock is not expected to remain at 1.2 during a financial crisis. Beta measures the systematic risk of a stock, which includes market-wide risks that cannot be diversified away. During a financial crisis, these risks are likely to increase, leading to an increase in beta.

Choice C is incorrect. The beta of the stock is not expected to decrease below 1.2 during a financial crisis. As mentioned above, beta measures systematic risk and this risk tends to increase during periods of financial instability or crises due to increased uncertainty and volatility in the market.

Choice D is incorrect. The beta of the stock will not necessarily equal 1 during a financial crisis. A beta value of 1 implies that the security's price will move with the market, but given that we are in a period of increased uncertainty and volatility due to the crisis, it's more likely for the beta value to be higher than 1 as it would reflect greater sensitivity and responsiveness towards changes in market conditions.

Q.2405 Which of the following are major risk factors for emerging market equities?

- I. Production output
- II. Political climate
- III. Growth rate
- IV. Consumption

- A. I and II
- B. II and IV
- C. I, II and III
- D. I, II, III and IV

The correct answer is **D**.

All four factors - production output, political climate, growth rate, and consumption - are major risk factors for emerging market equities. Production output refers to the total amount of goods or services produced by a company or an economy. A decrease in production output can lead to a decrease in the value of a company's equity. The political climate can also significantly impact the performance of emerging market equities. Political instability or unfavorable policies can deter investment, leading to a decrease in equity values. The growth rate of an economy is another major risk factor. A slow or negative growth rate can indicate a struggling economy, which can negatively impact the performance of equities. Lastly, consumption refers to the total amount of goods and services consumed in an economy. Increased consumption can lead to increased sales for companies, which can positively impact their equity values. Therefore, all four factors are major risk factors for emerging market equities.

Choice A is incorrect. While production output and political climate are indeed significant risk factors for emerging market equities, this choice omits the growth rate and consumption which are also major risk factors. The growth rate of an economy can indicate the potential for future profits, while consumption levels can reflect the demand for goods and services in that economy.

Choice B is incorrect. This choice only includes political climate and consumption as major risk factors. It neglects to consider production output which directly impacts company profits, and economic growth rate which provides a broader view of the economic health and potential profitability in an emerging market.

Choice C is incorrect. Although it correctly identifies production output, political climate, and growth rate as significant risk factors for emerging market equities, it fails to include consumption. Consumption levels can significantly impact company revenues in an emerging market as they indicate demand for goods or services.

Things to Remember

- Emerging markets are countries that are in the process of rapid industrialization and experiencing significant economic growth. These markets often present unique

opportunities and risks for investors.

- Political climate in emerging markets can be volatile and unpredictable, leading to sudden changes in policies, regulations, and government stability that can impact investments.
 - Growth rate is a key indicator of the economic health and potential profitability of an emerging market. Higher growth rates can attract investors seeking higher returns, but lower growth rates may signal economic challenges.
 - Consumption levels in emerging markets can provide insights into the demand for goods and services, which directly affects company revenues and profitability.
 - Other major risk factors for emerging market equities include currency fluctuations, interest rate changes, regulatory environment, and market liquidity.
-

Q.2406 The demographic details of two countries are presented in the table below:

Country	Avg. resident age
A	25
B	48

Considering the average age of the investors in country A and B, select the most appropriate option.

- A. $\text{Risk Premium}_A = \text{Risk Premium}_B$
- B. $\text{Risk Premium}_A > \text{Risk Premium}_B$
- C. $\text{Risk Premium}_A < \text{Risk Premium}_B$
- D. $\text{Risk Premium}_A = \text{Risk Premium}_B = 1$

The correct answer is **C**.

The risk premium in country A is less than the risk premium in country B. This is because studies have shown that risk aversion tends to increase as people age. In other words, older individuals are generally more cautious when it comes to taking on risk, and therefore require a higher return to compensate for the additional risk they are taking on. This is often referred to as the 'risk premium'. Given that the average age in country B is significantly higher than in country A, it can be inferred that the residents of country B would require a higher risk premium than those in country A. This is because the older residents of country B would be more risk averse, and therefore require a higher return to compensate for any additional risk they take on. Therefore, the risk premium in country B would be higher than in country A.

Choice A is incorrect. The risk premium cannot be equal in both countries because the average age of residents, which influences risk aversion, differs significantly between the two countries. Older individuals tend to be more risk averse and hence demand a higher risk premium.

Choice B is incorrect. This choice suggests that the risk premium in country A is greater than in country B, which contradicts the general understanding that older individuals are more risk averse and therefore require a higher risk premium.

Choice D is incorrect. This choice incorrectly assumes that the risk premiums in both countries are equal to 1, without considering any factors such as age or level of risk aversion among residents.

Q.2408 A class of real business cycle models developed in macroeconomics seeks to explain the movements of macro variables such as productivity and asset prices across the business cycle. Which of the following investors would least likely be concerned with productivity risk?

- A. Pension funds

- B. Sovereign wealth funds
- C. Investors in short-term bonds
- D. Family offices

The correct answer is **C**.

Investors in short-term bonds are least likely to be concerned with productivity risk. The reason for this is that productivity risk operates at business cycle frequencies and is therefore less relevant to investors with short investment horizons. In other words, the impact of productivity on the value of investments is more pronounced over longer periods. Therefore, for an investor in short-term bonds, changes in productivity are unlikely to have a significant impact on the value of their investment before it matures. This is why they are less concerned with productivity risk compared to other types of investors.

Choice A is incorrect. Pension funds are long-term investors who need to consider productivity risk as it can impact the returns on their investments over time. Changes in productivity can affect economic growth, inflation, and interest rates, all of which can have significant effects on the value of a pension fund's portfolio.

Choice B is incorrect. Sovereign wealth funds also need to consider productivity risk as they are typically invested in a wide range of asset classes across various countries. Changes in productivity can affect the relative attractiveness of different investments and thus influence the performance of a sovereign wealth fund's portfolio.

Choice D is incorrect. Family offices manage wealth for high-net-worth individuals or families and often have long-term investment horizons similar to pension funds and sovereign wealth funds. Therefore, they also need to consider productivity risk when making investment decisions.

Things to Remember

- Real Business Cycle (RBC) models are used in macroeconomics to explain the fluctuations in macroeconomic variables such as productivity, output, and employment over the business cycle.
- Productivity risk refers to the uncertainty or variability in the level of productivity in an economy, which can impact economic growth, asset prices, and investment returns.
- Investors with longer investment horizons, such as pension funds and sovereign wealth funds, are more likely to be concerned with productivity risk as it can affect the value of their portfolios over time.
- Short-term bond investors typically have shorter investment horizons and may be less concerned with productivity risk compared to long-term investors.

- Family offices are entities that manage the wealth of high-net-worth individuals or families and often have long-term investment horizons similar to pension funds and sovereign wealth funds.
-

Q.3019 In the Fama-French three-factor model, in addition to the market factor and the HML factor, what is the other factor and what does it refer to?

- A. The momentum factor, WML, which captures the outperformance of past winners in relation to past losers.
- B. The SMB factor, which captures the outperformance of high book-to-market stocks in relation to low book-to-market stocks.
- C. The SMB factor, which captures the outperformance of small firms in relation to the larger firms.
- D. The momentum factor, WML, which captures the outperformance of high-growth stocks in relation to low-growth stocks.

The correct answer is **C**.

The Fama-French three-factor model includes a factor known as SMB, which stands for Small Minus Big. This factor is designed to capture the outperformance of small firms in relation to larger firms. In other words, it measures the excess returns of small-cap stocks over large-cap stocks. The rationale behind this factor is that small-cap stocks, due to their higher risk, tend to provide higher returns than large-cap stocks. This factor is used to explain the size effect, which is the observed tendency for smaller firms to outperform larger firms, all else being equal. The SMB factor is calculated by subtracting the return of a portfolio of large-cap stocks from the return of a portfolio of small-cap stocks. Therefore, a positive SMB value indicates that small-cap stocks have outperformed large-cap stocks, while a negative SMB value indicates the opposite.

Choice A is incorrect. The momentum factor, WML (Winners Minus Losers), is not a part of the Fama-French three-factor model. This factor is used in Carhart's four-factor model, which extends the Fama-French model by adding a momentum factor.

Choice B is incorrect. While it correctly identifies SMB (Small Minus Big) as one of the factors in the Fama-French three-factor model, its explanation of what SMB represents is wrong. The SMB factor does not capture the outperformance of high book-to-market stocks over low book-to-market stocks; instead, it captures the excess returns of small-cap companies over large-cap companies.

Choice D is incorrect. Similar to choice A, this option incorrectly includes a momentum factor in the Fama-French three-factor model and misrepresents what this hypothetical WML would capture within this context. In reality, there isn't any such representation for high-growth and low-growth stocks within this particular model.

Things to Remember

- The Fama-French three-factor model was developed by Eugene Fama and Kenneth French to explain stock returns based on three factors: market risk, size (SMB), and value (HML).
 - The SMB factor in the Fama-French model captures the excess returns of small-cap stocks over large-cap stocks, reflecting the size effect in the market.
 - The HML factor in the Fama-French model captures the excess returns of value stocks over growth stocks, reflecting the value effect in the market.
 - The Fama-French three-factor model is widely used in asset pricing and portfolio management to better understand and predict stock returns.
 - Carhart's four-factor model extends the Fama-French model by adding a momentum factor (WML) to capture the outperformance of past winners over past losers in the market.
-

Q.3020 The size factor was introduced in 1981 by Banz. What does it imply?

- A. It refers to the fact that, after their beta values have been adjusted, large stocks tended to perform better as compared to their smaller counterparts.
- B. It refers to the risk of the size of a stock, in that, although value outperforms over the long run, larger stocks perform better than smaller ones during certain periods.
- C. It refers to the fact that the performance of smaller stocks is usually better in comparison to the larger ones after their betas have been adjusted.
- D. All the answers are correct.

The correct answer is **C**.

The size factor, as introduced by Banz in 1981, primarily refers to the observation that smaller stocks tend to outperform larger ones once their beta values have been adjusted. This phenomenon is often referred to as the 'size effect'. The beta value of a stock is a measure of its volatility in relation to the overall market. By adjusting for this volatility, Banz's research suggested that smaller stocks generally yield better returns than larger ones. However, it's important to note that this effect has been less pronounced since the mid-1980s, suggesting that

other factors may also play a significant role in determining stock performance.

Choice A is incorrect. Banz's size factor does not imply that large stocks perform better than smaller ones after adjusting for their beta values. In fact, it suggests the opposite - that smaller stocks tend to outperform larger ones when betas are adjusted.

Choice B is incorrect. While it's true that the performance of stocks can vary over different periods and under different conditions, this isn't what Banz's size factor primarily refers to. The size factor as introduced by Banz specifically refers to the tendency of smaller stocks to outperform larger ones after adjusting for beta values, not just during certain periods.

Choice D is incorrect. As explained above, choices A and B do not accurately represent the primary implication of the size factor as introduced by Banz.

Things to Remember

- Size effect: The size effect is a well-known anomaly in the stock market where smaller stocks tend to outperform larger stocks over time.
 - Beta: Beta is a measure of a stock's volatility in relation to the overall market. Stocks with betas greater than 1 are considered more volatile than the market, while those with betas less than 1 are considered less volatile.
 - Market anomalies: Market anomalies are phenomena that seem to contradict the efficient market hypothesis, suggesting that there are inefficiencies in the market that can be exploited for abnormal returns.
 - Factor investing: Factor investing involves targeting specific factors such as size, value, momentum, or quality to generate excess returns compared to the broader market.
-

Q.4547 Which one of the following factors is **NOT** an investment-style factor?

- A. Size factor
- B. GDP factor
- C. Value factor
- D. Market factor

The correct answer is **B**.

The Gross Domestic Product (GDP) factor is not considered an investment-style factor. Investment-style factors are specific characteristics or attributes that have been historically associated with higher returns. These factors include size, value, and market factors. The GDP factor, on the other hand, is a macroeconomic factor. Macroeconomic factors are those that affect the economy at a broad level, and they include indicators such as GDP, unemployment rates, and inflation rates. While these factors can certainly influence investment decisions, they are not considered investment-style factors because they do not pertain to specific investment styles or strategies.

Choice A is incorrect. The size factor, often referred to as the "small minus big" (SMB) factor, is a well-recognized investment-style factor. It suggests that smaller companies tend to outperform larger ones over time.

Choice C is incorrect. The value factor is another established investment-style factor. It indicates that stocks with lower price-to-book ratios and other similar metrics (i.e., "value" stocks) tend to outperform those with higher ratios (i.e., "growth" stocks).

Choice D is incorrect. The market factor, also known as the market risk premium, represents the expected return of the market above the risk-free rate. This too falls under investment-style factors as it explains differences in returns between different types of investments.

Things to Remember

- Investment-style factors are specific characteristics associated with higher returns
- Macroeconomic factors affect the economy at a broad level
- Size factor (SMB) suggests smaller companies outperform larger ones
- Value factor indicates value stocks outperform growth stocks
- Market factor (market risk premium) explains differences in returns between investments

Q.4548 Which of the following statements is *most likely* correct about momentum investing?

- A. It involves going long on stocks that have low prices and shorting stocks that have high prices
- B. It involves going long on stocks that have upward-trending returns over a specific period and short on stocks with downward-trending returns over the same period
- C. It involves going long on stocks that have small-cap stocks and shorting stocks large-cap stocks
- D. None of the above

The correct answer is **B**.

Momentum investing is a strategy that involves buying securities that have shown an upward trend in returns over a specific period and selling those that have shown a downward trend over the same period. The underlying belief is that these trends will continue in the future. This strategy is based on the principle of 'the trend is your friend', meaning that it is often more profitable to align one's positions with the current market trend rather than trying to predict future market movements. Momentum investors believe that stocks which have performed well in the past are likely to perform well in the future, and similarly, stocks which have performed poorly in the past are likely to continue doing so. This strategy is often used in conjunction with technical analysis and specific trading rules based on data like price and volume.

Choice A is incorrect. Momentum investing does not involve going long on stocks that have low prices and shorting stocks that have high prices. This is a description of value investing, not momentum investing. In value investing, investors seek out stocks they believe are undervalued by the market, which often means buying low-priced stocks in the hope they will increase in value.

Choice C is incorrect. Momentum investing does not involve going long on small-cap stocks and shorting large-cap stocks. This strategy might be part of a specific investment approach based on company size, but it's not inherent to momentum investing. In momentum investing, the focus is on the trend of stock returns regardless of company size.

Choice D is incorrect. As explained above, choice B accurately describes the core principle of momentum investing: going long on upward-trending returns over a specific period and shorting downward-trending returns over the same period.

Things to Remember

- Momentum investing is based on the belief that past trends in stock performance will continue into the future.
- Investors using momentum strategies often rely on technical analysis and specific trading rules.

- Value investing, in contrast to momentum investing, focuses on finding undervalued stocks rather than following trends.
- Market capitalization (small-cap vs. large-cap) is not a defining factor in momentum investing.

Q.4551 Maria Gilbert, FRM, is a portfolio manager who has decided to allocate foreign exchange markets in anticipation of high returns. She has identified a carry trade. An emerging market (SER) is offering relatively higher interest rates as compared to current USD rates. Applicable interest rates in the US are hovering at 2%. An emerging market is offering 7% on its government bonds. Maria has decided to allocate \$100 million in identified trade, which she intends to carry for a whole year. Maria has compiled the following additional data:

Variable	Value
Exchange Rate at T_0	SER2.050/US\$
Exchange Rate at T_1	SER2.071/US\$
$\beta_{i,FX}$	1.45

Calculate foreign carry return for the US portfolio manager.

- A. 0.0392
- B. 0.0592
- C. 0.0568
- D. 0.0706

The correct answer is **C**.

A carry trade involves a couple of foreign exchange transactions. An investor borrows at the lower interest rate and uses the proceeds to invest in a foreign country with higher interest rates. The difference between the exchange rates and the movement in corresponding currencies contributes to the overall return. Foreign exchange exposure can be hedged or left un-hedged. This lies in the risk manager's discretion (or overall policy guideline dictates) whether to hedge or not to depending on the forecast.

Formula:

$$E(FX_i) = \beta_{i,FX} E(HML_{FX})$$

Where:

- $E(FX_i)$ represents the expected foreign carry return of country i
- $\beta_{i,FX}$ is the factor load on the carry factor HML_{FX}
- HML_{FX} represents the return difference between foreign currency with a higher interest rate minus foreign currency with a lower interest rate effectively indicating $E(R_{FX,High\ interest} - R_{FX,Low\ interest})$

At T_0 , US\$100 million is converted to SER 205 million at an exchange rate of SER2.05/US\$. Remember, the primary motivation is to earn a relatively higher return in SER. The translated amount shall be invested at a rate of 7%:

$$\begin{aligned} & \Rightarrow 205,000,000(1 + 0.07)^1 \\ T_1 & \Rightarrow \text{SER } 219,350,000 \end{aligned}$$

Since we need US\$, SER shall be translated into US\$ at an exchange rate of SER2.071/US\$

$$\frac{219,350,000}{2.071} = \text{US } \$105,915,017$$

$$\text{Gross Return in US\$} = \frac{105,915,017}{100,000,000} = 5.92\%$$

$$\begin{aligned} \text{Portfolio Carry Trade Return: } E(FX_i) &= \beta_{i,FX} E(HML_{FX}) \\ E(FX_i) &= 1.45 (5.92\% - 2\%) \\ E(FX_i) &= 5.68\% \end{aligned}$$

Q.4553 John Edwin, a financial risk manager, is in the process of forecasting stock returns. Edwin intends to use the Fama-French three-factor model. He gathers the following information for a stock i:

Variable	Value	Factor beta	Value
Expected return on the market	9%	$\beta_{i,MKT}$	1.20
Value premium	3%	$\beta_{i,HML}$	1.25
Size premium	7%	$\beta_{i,SMB}$	1.10

If the risk-free rate is 3%, what is the expected rate of return of stock i?

- A. 0.2165
- B. 0.2225
- C. 0.2765
- D. 0.2825

The correct answer is **A**.

The Fama-French extension can be formulated as below.

$$E(R_i) = R_f + \beta_{i,MKT} E(R_m - R_f) + \beta_{i,SMB} E(SMB) + \beta_{i,HML} E(HML)$$

Calculation:

$$\begin{aligned}
 E(R_i) &= 3\% + 1.2(9\% - 3\%) + 1.1(7\%) + 1.25(3\%) \\
 &= 3\% + 7.2\% + 7.7\% + 3.75\% \\
 E(R_i) &= 21.65\%
 \end{aligned}$$

Q.4554 John Mathews, a portfolio risk manager at TM Partners, is evaluating the investment policy statement (IPS) of his clients. Due to structural shifts in the prevailing market variables, Mathews wants to re-assess the average investor's risk aversion based on the CAPM theory. Mathews' estimates are shown below:

Variable	Forecast
Risk-free rate	3%
Market risk premium	7%
Standard deviation of market returns	18%

Calculate the average investor's risk aversion coefficient.

- A. 2.16
- B. 0.39
- C. 0.22
- D. 1.23

The correct answer is **A**.

The market risk premium is a function of the average investor's risk aversion coefficient and the variance of market returns:

$$E(R_m) - R_f = \bar{\gamma} \sigma_m^2$$

$$\Rightarrow \text{Market risk premium} = \bar{\gamma} \sigma_m^2$$

Where $\bar{\gamma}$ represents the average investor's risk aversion based on the CAPM theory.

Calculation:

$$0.07 = \bar{\gamma} (0.18)^2$$

$$\frac{0.07}{0.0324} = \bar{\gamma}$$

$$\bar{\gamma} = 2.16$$

The average investor's risk aversion coefficient is, therefore, 2.16.

Q.4555 Mark Truman, FRM, is estimating the expected returns of a portfolio M. The risk-free rate is currently 4.5%, and some additional data gathered by Truman is shown below:

Factors	Value	Factor betas
Size premium	-3.30%	-1.10
Value premium	3.50%	1.77
Market risk premium	6.00%	1.30

If Truman is employing the Fama-French three-factor model, calculate the expected return of portfolio M.

- A. 0.1628
- B. 0.2213
- C. 0.0261
- D. Cannot be determined as the information is insufficient

The correct answer is **B**.

According to the Fama-French model, we have:

$$E(R_p) = R_f + \beta_{p,MKT}(\text{Market Risk Premium}) + \beta_{p,SMB}(\text{Size Premium}) + \beta_{p,HML}(\text{Value Premium})$$

Where:

- Market risk premium = $E(R_m) - R_f$
- Size premium = $E(SMB)$
- Value premium = $E(HML)$
- Market risk premium = 6%
- Size premium = $E(SMB) = -3.3\%$
- Value premium = $E(HML) = 3.50\%$

Thus,

$$E(R_p) = 0.045 + (1.3 \times 0.06) + (-1.10 \times -0.033) + (1.77 \times 0.035) = 22.13\%$$

Reading 148: Alpha (and the Low-Risk Anomaly)

Q.2411 The performance parameters of two fund managers (measured against the same benchmark) in an investment firm is shown in the table below:

Parameter	Fund Manager 1	Fund Manager 2
Return	12.35%	13.00%
α	2.35%	3.00%
Tracking error	25%	34%

Which of the following statements is correct?

- A. Fund manager 1 deviated more from the benchmark index.
- B. Fund manager 2 deviated more from the benchmark index.
- C. Fund manager 1 generated a higher excess return.
- D. Fund manager 2 has a higher information ratio.

The correct answer is **B**.

The tracking error is a measure of how much a fund manager's excess returns deviate over a benchmark index. It is calculated as the standard deviation of the excess returns. A large tracking error indicates a large variability of the excess return over the benchmark). In this case, Fund Manager 2 has a tracking error of 34%, which is higher than Fund Manager 1's tracking error of 25%. Therefore, Fund Manager 2 deviated more from the benchmark index.

Choice A is incorrect. The tracking error of Fund Manager 1 is less than that of Fund Manager 2, indicating that Fund Manager 1 deviated less from the benchmark index compared to Fund Manager 2.

Choice C is incorrect. The alpha (excess return) generated by Fund Manager 1 is lower than that of Fund Manager 2, indicating that the latter has generated a higher excess return.

Choice D is incorrect. The information ratio for manager 1 (i.e., $2.35/25=0.094$) is higher than that for manager 2 (i.e., $3/34=0.0882$).

Q.2412 Which of the following statements is not a characteristic of an appropriate benchmark? An appropriate benchmark should **not** be:

- A. Equally applied to all risky assets irrespective of their risk exposure
- B. Replicable

C. Tradeable

D. Risk-adjusted

The correct answer is **A**.

An appropriate benchmark should not be equally applied to all risky assets irrespective of their risk exposure. This is because different assets have different risk exposures and therefore, applying the same benchmark to all assets would not provide a fair and accurate comparison. For instance, a high-risk asset would naturally have a higher expected return than a low-risk asset. If the same benchmark is applied to both, the high-risk asset may appear to underperform when in fact it is simply taking on more risk. Therefore, it is crucial that the benchmark takes into account the risk exposure of the assets it is being compared to. This ensures that the comparison is fair and meaningful.

Choice B is incorrect. A benchmark should be replicable. This means that the performance of the benchmark can be replicated by a passive investment strategy. If a benchmark is not replicable, it would not provide a fair comparison for an active investment strategy.

Choice C is incorrect. A benchmark should be tradeable. This means that investors can buy and sell securities in the benchmark portfolio without any restrictions or limitations. If a benchmark is not tradeable, it would not reflect the true market conditions and hence, would not serve as an appropriate measure of performance.

Choice D is incorrect. A benchmark should be risk-adjusted. This means that the returns of the benchmark are adjusted for risk to provide a fair comparison with other investments which may have different levels of risk exposure. If a benchmark does not consider risk adjustment, it could potentially mislead investors about their relative performance.

Things to Remember

- Choosing an appropriate benchmark is crucial for evaluating the performance of an investment portfolio.
- Benchmarks help investors and portfolio managers assess how well their investments are performing relative to a standard or a market index.
- Replicability of a benchmark ensures that the benchmark's performance can be easily duplicated by passive investment strategies.
- Tradeability of a benchmark allows for accurate tracking of market conditions and ensures that the benchmark reflects real-world investment scenarios.
- Risk-adjustment of a benchmark is important to provide a fair comparison of investment performance, especially when comparing assets with different risk profiles.

Q.2414 A large investment management firm intends to launch a new hedge fund for its retail investors. The firm is in the process of identifying a benchmark which its fund would try to replicate. During a meeting of the investment management committee, the following remarks are made: Fund manager 1: *"Our firm has multiple active funds tracking different benchmarks. We must include the benchmark with the highest return as the benchmark for our new fund as this will also help us to attract new investors."* Fund manager 2: *"My research reveals that the returns for XYZ fund have been above the market return for several consecutive periods. I would, therefore, recommend the use of the market return as our benchmark, as it will clearly help us attract new investors."* Fund manager 3: *"There are many absolute return funds in the market which have consistently beaten the market. We must take one of those funds as the benchmark."* Which of the following is correct with respect to the statement made by Fund manager 3?

- A. Absolute return funds generate positive returns irrespective of market conditions and, therefore, it would be in order to use one of them as the benchmark.
- B. Absolute return funds take excessive risks, a characteristic which effectively nullifies them from being used as benchmarks.
- C. Absolute return funds employ different strategies which are usually not replicable, and hence one of them must not be used as the benchmark.
- D. Absolute return benchmarks are usually not diversified, and hence one of them must not be used as the benchmark.

The correct answer is C.

Absolute return funds employ a variety of strategies that are typically not replicable, making them unsuitable to be used as benchmarks. Absolute return funds aim to generate positive returns regardless of market conditions. They do this by employing a wide range of investment strategies, including long/short equity strategies, arbitrage strategies, and derivative strategies, among others. These strategies are often complex and require a high level of expertise to execute effectively. Furthermore, the strategies used by absolute return funds are often proprietary, meaning they are unique to the fund and not publicly available. This makes it virtually impossible for another fund to replicate the strategies used by an absolute return fund. Therefore, using an absolute return fund as a benchmark would be inappropriate because it would be impossible for the new fund to replicate the strategies used by the absolute return fund. This would make any comparison between the new fund and the absolute return fund meaningless.

Choice A is incorrect. While it's true that absolute return funds aim to generate positive returns irrespective of market conditions, this doesn't necessarily make them suitable benchmarks. Benchmarks should be representative of the investment strategy and risk profile of the fund they are measuring, and not all funds aim for absolute returns.

Choice B is incorrect. The assertion that absolute return funds take excessive risks is a

generalization and may not hold true for all such funds. Moreover, risk-taking in itself does not disqualify a fund from being used as a benchmark; rather, the benchmark should reflect the level of risk that is consistent with the fund's strategy.

Choice D is incorrect. The claim that absolute return benchmarks are usually not diversified may be misleading. Absolute return strategies can involve a wide range of asset classes and investment techniques, which can lead to diversification benefits. However, their lack of replicability makes them unsuitable as benchmarks.

Things to Remember

- Benchmarks are used to measure the performance of a fund against a standard or index.
 - Absolute return funds aim to generate positive returns regardless of market conditions.
 - Absolute return funds employ a variety of complex and often proprietary investment strategies.
 - Replicability is an important factor in selecting a benchmark, as the benchmark should be something that can be feasibly replicated by the fund being measured.
 - Benchmarks should be representative of the investment strategy and risk profile of the fund they are measuring.
-

Q.2415 The composition an index (ABC) is given below:

Assets	Proportion
Exotic Stones	10%
Real Estate	75%
Art	15%

Which of the following statements is accurate?

- A. ABC can be an ideal benchmark for real estate.
- B. ABC can be an ideal benchmark for hedge funds.
- C. ABC cannot be used as a benchmark index.
- D. ABC can be used as a benchmark for alternative investments.

The correct answer is **C**.

ABC cannot be used as a benchmark index. A benchmark index is a standard against which the performance of a security, mutual fund or investment strategy can be measured. Hence, an ideal benchmark should be investable, meaning that it is possible to replicate the performance of the index by investing in a portfolio that includes the index components. However, the assets included in the ABC index are all illiquid in nature. Illiquid assets are those that cannot be sold quickly without a substantial loss in value. Exotic Stones, Real Estate, and Art are all examples of such assets. Therefore, it would be difficult to trade the ABC index with ease and it may also require high transaction costs. This makes the ABC index unsuitable to be used as a benchmark index.

Choice A is incorrect. Although real estate makes up a significant proportion of the ABC index (75%), it is not an ideal benchmark for real estate because it also includes other types of assets, namely exotic stones and art. An ideal benchmark for real estate should be composed entirely or predominantly of real estate assets.

Choice B is incorrect. The ABC index cannot be an ideal benchmark for hedge funds as it does not reflect the diverse range of strategies and asset classes that hedge funds typically invest in. Hedge fund benchmarks usually consist of a broad-based market index or a peer group average.

Choice D is incorrect. While the ABC index does include alternative investments such as exotic stones and art, these only make up 25% of the total composition, making it less suitable as a benchmark for alternative investments which should ideally cover a wider range and higher proportion of such assets.

Q.2416 An investment management firm recruits multiple fund managers to manage its funds. In order to have some control over transaction costs, the firm allows each manager to make a maximum of 100 transactions per year. One of the fund managers wants an information ratio of 0.75. Assuming the “fundamental law” applies, her information coefficient must be equal to:

- A. 0.75
- B. 0.0075
- C. 0.075
- D. 7.5

The correct answer is **C**.

$$\text{Information Ratio} = \text{IC} \times \sqrt{\text{BR}}$$

IC = Information Coefficient

BR = Breadth of the strategy

$$\text{IC} = \frac{0.75}{\sqrt{100}} = 0.075$$

Q.2417 An investment management firm recruits multiple fund managers to manage its funds. In order to have some control over transaction costs, the firm allows each manager to make a maximum of 100 transactions per year. One of the fund managers alternatively calculates her information ratio by computing the excess return and the tracking error from the actual observed values. She uses the following formula:

$$IR = IC \times \sqrt{BR}$$

Where: IR is the information ratio.

IC is the information coefficient.

BR is the breadth of the strategy. Suppose that IC is assumed to be 7.5%. After computation, the fund manager observes that the information ratio is less than 0.75. This is most likely due to the:

- A. Correlation between the different bets of the fund manager.
- B. Cap on the number of transactions per year.
- C. Differences in portfolio weights.
- D. Differences in the number of transactions from one manager to another.

The correct answer is **A**.

The information ratio (IR) of a fund manager is influenced by the number of transactions they carry out, also known as the breadth of the strategy (BR). In this scenario, the firm's management has capped transactions at 100 per year. However, the 100 transactions executed by a fund manager may not all be distinct. For example, the fund manager might be investing in growth stocks and perform 20 transactions related to this. While these may appear as 20 separate transactions, they are all correlated. This correlation between the different bets of the fund manager reduces the breadth of the strategy, and consequently, the information ratio. Therefore, the correlation between the different bets of the fund manager is the most likely reason for the observed information ratio of less than 0.75.

Choice B is incorrect. The cap on the number of transactions per year does not directly affect the information ratio. The information ratio measures the excess return of a portfolio over its benchmark, relative to the consistency of that excess return (tracking error). While transaction costs can impact returns, they do not directly influence the calculation of an information ratio.

Choice C is incorrect. Differences in portfolio weights can lead to differences in returns and risk levels between portfolios, but they do not directly affect the calculation of an information ratio. The information ratio is a measure of risk-adjusted performance and does not depend on portfolio weights.

Choice D is incorrect. Differences in the number of transactions from one manager to another could potentially impact transaction costs and thus returns, but this would not directly influence an information ratio calculation which focuses on excess return over benchmark relative to tracking error.

Q.2418 An investment management firm recruits multiple fund managers to manage its funds. In order to have some control over transaction costs, the firm allows each manager to make a maximum of 100 transactions per year. The assets under management for the above fund increase from \$1million to \$20 million in 5 years. How does this affect the information ratio (IR)?

- A. It increases
- B. It remains unchanged
- C. It decreases
- D. It moves towards a value of 1

The correct answer is C.

The information ratio (IR) decreases as the assets under management increase. The IR is a measure of the risk-adjusted return of a financial portfolio. It is calculated as the portfolio's excess return over the benchmark, divided by the tracking error. The IR is influenced by two key variables according to the Fundamental Law of Active Management: the information coefficient (IC) and the breadth of investment opportunities. The IC is a measure of the portfolio manager's skill in selecting securities, while the breadth refers to the number of independent investment opportunities available. As the assets under management increase, the ability to generate a strong IC diminishes, leading to a decrease in the IR. This is because as funds grow larger, the performance tends to deteriorate, a phenomenon observed in mutual funds, hedge funds, and private equity. Therefore, as the assets under management for the fund increased from \$1 million to \$20 million, the IR would decrease due to the reduced ability to generate strong ICs.

Choice A is incorrect. The information ratio (IR) does not increase with the growth of assets under management. IR is a measure of portfolio returns above the returns of a benchmark, to the volatility of those returns. It does not directly correlate with the size of assets under management.

Choice B is incorrect. The IR does not remain unchanged either. As assets under management grow, transaction costs per dollar invested decrease due to economies of scale, but if the number of transactions remains constant, it may limit the manager's ability to take advantage of new investment opportunities and thus potentially reduce excess return and lower IR.

Choice D is incorrect. There's no rule or tendency for IR to move towards a value of 1 as assets under management increase. The value '1' has no specific significance in terms of information ratio.

Things to Remember

- Information Ratio (IR) is a measure of the risk-adjusted return of a financial portfolio.
- IR is calculated as the portfolio's excess return over the benchmark, divided by the tracking error.

- The Fundamental Law of Active Management states that the IR is influenced by the Information Coefficient (IC) and the breadth of investment opportunities.
 - IC measures the portfolio manager's skill in selecting securities.
 - The breadth of investment opportunities refers to the number of independent investment opportunities available.
 - As assets under management increase, the ability to generate a strong IC diminishes, leading to a decrease in the IR.
 - Performance tends to deteriorate as funds grow larger, observed in mutual funds, hedge funds, and private equity.
 - Transaction costs per dollar invested decrease with economies of scale, but a constant number of transactions may limit the manager's ability to take advantage of new opportunities.
-

Q.2419 A fund manager tracks the monthly return of ABC Corporation – an asset management company. She regresses the monthly return of ABC against the market returns for the past 5 years. The regression output is given in the table below:

Parameter	Coefficient
Alpha	0.82%
Beta	0.69
Asset under management	\$10 million

The stock generates a yearly active return of:

- A. 0.82% with approximately 70% of market risk.
- B. 8.20% with approximately 30% of market risk.
- C. 9.84% with approximately 70% of market risk.
- D. 0.82% with approximately 30% of market risk.

The correct answer is **C**.

The monthly active return generated by the stock = 0.82%

$$\text{Yearly Return} = 0.82\% \times 12 = 9.84\%$$

As Beta = 0.69, the stock carries approximately 70% of market risk.

Q.2421 A fund manager tracks the monthly return of ABC Corporation – an asset management company. She regresses the monthly return of ABC against the market returns for the past 5 years. The regression output is given in the table below:

Parameter	Coefficient
Alpha	0.82%
Beta	0.69
Asset under management	\$10 million

An investor wants to replicate the ABC Corporation's stock returns but she does not intend to hold the stock of ABC Corp. Which of the following portfolios mimics the stock of ABC Corp?

- A. 25% of T-Bills and 75% of the market portfolio
- B. 35% of T-Bills and 65% of the market portfolio
- C. 31% of T-Bills and 69% of the market portfolio
- D. 33% of T-Bills and 67% of the market portfolio

The correct answer is **C**.

The Capital Asset Pricing Model (CAPM) is a financial model that determines the expected return of an asset based on its beta and expected market returns. In this case, the beta of ABC Corporation's stock is 0.69, which means that the stock's return is 69% as volatile as the market's return. The remaining 31% of the return is not explained by the market's return, and can be considered as the risk-free rate. Therefore, a portfolio that consists of 31% T-Bills (which are considered risk-free) and 69% of the market portfolio would mimic the returns of ABC Corporation's stock. This is because the return of the portfolio would be 31% of the risk-free rate (from the T-Bills) plus 69% of the market return, which is the same as the return of ABC Corporation's stock according to the CAPM model.

Choice A is incorrect. This choice suggests a portfolio composition of 25% T-Bills and 75% of the market portfolio. However, this does not align with the beta coefficient from the regression analysis which is 0.69. The beta coefficient indicates that ABC Corporation's stock has a lower sensitivity to market movements than what this choice suggests.

Choice B is incorrect. This option proposes a portfolio composition of 35% T-Bills and 65% of the market portfolio, which again does not match with the beta value from our regression analysis (0.69). The proportion invested in the market portfolio should be equal to ABC Corporation's stock's sensitivity to market movements, as indicated by its beta.

Choice D is incorrect. As explained above, this option also fails to replicate ABC Corporation's returns because it suggests investing 33% in T-Bills and 67% in the market portfolio - proportions that do not align with our calculated beta value (0.69).

Q.2423 A large investment management firm manages multiple funds. It re-launches one of its best performing funds to attract new investors. John Grey, a prospective investor in the fund, goes through the fund factsheet. In the factsheet, the investment management firm states that the fund invests in large value stocks. To verify the correctness of this assertion, Grey plans to leverage the skills he's learned from his Finance undergraduate classes. Grey regresses the fund return against the market and a few other factors. The results are presented below:

$$r_{\text{fund}} - r_f = 0.82\% + 0.74(r_{\text{mkt}} - r_f) + 0.32 \text{ SMB} + 0.54 \text{ HML} + S_{\text{error}}$$

Which of the following statements is correct?

- A. The fund has more exposure towards large stocks.
- B. The fund has more exposure towards small stocks.
- C. The fund has equal exposure towards both small and large stocks.
- D. The fund has 100% exposure towards large stocks.

The correct answer is **B**.

The fund has more exposure towards small stocks. The regression equation provided in the question is a representation of the Fama-French three-factor model. This model is widely used in finance to describe stock returns. It includes three factors: the market risk premium ($r_{\text{mkt}} - r_f$), the size premium (SMB), and the value premium (HML). The size premium, SMB, is calculated by going long on small companies and short on large companies. Therefore, a positive SMB coefficient indicates that the fund has more exposure to small companies compared to large companies. In this case, the SMB coefficient is 0.32, which is positive. Therefore, it can be concluded that the fund has more exposure towards small stocks.

Choice A is incorrect. The fund does not have more exposure towards large stocks. The coefficient of SMB (Small Minus Big) in the regression equation is positive, indicating that the fund has more exposure to small-cap stocks rather than large-cap stocks.

Choice C is incorrect. The fund does not have equal exposure towards both small and large stocks. As explained above, the positive SMB coefficient suggests a greater tilt towards small-cap stocks.

Choice D is incorrect. The statement that the fund has 100% exposure towards large stocks is inaccurate as per the regression results which indicate a significant positive loading on SMB (small minus big), suggesting an inclination towards smaller companies rather than larger ones.

Q.2424 A large investment management firm manages multiple funds. It re-launches one of its best performing funds to attract new investors. John Grey, a prospective investor in the fund, goes through the fund factsheet. In the factsheet, the investment management firm states that the fund invests in large value stocks. To verify the correctness of this assertion, Grey plans to leverage the skills he's learned from his Finance undergraduate classes. Grey regresses the fund return against the market and a few other factors. The results are presented below:

$$r_{\text{fund}} - r_f = 0.82\% + 0.74(r_{\text{mkt}} - r_f) + 0.32 \text{ SMB} + 0.54 \text{ HML} + S_{\text{error}}$$

Which of the following statements is correct?

- A. The fund has more exposure towards value stocks.
- B. The fund has more exposure towards growth stocks.
- C. The fund has equal exposure towards both value and growth stocks.
- D. The fund is concentrated with growth stocks.

The correct answer is **A**.

The fund has more exposure towards value stocks. This conclusion is drawn from the positive coefficient of the HML (High Minus Low) factor in the regression equation. The HML factor is a measure of the spread in returns between value stocks and growth stocks. A positive HML coefficient indicates that the fund's returns move more closely with value stocks than with growth stocks. In other words, when value stocks perform well, the fund's returns are likely to increase, and when value stocks perform poorly, the fund's returns are likely to decrease. This suggests that the fund has a higher exposure to value stocks. The high positive coefficient in the HML means high book to market value, which indicates a value stock. Therefore, the assertion made by the investment management firm in the fund's factsheet is correct.

Choice B is incorrect. The fund does not have more exposure towards growth stocks. The positive coefficient of 0.54 for HML (High Minus Low) indicates that the fund has a higher exposure to value stocks, not growth stocks. Growth stocks typically have negative HML coefficients.

Choice C is incorrect. The fund does not have equal exposure towards both value and growth stocks. As mentioned above, the positive HML coefficient suggests a higher tilt towards value stocks.

Choice D is incorrect. The fund is not concentrated with growth stocks as the positive HML coefficient indicates a preference for value over growth.

Q.2753 You have been given the following information for a portfolio:

Portfolio return	10%
Beta	1.2
Standard deviation of the portfolio	5%
Benchmark return	8%
Tracking error	10%
Risk-free rate	3%

Calculate the Sharpe and the information ratios for the portfolio.

- A. Sharpe ratio: 1.4; Information ratio: 0.2
- B. Sharpe ratio: 1.4; Information ratio: 0.7
- C. Sharpe ratio: 0.4; Information ratio: 0.2
- D. Sharpe ratio: 0.4; Information ratio: 0.7

The correct answer is **A**.

$$\begin{aligned}\text{Sharpe Ratio} &= \frac{(R_p - R_f)}{\sigma} \\ &= \frac{(10 - 3)}{5} \\ &= 1.4\end{aligned}$$

$$\begin{aligned}\text{Information Ratio} &= \frac{(R_p - R_b)}{(\text{Tracking error})} \\ &= \frac{(10 - 8)}{10} \\ &= 0.2\end{aligned}$$

Q.2755 A portfolio manager invested \$1 million in a portfolio of equities. At that time the benchmark index against which the performance of the portfolio was to be tracked stood at 2000 points. One year later the value of the portfolio was \$1.12 million while the benchmark index was 2180 points. Calculate the information ratio for the portfolio if the tracking error is 4%.

A. 0.75

B. 0.6

C. 0.83

D. 1.0

The correct answer is **A**.

$$\text{Information Ratio} = \frac{\text{Alpha}}{\text{Tracking error}}$$

Alpha = Return on the portfolio - Return on the benchmark

$$\text{Alpha} = \left(\frac{1.12}{1}\right) - \left(\frac{2,180}{2,000} - 1\right)$$

$$\text{Alpha} = 3\%$$

$$\begin{aligned}\text{Information Ratio} &= \frac{0.03}{0.04} \\ &= 0.75\end{aligned}$$

Note that,

$$\text{IR} = \frac{(R_p - R_b)}{\sigma(R_p - R_b)} = \frac{(R_p - R_b)}{\text{Tracking error}} = \frac{\text{Active returns}}{\text{Tracking error}} = \frac{\text{Alpha}}{\text{Tracking error}}$$

Q.2761 Which of these managers will have the highest information ratio based on Grinold's law of active management?

- A. A manager with an information coefficient of 0.5 and a breadth of 20.
- B. A manager with an information coefficient of 0.05 and a breadth of 200.
- C. A manager with an information coefficient of 0.1 and a breadth of 100.
- D. A manager with an information coefficient of 0.8 and a breadth of 10.

The correct answer is **D**.

According to Grinold's law the maximum attainable information ratio can be calculated using the following formula:

$$IR = IC \times \sqrt{BC}$$

The four managers will have IRs of:

$$\begin{aligned} IR_A &= 0.5 \times \sqrt{20} = 2.24 \\ IR_B &= 0.05 \times \sqrt{200} = 0.71 \\ IR_C &= 0.1 \times \sqrt{100} = 1.00 \\ IR_D &= 0.8 \times \sqrt{10} = 2.53 \end{aligned}$$

Q.3021 James Dunn is a retail investor who expects his stock portfolio to return 21.2% in the following year. If the returns on risk-free Treasury notes are 11.6%, and the portfolio carries a standard deviation of 0.09, then which of the following is closest to the Sharpe ratio of Dunn's portfolio?

A. 1.5403

B. 2.9873

C. 1.0667

D. 2.4543

The correct answer is C.

Recall that the Sharpe ratio is given as:

$$\text{Sharpe Ratio} = \text{SR} = \frac{\overline{r_t - r_{ft}}}{\sigma}$$

$$r_t = 0.212, r_{ft} = 0.116, \sigma = 0.09$$

Therefore:

$$\text{Sharpe Ratio} = \text{SR} = \frac{0.212 - 0.116}{0.09} = 1.0667$$

A ratio of 1.0667 implies that the investor earns approx. 1.07 per unit of risk.

Q.3022 Tradability is one of the properties of a sound benchmark. Which of the following best explains the link between tradability and a sound benchmark?

A. If alphas are not evaluated relative to a tradable benchmark, the implementable returns on investment strategies are not a representation of the calculated alphas.

B. Both the asset owner and the fund manager are supposed to display the ability to trade the benchmark.

C. Produced by an independent asset provider, a tradable benchmark should be verifiable and not ambiguous about its contents.

D. Alphas must not be measured in relation to a tradable benchmark because if so, the computed alphas will not give the returns on investment strategies that can be implemented.

The correct answer is **A**.

The concept of alpha, in finance, refers to the excess return of an investment relative to the return of a benchmark index. If alphas are not evaluated relative to a tradable benchmark, the implementable returns on investment strategies are not a representation of the calculated alphas. This is because a tradable benchmark allows for the replication of the investment strategy, thereby providing a more accurate reflection of the potential returns. If the benchmark is not tradable, it may not accurately reflect the market conditions and the potential returns that can be achieved, leading to a discrepancy between the calculated alphas and the actual returns that can be implemented. Therefore, the tradability of a benchmark is crucial in ensuring that the calculated alphas accurately represent the potential returns of an investment strategy.

Choice B is incorrect. While it's true that both the asset owner and the fund manager should have a clear understanding of the benchmark, it's not necessary for both parties to be able to trade the benchmark. The ability to trade a benchmark is not a requirement for its effectiveness in reflecting potential returns of an investment strategy.

Choice C is incorrect. Although benchmarks should ideally be produced by an independent provider and be verifiable and unambiguous, these characteristics do not directly relate to tradability. A benchmark can still accurately reflect potential returns even if it isn't tradable.

Choice D is incorrect. This statement contradicts established financial theory which suggests that alphas should indeed be measured relative to a tradable benchmark. Doing so allows for more accurate reflection of implementable returns on investment strategies.

Things to Remember

- Alpha in finance refers to the excess return of an investment relative to a benchmark index.
- Tradable benchmarks allow for the replication of investment strategies, providing a more accurate reflection of potential returns.
- Market conditions and potential returns can be accurately reflected only if the benchmark is tradable.
- It is important for asset owners and fund managers to have a clear understanding of the benchmark, but trading the benchmark is not a requirement for its effectiveness.
- Benchmarks should ideally be produced by independent providers, verifiable, and unambiguous, but these characteristics do not directly relate to tradability.
- Measuring alphas relative to a tradable benchmark is consistent with established financial theory and allows for a more accurate reflection of implementable returns on

investment strategies.

Q.3023 The expected return of the market portfolio, $E(r_m)$, is 6.2% and the risk-free rate, r_f , is given as 3.9%. Let's also assume that the beta of the asset is given as 1.3. What is the asset's expected return, $E(r_i)$?

- A. 4.96%
- B. 2.99%
- C. 11.96%
- D. 6.89%

The correct answer is **D**.

Recall that:

$$E(r_i) - r_f = \beta_i(E(r_m) - r_f)$$

Therefore:

$$E(r_i) = 1.3(6.2\% - 3.9\%) + 3.9\% = 6.89\%$$

Q.4455 The standard deviation of a portfolio is 15%. If the portfolio's return is 22%, and the risk-free return is 6%, then what is the Sharpe ratio of the portfolio?

- A. 0.91
- B. 1.07
- C. 1.46
- D. 1.98

The correct answer is **B**.

Sharpe ratio = (Portfolio return - Risk-free return) / Standard deviation of the portfolio = (22% - 6%) / 15% = 1.07

Things to Remember

- The Sharpe ratio is a measure of risk-adjusted return, calculated by subtracting the risk-free rate of return from the portfolio's return and dividing the result by the standard deviation of the portfolio's returns.
 - A higher Sharpe ratio indicates better risk-adjusted performance.
 - Standard deviation is a measure of the dispersion of returns around the average return of a portfolio.
 - The risk-free rate is the return on an investment with zero risk, typically represented by government bonds.
 - Sharpe ratio helps investors understand the return they are receiving for the amount of risk they are taking.
-

Q.4456 The expected return of an investor's portfolio is 31% with a standard deviation of 19% while the expected return of the market is 22% with a standard deviation of 16%. Given that the risk-free rate is 5% and the portfolio's beta is 0.9, determine the difference between the Sharpe ratio of the portfolio and the Sharpe ratio of the market.

- A. 0.31
- B. 0.5
- C. 1.06
- D. 0.12

The correct answer is **A**.

Sharpe ratio of the investor's portfolio = $(E(R_p) - R_f)/\sigma_p = (0.31 - 0.05)/0.19 = 1.37$

Sharpe ratio of market portfolio = $(E(R_m) - R_f)/\sigma_m = (0.22 - 0.05)/0.16 = 1.06$

Difference = $1.37 - 1.06 = 0.31$

Things to Remember

- The Sharpe ratio is a measure of risk-adjusted return, calculated by subtracting the risk-free rate from the expected portfolio return and dividing the result by the standard deviation of the portfolio's excess return.
- A higher Sharpe ratio indicates a better risk-adjusted return for the investor.
- The Sharpe ratio helps investors understand the return they are receiving for the amount of risk they are taking.
- Beta measures the sensitivity of a portfolio's returns to the market returns. A beta of less than 1 indicates lower volatility compared to the market.
- The risk-free rate is the return on an investment with zero risk, typically represented by government bonds.

Q.4457 A 10-year research on 3 distinct portfolios and the market reveals the following information:

Portfolio	Average annual return	Standard deviation	Beta
1	14%	21%	1.15
2	16%	24%	1.00
3	20%	28%	1.25
S&P 500	12%	20%	

Given that the risk-free rate of return is 6%, use the Sharpe measure to rank the portfolios from the lowest to the highest.

- A. 1, 3, 2
- B. 2, 3, 1
- C. 2, 1, 3
- D. 1, 2, 3

The correct answer is **D**.

The formula for calculating the Sharpe ratio for a portfolio:

$$S_p = \frac{E(R_p) - R_f}{\sigma_p}$$

Hence,

$$S_1 = (14\% - 6\%)/21\% = 0.38$$

$$S_2 = (16\% - 6\%)/24\% = 0.42$$

$$S_3 = (20\% - 6\%)/28\% = 0.50$$

Q.4630 Which of the following statements *best explains* the volatility anomaly?

- A. High volatility stock earns higher risk-adjusted returns relative to the market portfolio
- B. Low volatility stock earns higher risk-adjusted returns relative to the market portfolio
- C. Low beta implies low volatility risk exposure
- D. None of the above

The correct answer is **B**.

The volatility anomaly refers to the empirical observation that low volatility stocks tend to earn higher risk-adjusted returns relative to the market portfolio. This contradicts the predictions of the Capital Asset Pricing Model (CAPM), which suggests that stocks with higher volatility should yield higher risk-adjusted returns. The CAPM is a model that describes the relationship between systematic risk and expected return for assets, particularly stocks. It is used in the pricing of risky securities, generating expected returns for assets given a certain level of systematic risk, and calculating costs of capital. However, the volatility anomaly suggests that the CAPM may not always accurately predict real-world outcomes. This anomaly has been observed in various studies and across different markets, suggesting that it is not just a result of specific market conditions or anomalies. The reasons for this anomaly are not entirely clear, but some theories suggest that it may be due to behavioral biases, market frictions, or risk factors not captured by the CAPM.

Choice A is incorrect. The volatility anomaly contradicts this statement. According to the Capital Asset Pricing Model (CAPM), high volatility stocks should yield higher risk-adjusted returns, but empirical evidence has shown that it's actually low volatility stocks that earn higher risk-adjusted returns relative to the market portfolio.

Choice C is incorrect. This statement is misleading as it implies a direct relationship between beta and volatility risk exposure. While beta does measure a stock's sensitivity to market movements, it doesn't necessarily equate to its overall volatility or risk exposure.

Choice D is incorrect. As explained above, there exists an observed trend in financial markets known as the 'volatility anomaly', which contradicts the predictions of CAPM and therefore cannot be classified as 'None of the above'.

Things to Remember

- Capital Asset Pricing Model (CAPM) is a model that describes the relationship between systematic risk and expected return for assets, particularly stocks.
- Systematic risk is the risk inherent to the entire market or an entire market segment.
- Empirical observations in finance can sometimes lead to anomalies that contradict traditional models like CAPM.
- Behavioral biases and market frictions are some factors that may contribute to anomalies like the volatility anomaly.
- Low volatility stocks are often referred to as defensive stocks due to their tendency to perform better during market downturns.

Q.4631 Which of the following is a potential reason for the low-risk anomaly?

- A. Leverage constraints
- B. Low alpha
- C. Inappropriate benchmark
- D. None of the above

The correct answer is **A**.

Leverage constraints can indeed be a potential reason for the low-risk anomaly. Most investors, especially those with a high-risk appetite, would prefer to take on more risk in their investment portfolios. However, due to various reasons such as regulatory restrictions or personal financial limitations, they might be unable to take on leverage, i.e., borrow additional funds to invest. As a result, these investors tend to hold stocks that have built-in leverage. These are typically high-beta stocks, which are more volatile and hence riskier than the market average. The high demand for such stocks can lead to their overpricing, which in turn results in lower returns. This phenomenon can explain the low-risk anomaly, where low-risk investments outperform high-risk ones.

Choice B is incorrect. Low alpha refers to a low level of excess return on an investment, after adjusting for market-related risk. While it could be a characteristic of certain investments, it does not explain the low-risk anomaly which suggests that lower-risk investments can sometimes yield higher returns than high-risk ones.

Choice C is incorrect. An inappropriate benchmark might lead to misinterpretation of performance but it doesn't cause the low-risk anomaly. The anomaly occurs due to inherent characteristics in the financial markets and not due to comparison with an inappropriate benchmark.

Choice D is incorrect. As explained above, there are plausible explanations for the occurrence of this low-risk anomaly such as leverage constraints (option A). Therefore, 'None of the above' cannot be correct.

Things to Remember

- Investors with a high-risk appetite may prefer to take on leverage in their investment portfolios to amplify returns.
- Regulatory restrictions or personal financial limitations can prevent investors from taking on leverage, leading them to hold stocks with built-in leverage.
- High-beta stocks are more volatile and riskier than the market average, making them attractive to investors seeking higher returns.
- The high demand for high-beta stocks can lead to their overpricing, resulting in lower

returns.

- The low-risk anomaly suggests that lower-risk investments can sometimes outperform higher-risk ones, contrary to traditional risk-return expectations.
-

Q.4632 John Walton, a portfolio manager, is evaluating the performance of a fund manager specializing in large-cap stocks. Given that the investment mandate is large-caps, he has compiled the following information:

Variable	Value
Annual benchmark return (LC) (Large-cap: 90%,Cash:10%)	8%
Annual benchmark return (Hybrid) (Large-cap: 50%,Small-cap:45%,Cash:5%)	10%
Portfolio Return	12%
Tracking Error	6%

Calculate the excess return per unit of excess risk.

- A. 0.85
- B. 0.67
- C. 1.17
- D. 0.87

The correct answer is **B**.

Alpha measures the average excess return generated by a portfolio relative to its appropriate benchmark carrying a comparable risk profile. In this case, we have a large-cap portfolio that can be most appropriately benchmarked against the benchmark (LC).

As per the given data, the information ratio is the best fit for the calculation of excess return per unit of excess risk. First, we need to find the alpha, which can be calculated as below:

$$\begin{aligned}\text{Alpha} &= \text{Annual portfolio return} - \text{Annual benchmark return} \\ &= 12\% - 8\% = 4\%\end{aligned}$$

$$\text{Excess Return} = \text{Alpha} = 4\%$$

Now, we can calculate the information ratio:

$$\text{IR} = \frac{\alpha}{\text{TE}} = \frac{4\%}{6\%} = 0.67$$

An information ratio of 0.67 indicated that the manager earned 67 basis points of excess return per unit of excess risk.

Q.4633 Mark Williams is a Chief Investment Officer at Berkeley Capital. He aims to achieve an annual information ratio of 0.83 on a firm-wide basis. The company places a bet every month. Calculate the forecast precision to attain the desired information ratio.

- A. 0.24
- B. 2.88
- C. 4.17
- D. Cannot be calculated with given information

The correct answer is **A**.

Information ratio (IR) is used in conjunction with the Information coefficient (IC) and Breadth of the strategy (BR). IC is the measure that tracks the accuracy of the forecast with the actual result. The higher accuracy leads to higher IC, which results in a higher IR. On the other hand, BR is the number of bets. In our scenario, the firm places 12 bets a year, so our BR equals 12.

To calculate IR, we use the formula:

$$\begin{aligned} \text{IR} &= \text{IC} \times \sqrt{\text{BR}} \\ 0.83 &= \text{IC} * \sqrt{12} \\ 0.83 &= \text{IC} * 3.46 \\ \{\text{IC}\} &\Rightarrow \frac{0.83}{3.46} = 0.24 \end{aligned}$$

William needs to produce an IC of 0.24 to reach an Information Ratio of 0.83, assuming 12 bets a year.

Q.4634 David Ken is a portfolio manager. He has identified a stock in the pharmaceutical industry, which he believes will outperform the benchmark. Ken does not want to take direct exposure in the stock; instead, he wants to replicate the returns by using a combination of available instruments, essentially the risk-free rate and the expected market return. Assuming that the corresponding beta of the underlying stock is 1.6, construct the appropriate portfolio that will replicate the returns of the stock.

- A. $0.6R_f + 1.6E(R_m)$
- B. $-0.6R_f + 1.6E(R_m)$
- C. $1.6E(R_m)$
- D. Cannot be determined due to insufficient information

The correct answer is **B**.

We can construct an asset's return mimicking portfolio by combining the risk-free portfolio and the market portfolio.

Formula.

We know that: $E(R_s) = R_f + \beta_{s,m} [E(R_m) - R_f]$

Rearranging the equation $\Rightarrow E(R_s) - R_f = \beta_{s,m} [E(R_m) - R_f]$

Since the stock has a beta of 1.6, we will use it into the equation:

$$\begin{aligned}\Rightarrow E(R_s) - R_f &= 1.6[E(R_m) - R_f] \\ \Rightarrow E(R_s) - R_f &= 1.6E(R_m) - R_f \\ \Rightarrow E(R_s) &= R_f + 1.6E(R) - 1.6R_f \\ \Rightarrow E(R) &= -0.6R_f + 1.6E(R_m)\end{aligned}$$

Hence, Ken will have to hold a short position in a risk-free portfolio of 60% and an extended position of 160% in the market portfolio.

Reading 149: Portfolio Construction

Q.2441 A fund manager makes the following comment:

“Active management should be easy with the right alphas.”

The manager is most likely:

- A. Incorrect because most active managers construct portfolios subject to certain constraints and therefore active management with the right alphas is not easy to achieve.
- B. Correct because active managers construct portfolios subject to various constraints, but these constraints do not make the portfolio less efficient.
- C. Incorrect because active managers construct portfolios without any constraints and, therefore, active management with the right alphas is not easy to achieve.
- D. Correct because most active managers do not take short positions and limit the amount of cash in the portfolio, thus making the portfolio more efficient.

The correct answer is **A**.

The statement made by the fund manager is indeed incorrect. Active management is a complex process that involves a lot of decision-making and strategizing. The 'right alphas' refer to the ability to generate excess returns over a benchmark index. However, achieving these 'right alphas' is not as easy as it sounds. Most active managers construct portfolios subject to certain constraints. These constraints could be restrictions on short positions, asset coverage, sector exposure, and many others. These constraints are put in place to manage risk and ensure diversification. Therefore, even with the 'right alphas', active management is not easy to achieve because of these constraints. The fund manager's statement oversimplifies the process and complexities involved in active management.

Choice B is incorrect. While it's true that active managers construct portfolios subject to various constraints, these constraints can indeed make the portfolio less efficient. Constraints such as limits on short selling, sector concentration, and tracking error can limit the manager's ability to fully exploit their alpha predictions.

Choice C is incorrect. The statement is false because active managers do construct portfolios with certain constraints. These constraints could be regulatory or self-imposed and they often make active management more challenging even with the right alphas.

Choice D is incorrect. Although some active managers may avoid taking short positions or limit cash holdings in their portfolio, this does not necessarily make the portfolio more efficient. In fact, these restrictions could potentially limit the manager's flexibility in responding to market changes and thus reduce efficiency.

Things to Remember

- Active management involves making investment decisions with the goal of

outperforming a specific benchmark index.

- Alpha is a measure of the active return on an investment, relative to a benchmark index.
 - Constraints in portfolio construction can include limits on sector exposure, position sizes, leverage, and risk management guidelines.
 - Short selling involves selling borrowed securities with the expectation of buying them back at a lower price in the future.
 - Cash holdings in a portfolio can impact liquidity, risk management, and overall portfolio performance.
-

Q.2442 Which of the following statements is (are) correct?

- I. Any complex portfolio construction process can be replaced by a process which first refines the alphas and uses a simple unconstrained mean/variance optimization to determine the active positions
 - II. Simple models are always better than complicated implementation schemes.
- A. Both I and II
 - B. None of the above
 - C. Only I
 - D. Only II

The correct answer is **C**.

The first statement is accurate. In portfolio management, regardless of the complexity of the portfolio construction process, it can be replaced by a process that first refines the alphas and then uses a simple unconstrained mean/variance optimization to determine the active positions. This is because the alpha refinement process helps to improve the accuracy of the expected returns, which is a critical input in the mean/variance optimization process. The mean/variance optimization process then uses these refined alphas to determine the optimal weights for the assets in the portfolio, thereby determining the active positions. This process is simple and straightforward, yet it is effective in constructing a portfolio that maximizes expected return for a given level of risk.

Choice A is incorrect. While Statement I is true, Statement II is not universally accurate. Simple models do not always outperform complex implementation schemes. The performance of a model depends on various factors such as the nature of the data, the complexity of the problem at hand, and how well the model captures the underlying patterns in the data.

Choice B is incorrect. This choice suggests that both statements are false which isn't correct as Statement I holds true.

Choice D is incorrect. As explained above, Statement II isn't universally accurate hence this choice which asserts only statement II to be true is incorrect.

Things to Remember

- Alpha refinement process in portfolio management
- Mean/variance optimization in portfolio construction
- Factors influencing model performance
- Complexity vs. simplicity in model implementation

Q.2443 Which of the approaches explained below can be used to make alphas industry-neutral?

- A. Calculate the capitalization-weighted alpha for each industry, and then subtract the industry average alpha from each alpha in that industry.
- B. Calculate the industry average alpha and subtract it from each alpha in that industry.
- C. Calculate the capitalization-weighted alpha for each industry, and then add the industry average alpha to each alpha in that industry.
- D. Calculate the industry average alpha and add it to each alpha in that industry.

The correct answer is **A**.

The process of making alphas industry-neutral involves several steps. First, the exposures are segregated industry-wise. This means that the investments or assets are categorized based on the industry they belong to. Next, the capitalization-weighted alpha for each industry is calculated. The capitalization-weighted alpha is a measure of the performance of a portfolio where each component stock is weighted according to its market capitalization. This gives a more accurate representation of the industry's performance as it takes into account the size of the companies within the industry. After this, the industry average alpha is worked out. The industry average alpha is the average of the alphas of all the companies within a particular industry. Finally, the industry average alpha is subtracted from each alpha in that industry. This adjustment ensures that the alphas are industry-neutral, meaning that they do not contain any active bets on industries. Therefore, choice A accurately describes the process of making alphas industry-neutral.

Choice B is incorrect. This method suggests subtracting the industry average alpha from each alpha in that industry. However, it does not take into account the capitalization-weighted alpha for each industry which is a crucial step in ensuring that portfolios contain no active bets on industries.

Choice C is incorrect. This method suggests adding the industry average alpha to each alpha in that industry after calculating the capitalization-weighted alpha for each industry. However, this would not make alphas industry-neutral as required but rather increase their value by a constant factor (the average), which does not align with the process described.

Choice D is incorrect. Similar to choice B, this method also fails to consider the capitalization-weighted alpha for each industry and instead proposes adding an arbitrary constant (the average) to all alphas within an industry, which would distort their relative values and fail to achieve an active bet-free portfolio.

Things to Remember

- Alpha: Alpha is a measure of the active return on an investment, relative to a benchmark index. It represents the performance of a portfolio after adjusting for

market risk.

- Industry Neutral: Making alphas industry-neutral involves removing any active bets on industries from a portfolio, ensuring that the performance is driven solely by stock selection and not industry allocation.
 - Capitalization-Weighted Alpha: This is a method of calculating alpha where each stock's contribution is weighted based on its market capitalization, giving more weight to larger companies.
 - Industry Average Alpha: The average alpha of all the companies within a particular industry, used to compare individual alphas within that industry.
 - Active Bets: Active bets refer to intentional deviations from a benchmark index in an attempt to outperform the market. Making alphas industry-neutral helps remove these active bets on industries.
-

Q.2444 All the following are procedures for refining alphas, EXCEPT:

- A. Scaling
- B. Trimming
- C. Transformation
- D. Neutralization

The correct answer is C.

Transformation is not a recognized procedure in the refinement of alphas. The process of refining alphas typically involves scaling, trimming, and neutralization. Scaling involves adjusting the alphas to have a mean of 0 and a standard deviation that is a product of the volatility and the Information Coefficient (IC). Trimming involves eliminating very large positive or negative alphas that could unduly influence the results. Neutralization involves adjusting the alphas to be neutral with regards to certain factors such as the benchmark, cash, industry, risk factor, etc. However, transformation, which involves changing the form or character of something, is not a recognized procedure in the refinement of alphas.

Choice A is incorrect. Scaling is a recognized procedure in the refinement of alphas. It involves adjusting the size of alpha estimates to ensure they are comparable across different strategies or time periods.

Choice B is incorrect. Trimming is also a recognized procedure in refining alphas. It involves removing extreme alpha estimates that may be due to outliers or errors, thereby improving the reliability of the results.

Choice D is incorrect. Neutralization is another recognized procedure in refining alphas. This process involves adjusting alpha estimates to remove any influence from common risk factors, ensuring that they represent unique sources of return.

Things to Remember

- Scaling involves adjusting alphas to have a mean of 0 and a standard deviation based on the volatility and Information Coefficient (IC).
 - Trimming involves removing extreme alpha estimates that could skew the results.
 - Neutralization adjusts alphas to be neutral with respect to certain factors like benchmarks, industry, or risk factors.
 - Alphas are refined to improve the accuracy and reliability of investment strategies.
 - Transformation, in the context of refining alphas, is not a recognized procedure and involves changing the form or character of something.
-

Q.2445 A portfolio manager is evaluating an active position in a stock with the following characteristics: Active position alpha: 0.6%

Active position beta: 1.3

Benchmark alpha: 0.020%

Benchmark beta: 1.0

The manager wants to neutralize the active position with respect to the benchmark beta. After the neutralization, what is the modified benchmark-neutral alpha of the active position?

- A. 0.54%
- B. 0.57%
- C. 0.62%
- D. 0.66%

The correct answer is **B**.

To compute the modified benchmark-neutral alpha, we need to subtract the product of the

benchmark alpha and the active position beta from the alpha of the active position. This neutralizes the active position's beta relative to the benchmark beta, ensuring that the active position doesn't make a bet on the market (and benchmark portfolio) returns.

Using the given information, we can calculate the modified benchmark-neutral alpha as follows:

$$\begin{aligned}\text{Modified benchmark-neutral alpha} &= \text{Active position alpha} - (\text{Benchmark alpha} \times \text{Active position beta}) \\ &= 0.6\% - (0.020\% \times 1.3) \\ &= 0.6\% - 0.026\% \\ &= 0.574\%\end{aligned}$$

Things to Remember

- Active position alpha represents the excess return of the portfolio manager's active position compared to the benchmark.
 - Active position beta measures the sensitivity of the portfolio manager's active position to movements in the benchmark.
 - Benchmark alpha is the expected return of the benchmark index.
 - Benchmark beta is a measure of the benchmark's systematic risk compared to the market.
 - Neutralizing the active position with respect to the benchmark beta involves adjusting the active position to have the same beta as the benchmark, effectively removing the market risk factor.
 - Modified benchmark-neutral alpha is the alpha of the active position after neutralizing its beta exposure to match that of the benchmark.
-

Q.2446 A fund manager makes the following comments:

- I. The correct way to compare transactions costs incurred on the annual rate of gain from alpha and the annual rate of loss from active risk is to amortize the transactions costs where the rate of amortization depends on the anticipated holding period
- II. The annualized transactions cost is the round trip cost divided by the holding period in years

Which of the fund manager's comments are accurate?

- A. Only I
- B. Only II
- C. Both I and II
- D. None of the above

The correct answer is **C**.

Both statements made by the fund manager are accurate. The first statement correctly identifies the need to amortize transaction costs in order to compare them with the annual rate of gain from alpha and the annual rate of loss from active risk. This is because transaction costs are incurred at the point of rebalancing a portfolio, while alpha and active risk are expected over the next year. Therefore, to make a fair comparison, transaction costs must be allocated over the one-year period, with the rate of amortization depending on the anticipated holding period. The second statement is also correct in stating that the annualized transaction cost is calculated by dividing the round trip cost by the holding period in years. This is a standard method of calculating annualized transaction costs, allowing for a consistent comparison across different holding periods.

Choice A is incorrect. The first statement made by the fund manager is indeed accurate. Transaction costs are typically amortized over the anticipated holding period, which means they are spread out over that time frame rather than being accounted for all at once. This allows for a more accurate comparison of transaction costs to the annual rate of gain from alpha and the annual rate of loss from active risk.

Choice B is incorrect. The second statement made by the fund manager is also correct. The annualized transaction cost can be calculated by dividing the round trip cost (i.e., the total cost of buying and then selling an asset) by the holding period in years. This provides a yearly estimate of transaction costs, which can be compared to other yearly rates such as those from alpha or active risk.

Choice D is incorrect. As explained above, both statements made by the fund manager are accurate, so this choice cannot be correct.

Things to Remember

- Transaction costs are incurred at the point of rebalancing a portfolio, while alpha and active risk are expected over a future period.
- Amortizing transaction costs involves spreading them out over the anticipated holding period for a fair comparison.
- Round trip cost is the total cost of buying and then selling an asset.
- Annualized transaction cost is calculated by dividing the round trip cost by the holding

period in years.

- Comparing transaction costs to annual rates of gain and loss helps in evaluating the overall performance and efficiency of a fund manager.
-

Q.2447 A fund manager desires an active risk of 5%. If the information ratio of the portfolio is 0.5, then the active risk aversion parameter is:

- A. 0.5
- B. 5
- C. 0.05
- D. 0.005

The correct answer is **C**.

The relation between active risk, information ratio, and active risk aversion is as follow:

$$\lambda_a = \frac{IR}{(2 * \psi_p)}$$
$$\lambda_a = \frac{0.5}{(2 * 5)} = 0.05$$

It is important to bear in mind that in the risk aversion equation above, the desired level of active risk is measured in percentage points rather than in decimal form. Remember that active risk is just another name for tracking error.

Q.2452 All the following are shortcomings of screen technique, EXCEPT:

- A. It ignores all information in the alphas apart from rankings.
- B. It does not protect against biases in the alphas.
- C. It may result in more risky portfolios.
- D. It is hard to code.

The correct answer is **D**.

The statement 'It is hard to code' is not a shortcoming of the screen technique. In fact, one of the advantages of the screen technique is its simplicity in implementation, which makes its coding relatively easy. The screen technique involves ranking stocks based on their alpha, a process that can be easily coded and automated. Therefore, the assertion that the screen technique is hard to code is incorrect and does not represent a limitation of this method.

Choice A is incorrect. The screen technique indeed ignores all information in the alphas apart from rankings. This is a limitation because it does not consider the magnitude of alpha, only its relative position.

Choice B is incorrect. The screen technique does not protect against biases in the alphas. If there are any systematic errors or biases in the calculation of alpha, these will be carried forward into the portfolio selection process.

Choice C is incorrect. The screen technique may result in more risky portfolios as it focuses on stocks with high alpha without considering their associated risks or correlations with other assets in the portfolio.

Things to Remember

- Screen technique involves ranking stocks based on their alpha, ignoring all information in the alphas apart from rankings.
- It does not protect against biases in the alphas, so any systematic errors or biases in alpha calculations can affect the portfolio selection process.
- Using the screen technique may lead to more risky portfolios as it focuses solely on high alpha stocks without considering associated risks or correlations.
- One advantage of the screen technique is its simplicity in implementation, making coding relatively easy and automation feasible.

Q.2455 Quadratic programming is a type of mathematical optimization that deals with certain constraints. Which of the following options does NOT accurately describe the characteristics or considerations of quadratic programming?

- A. It includes the linear program as a special case.
- B. It considers only two elements – risk and transactions costs.
- C. It requires a large number of inputs.
- D. The use of a large number of inputs increase the noise in the portfolio construction process.

The correct answer is **B**.

Quadratic programming explicitly considers each of the three elements – risk, transaction costs, and alpha.

Things to Remember

- Quadratic programming is a type of mathematical optimization that deals with quadratic objective functions and linear constraints.
- It is commonly used in portfolio optimization to minimize risk while achieving a certain level of return.
- Quadratic programming can handle non-linear relationships between variables, making it more flexible than linear programming.
- It is important to consider the number of inputs in quadratic programming as a large number of inputs can lead to computational challenges and increased noise in the optimization process.
- Transaction costs are an important consideration in portfolio optimization as they can impact the overall performance of the portfolio.

Q.2457 Which of the following statements best describes the primary purpose of scaling alpha in the context of portfolio optimization?

- A. To convert the alpha values of different securities into a standardized measure for comparison purposes.

- B. To adjust the alpha values of portfolio holdings based on their historical volatility.
- C. To modify the portfolio's overall risk by increasing or decreasing the portfolio's alpha.
- D. To account for the impact of transaction costs on the portfolio's alpha generation.

The correct answer is **A**.

Scaling alpha is a technique used in portfolio optimization to standardize the alpha values of different securities. Alpha, in the context of finance, is a measure of the excess return or active return of an investment or a portfolio. However, the alpha values of different securities can vary widely due to differences in their risk-return profiles, market conditions, and other factors. This can make it difficult for portfolio managers to compare the performance of different securities on an equal footing. By scaling alpha, portfolio managers can convert these disparate alpha values into a standardized measure. This allows them to compare the risk-adjusted performance of various securities more effectively and make more informed decisions about their portfolio allocations. The standardized measure provided by scaling alpha also enables a more accurate assessment of the portfolio's risk-return profile, which is crucial for optimizing the portfolio's composition.

Choice B is incorrect. While adjusting the alpha values of portfolio holdings based on their historical volatility can be a part of risk management, it is not the primary objective of scaling alpha. Scaling alpha primarily aims to standardize the measure of different securities' alphas for comparison purposes, not to adjust them based on volatility.

Choice C is incorrect. Modifying the portfolio's overall risk by increasing or decreasing the portfolio's alpha might be a result of scaling alpha but it is not its primary purpose. The main goal of scaling alpha is to convert disparate alphas into a standardized measure for easier comparison and analysis.

Choice D is incorrect. Accounting for transaction costs in generating portfolio's alpha could be an important consideration in managing portfolios, but it does not represent the main objective behind scaling alpha. The process primarily focuses on standardizing alphas across different securities for comparative analysis.

Things to Remember

- Alpha: A measure of the excess return or active return of an investment or a portfolio.
- Portfolio Optimization: The process of constructing a portfolio of assets that maximizes returns for a given level of risk.
- Risk-Adjusted Performance: Evaluating an investment's performance by considering the level of risk taken to achieve that return.
- Volatility: A measure of the dispersion of returns for a given security or market index.

- Transaction Costs: The costs associated with buying or selling securities within a portfolio.
-

Q.2458 Dispersion is caused by all the following, EXCEPT:

- A. Different betas and factor exposures.
- B. The different number of stocks the portfolios have.
- C. Identical holdings in each portfolio.
- D. The overall number of portfolio under management.

The correct answer is **C**.

Dispersion in portfolio returns is caused by the differences in the holdings of each portfolio. If all portfolios have identical holdings, then the returns from all these portfolios will also be identical, eliminating any dispersion. This is because the performance of each portfolio will be influenced by the same set of assets, leading to the same returns. Therefore, identical holdings in each portfolio do not cause dispersion.

Choice A is incorrect. Different betas and factor exposures can contribute to dispersion as they reflect the sensitivity of a portfolio's returns to changes in the market or specific factors. Portfolios with different betas and factor exposures will respond differently to market movements, leading to variation in returns.

Choice B is incorrect. The number of stocks a portfolio has can also influence dispersion. Portfolios with more stocks are likely to have more diversified risk, which could lead to less variation in returns compared to portfolios with fewer stocks.

Choice D is incorrect. The overall number of portfolios under management does not directly influence the dispersion of returns among those portfolios. It's not the quantity but rather the composition and management strategy of each portfolio that contributes to dispersion.

Things to Remember

- Dispersion in portfolio returns is a measure of the variability or spread of returns among different portfolios.
- Betas and factor exposures reflect the sensitivity of a portfolio's returns to market movements and specific factors, impacting dispersion.
- Diversification plays a key role in reducing dispersion as it helps spread risk across

different assets.

- Active management strategies can also contribute to dispersion as different managers may have varying investment approaches and decisions.
 - Monitoring and managing dispersion is important for risk management and performance evaluation in portfolio management.
-

Q.2459 Why do managers find certain levels of dispersion optimal?

- A. It results in higher average returns.
- B. It remains constant over time.
- C. Transactions cost incurred to reduce dispersion is very low.
- D. It does not affect the average returns.

The correct answer is **A**.

Dispersion in the context of financial management refers to the degree of variability or spread in a set of values. In the case of securities, it is often measured by the standard deviation of returns. A higher standard deviation indicates a greater dispersion of returns, which in turn implies a higher level of risk associated with the investment. According to modern portfolio theory (MPT), this volatility creates risk that is associated with the degree of dispersion of returns around the average. Therefore, a higher dispersion of returns can result in higher average returns, as investors are compensated for taking on more risk. This is why managers often find certain levels of dispersion to be optimal - it allows for the potential of higher average returns, albeit at a higher level of risk.

Choice B is incorrect. The level of dispersion does not remain constant over time. It varies based on the volatility of the market, changes in economic conditions, and other factors that influence the returns on an investment.

Choice C is incorrect. The transaction cost incurred to reduce dispersion can be significant depending on the size and frequency of transactions required to rebalance a portfolio. Therefore, it's not accurate to say that these costs are always very low.

Choice D is incorrect. Dispersion does affect average returns as it represents the risk associated with an investment. Higher dispersion indicates higher risk which could potentially lead to higher returns but also greater potential for loss.

Things to Remember

- Dispersion is often measured by the standard deviation of returns in the context of

financial management.

- Modern portfolio theory (MPT) suggests that higher dispersion of returns can lead to higher average returns, as investors are compensated for taking on more risk.
 - The level of dispersion is not constant and can vary based on market volatility and economic conditions.
 - Transaction costs incurred to reduce dispersion can be significant depending on the portfolio rebalancing required.
 - Dispersion does affect average returns as it represents the risk associated with an investment.
-

Q.2744 Which of these is not a method of refining alphas?

- A. Trimming
- B. Interpolation
- C. Scaling
- D. Neutralization

The correct answer is **B**.

The three methods of refining alphas include trimming, scaling and neutralization. Trimming involves removing extreme values while scaling is performed to make the alphas have the proper scale for the portfolio construction process. The removal of biases from alphas is known as neutralization.

Things to Remember

- Alpha: Alpha is a measure of the active return on an investment, the return in excess of a benchmark index. It is often used to assess the skill of a portfolio manager.
 - Trimming: Trimming involves removing extreme values or outliers from the alpha signal to improve its quality and reduce noise.
 - Scaling: Scaling is the process of adjusting the magnitude of alphas to ensure they are suitable for the portfolio construction process.
 - Neutralization: Neutralization refers to the process of removing biases from alphas, such as sector or style biases, to ensure that the alpha signal is truly independent and reflects stock selection skill.
-

Q.2745 A portfolio manager has a specific approach to building a portfolio. The manager first ranks stocks based on their alpha, then selects the top ten stocks from this ranking. These selected stocks are then used to create a portfolio where each stock is given equal weight. Which portfolio construction technique is the manager most likely using?

- A. Stratification
- B. Linear programming
- C. Quadratic programming

D. Screens

The correct answer is **D**.

There are four commonly used classes of portfolio construction techniques. These include,

- Screens: which involves simple ranking of the assets and composing an equal-weighted or capitalization-weighted portfolio.
- Stratification: builds on the screens method by ensuring that each category of assets is present in the portfolio.
- Linear programming: is used to choose a portfolio that closely resembles the benchmark portfolio using stratification based on industry characteristics.
- Quadratic programming: this technique models alpha, risk and transaction costs and can also incorporate constraints.

Things to Remember

- Alpha: A measure of the active return on an investment, the return in excess of a benchmark index.
 - Portfolio Construction: The process of selecting the appropriate mix of assets to achieve the investment objectives within the constraints of risk tolerance, time horizon, and investment goals.
 - Equal Weighting: A portfolio construction technique where each asset in the portfolio is given the same weight or allocation.
 - Capitalization Weighting: A portfolio construction technique where the weights of assets are based on their market capitalization.
 - Benchmark Portfolio: A portfolio that represents a standard against which the performance of an investment portfolio can be measured.
-

Reading 150: Portfolio Risk: Analytical Methods

Q.2462 Consider a hedge fund which is \$1 billion long in corporate bonds and \$1 billion short in Treasury bonds. Which of the following statements is (are) correct?

- I. If zero correlation exists between corporate and treasury bonds, then the portfolio VaR is zero
 - II. If the correlation between corporate and Treasury bonds is equal to -1, the risk will be amplified
- A. Only I
- B. Only II
- C. Both I and II
- D. Neither I nor II

The correct answer is **B**.

Statement 1 is incorrect. If the correlation between two assets is zero, then the portfolio VaR is given by the square root of the sum of the squared individual VaRs. i.e

$$\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$$

Statement II is correct. A correlation of -1 indicates a perfect negative correlation between corporate and Treasury bonds. A correlation of -1 normally implies that the assets move in opposite directions, which can potentially provide some level of diversification. In this case, however, since the hedge fund comprises \$1 billion long corporate bonds and short in Treasury bonds, there are no diversification benefits, and as such, risk is amplified.

Note that the portfolio VaR equals the undiversified VaR (the sum of individual VaRs) only if the correlations are unity, and that short sales are not allowed.

Things to Remember

- Correlation measures the degree to which two assets move in relation to each other. A correlation of -1 indicates a perfect negative correlation, 0 indicates no correlation, and +1 indicates a perfect positive correlation.
- Portfolio VaR is the measure of potential loss in the value of a portfolio over a specified time horizon for a given confidence level. It takes into account the individual VaRs of the assets in the portfolio as well as their correlations.

- When assets have a correlation of -1, they move in opposite directions. While this can provide diversification benefits in some cases, in the context of a hedge fund with a long position in corporate bonds and a short position in Treasury bonds, the risk can be amplified due to the specific positions taken.
 - Short selling involves selling an asset that the seller does not own. In the context of the hedge fund scenario, being short in Treasury bonds means the fund has sold Treasury bonds it does not own, with the expectation of buying them back at a lower price in the future.
-

Q.2463 A portfolio manager intends to calculate the variance of his portfolio which consists of n different securities. The number of different covariance terms required to compute the portfolio variance is:

- A. $\frac{n(n-1)}{2}$
- B. $\frac{n(n+1)}{2}$
- C. $\frac{n}{2}$
- D. $\frac{(n+1)}{2}$

The correct answer is **A**.

The number of ways in which 2 numbers can be selected out of n numbers =

$${}^nC_2 = \frac{n!}{((n-2)! \times 2!)} = \frac{n(n-1)}{2}$$

When computing portfolio variance, the covariance for each pair of securities is required. The number of pairs of securities that can be formed out of n securities is computed by the expression $\frac{n(n-1)}{2}$.

Things to Remember

- Portfolio variance is a measure of the dispersion of returns of a portfolio. It takes into account the weights of each security in the portfolio as well as the covariance between the securities.

- Covariance is a measure of how two variables change together. In the context of portfolio management, it measures the relationship between the returns of two securities.
 - When calculating the variance of a portfolio with n different securities, the covariance terms between each pair of securities are needed to fully capture the risk of the portfolio.
 - The formula to calculate the number of ways to select 2 items out of n items is given by
$${}_n C_2 = \frac{n!}{(n-2)! \times 2!} = \frac{n(n-1)}{2}.$$
 - Having a larger number of covariance terms in the portfolio variance calculation allows for a more accurate assessment of the overall risk and diversification benefits of the portfolio.
-

Q.2464 A fund manager is studying the relationship between covariance and diversification. Which of the following statements would be correct for the manager to make in this context?

- A. A higher covariance implies a stronger relationship between the assets' returns, indicating more diversification benefits.
- B. We can still achieve diversification with positive covariance.
- C. Diversification is not possible with positive covariance.
- D. Covariance is not relevant to diversification.

The correct answer is **B**.

While it is generally more desirable to have assets with negative or low covariance for better diversification benefits, positive covariance does not necessarily mean that diversification is impossible. Even with positive covariance, it is still possible to achieve some level of diversification by carefully selecting assets with different risk-return characteristics and correlation structures.

A is incorrect: The higher the covariance, the lesser the diversification benefits.

C is incorrect. Diversification is indeed possible with positive covariance, although its benefits might be lower compared to assets with negative or low covariance.

D is incorrect. Covariance is a crucial factor in portfolio diversification. It helps investors

understand how the returns of different assets move concerning each other, enabling them to construct a diversified portfolio that balances risk and return effectively.

Things to Remember

- Covariance measures the relationship between the returns of two assets. A positive covariance indicates that the assets tend to move in the same direction, while a negative covariance indicates they move in opposite directions.
 - Diversification is a risk management technique that involves spreading investments across different assets to reduce overall risk.
 - Correlation is another important concept related to covariance and diversification. It measures the strength and direction of the linear relationship between two variables, such as asset returns.
 - Efficient frontier is a concept in portfolio theory that represents a set of optimal portfolios that offer the highest expected return for a given level of risk or the lowest risk for a given level of expected return.
 - Modern Portfolio Theory (MPT) is a framework for constructing efficient portfolios that maximize returns for a given level of risk, taking into account the correlation and covariance between assets.
-

Q.2465 Which of the following factors reduce(s) portfolio risk?

- I. A large number of assets
 - II. Low correlations among assets
-
- A. I only
 - B. II only
 - C. Both I and II
 - D. None of the above

The correct answer is **C**.

Both a large number of assets and low correlations among assets reduce portfolio risk. The concept of diversification in portfolio management is based on the idea that spreading investments across a wide range of assets can reduce the risk associated with any single asset. The more assets in a portfolio, the less impact any single asset's performance will have on the overall portfolio. This is because the individual asset risks are spread out and hence, the overall portfolio risk is reduced. This is known as the diversification effect.

Similarly, low correlations among assets also contribute to reducing portfolio risk. If the assets in a portfolio are perfectly correlated, they would all move in the same direction at the same time. This means that if one asset performs poorly, all assets are likely to perform poorly, leading to a high portfolio risk. On the other hand, if the assets are not correlated, they are likely to move independently of each other. This means that poor performance by one asset may be offset by good performance by another asset, leading to a lower portfolio risk. Therefore, both a large number of assets and low correlations among assets contribute to reducing portfolio risk.

Choice A is incorrect. While it's true that a large number of assets can help in diversification, it alone does not guarantee a reduction in portfolio risk. The type of assets and their individual risks also play a significant role. Therefore, having a large number of assets is not the only factor that reduces portfolio risk.

Choice B is incorrect. Low correlations among assets can indeed reduce portfolio risk as it ensures that not all assets will move in the same direction at the same time. However, this alone cannot ensure reduced portfolio risk as the individual risks associated with each asset also matter.

Choice D is incorrect. As explained above, both factors - a large number of assets and low correlations among them - are important for reducing portfolio risk.

Things to Remember

- Asset Diversification: Spreading investments across a wide range of assets to reduce risk.
- Correlation: Measure of how assets move in relation to each other, with low correlation indicating independence.
- Portfolio Risk: The overall risk associated with a portfolio of investments.
- Diversification Effect: The reduction in risk achieved by spreading investments across different assets.
- Individual Asset Risk: The risk associated with each individual asset in a portfolio.

Q.2466 Consider the following expressions in relation to a portfolio which consists of two assets:

- I. $\text{VaR}_p < \text{VaR}_1 + \text{VaR}_2$
- II. $\text{VaR}_p = \text{VaR}_1 + \text{VaR}_2$

Which of the following statements is correct?

- A. When assets are perfectly correlated, then expression I is correct.
- B. When assets are perfectly correlated, then expression II is correct.
- C. All of the above.
- D. None of the above.

The correct answer is **B**.

When assets are perfectly correlated, then expression II is correct. This is because the Value at Risk (VaR) of a portfolio is calculated using the formula: $\text{VaR}_p = \sqrt{(\text{VaR}_1^2 + \text{VaR}_2^2 + 2 * \rho * \text{VaR}_1 * \text{VaR}_2)}$. In this formula, ρ represents the correlation between the two assets. When the assets are perfectly correlated, ρ equals 1. Substituting $\rho = 1$ into the formula gives us $\text{VaR}_p = \text{VaR}_1 + \text{VaR}_2$, which is expression II. Therefore, when the assets in a portfolio are perfectly correlated, the VaR of the portfolio is equal to the sum of the VaRs of the individual assets.

Choice A is incorrect. When assets are perfectly correlated, expression I is not correct. In fact, when the correlation between two assets is perfect (either +1 or -1), the VaR of the portfolio equals to the sum of individual VaRs, which aligns with expression II and not I.

Choice C is incorrect. As explained above, only one of these statements (expression II) holds true when assets are perfectly correlated. Therefore, it's not accurate to say that all of the above statements are correct.

Choice D is incorrect. This choice suggests that none of the provided options accurately reflect the relationship between these expressions and asset correlation in a portfolio. However, as we've established in Choice B explanation, when assets are perfectly correlated then expression II does hold true.

Things to Remember

- Value at Risk (VaR) is a measure used to assess the potential loss in value of a portfolio over a specific time period and with a certain level of confidence.
- Correlation between assets in a portfolio plays a crucial role in determining the overall risk and diversification benefits of the portfolio.
- Perfect correlation (correlation coefficient of +1) between assets implies that they move in the same direction all the time, while perfect negative correlation (correlation

coefficient of -1) implies they move in opposite directions.

- When assets are perfectly correlated, the diversification benefits are reduced as the movements of the assets are highly synchronized.
 - When assets are perfectly negatively correlated, the diversification benefits are maximized as the movements of the assets offset each other.
-

Q.2467 A portfolio manager holds a portfolio which consists only of IT stocks. Borrowing some knowledge of the CAPM, he argues that since β measures the contribution of each security to total portfolio risk, he has to include securities with low β in his portfolio in order to reduce his portfolio risk.

Is the manager's assertion correct?

- A. Yes because β measures the contribution of one security to total portfolio risk.
- B. No because the portfolio held by the manager is not well diversified, β is not an appropriate risk parameter.
- C. No because β measures non-systematic risk.
- D. Yes because β measures systematic risk.

The correct answer is **B**.

According to the capital asset pricing model (CAPM), well-diversified investors only need to be compensated for the systematic risk of securities relative to the market. β measures the systematic risk of a security. As the portfolio held by the manager is not well diversified, β is not an accurate, reliable risk parameter.

Things to Remember

- Systematic risk is the risk that is inherent to the entire market or an entire market segment. It cannot be diversified away.
- Non-systematic risk, also known as idiosyncratic risk, is the risk that is specific to an individual security and can be diversified away by holding a diversified portfolio.
- The beta coefficient measures the sensitivity of a security's returns to the market returns. A beta of 1 indicates that the security's price will move with the market, while a beta greater than 1 indicates higher volatility and a beta less than 1 indicates lower volatility than the market.
- Well-diversified portfolios can reduce non-systematic risk through diversification, but systematic risk, as measured by beta, remains.

Q.2468 An investor wants to lower the portfolio VaR and has the choice to reduce his position by a fixed amount of \$100,000. How best should the investor go about it?

- A. He must rank all marginal VaR numbers and select the asset with the smallest VaR because selecting that asset will have the largest hedging effect.
- B. He must rank all marginal VaR numbers and select the asset with the largest VaR because selecting that asset will have the smallest hedging effect.
- C. He must rank all marginal VaR numbers and select the asset with the largest marginal VaR because selecting that asset will have the largest hedging effect.
- D. Not possible to determine

The correct answer is **C**.

The investor should rank all marginal VaR numbers and select the asset with the largest marginal VaR because selecting that asset will have the largest hedging effect. Marginal VaR is the change in portfolio VaR resulting from taking an additional dollar of exposure to a given component. In this context, the asset with the largest VaR would contribute the most to the overall portfolio risk. Therefore, reducing the position in this asset would have the largest impact on reducing the portfolio's overall risk. This strategy is based on the principle that the assets that contribute the most to portfolio risk should be the first ones to be reduced when the goal is to lower portfolio risk.

Choice A is incorrect. Selecting the asset with the smallest VaR will not have the largest hedging effect. In fact, it would be less effective in reducing portfolio risk because assets with smaller VaR contribute less to overall portfolio risk.

Choice B is incorrect. While it's true that an asset with a larger VaR contributes more to overall portfolio risk, selecting such an asset will not have the smallest hedging effect. On contrary, reducing position in this asset would result in a larger reduction of portfolio risk.

Choice D is incorrect. As explained above, ranking all marginal VaR numbers and selecting the asset with the largest VaR for reduction will indeed have the largest hedging effect and thus effectively decrease portfolio's Value at Risk (VaR).

Things to Remember

- Value at Risk (VaR) is a measure used to assess the potential loss in value of a portfolio due to market risk over a specified time period and confidence level.
- Marginal VaR is the change in portfolio VaR resulting from taking an additional dollar of exposure to a given component.
- Hedging involves taking an offsetting position in a related security to reduce the risk of adverse price movements in an asset.
- Portfolio risk management involves analyzing the contribution of each asset to the

overall risk of the portfolio and making strategic decisions to reduce risk.

Q.2469 A portfolio manager intends to add a new position to the portfolio. He argues that the use of either marginal or incremental VaR to measure the change in risk will provide the same result, as both tools measure the change in VaR due to the addition of a new position.

Which of the following statements is accurate?

- A. The portfolio manager is correct.
- B. The portfolio manager must use marginal VaR to compute the change in VaR since, in this case, VaR changes in a linear fashion.
- C. The portfolio manager must use incremental VaR to compute the change in VaR since, in this case, VaR changes in a non-linear fashion.
- D. The portfolio manager must use marginal VaR to compute the change in VaR since, in this case, VaR changes in a non-linear fashion.

The correct answer is **C**.

The portfolio manager should use incremental VaR to compute the change in VaR in this scenario because VaR changes in a non-linear fashion. Incremental VaR is the change in VaR from the addition of a new position in a portfolio. Since it applies to an entire position, it is generally larger than marginal VaR and may include nonlinear relationships. For this reason, incremental VaR can be viewed as the more accurate VaR measure especially when the anticipated positions are large in magnitude and size. In the question, the manager intends to change the composition of his portfolio. That implicitly means the changes would be significant, and the change in VaR would be best measured using the incremental VaR.

Choice A is incorrect. The portfolio manager's assertion that the change in VaR due to the addition of a new asset can be measured equally well by either marginal or incremental VaR is not accurate. Marginal and incremental VaR measure different aspects of risk and are used in different contexts.

Choice B is incorrect. The statement suggests that marginal VaR should be used because it assumes changes in VaR are linear, which is not always true. Marginal VaR measures the effect on overall portfolio risk when a small amount of an asset is added to the portfolio, assuming all other factors remain constant. However, this does not necessarily mean that changes in VaR will always follow a linear pattern.

Choice D is incorrect. This choice incorrectly suggests that marginal VaR should be used when changes in Value at Risk (VaR) occur non-linearly. In reality, incremental Value at Risk (IVaR), rather than marginal Value at Risk (MVaR), would be more appropriate for measuring non-linear changes as it considers the change in total portfolio risk when an entire position or asset class is added or removed from the portfolio.

Things to Remember

- Value at Risk (VaR) is a measure used to quantify the level of financial risk within a portfolio over a specific time frame.
 - Marginal VaR measures the change in overall portfolio risk when a small amount of an asset is added to the portfolio, assuming all other factors remain constant.
 - Incremental VaR measures the change in VaR from the addition of a new position in a portfolio and is generally larger than marginal VaR, especially in cases with nonlinear relationships.
 - When making significant changes to a portfolio composition, such as adding a new position, incremental VaR is considered a more accurate measure of the change in risk compared to marginal VaR.
-

Q.2470 A portfolio manager intends to decompose the portfolio risk in order to compute the contribution of each asset to the overall portfolio VaR. The manager argues that taking individual VaRs, adding them up, and computing their individual percentages will provide the component VaR of each asset.

Which of the following statements is accurate?

- A. The manager is correct.
- B. The manager is incorrect as the method adopted completely ignores the diversification effects.
- C. The manager is incorrect as the method adopted assumes a non-linear VaR.
- D. The manager is incorrect as the method adopted assumes a linear VaR.

The correct answer is **B**.

The manager's approach is incorrect because it completely disregards the effects of diversification. In a portfolio, assets are usually correlated with each other. This correlation leads to a diversification effect when these assets are combined in a portfolio. Diversification effect refers to the reduction in risk that can be achieved by combining different kinds of investments within a portfolio. The risk reduction is possible because different investments will respond differently to the same event; while some investments might suffer, others might benefit. Therefore, the total portfolio VaR cannot be calculated by simply adding up the individual VaRs of the assets. Doing so would completely ignore the diversification effect, which is a crucial aspect of portfolio risk management. Therefore, the manager's proposed method is flawed and does not accurately reflect the true risk of the portfolio.

Choice A is incorrect. The manager's proposed method of calculating individual VaRs and summing them to determine their individual percentages does not accurately reflect the component VaR of each asset. This approach fails to account for the diversification effects that can significantly impact portfolio risk.

Choice C is incorrect. The statement made by the manager does not necessarily assume a non-linear VaR. Non-linear VaR would imply that changes in portfolio value do not correspond proportionally to changes in risk factors, which is not suggested by the manager's method.

Choice D is incorrect. While it may seem like the manager's approach assumes a linear VaR, this isn't entirely accurate either. A linear VaR would suggest that changes in portfolio value correspond directly with changes in risk factors, but this isn't necessarily implied by simply summing individual asset VaRs.

Things to Remember

- Portfolio risk decomposition
 - Portfolio VaR (Value at Risk)
 - Correlation among assets in a portfolio
 - Diversification effect in portfolio risk management
 - Linear VaR vs. Non-linear VaR
-

Q.2471 A fund manager wants to create a hedged position using two securities – A and B – and, upon further examination, it is observed that both the securities are negatively correlated. To create a hedged position, the manager must:

- A. Short A and go long on B.
- B. Long A and go short on B.
- C. Go long on both A and B.
- D. None of the above.

The correct answer is **C**.

Going long on both A and B is the correct strategy when the securities are negatively correlated. In the context of financial markets, correlation is a statistical measure that expresses the extent to which two securities move in relation to each other. A negative correlation implies that the securities move in opposite directions. If one security increases in value, the other security decreases in value, and vice versa. Therefore, by going long on both securities, the fund manager

can create a hedged position. This is because the gain from one security will offset the loss from the other, thereby reducing the overall risk of the portfolio. This strategy is particularly effective in volatile markets where prices can fluctuate widely in a short period of time. By going long on both securities, the fund manager can take advantage of price movements in both directions, thereby maximizing potential returns while minimizing potential losses.

Choice A is incorrect. Shorting security A and going long on B would not establish a hedged position in this case. This strategy would be more appropriate if the securities were positively correlated, as gains from one could offset losses from the other.

Choice B is incorrect. Going long on security A and short on B also does not create a hedged position given the negative correlation between the two securities. Similar to choice A, this strategy would be more suitable for positively correlated securities.

Choice D is incorrect. The statement "None of the above" implies that none of the strategies listed can establish a hedged position, which contradicts with our correct answer (choice C). Therefore, it cannot be correct.

Things to Remember

- Correlation is a statistical measure that expresses the extent to which two securities move in relation to each other.
 - Negative correlation implies that the securities move in opposite directions.
 - Going long on a security means buying it with the expectation that its price will rise.
 - Shorting a security involves borrowing and selling it with the expectation that its price will decrease, allowing the short seller to buy it back at a lower price.
 - A hedged position is a strategy that aims to reduce risk by taking offsetting positions in different securities or assets.
-

Q.2472 A fund manager has the following currency portfolio:

Asset	W_1	Marginal VaR
CAD	66.67%	0.0528
EUR	33.33%	0.1521
Total	100.00%	

To minimize portfolio risk, the fund manager must:

- A. Cut CAD position and add EUR position
- B. Cut EUR position and add CAD position
- C. Add both CAD and EUR positions
- D. None of the above

The correct answer is **B**.

The fund manager should cut the EUR position and add the CAD position. The reason for this is that the marginal Value at Risk (VaR) for EUR is higher than the marginal VaR for CAD. Marginal VaR is a measure of the risk of loss in the value of a portfolio due to a small change in a particular asset's position. In this case, the marginal VaR for EUR is 0.1521, while the marginal VaR for CAD is 0.0528. This means that a small change in the EUR position would result in a larger potential loss compared to a similar change in the CAD position. Therefore, by reducing the EUR position and increasing the CAD position, the fund manager can reduce the overall risk of the portfolio. This strategy is based on the principle of diversification, which aims to reduce risk by spreading investments across different assets. By allocating more of the portfolio to the asset with the lower marginal VaR (CAD), the fund manager can achieve a more balanced and less risky portfolio.

Choice A is incorrect. Cutting the CAD position and adding to the EUR position would increase the portfolio's risk, not decrease it. This is because the marginal VaR of EUR (0.1521) is higher than that of CAD (0.0528), meaning that each additional unit of EUR added to the portfolio increases its risk more than an equivalent unit of CAD.

Choice C is incorrect. Adding both CAD and EUR positions would also increase the portfolio's risk, as both assets have a positive marginal VaR. The objective here is to minimize risk, not increase it.

Choice D is incorrect. Doing nothing or adopting a strategy other than cutting down on higher-risk assets (in this case, EUR) and increasing lower-risk ones (CAD) will not help in minimizing portfolio risk.

Q.2473 Consider the following currency portfolio:

Asset	E_i	W_i	β_i	$\frac{E_i}{\beta_i}$
CAD	8.00%	66.67%	0.615	0.1301
EUR	5.00%	33.33%	1.770	0.0282
Total		100%		
Diversified VaR		\$257,738		
Standard Dev		15.62%		
Expected return		?		
Sharpe Ratio		?		

The expected return of the currency portfolio is:

- A. 0.06
- B. 0.07
- C. 0.08
- D. 0.09

The correct answer is **B**.

Expected return is equal to:

$$E_i = \sum_{i=1}^k w_i R_i$$

$$E_i = 8\% \times 66.67\% + 5\% \times 33.33\% = 7\%$$

Q.2474 Consider the following currency portfolio:

Asset	E_i	W_i	β_i	$\frac{E_i}{\beta_i}$
CAD	8.00%	66.67%	0.615	0.1301
EUR	5.00%	33.33%	1.770	0.0282
Total		100%		
Diversified VaR		\$257,738		
Standard Dev		15.62%		
Expected return		?		
Sharpe Ratio		?		

Given that the current risk free rate is 2%, which of the following is closest to the Sharpe ratio of the portfolio?

- A. 0.384
- B. 0.32
- C. 0.564
- D. 0.621

The correct answer is **B**.

The expected return is equal to:

$$\sum_{i=1}^k w_i R_i = 8\% \times 66.67\% + 5\% \times 33.33\% = 7\%$$

The Sharpe Ratio is:

$$SR = \frac{E_i - r}{s} = \frac{7\% - 2\%}{15.62\%} = 0.32$$

Where,

SR= Sharpe ratio

r=Current risk-free rate

s=Standard deviation

Q.2476 XYZ Bank has an advanced credit risk management system that's capable of pricing

derivatives and supporting the generation of VaR reports. The bank has a proper VaR system in place with daily reporting to the Risk Management Committee (RMC). The bank has delegated joint responsibilities for front- and back-office functions to its few employees. In light of this information, which of the following is most likely true?

- A. Since the bank has an advanced credit risk management system in place, its risk management practices are satisfactory.
- B. Since the bank has the capability to generate VaR reports, its risk management practices are satisfactory.
- C. The reporting frequency to the RMC must be increased.
- D. The joint responsibility for front- and back-office functions makes the bank more vulnerable to trading frauds.

The correct answer is **D**.

The joint responsibility for front- and back-office functions makes the bank more vulnerable to trading frauds. This is because the front office is typically responsible for generating business (i.e., making trades), while the back office is responsible for the administration and support functions, such as trade confirmation and settlement. When these two functions are combined, it can create a conflict of interest and increase the risk of fraudulent activities. This is because the same person who is making the trades is also responsible for confirming and settling those trades. This lack of segregation of duties can allow fraudulent activities to go undetected, as the person could potentially manipulate the records to hide trading losses. This was the case with Barings Bank, where Nick Leeson was given joint responsibility for front- and back-office functions, which allowed him to hide trading losses in a special 'error' account.

Choice A is incorrect. While having an advanced credit risk management system is a positive aspect, it does not necessarily mean that the bank's risk management practices are satisfactory. The effectiveness of the system depends on how well it is implemented and used in practice. It also depends on other factors such as the quality of data input into the system, the skills and knowledge of those using it, and whether there are adequate checks and balances in place.

Choice B is incorrect. The ability to generate VaR reports alone does not guarantee satisfactory risk management practices. VaR reports provide a measure of potential losses under normal market conditions but they do not capture tail risks or extreme events which can lead to significant losses for a bank. Furthermore, these reports need to be interpreted correctly by skilled personnel for effective risk management.

Choice C is incorrect. The frequency of reporting to RMC should be determined based on various factors such as complexity of operations, volatility in markets etc., rather than being increased arbitrarily without considering these factors.

Things to Remember

- VaR (Value at Risk) is a widely used risk management measure that quantifies the potential loss in value of a risky asset or portfolio over a specified time period under

normal market conditions.

- Segregation of duties is a key internal control principle that aims to prevent fraud and errors by dividing responsibilities among different individuals or departments.
 - Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events.
 - Market risk is the risk of losses in on and off-balance sheet positions arising from adverse movements in market prices.
 - Credit risk is the risk of loss due to a borrower's failure to make payments on any type of debt.
-

Q.2477 Portfolio X has a VaR of \$100,000. The portfolio is made up of 5 assets, A, B, C, D, and E. These assets are equally weighted. If Asset C has a β of 1.5, then the component VaR of Asset C is equal to:

- A. \$30,000
- B. \$20,000
- C. \$40,000
- D. \$10,000

The correct answer is **A**.

$$\text{Component VaR}_C = \text{VaR}_p \times \beta_c \times w_c = \$100,000 \times 1.5 \times \left(\frac{1}{5}\right) = \$30,000$$

Q.2478 Portfolio X has a VaR of \$100,000. The portfolio is made up of 5 assets, A, B, C, D, and E each valued at \$2,000,000. These assets are equally weighted. If Asset C has a β of 1.5, then the marginal VaR of Asset C is equal to:

- A. 0.075
- B. 0.015
- C. 0.02
- D. 0.025

The correct answer is **B**.

$$\begin{aligned}\text{Marginal VaR}_C &= \frac{\text{VaR}_p}{\text{Value of the portfolio}} \times \beta_C \\ &= \frac{100,000}{5 \times 2,000,000} \times 1.5 \\ &= 0.015\end{aligned}$$

Q.2746 The VaR of a portfolio will be maximized when the pairwise correlations between the returns of the underlying assets are:

- A. Perfectly positive
- B. Perfectly negative
- C. Uncorrelated
- D. Either positive or negative but not perfectly negative or positive

The correct answer is **A**.

The VaR of a portfolio will be maximized when the pairwise correlations between the returns of the underlying assets are perfectly positive. This is because when the returns of the assets in a portfolio are positively correlated, the portfolio is less diversified than when the returns are less correlated or negatively correlated. The extreme case situation occurs when the assets are perfectly positively correlated. In such a situation, the VaR of the portfolio will be at its highest. This is due to the fact that positive correlation implies that the assets tend to move in the same direction. Therefore, if one asset experiences a loss, it is likely that the other assets will also experience a loss, leading to a larger overall loss for the portfolio. This increases the risk of the portfolio, and hence, the VaR is maximized.

Choice B is incorrect. Perfectly negative correlation between the returns of the underlying assets in a portfolio would not maximize the VaR. In fact, it would minimize it because one asset's

gain would offset another's loss, thereby reducing overall portfolio risk.

Choice C is incorrect. Uncorrelated returns of the underlying assets in a portfolio also wouldn't maximize VaR. While uncorrelated assets can help diversify and reduce risk, they do not necessarily lead to maximum potential losses as measured by VaR.

Choice D is incorrect. Having either positive or negative correlations but not perfectly so will not result in maximum VaR either. The degree of correlation impacts how much an asset's return might offset another's within a portfolio, but without perfect positive correlation, there isn't maximization of potential losses as measured by VaR.

Things to Remember

- VaR (Value at Risk) is a measure used to assess the potential loss in value of a portfolio over a specific time period under normal market conditions at a given confidence level.
 - Correlation measures the relationship between the returns of two assets. A correlation of +1 indicates a perfect positive relationship, -1 indicates a perfect negative relationship, and 0 indicates no relationship.
 - Diversification is the strategy of spreading investments across different assets to reduce risk. It is based on the principle that different assets react differently to the same event, thus reducing overall portfolio risk.
 - Portfolio risk is influenced by the individual risks of the assets in the portfolio as well as the correlations between them. Understanding these relationships is crucial for effective risk management.
-

Q.2747 A portfolio consists of five equally weighted assets, each having a value of \$200,000. If the standard deviation of returns for the assets is 15%, and the correlation between each pair of assets is 0.4, calculate the portfolio VaR at a 95% confidence level.

A. \$152,880

B. \$177,989

C. \$128,310

D. \$212,072

The correct answer is **B**.

$$\text{VaR} = Z \times \sigma_p \times V$$

Where

$$\begin{aligned}\sigma_p &= \sigma \sqrt{\frac{1}{N} + (1 - \frac{1}{N})\rho} \\ &= 0.15 \sqrt{\frac{1}{5} + (1 - \frac{1}{5})0.4} \\ &= 10.82\%\end{aligned}$$

Thus,

$$\text{VaR} = 1.645 \times 0.1082 \times 200,000 \times 5 = \$177,989$$

Q.2748 A portfolio of four equal weighted assets has a total value of \$2 million. The VaR of the portfolio is \$450,000. If one of the assets in the portfolio has a beta of 1.2, calculate the marginal VaR of this asset.

- A. 0.19
- B. 0.27
- C. 1.47
- D. 0.93

The correct answer is **B**.

$$\begin{aligned}\text{Marginal VaR} &= \left(\frac{\text{VaRp}}{\text{Portfolio value}} \right) \times \beta_{\text{asset}} \\ &= \frac{450,000}{(2,000,000)} \times 1.2 \\ &= 0.27\end{aligned}$$

Q.2749 A portfolio consists of five equally weighted assets each having a value of \$200,000. The VaR of the portfolio is \$350,000. If the beta of one of the assets is 0.75, calculate the component VaR for this asset.

- A. \$52,500
- B. \$55,000
- C. \$50,000
- D. \$57,500

The correct answer is **A**.

$$\begin{aligned}\text{Component VaR} &= \text{VaRp} \times \beta_{\text{asset}} \times \text{asset weight} \\ \text{Component VaR} &= 350,000 \times 0.75 \times \frac{200,000}{(200,000 \times 5)} \\ \text{Component VaR} &= \$52,500\end{aligned}$$

Q.3026 We are given a portfolio having two foreign currencies, namely the Australian Dollar (AUD) and the Sterling Pound (GBP). These two currencies are uncorrelated, with standard deviations against the dollar of 6.5% and 10%, respectively. The portfolio has USD 4 million invested in the AUD and USD 3 million invested in the GBP. Compute the portfolio VaR at the 95% confidence level, assuming that $\alpha=1.65$.

- A. USD 675,218
- B. USD 546,889
- C. USD 655,032
- D. USD 586,274

The correct answer is **C**.

Recall that:

$$\text{Portfolio VaR} = \text{VaR}_P = \alpha \sigma_P W = \alpha \sqrt{x' \Sigma x}$$

The first step is to calculate the variance of the portfolio dollar returns. Let x be the dollar amounts allocated to each risk factor, in millions. We first calculate the product:

$$\Sigma x = \begin{bmatrix} 0.065^2 & 0 \\ 0 & 0.1^2 \end{bmatrix} \begin{bmatrix} \$4 \\ \$3 \end{bmatrix} = \begin{bmatrix} 0.065^2 \times \$4 + 0 \times \$3 \\ 0 \times \$4 + 0.1^2 \times \$3 \end{bmatrix} = \begin{bmatrix} \$0.0169 \\ \$0.03 \end{bmatrix}$$

Therefore, the portfolio value in dollar returns is:

$$\sigma_P^2 W^2 = x' \Sigma x = \begin{bmatrix} \$4 & \$3 \end{bmatrix} \begin{bmatrix} \$0.0169 \\ \$0.03 \end{bmatrix} = \$4 \times 0.0169 + \$3 \times 0.03 = 0.1576$$

The dollar volatility is:

$$\sqrt{0.1576} = \$0.396989 \text{ million}$$

With $\alpha = 1.65$, the portfolio VaR is given as:

$$\text{VaR}_P = 1.65 \times \$396989 = \$655,032$$

Q.3141 Bob William is an equity strategist at Hambantota Investments, a large asset management company in East Asia. He is evaluating the risk of a portfolio and computes the VaR for the two positions in his portfolio as follows: $\text{VaR}_1 = \$3.6$ million; and $\text{VaR}_2 = \$1.2$ million. Based on this information, the VaR of the portfolio VaR_p , if the returns of the two assets are uncorrelated, is closest to:

- A. \$3.794 million
- B. \$14.40 million
- C. \$4.80 million
- D. \$10.923 million

The correct answer is **A**.

The portfolio VaR reduces to the following expression in case the correlation is zero:

$$\sqrt{\text{VaR}_1^2 + \text{VaR}_2^2} = [3.6^2 + 1.2^2]^{1/2} = 3.794$$

Q.3144 Jack Wills is a portfolio manager at a sovereign wealth fund. His portfolio has a VaR of \$8,000,000. The portfolio is made up of investments in four companies: Sahara Petroleum, Burki Refinery, Dumsim Drillers, and RedSea Exploration. The four company assets are equally weighted within the portfolio with each position valued at \$10,000,000:

- Sahara Petroleum has a beta of 1.2
- Burki Refinery has a beta of 1.4.
- Dumsim Drillers has a beta of 0.8
- RedSea Exploration has a beta of 0.6

Based on the above information, what is the marginal VaR of Burki Refinery?

- A. 0.8
- B. 0.14
- C. 1.12
- D. 0.28

The correct answer is **D**.

$$\Delta \text{VaR}_i = \frac{\partial \text{VaR}}{\partial X_i} = \alpha (\beta_i \times \sigma_p) = \frac{\text{VaR}}{W} \times \beta_i$$

Therefore,

$$\text{VaR}_{\text{Burki Refinery}} = \left(\frac{8,000,000}{40,000,000} \right) \times 1.4 = 0.28$$

Q.3145 Jack Wills is a portfolio manager at a sovereign wealth fund. His portfolio has a VaR of \$8,000,000. The portfolio is made up of investments in four companies: Sahara Petroleum, Burki Refinery, Dumsim Drillers, and RedSea Exploration. The four company assets are equally weighted within the portfolio with each position valued at \$10,000,000:

- Sahara Petroleum has a beta of 1.2
- Burki Refinery has a beta of 1.4.
- Dumsim Drillers has a beta of 0.8
- RedSea Exploration has a beta of 0.6

Based on the above information, the component VaR for Sahara Petroleum is closest to:

- A. \$2,400,000
- B. \$1,920,000
- C. \$3,000,000
- D. \$4,320,000

The correct answer is **A**.

$$\begin{aligned}
 \text{Component VaR}_i &= (\Delta \text{VaR}_i) \times w_i W \\
 &= \frac{\text{VaR} \beta_i}{W} \times w_i W = \text{VaR} \beta_i w_i \\
 &= 8,000,000 * 1.2 * \left(\frac{10,000,000}{40,000,000} \right) \\
 &= 2,400,000
 \end{aligned}$$

Q.3146 A portfolio consists of two positions. The VaR of the two positions are \$10 million and \$40 million. Given that the returns of the two positions are not correlated, the VaR of the portfolio would be closest to:

- A. \$100 million
- B. \$300 million
- C. \$20 million
- D. \$41.23 million

The correct answer is **D**.

When positions are uncorrelated positions, the VaR of the portfolio is the square root of the sum of the squared VaRs:

$$\text{VaR}_p = \sqrt{10^2 + 40^2} \times (\$ \text{ million}) = \$41.231 \text{ million}$$

Q.3147 Debra Jeniffer is a commodity trader at Venus Investment, a large hedge fund domiciled in the Cayman Islands. While analyzing the trade positions for the fund, she computes the VaR of gold futures contract at \$67 million and the VaR of palladium contract at \$32 million. If the correlation between changes in commodity prices is 0.4, then what is the combined VaR?

- A. \$99.00 million
- B. \$85.02 million
- C. \$74.24 million
- D. \$35.00 million

The correct answer is **B**.

$$\text{VaR}_p = \sqrt{(\text{VaR}_A^2 + \text{VaR}_B^2 + 2 * \rho_{AB} * \text{VaR}_A * \text{VaR}_B)}$$

$$\text{VaR} = \sqrt{[\$67^2 + \$32^2 + (2 \times \$67 \times \$32 \times 0.4)]} = \sqrt{7228.2} = \$85.02 \text{ million}$$

Q.3148 Graham Bell is a currency strategist at the currency trading desk of Himilton Commercial Bank. His portfolio has a current exposure of \$26 million in British Pound and \$40 million in Turkish Lira. The portfolio beta of British Pound is 0.80 while that of the Lira is 1.35. The diversified portfolio VAR is \$7 million. Based on this information, what are the marginal VaR and the component VaR of the British Pound position?

- A. $MVaR_{\text{British Pound}} : 0.085$; $CVaR_{\text{British Pound}} : \2.2 million
- B. $MVaR_{\text{British Pound}} : 2.2$; $CVaR_{\text{British Pound}} : \0.85 million
- C. $MVaR_{\text{British Pound}} : 0.22$; $CVaR_{\text{British Pound}} : \8.5 million
- D. $MVaR_{\text{British Pound}} : 0.45$; $CVaR_{\text{British Pound}} : \2.95 million

The correct answer is **A**.

$$\begin{aligned} \text{Marginal VaR of British Pound} &= \left(\frac{VaR_{\text{Portfolio}}}{\text{Portfolio Value}} \right) \times \beta_{\text{British Pound}} \\ &= \left(\frac{\$7}{\$66} \right) \times 0.80 = 0.085 \end{aligned}$$

$$\begin{aligned} \text{Component VaR of British Pound} &= (VaR_{\text{Portfolio}} \times \beta \times \left(\frac{\text{Value in British Pound}}{\text{Portfolio Value}} \right)) \\ &= \$7 \text{ million} \times 0.80 \times \left(\frac{\$26}{\$66} \right) = \$2.2 \text{ million} \end{aligned}$$

Q.3150 Sasha Lamb is a portfolio manager at Lizard Financial. She is evaluating the risks of a portfolio of stocks. The portfolio's net asset value (NAV) is \$950 million and contains \$98 million invested in the common stocks of Bambino Entertainment. The annual standard deviation of returns for Bambino Entertainment is 24% and the annual standard deviation of the overall portfolio is 19%. The correlation of returns between Bambino Entertainment and the portfolio is 0.6. Using a 1-year 95% VaR and assuming that the returns are normally distributed, the component VaR of Bambino Entertainment is closest to:

- A. \$62.09 million
- B. \$23.28 million
- C. \$38.81 million
- D. \$15.52 million

The correct answer is **B**.

$$\text{VaR}_{\text{Bambino}} = 98 \text{ million} \times 0.24 \times 1.65 = 38.81 \text{ million}$$

The fact that β_i equals the product of the correlation ρ_i and σ_i divided by the portfolio σ_P is taken into account to further simplify the portfolio VaR:

$$\text{CVaR}_i = \text{VaR} w_i \beta_i = (\alpha \sigma_P W) w_i \beta_i = (\alpha \sigma_i w_i W) \rho_i = \text{VaR}_i \rho_i$$

The component VaR for Bambino Entertainment ($\text{CVaR}_{\text{Bambino}}$) can be presented as=
 $\text{VaR}_{\text{Bambino}} \times \text{Correlation of Bambino With the portfolio} = 38.81 \times 0.6 = 23.28 \text{ million}$

Q.3151 Kinetic Financial manages portfolios of large multinational companies consisting of low-beta stock. One of their fund invested in Goodman fertilizers and HiEdge Autos. The risk analyst of Kinetic recently estimated that the annual VaR at a 95% confidence level, assuming a 250-day year for the entire portfolio, was \$7.2 million based on the portfolio's market value of \$65 million and a correlation coefficient between Goodman and HiEdge of zero. If the annual loss in Goodman's stock is only expected to exceed \$5 million five percent of the time, then what is the daily expected loss for HiEdge's stock that will be exceeded five percent of the time?

- A. \$2,500,670
- B. \$327,612
- C. \$5,180,000
- D. \$107,360

The correct answer is **B**.

$$\begin{aligned} \text{VaR}_p &= \sqrt{\alpha^2 w_1^2 \sigma_1^2 + \alpha^2 w_2^2 \sigma_2^2 \sigma_1^2} = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2} \\ 7.2^2 &= 5^2 + \text{VaR}_{\text{HiEdge}}^2 \\ 51.84 &= 25 + \text{VaR}_{\text{HiEdge}}^2 \\ \text{VaR}_{\text{HiEdge}}^2 &= 51.84 - 25 = 26.84 \\ \text{VaR}_{\text{HiEdge}} &= \sqrt{26.84} = 5.180 \text{ million} \end{aligned}$$

$$\text{VaR}(T \text{ days}) = \text{VaR}(1 \text{ day}) * \text{square root of } T$$

Thus

$$\begin{aligned} \text{VaR}(1 \text{ day}) &= \frac{\text{VaR}(T \text{ days})}{(\text{square root of } T)} \\ \text{VaR}(1 \text{ day}) &= \frac{5,180,000}{250^{0.5}} = 327,612 \end{aligned}$$

Q.4710 Which one of the following correctly describes individual VaR?

- A. It explains VaR in terms of change in position in a portfolio.
- B. Every component is taken separately under the approach.
- C. It takes to consideration diversification benefits between components.
- D. It is to the additional amount of risk that new security adds to an existing portfolio.

The correct answer is **B**.

Individual VaR refers to the Value at Risk of an individual position in isolation. This means that each component or position in a portfolio is evaluated separately under this approach. The individual VaR does not take into account the correlations between different positions or the overall portfolio context. Instead, it focuses on the potential loss that could occur from a single position, assuming a certain level of confidence and a specific time horizon. This approach is particularly useful for identifying the riskiest positions within a portfolio and for making decisions about whether to hold, sell, or hedge these positions.

Choice A is incorrect. Individual VaR does not explain VaR in terms of change in position in a portfolio. Rather, it quantifies the risk associated with each individual component or security within the portfolio.

Choice C is incorrect. Individual VaR does not take into consideration diversification benefits between components. This concept refers to the risk of each individual component without considering its correlation with other components.

Choice D is incorrect. The statement describes incremental VaR, not individual VaR. Incremental VaR measures the additional amount of risk that a new security adds to an existing portfolio, while individual VaR focuses on the risk associated with each separate component within a portfolio.

Things to Remember

- **Portfolio VaR:** This is the total VaR of the entire portfolio, taking into account the correlations between different positions.
- **Systemic Risk:** The risk that affects the entire market or economy, rather than just a specific security or portfolio.
- **Conditional VaR (CVaR):** Also known as Expected Shortfall, CVaR measures the expected loss beyond the VaR level.
- **Backtesting:** A process of comparing actual losses to VaR estimates to assess the accuracy of the VaR model.
- **Stress Testing:** A risk management technique that involves testing a portfolio's performance under extreme and adverse market conditions.

Q.4711 Consider an investment company portfolio made up of two foreign currencies, the Japanese Yen (JPY) and the Euro (EUR). Assume that the currencies are uncorrelated and that their volatilities are 6% and 10%, respectively. The portfolio has US\$2.5 million invested in the JPY and US\$1 million invested in EUR. Calculate the dollar volatility of the portfolio.

- A. \$0.0325 million
- B. \$0.4750 million
- C. \$0.1803 million
- D. \$0.009 million

The correct answer is **C**.

To find the dollar volatility of the portfolio, we use the formula:

$$\begin{aligned}\text{Dollar volatility} &= \sqrt{\sigma_p^2 W^2} = \sqrt{X' \Sigma x} \\ \Sigma x &= \begin{bmatrix} 0.06^2 & 0 \\ 0 & 0.1^2 \end{bmatrix} \begin{bmatrix} 2.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.06^2 \times 2.5 + 0 \times 1 \\ 0 \times 2.5 + 0.1^2 \times 1 \end{bmatrix} = \begin{bmatrix} 0.009 \\ 0.01 \end{bmatrix} \\ X' \Sigma x &= [\$2.5 \quad \$1] \begin{bmatrix} 0.009 \\ 0.01 \end{bmatrix} = \$0.0325 \text{ million} \\ \text{Dollar volatility} &= \sqrt{0.0325} = \$0.1803 \text{ million}\end{aligned}$$

Q.4712 A portfolio is made up of two uncorrelated positions. The VaR of position one (VaR_1) is \$3 million while that of position two (VaR_2) is \$2 million. Calculate the portfolio VaR.

- A. \$3.606 million
- B. \$5.000 million
- C. \$1.198 million
- D. \$2.504million

The correct answer is **A**.

Since the positions are uncorrelated,

$$\begin{aligned}VaR_p &= \sqrt{VAR_1^2 + VAR_2^2} \\ &= \sqrt{3^2 + 2^2} = 3.606 \text{ million}\end{aligned}$$

Q.4715 James Wit is a portfolio manager at ABC Investment Ltd. His goal is to create a new pool of investments comprising of different assets. Wit begins the investment process by adding two assets A and B in a new portfolio. The assets are perfectly correlated and have a volatility of 6% and 10%, respectively. The amount invested in asset A is \$2.5 million, while that invested in B is \$1 million. What are the individual VaRs for this portfolio at a 95% confidence level ?

- A. Asset A VaR: \$247,500; Asset B VaR: \$165,000
- B. Asset A VaR: \$412,500; Asset B VaR: \$107,692
- C. Asset A VaR: \$1,269,200; Asset B VaR: \$285,500
- D. Asset A VaR: \$285,500; Asset B VaR: \$412,500

The correct answer is **A**.

The formula used for calculating Individual VaR is:

$$\text{Individual VaR} = \text{Value Invested} \times \text{Volatility} \times Z_{\alpha}$$

where $Z_{\alpha} = 1.65$ for a 95% confidence level.

Value Invested in Asset A: \$2,500,000

Volatility of Asset A: 6%

$$\text{VaR}_A = \$2,500,000 \times 0.06 \times 1.65 = \$247,500$$

Value Invested in Asset B: \$1,000,000

Volatility of Asset B: 10%

$$\text{VaR}_B = \$1,000,000 \times 0.10 \times 1.65 = \$165,000$$

The VaRs for assets A and B are \$247,500 and \$165,000, respectively.

Reading 151: VaR and Risk Budgeting in Investment Management

Q.2479 The daily application of VaR measures has become a requirement of bank trading portfolios due to all the following factors, EXCEPT:

- A. Short horizon
- B. Rapid turnover
- C. Historical returns
- D. High leverage

The correct answer is **C**.

Value at Risk (VaR) is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. This metric is most commonly used by investment and commercial banks to determine the extent and occurrence rate of potential losses in their institutional portfolios. VaR modeling determines the potential for loss in the entity being assessed, and the probability of occurrence for the defined loss. Unlike historical returns, VaR is a forward-looking measure of risk. Historical returns, on the other hand, are purely retrospective and do not provide a timely measurement of risk. Therefore, while VaR measures are appropriate for bank trading portfolios characterized by a short horizon, rapid turnover, and high leverage, historical returns are not. This is because historical returns do not account for the current positions and the potential future losses that could occur in the portfolio.

Choice A is incorrect. The short horizon of trading portfolios necessitates the daily application of VaR measures. This is because the value of these portfolios can change significantly in a short period, and therefore, it's crucial to assess the risk on a daily basis.

Choice B is incorrect. Rapid turnover also requires daily application of VaR measures. Trading portfolios often involve frequent buying and selling of securities, which can lead to significant changes in portfolio value from day to day. Therefore, it's important to measure risk on a daily basis.

Choice D is incorrect. High leverage implies that small changes in asset prices can have large effects on the value of the portfolio due to borrowed funds being used for investment purposes. This makes it necessary for banks' trading portfolios to apply VaR measures daily.

Things to Remember

- Value at Risk (VaR) is a measure of the potential loss in an investment portfolio over a specific time frame.
- VaR is commonly used by banks to assess and manage risk in their trading portfolios.
- VaR is a forward-looking measure of risk, unlike historical returns which are

retrospective.

- Short horizon in trading portfolios necessitates daily application of VaR measures.
 - Rapid turnover in trading portfolios also requires daily application of VaR measures.
 - High leverage in trading portfolios makes it necessary to apply VaR measures daily.
-

Q.2480 A VaR risk management system is suited to the investment management industry due to the following reasons, EXCEPT:

- A. The need for risk measures which take diversification into account.
- B. Complex financial instruments which create a need for stronger, centralized risk management systems.
- C. The dynamic nature of investment portfolios.
- D. The presence of highly illiquid assets such as convertible bonds, traded infrequently.

The correct answer is **D**.

The presence of highly illiquid assets such as convertible bonds, traded infrequently, does not make VaR risk management systems suitable for the investment management industry. VaR is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. This measure is most commonly used by investment and commercial banks to determine the extent and occurrence ratio of potential losses in their institutional portfolios. However, VaR is not well-suited to handle illiquid assets like convertible bonds that are traded infrequently. The reason for this is that VaR relies on historical data to estimate future risk, and illiquid assets that are traded infrequently do not provide a sufficient amount of data for accurate risk estimation. Furthermore, the price volatility of illiquid assets can be extreme, which can lead to significant underestimation of risk by VaR models. Therefore, while VaR is a useful tool for managing risk in many contexts, it is not ideally suited for portfolios containing a significant proportion of illiquid assets.

Choice A is incorrect. The global nature of investments necessitates risk measures that account for diversification. VaR is particularly well-suited to this as it provides a single, consolidated view of risk across all types of investments and geographies, thereby taking into account the benefits of diversification.

Choice B is incorrect. The complexity of financial instruments requires robust and centralized risk management systems. VaR systems are capable of handling complex financial instruments and providing a comprehensive view of the risks associated with them.

Choice C is incorrect. The dynamic nature of investment portfolios demands forward-looking risk measures. VaR systems are designed to be forward-looking, using historical data to predict

future risks.

Things to Remember

- VaR (Value at Risk) is a statistical measure used to quantify the level of financial risk within a firm or investment portfolio over a specific time frame.
 - VaR is commonly used by investment and commercial banks to determine the extent and occurrence ratio of potential losses in their institutional portfolios.
 - VaR relies on historical data to estimate future risk, which can be a limitation when dealing with highly illiquid assets or assets with extreme price volatility.
 - VaR is a useful tool for managing risk in many contexts, providing a consolidated view of risk across different types of investments and geographies.
 - Risk management systems need to account for diversification, complexity of financial instruments, and the dynamic nature of investment portfolios to be effective in the investment management industry.
-

Q.2481 A fund manager manages a fund which mainly invests in illiquid assets such as convertible bonds, which are traded infrequently. The manager computes the fund volatility and finds it to be on the lower side. He concludes that the fund has low total risk due to low volatility. Is the manager's conclusion correct?

- A. Yes, because volatility indicates the riskiness of an asset or portfolio. Therefore, low volatility leads to low risk.
- B. No, because correlations with other asset classes will be artificially lowered, giving the appearance of low systematic risk.
- C. No, because volatility will be artificially lowered, giving the appearance of low total risk.
- D. Both B and C.

The correct answer is **D**.

Both statements B and C are accurate. The fund manager's conclusion is incorrect because the low volatility of the fund, which primarily invests in illiquid assets, can give a misleading picture of the fund's total risk. This is due to two types of biases that can occur when risk measures are

based on monthly returns of infrequently traded assets.

Firstly, the correlations with other asset classes can be artificially lowered, giving the appearance of low systematic risk. This bias can be corrected using enlarged regressions with additional lags of the market factors and summing the coefficients across lags. This means that the systematic risk of the fund might be higher than what is indicated by the low volatility.

Secondly, the volatility itself can be artificially lowered, giving the appearance of low total risk. This bias can be corrected by taking into account the positive serial autocorrelation in returns when extrapolating risk to longer horizons. This means that the total risk of the fund might be higher than what is indicated by the low volatility.

Therefore, the fund manager's conclusion that the fund has low total risk due to low volatility is incorrect because the low volatility might be giving a misleading picture of the fund's total and systematic risk.

Choice A is incorrect. While it's true that volatility can indicate the riskiness of an asset or portfolio, it's not the only factor to consider. In this case, because the fund primarily invests in illiquid assets like convertible bonds which are not frequently traded, the low volatility may be misleading and does not necessarily mean that the fund has a low total risk.

Choice B is incorrect. Although correlations with other asset classes might be artificially lowered due to illiquidity, giving an appearance of low systematic risk, this doesn't necessarily mean that the total risk of the fund is low. Systematic risk is just one component of total risk; specific or unsystematic risks associated with illiquid assets could still be high.

Choice C is incorrect. While it's true that volatility might be artificially lowered due to infrequent trading of illiquid assets like convertible bonds, giving an appearance of low total risk, this doesn't necessarily mean that actual total risk is low. The inherent risks associated with these types of assets could still result in high levels of actual total risk despite appearances.

Things to Remember

- Volatility is a measure of the dispersion of returns for a given security or market index. It is often used as a proxy for risk, but it is not the only measure of risk.
- Systematic risk, also known as market risk, is the risk inherent to the entire market or an entire market segment. It cannot be diversified away through portfolio management.
- Specific risk, also known as unsystematic risk, is the risk associated with individual assets or small groups of assets. It can be diversified away through proper portfolio management.
- Illiquid assets are assets that cannot be easily sold or converted into cash without a substantial loss in value. They typically have higher specific risk due to their lack of

marketability.

- Serial autocorrelation in returns refers to the relationship between the current return and past returns. Ignoring this autocorrelation can lead to biased estimates of risk measures.
-

Q.2482 A hedge fund manager makes the following comment:

“Hedge funds are transparent and reveal all information about their positions. However, due to group heterogeneity, risk measurement problems arise.”

Is the manager’s comment correct?

- A. Yes.
- B. No, because hedge funds are usually homogenous.
- C. No, because hedge funds lack transparency and often refuse to reveal information about their positions.
- D. No, because the law requires hedge funds to be discrete and conceal investment-related information in line with the principle of client confidentiality.

The correct answer is **C**.

Hedge funds are known for their lack of transparency. This is primarily because many hedge funds are reluctant to disclose information about their positions. The reason behind this reluctance is the fear that others may exploit this information to their advantage. This lack of transparency poses a significant challenge in risk measurement. Risk measurement is a critical aspect of managing hedge funds as it helps in understanding the potential losses that could occur due to adverse market movements. However, the lack of transparency makes it difficult to accurately measure the risk associated with hedge funds. Therefore, the statement made by the hedge fund manager is incorrect as hedge funds are not transparent and often refuse to reveal information about their positions.

Choice A is incorrect. While it is true that hedge funds can be heterogeneous, implying a variety of investment strategies and risk profiles, the assertion that they are transparent and disclose all information about their positions is not accurate. Hedge funds are known for their lack of transparency due to the proprietary nature of their trading strategies.

Choice B is incorrect. The statement contradicts the manager's claim about heterogeneity among hedge funds. Hedge funds are not usually homogenous; they employ a wide range of investment strategies and have varying degrees of risk tolerance, which contributes to their heterogeneity.

Choice D is incorrect. Although client confidentiality is important in the financial industry, it does not legally require hedge funds to conceal all investment-related information. However, due to competitive reasons or strategic considerations, many hedge funds choose not to disclose specific details about their positions or trading strategies.

Things to Remember

- Hedge funds are alternative investment vehicles that typically pool capital from accredited investors and employ a variety of strategies to generate returns.
 - Hedge funds are known for their flexibility in investment strategies, which can range from long/short equity, event-driven, global macro, to quantitative trading.
 - Hedge funds often charge both management fees (typically 1-2% of assets under management) and performance fees (typically 20% of profits).
 - Risk management is a crucial aspect of hedge fund operations, as these funds often use leverage and derivatives to enhance returns, which can also amplify risks.
 - Due to their complex strategies and lack of transparency, hedge funds are typically only available to accredited investors who meet certain income or net worth requirements.
-

Q.2485 A fund manager decomposes the fund's total VaR as under:

Source of Risk	VaR
Policy-mix VaR	19.6%
Active management VaR	1.6%
Asset VaR	19.3%

Upon review of the report above, the fund manager concludes that they can take greater deviations from the benchmark without affecting the plan's total VaR. Is the manager's conclusion correct?

- A. No.
- B. Yes, because the policy-mix VaR and the active management VaR do not add up to the asset VaR.
- C. Yes, because the policy-mix VaR is greater than the active management VaR.
- D. Yes, because the active management VaR is greater than the policy-mix VaR.

The correct answer is **B**.

The fund manager's conclusion is correct because the Policy-mix VaR and the Active management VaR do not add up to the Asset VaR. This indicates that there is a negative correlation between the Policy-mix VaR and the Active management VaR, which results in a lower overall Asset VaR. The negative correlation implies that when one risk increases, the other decreases, and vice versa. This phenomenon allows for greater deviations from the benchmark without affecting the total VaR of the plan. Therefore, the fund manager can take more risks in active management (deviating from the benchmark) without increasing the total VaR. This is because the increase in Active management VaR would be offset by a decrease in Policy-mix VaR due to their negative correlation.

Choice A is incorrect. The fund manager's conclusion is accurate because the policy-mix VaR and the active management VaR do not add up to the asset VaR. Therefore, increasing deviations from the benchmark would not necessarily impact the total VaR of the plan.

Choice C is incorrect. The fact that policy-mix VaR is greater than active management VaR does not justify that increasing deviations from benchmark will not impact total plan's VAR. These two components are independent and their relative sizes do not determine their effect on total VAR.

Choice D is incorrect. Similarly, whether active management VAR is greater than policy-mix VAR has no bearing on whether increasing deviations from benchmark will affect total plan's VAR or not.

Q.2487 A pension fund manager makes the following comments:

- I. Volatility of assets is the appropriate risk parameter for a pension fund.
- II. The value of pension fund assets may not be sufficient to cover the liabilities of the fund. Thus, an asset/liability management (ALM) framework is desirable.
- III. Pension funds are exposed to funding risk, hence an ALM framework is desirable.

Which of the pension fund manager's comments is (are) accurate?

- A. Only I
- B. Both II and III
- C. Both I and III
- D. Only II

The correct answer is **B**.

The second and third statements made by the pension fund manager are accurate. The value of pension fund assets may not always be sufficient to cover the liabilities of the fund. This is known as funding risk. A pension fund with defined benefits promises a stream of fixed payments to retirees. If the assets are not sufficient to cover these liabilities, the shortfall will have to be made up by the fund's owner. Therefore, risk should be viewed in an asset/liability management (ALM) framework, to ensure availability of sufficient assets to service the liabilities. An ALM framework is indeed desirable for managing this risk, as it provides a structured approach to managing the risks associated with the mismatch between assets and liabilities.

Choice A is incorrect. While volatility of assets is a risk parameter for a pension fund, it is not the only one. Other risks such as funding risk, interest rate risk, and longevity risk are also important considerations in managing a pension fund.

Choice C is incorrect. This choice suggests that both volatility of assets and exposure to funding risk make an ALM framework desirable for pension funds. However, this ignores the fact that the value of pension fund assets may not be sufficient to cover liabilities - another key reason why an ALM framework is necessary.

Choice D is incorrect. Although it's true that an ALM framework can help ensure that the value of pension fund assets covers its liabilities, this choice fails to acknowledge the importance of managing funding risks which also makes an ALM framework desirable.

Things to Remember

- Asset/Liability Management (ALM) framework is essential for pension funds to manage risks associated with the mismatch between assets and liabilities.
- Funding risk is a key risk faced by pension funds, where the value of assets may not be sufficient to cover the liabilities.

- Volatility of assets is an important risk parameter for pension funds, but not the only one. Other risks include interest rate risk and longevity risk.
 - An ALM framework helps ensure that pension fund assets are adequate to meet the liabilities, providing a structured approach to risk management.
-

Q.2488 A pension fund manager manages a fund with liabilities consisting mainly of nominal payments. The manager predicts the interest rate to decrease going forward. If that happens, what will be the impact on the portfolio?

- A. The value of equities on the asset side will decrease.
- B. The value of liabilities will decrease.
- C. The value of equity will increase, but the increase in the value of liabilities will exceed the increase in the value of equity.
- D. The value of equity will decrease, but the decrease in the value of liabilities will exceed the decrease in the value of equity.

The correct answer is **C**.

The value of equity will increase, but the increase in the value of liabilities will exceed the increase in the value of equity. This is because, for pension funds, when liabilities consist mainly of nominal payments, their value, in general, will behave like a short position in a long-term bond. Therefore, decreases in interest rates, while beneficial for equities on the asset side, can increase even more the value of liabilities, thereby negatively affecting the surplus. This means that even though the value of equities might increase due to the decrease in interest rates, the increase in the value of liabilities due to the same decrease in interest rates will be greater. This is because the liabilities are behaving like a short position in a long-term bond, and the value of a bond increases when interest rates decrease. Therefore, the overall effect on the portfolio will be negative, as the increase in the value of liabilities will exceed the increase in the value of equity.

Choice A is incorrect. The value of equities on the asset side will not necessarily decrease with a downward trend in interest rates. In fact, lower interest rates often lead to an increase in equity prices as borrowing costs are reduced, making investments more attractive.

Choice B is incorrect. The value of liabilities will not decrease with a downward trend in interest rates. Since the fund's liabilities are primarily composed of nominal payments, they are fixed and do not change with fluctuations in interest rates.

Choice D is incorrect. This choice incorrectly suggests that both equity values and liability values would decrease with falling interest rates. As explained above, lower interest rates typically lead to increased equity prices while having no impact on the fixed nominal payments that make up the majority of the fund's liabilities.

Things to Remember

- Interest Rate Risk: Pension funds are exposed to interest rate risk, which refers to the impact of changes in interest rates on the value of the fund's assets and liabilities.
 - Duration Matching: Pension funds often use duration matching to manage interest rate risk by aligning the durations of their assets and liabilities.
 - Equity Investments: Pension funds typically hold a portion of their assets in equities to generate higher returns over the long term.
 - Liability-Driven Investing (LDI): LDI is an investment strategy that focuses on managing a pension fund's assets and liabilities together to meet specific funding goals.
 - Fixed vs. Floating Liabilities: Understanding the composition of a pension fund's liabilities is crucial in assessing the impact of interest rate changes on the fund's overall value.
-

Q.2490 A pension fund has \$100,000 in assets and \$90,000 in liabilities. Assume that the expected return on the surplus is 5%, and the annual VaR of the surplus is 22% at the 99% confidence level.

The initial surplus of the fund is equal to:

- A. \$10,000
- B. \$20,000
- C. \$5,000
- D. \$1,000

The correct answer is **A**.

$$\begin{aligned}\text{Surplus} &= V_{\text{assets}} - V_{\text{liabilities}} \\ &= \$100,000 - \$90,000 = \$10,000\end{aligned}$$

Q.2491 A pension fund has \$100,000 in assets and \$90,000 in liabilities. Assume that the surplus is expected to grow at a rate of 5% per year, and the annual VaR of the surplus is 22% at the 99% confidence level.

The expected surplus after one year is equal to:

- A. \$10,000
- B. \$10,500
- C. \$5,000
- D. \$25,000

The correct answer is **B**.

$$\begin{aligned}\text{Surplus}_{\text{initial}} &= \$10,000 \\ \text{Surplus}_{\text{after one year}} &= \text{Surplus}_{\text{initial}} \times (1 + \text{expected growth}) = \$10,000 \times (1 + 5\%) = \$10,500\end{aligned}$$

Q.2492 A pension fund has \$100,000 in assets and \$90,000 in liabilities. Assume that the expected return on the surplus is 50%, and the annual VaR of the surplus is 220% at the 99% confidence level.

Which of these statements is (are) accurate?

- I. There is a 99% probability that, over the next year, the surplus will turn into a deficit of \$7,000 or more
 - II. There is a 1% probability that, over the next year, the surplus will turn into a deficit of \$7,000 or more
 - III. The surplus VaR is equal to \$22,000
 - IV. The surplus VaR is equal to \$5,000
- A. Only I
 - B. Both II and III
 - C. Both II and IV
 - D. Both I and IV

The correct answer is **B**.

The Value at Risk (VaR) is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. In this case, the

annual VaR of the surplus is given as 220%. This means that there is a 1% chance that the surplus (assets - liabilities) will decrease by 220% of its current value over the next year. The current surplus is \$10,000 (assets of \$100,000 - liabilities of \$90,000). Therefore, the dollar value of the VaR is 220% of \$10,000, which equals \$22,000. This is statement III. Furthermore, since VaR is calculated at a 99% confidence level, this means there is a 1% probability (100% - 99%) that losses will exceed the VaR. Therefore, there is a 1% chance that the surplus will decrease by more than \$22,000 over the next year. Given that the expected surplus after one year is \$15,000 ($\$10,000 * (1 + 50\%)$), the surplus could turn into a deficit of \$7,000 ($\$22,000 - \$15,000$). This is statement II. Therefore, both statements II and III are accurate.

Choice A is incorrect. The statement suggests that there is a 99% probability that the surplus will turn into a deficit of \$7,000 or more. However, this interpretation of VaR is incorrect. VaR at a 99% confidence level means that there is a 1% chance (not 99%) that losses will exceed the VaR amount.

Choice C is incorrect. While it correctly identifies the probability associated with the VaR (1%), it incorrectly states that the surplus VaR equals \$5,000. The surplus in this case equals to assets minus liabilities which equals to \$10,000 and its annual VaR at a 99% confidence level has been estimated to be 220%, which equates to \$22,000 not \$5,000.

Choice D is incorrect. This choice includes both an incorrect interpretation of the probability associated with the VaR (as explained in Choice A) and an inaccurate calculation of the surplus's Value at Risk (as explained in Choice C).

Things to Remember

- Value at Risk (VaR) is a measure of the potential loss on an investment or portfolio over a specific time period for a given confidence interval.
- VaR is typically expressed as a percentage of the current value of the portfolio or assets.
- The confidence level associated with VaR represents the probability that the actual loss will not exceed the VaR amount.
- Expected return is the anticipated return on an investment based on historical data or projections.
- Surplus in the context of a pension fund refers to the excess of assets over liabilities.

Q.2493 The ultimate responsibility for the pension fund rests with:

A. Employees

- B. The fund manager
- C. The plan sponsor
- D. None of the above

The correct answer is **C**.

The plan sponsor is the entity that establishes the pension plan. This could be an employer or a labor union. The plan sponsor is responsible for setting up the plan, deciding on the benefits it will provide, and ensuring that it complies with all relevant laws and regulations. They are also responsible for the financial risk associated with the plan. This includes the risk that the plan's investments will not perform as expected, and the risk that the plan will not have enough money to pay all promised benefits. In the event of a shortfall, the plan sponsor may be required to make additional contributions to the plan. Therefore, the ultimate responsibility for the pension fund rests with the plan sponsor.

Choice A is incorrect. While employees contribute to the pension fund, they are not responsible for its management or ultimate accountability. Their role is primarily to provide the funds that will be invested and managed by others.

Choice B is incorrect. The fund manager oversees the investments of the pension fund, but they do not hold ultimate responsibility for it. They are tasked with making investment decisions in line with the objectives of the pension plan, but accountability lies with those who establish and maintain these objectives.

Choice D is incorrect. There must be an entity held accountable for a pension fund's operation and growth; it cannot be 'none of the above'. In this case, it's established that this responsibility falls on the plan sponsor.

Things to Remember

- **Pension Fund:** A fund that is established by an employer to provide for the retirement benefits of its employees.
- **Plan Sponsor:** The entity, which could be an employer or a labor union, that establishes and maintains the pension plan.
- **Investment Management:** The process of managing the assets of a pension fund to achieve the fund's objectives while considering risk and return.
- **Fiduciary Responsibility:** The legal obligation of the plan sponsor and other fiduciaries to act in the best interest of the plan participants and beneficiaries.
- **Regulatory Compliance:** Pension plans must comply with various laws and regulations

set forth by government authorities to protect the interests of plan participants.

Q.2494 Consider the following comment made by a pension fund manager: “Cash flow risk is the risk of year-to-year fluctuations in contributions to the pension fund. It restricts the plan sponsor from adopting a more volatile risk profile.” Is the statement true?

- A. Yes, the statement is true for all plan sponsors.
- B. No, because a plan sponsor who can absorb greater variations in funding costs can adopt a more volatile risk profile.
- C. No, because a plan sponsor who cannot absorb greater variations in funding costs can adopt a more volatile risk profile.
- D. No, because cash flow risk can be mitigated by restricting the sponsor’s funding options.

The correct answer is **B**.

The statement made by the pension fund manager is not universally true. It is important to understand that the ability of a plan sponsor to adopt a more volatile risk profile is not solely dependent on cash flow risk. Rather, it is also significantly influenced by the plan sponsor's capacity to absorb greater variations in funding costs. If a plan sponsor has the financial strength to withstand larger fluctuations in funding costs, they have the flexibility to adopt a more volatile risk profile. This is because they can afford to take on higher-risk investments that may potentially yield higher returns, despite the increased risk of year-to-year fluctuations in contributions to the pension fund. Therefore, the statement made by the pension fund manager is not true for plan sponsors who can absorb greater variations in funding costs.

Choice A is incorrect. The statement is not universally true for all plan sponsors. The ability to adopt a volatile risk profile depends on the specific circumstances of each plan sponsor, including their financial strength and risk tolerance.

Choice C is incorrect. This statement contradicts the basic principles of risk management. A plan sponsor who cannot absorb greater variations in funding costs should ideally adopt a less volatile risk profile to minimize cash flow risks.

Choice D is incorrect. While it's true that cash flow risk can be mitigated by restricting the sponsor’s funding options, this does not necessarily enable them to adopt a more volatile risk profile. In fact, restricting funding options may limit the potential for higher returns associated with more risky investments.

Things to Remember

- Cash flow risk is the risk associated with the variability in cash flows, which can impact the ability of a plan sponsor to meet its funding obligations.

- Plan sponsors need to consider their risk tolerance, financial strength, and funding options when determining the appropriate risk profile for their pension fund.
 - Managing cash flow risk involves strategies such as asset-liability matching, liquidity management, and stress testing to ensure the fund can meet its obligations even during periods of fluctuating cash flows.
 - Plan sponsors may use techniques like cash flow matching, duration matching, and immunization to mitigate the impact of cash flow risk on their pension fund.
 - Understanding the relationship between funding costs, investment returns, and risk tolerance is crucial for determining the optimal risk profile for a pension fund.
-

Q.2495 A firm enjoys greater operating profits. The firm is a sponsor of a defined benefits pension plan. The firm's pension fund manager predicts that going forward, the expected surplus growth may fall, but the firm will continue enjoying greater operating profits. Which of the following is most likely correct?

- A. The firm must act urgently to restrict the fall in expected surplus growth.
- B. The fall in expected surplus growth will affect the operating profit of the firm.
- C. As the firm will continue enjoying greater operating profits, the fall in expected surplus growth may be less of a concern.
- D. As the firm will continue enjoying greater operating profits, the fall in expected surplus growth is of grave concern.

The correct answer is C.

The fall in expected surplus growth may be less of concern because it has been predicted that the firm will continue enjoying greater operating profits. This is because the firm's increased operating profits can potentially offset the decrease in the surplus growth of the pension fund. In other words, the firm's strong financial performance can provide a buffer against the negative impact of the falling surplus growth. This is a concept known as economic risk, which refers to the risk of variation in the total economic earnings of the plan sponsor. In this case, the firm's high operating profits can help mitigate the economic risk associated with the decrease in the surplus growth of the pension fund.

Choice A is incorrect. While it may seem intuitive to act urgently in response to a potential decrease in expected surplus growth, this is not necessarily the best course of action. The firm's increased operating profits could provide a buffer against the potential decrease, and rash

actions could potentially exacerbate the situation.

Choice B is incorrect. The fall in expected surplus growth of the pension fund does not directly affect the operating profit of the firm. Operating profit is derived from core business operations, while pension fund performance relates to employee benefits and obligations.

Choice D is incorrect. Although any potential decrease in expected surplus growth should be monitored closely, it does not necessarily constitute a grave concern for a company experiencing increased operating profits. These additional profits can help mitigate any negative impacts on overall financial health.

Things to Remember

- **Defined benefits pension plan:** A type of pension plan where an employer promises a specified pension payment to employees upon retirement, based on a formula that typically considers factors such as salary history and years of service.
 - **Surplus growth:** Refers to the increase in the surplus of a pension fund, which occurs when the assets of the fund exceed its liabilities.
 - **Economic risk:** The risk of variation in the total economic earnings of a plan sponsor, which can be influenced by factors such as market conditions, economic performance, and investment returns.
 - **Operating profits:** The profits generated from a company's core business operations, excluding non-operating income and expenses.
 - **Financial performance:** Refers to how well a firm can generate profits and grow its assets over time, indicating its overall health and stability.
-

Q.2496 A pension fund has heavily invested in stocks. The fund manager predicts a slowdown in the economy going forward. Which of the following is an accurate statement?

- A. There will be no adverse effect on the pension fund.
- B. The value of fund's assets will increase, thus increasing the surplus.
- C. The value of fund's assets will decrease, and the economic downturn will severely restrict the sponsor's ability to make contributions.
- D. The value of fund's assets will increase, but the economic downturn will severely restrict the sponsor's ability to make contributions.

The correct answer is **C**.

When a pension fund is heavily invested in equities, an economic downturn can have a significant impact on the value of the fund's assets. This is because the performance of equities is closely tied to the performance of the economy. In a downturn, companies may see reduced profits or even losses, which can lead to a decrease in stock prices. As a result, the value of the fund's assets, which are heavily invested in these stocks, will decrease. Furthermore, an economic downturn can also severely restrict the sponsor's ability to make contributions to the fund. This is because the sponsor, which is typically a company, may also be negatively affected by the downturn. They may see reduced profits or even losses, which can limit their ability to contribute to the pension fund. Therefore, both the value of the fund's assets and the sponsor's ability to make contributions are likely to be negatively affected in an economic downturn.

Choice A is incorrect. This statement is not accurate because an economic downturn generally has a negative impact on equities, which would likely lead to a decrease in the value of the pension fund's assets. Therefore, there will be an adverse effect on the pension fund.

Choice B is incorrect. This statement contradicts general market behavior during an economic downturn. Typically, equities lose value during such periods, so it's unlikely that the value of the fund's assets will increase.

Choice D is incorrect. Similar to choice B, this option incorrectly suggests that asset values would increase in an economic downturn. While it correctly identifies that sponsor contributions may be restricted due to poor economic conditions, it fails to accurately predict asset performance under these circumstances.

Things to Remember

- **Asset Allocation:** Pension funds often have a specific asset allocation strategy that includes investments in stocks, bonds, and other asset classes to meet their long-term obligations.
- **Market Risk:** Economic downturns can lead to increased market risk, affecting the value of investments in a pension fund.
- **Contributions:** The sponsor's ability to make contributions to the pension fund is crucial for meeting future obligations to pensioners.
- **Diversification:** Diversifying investments can help reduce the impact of economic downturns on a pension fund's overall performance.
- **Actuarial Assumptions:** Pension funds rely on actuarial assumptions to forecast future liabilities and funding requirements.

Q.2497 AlphaStop sponsors a defined benefits pension plan. The company's pension plan risk management system generates reports based on movements in the assets and the surplus. The pension management plan is not integrated with the firm's financial goals. The management has access to daily risk reports, and the board receives monthly risk reports. Which of the following is an accurate statement?

- A. The company's pension plan management is satisfactory.
- B. The reporting frequency to the board must be increased.
- C. The company must integrate its pension plan management with its overall financial goals.
- D. The company's pension plan must not generate reports based on movements in the assets.

The correct answer is C.

From the viewpoint of the plan sponsor, risk is measured not only by movements in the assets, or even the surplus, but also by the ultimate effect on the economic value of the firm. Thus, pension plan management should be integrated with the overall financial goals of the plan sponsor. This is in line with the trend toward enterprise-wide risk management.

Things to Remember

1. Pension plan management should be integrated with the overall financial goals of the plan sponsor. This can facilitate better decision-making and risk management, and ensure that the pension plan contributes positively to the firm's financial health and stability.
 2. Risk reports are an important tool for managing pension plan risk. They provide valuable information about the pension plan's financial situation and risk exposure, which can help the company to manage the pension plan more effectively.
 3. The frequency of risk reporting should be balanced with the quality and relevance of the information. While it is important for the board to be informed about the pension plan's risk situation, increasing the reporting frequency might not be the most effective solution, and could even lead to information overload.
 4. Enterprise-wide risk management is a holistic and integrated approach to managing all risks that a firm might face. It aligns with the trend towards integrating pension plan management with the firm's financial goals.
-

Q.2498 VaR systems, in the context of the investment management industry, can be used to measure the following risks, EXCEPT:

- A. Liquidity risk

- B. Market risk
- C. Credit risk
- D. Operational risk

The correct answer is **D**.

Operational risk cannot be measured by VaR systems in the context of the investment management industry. Operational risk refers to the risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events. This includes legal risk, but excludes strategic and reputational risk. Operational risk is inherently difficult to quantify and model due to its complex and idiosyncratic nature. VaR systems, which are designed to measure and control market and credit risks, are not equipped to handle the unique characteristics of operational risk. Therefore, operational risk is typically managed through other means, such as internal controls, audits, and risk management frameworks.

Choice A is incorrect. Liquidity risk can be measured by VaR systems. This type of risk refers to the possibility that an investor may not be able to buy or sell assets quickly enough at a price close to their true underlying value due to lack of market liquidity. VaR systems can estimate this risk by simulating various market conditions and observing the impact on asset prices.

Choice B is incorrect. Market risk, which refers to the potential for losses in positions arising from movements in market prices, is one of the primary risks that VaR systems are designed to measure. They do this by simulating changes in various factors such as interest rates, exchange rates, equity prices and commodity prices.

Choice C is incorrect. Credit risk can also be measured by VaR systems. Credit risk refers to the potential loss resulting from a borrower's failure to meet its obligations as agreed upon in a contract. VaR models can estimate this type of risk using credit spreads and default probabilities.

Things to Remember

- VaR (Value at Risk) is a widely used risk management measure that quantifies the potential loss in value of a portfolio over a specified time horizon and at a given confidence level.
- Liquidity risk is the risk that an investor may not be able to buy or sell an asset quickly enough at a price close to its true underlying value.
- Market risk is the risk of losses in positions due to movements in market prices, such as interest rates, exchange rates, equity prices, and commodity prices.
- Credit risk is the risk of loss resulting from a borrower's failure to meet its obligations as agreed upon in a contract.

- Operational risk is the risk of loss resulting from inadequate or failed internal processes, people, systems, or external events, excluding strategic and reputational risk.
 - VaR systems are not typically used to measure operational risk due to its complex and idiosyncratic nature.
-

Q.2499 As a fund investor, you notice a sudden jump in the reported VaR of the fund. All the following may provide an explanation for the sudden jump, EXCEPT:

- A. The manager may be taking more risk.
- B. Different managers may be taking similar bets.
- C. Managers may be taking different bets.
- D. The volatility of the markets may have increased.

The correct answer is C.

The statement that managers may be taking different bets does not provide a plausible explanation for a sudden increase in the reported VaR of a fund. VaR is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame. It provides a worst-case scenario prediction with a specific confidence level. If managers are taking different bets, it implies that they are diversifying the portfolio, which should, in theory, reduce the risk and hence the VaR. Diversification is a risk management strategy that mixes a wide variety of investments within a portfolio. The rationale behind this technique is that a portfolio constructed of different kinds of investments will, on average, yield higher returns and pose a lower risk than any individual investment found within the portfolio. Therefore, if managers are taking different bets, it should not cause a sudden jump in the reported VaR of the fund.

Choice A is incorrect. The manager taking more risk can indeed lead to a surge in the reported VaR of the fund. This is because VaR measures the potential loss in value of a risky asset or portfolio over a defined period for a given confidence interval. Therefore, if the manager starts taking more risk, it would naturally increase the potential loss and hence increase the VaR.

Choice B is incorrect. Different managers taking similar bets could also potentially explain an increase in VaR. If different managers are making similar bets, they are likely to be exposed to similar risks which could lead to higher aggregate risk at fund level and hence higher VaR.

Choice D is incorrect. The volatility of markets increasing can certainly cause an increase in reported VaR as well. Higher market volatility means greater uncertainty or risk associated with price changes in a security, which directly impacts its potential loss and therefore increases its Value at Risk (VaR).

Things to Remember

- VaR (Value at Risk) is a statistical technique used to measure and quantify the level of financial risk within a firm or investment portfolio over a specific time frame.
 - Diversification is a risk management strategy that mixes a wide variety of investments within a portfolio to reduce risk.
 - Higher market volatility can lead to an increase in reported VaR as it increases the uncertainty and risk associated with price changes in securities.
 - Managers taking more risk can lead to a surge in reported VaR as it increases the potential loss in value of a risky asset or portfolio.
 - Similar bets by different managers can lead to higher aggregate risk at the fund level and hence higher VaR.
-

Q.2501 Which of the following statements is (are) correct?

- I. Risk cannot be measured easily for some series which have very short histories, such as emerging markets, initial public offerings, etc.
 - II. In some cases, the missing series can be replaced by a proxy, using a mapping approach
- A. Only I
 - B. Only II
 - C. Both I and II
 - D. None of the above

The correct answer is **C**.

Both statements I and II are accurate in the context of risk management. Statement I correctly points out that risk measurement can be challenging for certain asset classes due to their illiquidity. This is particularly true for real estate, venture capital, and some categories of hedge funds. Additionally, some series, such as emerging markets and initial public offerings, have very short histories, making risk measurement difficult. Statement II correctly suggests that in some cases, the missing series can be replaced by a proxy using a mapping approach. This is a common practice in risk management when dealing with illiquid assets or series with short

histories. Therefore, both statements I and II are correct, making choice C the right answer.

Choice A is incorrect. This choice suggests that only statement I accurately reflects the scenarios described in the question. However, both statements I and II are accurate. Statement I correctly identifies that certain asset classes like real estate, venture capital, and some categories of hedge funds can be challenging to measure due to their illiquidity. Statement II also correctly states that some series may have very short histories or none at all, but missing series can sometimes be substituted with a proxy using a mapping approach.

Choice B is incorrect. This choice suggests that only statement II accurately reflects the scenarios described in the question. However, as explained above both statements I and II are accurate.

Choice D is incorrect. This choice suggests that neither statement I nor statement II accurately reflect these scenarios which is not correct as both statements do indeed provide an accurate reflection of these scenarios in risk management.

Things to Remember

- Illiquidity of assets can pose challenges in measuring risk accurately, especially for asset classes like real estate, venture capital, and certain hedge funds.
 - Short histories of some series, such as emerging markets and initial public offerings, can make risk measurement difficult.
 - Proxy mapping is a common approach used in risk management to replace missing series with a proxy when dealing with illiquid assets or series with short histories.
 - Risk measurement for alternative investments often requires specialized techniques due to their unique characteristics and data limitations.
 - Historical data quality and availability play a crucial role in the accuracy of risk measurement and management.
-

Q.2502 All the following roles are performed by a global custodian, EXCEPT:

- A. Providing centralized risk management system.
- B. Generating a report that provides a consolidated picture of the total exposure of the fund.
- C. Providing market data and maintaining information on positions.
- D. Providing risk indicators based on historical data.

The correct answer is **D**.

Providing risk indicators based on historical data is not typically a role performed by a global custodian. While global custodians do play a significant role in risk management by providing a centralized system and maintaining position information, they do not typically provide risk indicators based on historical data. This task is usually performed by risk management teams or specialized risk management software. These teams or software analyze historical data to identify patterns and trends that can be used to predict future risks. This information is then used to develop strategies to mitigate these risks. Therefore, while global custodians do play a role in risk management, providing risk indicators based on historical data is not typically part of their responsibilities.

Choice A is incorrect. Global custodians do provide centralized risk management systems. This service is essential for managing and mitigating risks associated with global investments.

Choice B is incorrect. Generating a report that provides a consolidated picture of the total exposure of the fund is indeed one of the services provided by global custodians. These reports help in understanding the overall risk profile and exposure of an investment portfolio.

Choice C is incorrect. Providing market data and maintaining information on positions are also typical services offered by global custodians. They play a crucial role in providing accurate, timely, and comprehensive market data to their clients.

Things to Remember

- **Global Custodian:** A global custodian is a financial institution that is responsible for safeguarding and administering financial assets on behalf of institutional investors. They play a crucial role in providing various services related to asset management and investment operations.
- **Risk Management:** Risk management is the process of identifying, assessing, and prioritizing risks followed by coordinated and economical application of resources to minimize, monitor, and control the probability and impact of unfortunate events.
- **Historical Data Analysis:** Historical data analysis involves the examination of past data to identify trends, patterns, and relationships that can help in making informed decisions about the future. It is a key component of risk management and decision-making processes.
- **Risk Indicators:** Risk indicators are metrics or signals that provide information about the level of risk associated with a particular investment or portfolio. These indicators help in assessing and monitoring risks to make informed decisions.

Q.2503 A pension fund wants to allocate \$300 million to a pool of active managers so as to maximize the information ratio of the fund subject to an overall tracking error volatility (TEV) of 4%. The table below provides more information:

	TEV	Information Ratio
Manager 1	6.0%	0.60
Manager 2	8.0%	0.40
Index	0.0%	0.00
Portfolio	4.0%	0.72

Assuming that the excess returns of the managers are independent of each other, the optimal allocation for Manager 1 is equal to:

- A. \$166.67 million
- B. \$240 million
- C. \$450 million
- D. \$83.33 million

The correct answer is **A**.

Under the given conditions, the optimal allocation for Manager 1 is calculated as:

$$\begin{aligned}\text{Weight of portfolio(i)} &= \frac{IR_i}{IR_P} \times \frac{\text{Portfolio's TEV}}{\text{Manager's TEV}} \\ &= \frac{0.60}{0.72} \times \frac{4\%}{6\%} = 56\%\end{aligned}$$

$$\text{Optimal allocation} = \$300 \text{ million} \times 56\% = \$166.67 \text{ million}$$

Q.2504 A pension fund wants to allocate \$300 million to a pool of active managers so as to maximize the information ratio of the fund subject to an overall tracking error volatility (TEV) of 4%. The table below provides more information:

	TEV	Information Ratio
Manager 1	6.0%	0.60
Manager 2	8.0%	0.40
Index	0.0%	0.00
Portfolio	4.0%	0.72

Assuming that the excess returns of the managers are independent of each other, the optimal allocation for Manager 2 is equal to:

- A. \$80,000,000
- B. \$83,333,333
- C. \$270,000,000
- D. \$33,330,000

The correct answer is **B**.

$$\text{Weight of the portfolio managed by manager } i = \frac{IR_i}{IR_P} \times \frac{\text{portfolio's TEV}}{\text{manager's TEV}}$$

For manager 2,

$$\text{Weight of portfolio} = \frac{0.4 \times 0.04}{0.72 \times 0.08}$$

Thus, the optimal allocation for Manager 2 is:

$$\frac{0.4 \times 0.04}{0.72 \times 0.08} \times \$300,000,000 = \$83,333,333.33$$

Q.2762 A pension fund needs to allocate \$20 million to three different asset classes. The details of the asset classes are shown in the following table:

Asset class	Volatility
US stocks	12%
US bonds	5%
Non-US stocks	18%

The total volatility profile for the fund has been decided to be 10%. If the optimal asset allocation between the three classes is 30%, 40% and 30% respectively, what will be the total VaR budget for the fund and the undiversified VaR at a 99% confidence level?

- A. Total VaR: \$3.30 million; Undiversified VaR: \$6.20 million
- B. Total VaR: \$4.6 million; Undiversified VaR: \$6.20 million
- C. Total VaR: \$4.66 million; Undiversified VaR: \$5.13 million
- D. Total VaR: \$3.30 million; Undiversified VaR: \$5.13 million

The correct answer is **C**.

$$\begin{aligned}
 \text{The total VaR budget will be} &= Z \times \sigma \times V \\
 \text{Total VaR budget} &= 2.33 \times 0.10 \times 20 \\
 &= \$4.66 \text{ million}
 \end{aligned}$$

For the three asset classes, the VaR will be:

$$\begin{aligned}
 \text{US Stocks} &= 2.33 \times 0.12 \times 0.3 \times 20 = \$1.68 \text{ million} \\
 \text{US Bonds} &= 2.33 \times 0.05 \times 0.4 \times 20 = \$0.93 \text{ million} \\
 \text{Non-US Stocks} &= 2.33 \times 0.18 \times 0.3 \times 20 = \$2.52 \text{ million}
 \end{aligned}$$

$$\text{Total undiversified VaR} = \$5.13 \text{ million}$$

Undiversified VaR is the sum of individual VaRs, or the portfolio VaR when there is no short position.

Q.2763 You have been provided the following information about two portfolio managers, and the overall portfolio:

	Tracking error volatility (TEV)	Information ratio
Manager 1	5%	0.5
Manager 2	4%	0.3
Portfolio	3%	0.6

Assuming the excess returns of the two managers are independent of each other, what will be the optimal asset allocation?

- A. 50% for Manager 1 and 38% for Manager 2
- B. 38% for Manager 1 and 50% for Manager 2
- C. 40% for Manager 1 and 60% for Manager 2
- D. 60% for Manager 1 and 40% for Manager 2

The correct answer is **A**.

$$\text{Optimal weight} = \frac{IR_i}{IR_p} \times \frac{\text{Portfolio TEV}}{\text{Manager's TEV}}$$

$$\text{Optimal weight Manager 1} = \frac{(0.5 \times 0.03)}{(0.6 \times 0.05)} = 50\%$$

$$\text{Optimal weight Manager 2} = \frac{(0.3 \times 0.03)}{(0.6 \times 0.04)} = 38\%$$

Q.3028 Differentiate between absolute risk and relative risk.

- A. Absolute risk is the risk of dollar loss owing to the policy mix selected by the fund; relative risk is the risk of dollar loss over the horizon and may also be called asset risk.
- B. Absolute risk is the risk of a dollar loss over the horizon and is sometimes called asset risk; relative risk is the risk of a dollar loss in a fund relative to its benchmark.
- C. Absolute risk is the risk of dollar loss in a fund relative to its benchmark; relative risk is the risk of dollar loss owing to the policy mix selected by the fund.
- D. Absolute risk is the risk of dollar loss in a fund relative to its benchmark and sometimes is called asset risk; relative risk is the risk of a dollar loss over the horizon.

The correct answer is **B**.

Absolute risk is the risk of a dollar loss over the horizon and is sometimes called asset risk; relative risk is the risk of a dollar loss in a fund relative to its benchmark. Absolute risk, also known as asset risk, refers to the potential for loss in the value of an asset over a specific time horizon. This risk is inherent in all investments and is independent of any benchmark. On the other hand, relative risk is the risk that a particular investment will underperform its benchmark or that it will not meet its goals, regardless of whether the goals are to outperform a benchmark or to meet certain predetermined objectives. The benchmark could be a market index or a certain level of return. Therefore, relative risk is the risk of underperformance or not meeting objectives, relative to the benchmark.

Choice A is incorrect because it incorrectly defines both absolute risk and relative risk. Absolute risk is not the risk of dollar loss owing to the policy mix selected by the fund; it is the risk of a dollar loss over the horizon and is sometimes called asset risk. Similarly, relative risk is not the risk of dollar loss over the horizon; it is the risk of a dollar loss in a fund relative to its benchmark. The policy mix of a fund can contribute to its relative risk, but it is not the defining characteristic of absolute risk.

Choice C is incorrect because it reverses the definitions of absolute risk and relative risk. Absolute risk is not the risk of dollar loss in a fund relative to its benchmark; it is the risk of a dollar loss over the horizon and is sometimes called asset risk. Conversely, relative risk is not the risk of dollar loss owing to the policy mix selected by the fund; it is the risk of a dollar loss in a fund relative to its benchmark. The policy mix of a fund can contribute to its relative risk, but it is not the defining characteristic of absolute risk.

Choice D is incorrect because it incorrectly defines both absolute risk and relative risk. Absolute risk is not the risk of dollar loss in a fund relative to its benchmark; it is the risk of a dollar loss over the horizon and is sometimes called asset risk. Similarly, relative risk is not the risk of dollar loss over the horizon; it is the risk of a dollar loss in a fund relative to its benchmark. The policy mix of a fund can contribute to its relative risk, but it is not the defining characteristic of absolute risk.

Things to Remember

- Absolute risk is also known as asset risk and refers to the potential for loss in the value of an asset over a specific time horizon.
- Relative risk is the risk that a particular investment will underperform its benchmark or not meet its goals, regardless of whether the goals are to outperform a benchmark or meet certain predetermined objectives.
- The benchmark in relative risk could be a market index or a certain level of return.
- Absolute risk is independent of any benchmark, while relative risk is defined in comparison to a benchmark.

- The policy mix of a fund can contribute to its relative risk, but it is not the defining characteristic of absolute risk.
-

Q.3029 PERF has allocated \$650 million in U.S. stocks to two active managers who are equally good, with each manager receiving the same amount of \$325 million. The managers created two distinct portfolios but both have a volatility of 15.8% and a correlation of 0.81 with each other. Compute the total risk budget, assuming that $\alpha = 2.33$.

- A. \$261.947
- B. \$307.148
- C. \$227.641
- D. \$321.049

The correct answer is **C**.

We first determine the risk budget which is given as:

$$\begin{aligned}\alpha\sigma W &= 2.33 \times 0.158 \times \$325 \\ &= \$119.6455 \text{ million each.}\end{aligned}$$

Therefore, the total risk budget is given as:

$$\sqrt{\$119.6455^2 + \$119.6455^2 + 2 \times 0.81 \times \$119.6455 \times \$119.6455} = \$227.641 \text{ million}$$

Q.3157 Anthony James has a portfolio with only a single position of \$700 million invested in shares of Grinold Bank. The manager is considering adding a \$300 million position in Shares of Fujarah Bank or Yamama Bank to the portfolio. The current volatility of Grinolds is 12%. In addition, the shares of Fujarah Bank have a return volatility of 9% and a correlation with Grinold equal to 0.8, while the shares of Yamama Bank have a return volatility of 12% and a correlation with Grinold equal to zero. What will be the VaR at the 99% level of confidence of Anthony's portfolio if he further invests \$300 million in the shares of Fujarah Bank?

- A. \$248.92 million
- B. \$195.72 million
- C. \$443.92 million
- D. \$53.2 million

The correct answer is **A**.

$$\text{The current VaR}_{\text{Grinolds}} = 2.33 * 0.12 * 700 = 195.72$$

$$\text{Standard deviation of Grinold + Fujara} = \sqrt{((0.7)^2(0.12)^2 + (0.3)^2(0.09)^2 + (2)(0.7)(0.3)(0.12)(0.09))}$$

$$= \sqrt{0.0114138} = 0.1068 \text{ or } 10.68\%$$

$$\text{VaR}_{\text{Grinold + Fujarah}} = 2.33 * 0.1068 * 1000 = \$248.92 \text{ million}$$

Q.3159 Donald Clayton is an investment analyst at XYZ Corporation. Clayton wants to construct a portfolio made up of two stocks, A and B. Given $\text{VaR}_A = \$7.9$ million and $\text{VaR}_B = \$5.7$ million. Calculate the portfolio VaR assuming that the two stocks are uncorrelated.

- A. \$6.3 million
- B. \$10.52 million
- C. \$13.6 million
- D. \$9.74 million

The correct answer is **D**.

Since the correlation between A and B IS 0, the formula for Portfolio VaR reduces to:

$$\text{VaR}_p = \sqrt{\alpha^2 w_1^2 \sigma_1^2 + \alpha^2 w_2^2 \sigma_2^2} = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2}$$

VaR of portfolio at a zero correlation is

$$\sqrt{[(7.9)^2 + (5.7)^2]} = 9.74 \text{ million}$$

Q.3162 Consider a two-asset portfolio. The portfolio weight of Asset A is 0.6 and the portfolio weight of Asset B is 0.4. The value of the total portfolio is USD1 million and the standard deviation of its decimal return is 0.060606. If the betas of Asset A and Asset B are 0.8 and 1.3 respectively, the respective marginal VAR of Asset A and the component VAR of Asset B at a 95% confidence level are closest to:

- A. USD 0.80 and USD 32,000
- B. USD 0.80 and USD 99,999.9
- C. USD 0.08 and USD 51,842.37
- D. USD 0.08 and USD 48,000

The correct answer is **C**.

$$\begin{aligned}\text{Diversified VAR (DVAR)} &= z \times \text{Std Dev} \times \text{Portfolio Value} \\ &= 1.645 \times 0.060606 \times 1000000 = 99696.87\end{aligned}$$

$$\begin{aligned}\text{Marginal VAR of Asset A} &= \text{DVAR} \times \frac{\beta_A}{\text{Portfolio Value}} \\ &= 99696.87 \times \frac{0.8}{1000000} = \$0.079757496 \approx 0.08\end{aligned}$$

$$\begin{aligned}\text{Component VAR of Asset B} &= \text{DVAR} \times \beta_B \times \text{Weight of Asset B} \\ &= 99696.87 \times 1.3 \times 0.4 = 51,842.37\end{aligned}$$

Q.3165 Abraham Bell is a portfolio manager at Tipu Investment. He has \$40 million invested in JetAero Engineering and \$25 million in Indigo Engineering. The 95% 1-day VaR are \$2.3 million and \$1.7 million for JetAero and Indigo, respectively. The correlation between the returns of both shares is 0.6. The investment committee has finally decided to rebalance their position between these stock by selling \$7 million of JetAero and using the proceeds to buy more of Indigo. Assuming that returns are normally distributed and that the rebalancing does not affect the volatility of the individual stocks, what effect will this have on the 95% 1-day portfolio VaR?

- A. The portfolio VaR will remain unchanged
- B. The portfolio VaR will increase by \$60,000
- C. The portfolio VaR will decrease by \$34,000
- D. The portfolio VaR will increase by \$34,000

The correct answer is **B**.

$$VaR_p = \sqrt{(VaR_A^2 + VaR_B^2 + 2 \times \rho_{AB} \times VaR_A \times VaR_B)}$$

$$\text{Initial portfolio } VaR_p = \sqrt{2.3^2 + 1.7^2 + 2 \times 0.6 \times 2.3 \times 1.7} = 3.59 \text{ million}$$

Then, we calculate readjusted after the rebalancing of individual stock:

$$\text{Readjusted daily VaR of JetAero} = \left[\frac{(40 - 7)}{40} \right] \times 2.3 = 1.8975 \text{ million}$$

$$\text{Readjusted daily VaR of Indigo} = \left[\frac{(25 + 7)}{25} \right]^* 1.7 = 2.176 \text{ million}$$

$$\text{New VaR of the portfolio is} = \sqrt{1.8975^2 + 2.176^2 + 2 \times 0.6 \times 1.8975 \times 2.176} = 3.65 \text{ million}$$

Therefore, the VaR will increase by $(3,650,000 - 3,590,000)$, or \$60,000.

Q.3166 MaYong Li is a portfolio manager at Xing Nan Investment, a large asset management company in mainland China. He has a mandate to manage \$100 million to be invested in only those shares whose business are related to the Chinese shipping industry. Due to this limited scope, he has selected only two stocks worth investing in. He has \$60 million invested in BeiXin Shipping and \$40 million invested in TongChi Logistics with a 95% 1-day VaR of \$6.3 million and \$4.1 million, respectively. Assuming that returns are normally distributed and that individual VaRs also remains unchanged, what effect will this have on the 95% 1-day portfolio VaR if the correlation between the returns of both shares change from zero in 2017 to perfectly correlated in 2018?

- A. The portfolio VaR will remain unchanged
- B. The portfolio VaR will increase by \$2.88 million
- C. The portfolio VaR will decrease by \$2.88 million
- D. The portfolio VaR will decrease by 3.4 million

The correct answer is **B**.

The portfolio VaR reduces to the following expression in case the correlation ρ is zero:

$$\begin{aligned} \text{VaR}_p &= \sqrt{\alpha^2 w_1^2 \sigma_1^2 + \alpha^2 w_2^2 \sigma_1^2 \sigma_2^2} = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2} \\ \text{VaR in 2017} &= \sqrt{(6.3^2 + 4.1^2)} = \$7.52 \text{ million} \end{aligned}$$

w_1 and w_2 are both positive if the correlation is exactly 1, and thus the equation becomes:

$$\begin{aligned} \text{VaR}_p &= \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\text{VaR}_1 \times \text{VaR}_2} \\ &= \text{VaR}_1 + \text{VaR}_2 \\ \text{VaR in 2018} &= 6.3 + 4.1 = \$10.4 \text{ million} \end{aligned}$$

The portfolio VaR increased by $10.4 - 7.52 = \$2.88$ million

Q.3167 Alan Green is an equity strategist at Platinum Investments. The firm has a large cap equity fund that has a net asset value (NAV) of \$5.6 billion and is fully invested in equities. The returns are normal distributed with volatility of 12% per annum. The fund measures absolute risk with a 95%, one-year VAR at \$1.4 billion. The fund wants to allocate this risk to two equity managers, each with the same VAR budget. Given that the correlation between the returns generated by the two managers is 1.0, the VAR budget for each fund manager should be:

- A. \$0.41 billion
- B. \$1.06 billion
- C. \$0.72 billion
- D. \$0.70 billion

The correct answer is **D**.

$$\text{VaR}_p = \sqrt{(\text{VaR}_A^2 + \text{VaR}_B^2 + 2 \cdot \rho_{AB} \cdot \text{VaR}_A \cdot \text{VaR}_B)}$$

Assume X be the risk budget allocation to each manager. This should be such that:

$$X^2 + X^2 + 2\rho_{XX} = \$1.4^2$$

Solving for X, we get $= X^2[1 + 1 + 2(1)] = \$1.4^2$

$$4x^2 = 1.4^2$$

$$X = 0.7 \text{ billion}$$

Q.4702 Which of the following statements differentiates between absolute risk and relative risk?

- A. Absolute risk is the possibility of loss caused by an insufficient cover to the liabilities of the fund, while relative risk is where the surplus ultimately applies to the risk of the plan owner.
- B. Absolute risk arises from yearly changes in the contribution to the pension fund, while relative risk stems from a fluctuation of financial earnings of the plan sponsor.
- C. Absolute risk is the possibility of a threat from base currency loss over the horizon, while relative risk is the base currency difference between the return and that invested in the benchmark.
- D. Absolute risk is the base currency loss regarding the benchmark selected by the fund,

while relative risk is the sum of profit or losses from active managers concerning the baseline.

The correct answer is **C**.

Absolute risk, also referred to as asset risk, is the possibility of a dollar loss over the horizon, and its rate of return is R_{asset} . On the other hand, relative risk is the dollar difference between the fund return and that of the amount invested in the benchmark. The tracking error is the appropriate return for this risk, given by $E = R_{\text{asset}} - R^b$. This choice accurately differentiates between absolute risk and relative risk by correctly defining each term and their implications in financial risk management.

Choice A is incorrect. Absolute risk does not refer to the possibility of loss caused by an insufficient cover to the liabilities of the fund. Instead, it refers to the potential for a dollar loss over a certain period. Similarly, relative risk does not pertain to where surplus ultimately applies but rather refers to the dollar difference between the return of a fund and that invested in a benchmark.

Choice B is incorrect. Absolute risk does not arise from yearly changes in contributions to pension funds; it pertains more broadly to potential losses over time. Relative risk, on the other hand, doesn't stem from fluctuations in financial earnings of plan sponsors but rather from differences between returns on investments and benchmarks.

Choice D is incorrect. Absolute risk isn't about base currency loss regarding any benchmark selected by a fund; it's about possible losses over time irrespective of any benchmark comparison. Relative risk isn't about summing up profits or losses from active managers concerning baseline; instead, it's about comparing returns against those achieved by investing in benchmarks.

Things to Remember

- Absolute risk is the risk of a loss occurring over a certain time horizon, typically measured in dollar terms.
- Relative risk is the difference in returns between a fund and its benchmark, often measured by tracking error.
- Tracking error is the standard deviation of the difference between the portfolio return and the benchmark return.
- Absolute risk is more concerned with the potential for losses, while relative risk focuses on how a fund performs compared to a benchmark.
- A common measure of relative risk is the Sharpe ratio, which considers both risk and return in comparing a fund to its benchmark.

Q.4704 Which one of the following is true about the “buy side” of the investment management industry? Investors:

- A. have a shorter horizon.
- B. have a slow turnover.
- C. have higher leverage.
- D. control risks by utilizing position limits and stop-loss rules.

The correct answer is **B**.

'Buy side' investors in the investment management industry are characterized by slow turnover. This is primarily due to the nature of the assets they deal with. These assets are often bulky and come with high costs. As a result, the process of buying and selling these assets is not as frequent or fast-paced as it might be in other sectors of the industry. This slow turnover is a defining characteristic of 'buy side' investors and sets them apart from other participants in the investment management industry.

Choice A is incorrect. Buy-side investors typically have a longer-term investment horizon, not shorter. They are primarily concerned with generating returns over the long term and therefore tend to hold positions for extended periods.

Choice C is incorrect. Leverage can vary greatly among buy-side investors and it's not a defining characteristic of them. Some may use high leverage while others may use none at all, depending on their investment strategy and risk tolerance.

Choice D is incorrect. While it's true that buy-side investors control risks by utilizing position limits and stop-loss rules, this is not unique to them as sell-side participants also employ these risk management techniques.

Things to Remember

- 'Buy side' investors typically include entities such as mutual funds, pension funds, insurance companies, and hedge funds.
- They are focused on managing the assets of their clients or shareholders, as opposed to executing trades for clients like the 'sell side'.
- Buy-side investors often conduct in-depth research and analysis to make investment decisions, aiming to generate returns for their clients over the long term.
- They may use various strategies such as long-only, long-short, event-driven, or

quantitative strategies to achieve their investment objectives.

- Risk management is a crucial aspect of buy-side investing, and they employ various tools and techniques to control and mitigate risks in their portfolios.
-

Q.4705 A pension fund allocates \$ 4,550 million to different classes of assets. The volatility of the investment is 7%. At a 95% level of confidence, what is the value-at-risk?

- A. \$ 318.5 million
- B. \$ 742.11 million
- C. \$ 525.53 million
- D. \$ 382.20 million

The correct answer is **C**.

$$\text{VaR} = \alpha \sigma w$$

At 95% confidence level, $\alpha=1.65$

$$\text{VaR} = 1.65 \times 0.07 \times \$4,550 = \$525.53 \text{ million}$$

Q.4706 Consider a company with three active managers. Their expected alpha and tracking error are 5% and 8.4%, respectively. Calculate the total information ratio. **Assumptions**

- The information ratio is the same for each manager.
- All the strategies have equal weights.
- The trading strategies are independent.

A. 1.031

B. 0.595

C. 1.68

D. 0.007

The correct answer is **A**.

When IRs are equal,

$$\begin{aligned}\text{Total IR} &= \text{IR}_i \sqrt{N} \\ \text{IR} &= \frac{\alpha}{\text{TE}} = \frac{0.05}{0.084} = 0.595 \\ \therefore \text{Total IR} &= 0.595 \sqrt{3} = 1.031\end{aligned}$$

Note that the law of active management, also known as the total information ratio, states that the overall information ratio for a group of N managers can be calculated as IR multiplied by the square root of N. As N increases, the theoretical information ratio of a fund would approach infinity. However, in reality, it is challenging to find a significant number of completely unrelated trading strategies.

Q.4707 Consider a pension fund that wants to allocate \$ 4,000 million to a pool of 4 active managers to maximize the information ratio of the fund subject to the overall volatility of 10.1%. Suppose that they each receive the same amount. Using a 95% confidence level, what is the risk budget for each manager?

- A. \$ 404.00 million
- B. \$ 166.65 million
- C. \$ 666.60 million
- D. \$ 941.32 million

The correct answer is **B**.

Since each receives the same amount, the allocation for each is:

$$\frac{\$4,000}{4} = \$1,000 \text{ million}$$

The risk budget is then:

$$\begin{aligned} \text{VaR} &= \alpha \sigma W \\ &= 1.65 \times 0.101 \times \$1,000 = \$166.65 \text{ million} \end{aligned}$$

Q.4708 Consider a company that allocates \$ 6,200 million to two asset classes, machinery, and vehicles.

Asset Class	Weights	Volatility
Machinery	68%	14.2%
Vehicles	32%	18.8%

Which of the following gives the capital allocation for each asset?

- A. Machinery: \$880; Vehicles: \$1,165
- B. Machinery: \$4,622; Vehicles: \$9823
- C. Machinery: \$4,216; Vehicles:\$1,984
- D. Machinery: \$1,452; Vehicles: \$6,956

The correct answer is **C**.

The allocation for each asset is given by $w_i W$.

For machinery= $0.68 \times \$6,200 = \$4,216$ million

For vehicles= $0.32 \times \$6,200 = \$1,984$ million

Q.4709 Consider a company that allocates \$ 6,200 million to two asset classes, machinery, and vehicles.

Asset	Weights	Volatility
Machinery	68%	14.2%
Vehicles	32%	18.8%

What is the risk budget for each asset? Take $\alpha=1.65$.

A. Machinery: \$4,216; Vehicles: \$1,984

B. Machinery: \$988; Vehicles:\$615

C. Machinery: \$1,394; Vehicles: \$869

D. Machinery: \$4,730; Vehicles: \$1,048

The correct answer is **B**.

The allocation for each asset is given by $w_i W$.

For machinery = $0.68 \times \$6,200 = \$4,216$ million

For vehicles = $0.32 \times \$6,200 = \$1,984$ million

Risk budget = $\alpha \sigma w$

For machinery = $1.65 \times 0.142 \times \$4,216 = \$987.81$ million

For vehicles = $1.65 \times 0.188 \times \$1,984 = \$615.44$ million

Reading 152: Risk Monitoring and Performance Measurement

Q.2507 You are a risk management consultant working with a prestigious financial firm, tasked with developing a comprehensive risk plan for a highly complex investment venture. The investment involves launching an innovative product that incorporates cutting-edge technology in an emerging market. Your team has identified numerous potential risks that could impact the venture's success, and now you must focus on risk planning to mitigate and manage these uncertainties effectively. As a risk management consultant, your primary objective in risk planning for this intricate investment venture is to:

- A. Eliminate all identified risks completely.
- B. Transfer all risks to external parties.
- C. Maximize the potential returns of the investment venture.
- D. Reduce the impact of identified risks to an acceptable level.

The correct answer is **D**.

In risk planning, the main objective is not to eliminate all risks entirely, as this is often impractical and could lead to excessive precautionary measures that hinder potential opportunities. Instead, risk planning aims to identify, assess, and mitigate risks, reducing their impact to an acceptable level. This allows the investment venture to progress with a calculated approach, acknowledging that risks exist but are managed appropriately.

A is incorrect This option is not feasible in most real-world scenarios, especially in a complex investment venture involving emerging markets and innovative technology. Attempting to eliminate all identified risks completely would likely result in an over-constrained project, stalling progress and stifling potential growth.

B is incorrect While risk transfer is a common risk management strategy, it is not the primary objective in risk planning. Transferring all risks to external parties (such as insurance companies or third-party vendors) may alleviate some burden, but it does not address the core responsibility of the investment firm to actively manage and mitigate risks internally.

C is incorrect While maximizing potential returns is a desirable goal, it is not the primary objective of risk planning. Risk planning primarily focuses on identifying and managing potential risks to safeguard the investment's success and sustainability. It involves assessing risk-reward trade-offs and ensuring the venture's viability amidst uncertainties, but the primary focus remains on risk management, not returns.

Things to Remember

- In risk planning, the key goal is not to eradicate all risks, but rather to reduce their impact to a level deemed acceptable. This strategy allows room for calculated growth, while ensuring risks are appropriately managed.

- Risks are inherent in any investment venture, especially those involving emerging markets and innovative technologies. The existence of risks is acknowledged, but they are not viewed as roadblocks; instead, they are assessed, prioritized, and managed within the risk plan.
 - Risk transfer, while a valid risk management strategy, is not the primary goal in risk planning. Risks can be mitigated but they can't always be completely transferred, emphasizing the importance of internal risk management.
 - The ultimate aim of risk planning is to safeguard the investment's success and sustainability, considering both risk and reward aspects, but it doesn't necessarily seek to maximize returns.
-

Q.2508 SolarTech Inc., a company specializing in the production of solar panels, has recently been acquired by a larger energy corporation, Delta Energy Group. The merger has introduced a greater complexity in the company's operations, thereby increasing its risk profile. Olivia Ross, Delta Energy's Chief Risk Officer, is tasked with aligning SolarTech's existing internal control system with Delta Energy's more robust risk management framework. SolarTech's risk monitoring activities have, until now, primarily focused on financial risk controls, while Delta Energy Group's approach encompasses a broader range of risks. Olivia knows that risk monitoring is crucial in an internal control environment, but she faces resistance from SolarTech's management, who find it difficult to understand the broader role of risk monitoring. In order to convince SolarTech's management of the importance of a comprehensive risk monitoring process, which of the following statements should Olivia emphasize?

- A. Risk monitoring primarily helps to identify and manage financial risks.
- B. Risk monitoring reduces the need for internal audits.
- C. Risk monitoring ensures that deviations from risk thresholds are identified and reported.
- D. Risk monitoring eliminates all risk associated with business operations.

The correct answer is **C**.

Risk monitoring is a continuous process that allows an organization to observe risk exposures in real time. It enables an organization to identify when actual outcomes deviate from expected outcomes and thus breach established risk thresholds. When deviations occur, a robust risk monitoring system will ensure that these are promptly reported to relevant stakeholders,

allowing for timely decision-making and action.

A is incorrect. Risk monitoring is not limited to financial risks. It encompasses all types of risks that an organization could face, including operational, reputational, strategic, and compliance risks.

B is incorrect. While a robust risk monitoring system can enhance the effectiveness of internal audits, it does not replace the need for them. Internal audits provide a more comprehensive and detailed review of an organization's controls, operations, and processes.

D is incorrect. Risk monitoring does not eliminate all risks associated with business operations. Its role is to facilitate the early detection of risks and assist in managing them effectively. It's important to remember that not all risks can be eliminated; some need to be managed or accepted depending on the organization's risk appetite.

Things to Remember

- Risk monitoring is a continuous, real-time process aimed at observing risk exposures within an organization. It serves as a critical tool for identifying deviations from expected outcomes or breaches of established risk thresholds.
- When such deviations are observed, a comprehensive risk monitoring system promptly reports these to the relevant stakeholders. This facilitates timely decision-making and action to manage the identified risks effectively.
- Risk monitoring covers a broad spectrum of risks, not just financial ones. Operational, reputational, strategic, and compliance risks are among the categories that should be continuously monitored.
- While robust risk monitoring enhances internal audits' effectiveness, it does not replace the need for them. Audits offer a more in-depth review of an organization's operations, controls, and processes.
- Risk monitoring is not designed to eliminate all risks, but rather to facilitate early detection and effective management of risks in line with the organization's risk appetite.

Q.2509 DigitalSpace Inc. is a fast-growing technology company that recently completed an Initial Public Offering (IPO). With its enhanced capital structure, the company is planning to enter new markets and invest in other ambitious ventures. Although the company has achieved impressive growth over the past few years, the management team, led by CEO Emma Thompson, is

relatively inexperienced when it comes to dealing with a diverse group of stakeholders in a public company setting. To address this, the company has hired Alex Martinez, an experienced Chief Risk Officer, to help develop a comprehensive risk management framework. Alex understands that cultivating a sense of risk consciousness within DigitalSpace is a critical first step in this process. He starts by identifying the potential sources of risk consciousness within the company. Based on the situation at DigitalSpace Inc., which of the following is the most likely source of risk consciousness that Alex Martinez could leverage?

- A. The risk management practices followed by direct competitors.
- B. The personal risk tolerance of individual employees.
- C. The concern of lenders about the destinations of their funds.
- D. The coding practices followed in the company's software development process.

The correct answer is **C**.

Having recently completed an IPO, DigitalSpace is likely to be scrutinized by banks and other lenders who are interested in knowing how their funds will be invested. The company will need to demonstrate a solid understanding of risk management to these stakeholders to maintain access to funding. This concern by lenders can be a significant driver of risk consciousness within the organization.

A is incorrect. While the risk management practices of direct competitors may be of interest to DigitalSpace, they are not a direct source of risk consciousness. The practices of competitors can influence industry norms, but risk consciousness typically emerges more directly from stakeholders such as lenders, management, and investors.

B is incorrect. While the personal risk tolerance of individual employees might influence their behavior at work, it's not typically a primary source of risk consciousness within an organization. Organizational risk consciousness tends to be driven more by institutional stakeholders and their requirements and expectations.

D is incorrect. While the coding practices followed in the software development process may introduce operational risk to the company, they are not a source of risk consciousness. Risk consciousness refers to the awareness and understanding of risk at an organizational level and is often driven by the concerns of stakeholders such as lenders, management, and investors.

Things to Remember

- There is a growing sense of risk consciousness within organizations, driven by increased scrutiny from banks, investors, and other stakeholders who require more transparency in asset placements and management.
- Investors are expected to possess more firsthand knowledge about their investment choices, influenced by several high-profile losses. This has led to a greater demand for

asset managers to predict and explain potential performance in high-risk scenarios.

- The increased risk consciousness has resulted in the formation of independent risk management units (RMUs) in many organizations. These units monitor risk exposures of portfolios to ensure they align with risk budgets.
 - The establishment of RMUs was largely influenced by a 1996 paper from the Working Group, advocating for the independence of these units as a core part of effective internal control environments.
 - The Working Group's Third Standard mandates independent internal groups to perform various oversight functions including overseeing investment activity, limit and exception monitoring, stress and back tests, and ensuring managers conduct independent risk oversight.
-

Q.2510 CapitalStrong is an investment management company that offers a range of services to high net worth individuals, pension funds, and corporate clients. Recently, CapitalStrong's internal Risk Management Unit (RMU) has noticed some discrepancies in the performance of one of its portfolios managed by a senior portfolio manager, Mr. John Haynes. The RMU had set predetermined guidelines on acceptable variance and specified actions to be taken if these were exceeded. They had also decomposed tracking risk into several subsections to ensure consistent risk allocation according to the agreed investment philosophy. However, the RMU has noticed a consistent variation from these guidelines, raising questions about Mr. Haynes' investment activities. Given this scenario, which of the following would be the most effective way for the RMU at CapitalStrong to confirm whether Mr. Haynes' investment activities are consistent with the established expectations?

- A. Conduct a complete audit of Mr. Haynes' personal finances.
- B. Initiate a company-wide review of all portfolio managers' activities.
- C. Compare the forecasted tracking error with the budget and examine risk allocation by subsection.
- D. Replace Mr. Haynes with a portfolio manager with a better track record.

The correct answer is **C**.

The RMU can confirm the consistency of Mr. Haynes' investment activities by comparing his forecasted tracking error with the predetermined budget. Significant variances may indicate

discrepancies in risk management, which may require further investigation or immediate action. Furthermore, by decomposing tracking risk into subsections, the RMU can ensure that risk is allocated in accordance with expectations, detecting any style drift in the process.

A is incorrect. Conducting a complete audit of Mr. Haynes' personal finances is irrelevant in this context. His personal finances do not directly influence the performance of the portfolios he manages at CapitalStrong.

B is incorrect. While a company-wide review could potentially unearth other issues, it is not the most effective or immediate way to assess whether Mr. Haynes' activities are consistent with expectations. The problem is specifically with Mr. Haynes' portfolio, and thus the focus should be on this specific situation.

D is incorrect. Replacing Mr. Haynes is a drastic action that does not necessarily confirm whether his investment activities were consistent with the established expectations. It would be more prudent to investigate the discrepancy and make informed decisions based on the findings.

Things to Remember

- Risk Management Units (RMUs) can confirm the consistency of a portfolio manager's activities by comparing the forecasted tracking error with the predetermined budget. Significant variances may indicate discrepancies in risk management, necessitating further investigation or immediate action.
 - Decomposing tracking risk into subsections aids in ensuring risk allocation aligns with the agreed investment philosophy. This allows for the detection of any style drift, further informing the consistency of investment activities.
-

Q.2511 SecureFuture Investments, a prominent global asset management firm, is currently managing a high-profile retirement fund with a long-term investment horizon. Recently, the Chief Risk Officer (CRO), Allison, noted a high level of short-term investment volatility in the fund which contradicted the expected risk profile. Allison decided to conduct a detailed risk monitoring exercise to ensure that the fund's activities are aligned with the fund's expectations and risk profile. In the context of the scenario, which of the following risk monitoring strategies would Allison most likely find useful in determining if the retirement fund's activities are consistent with the fund's expectations?

- A. Evaluate the diversification of the fund's investment portfolio.
- B. Investigate the personal investment strategies of the portfolio managers.
- C. Rely solely on historic performance data for risk evaluation.
- D. Consult with fund investors on their risk tolerance.

The correct answer is **A**.

Risk monitoring can validate the alignment of investment activities with set expectations by evaluating the diversification of the fund's investment portfolio. This analysis can provide insights into the risk-return tradeoff, investment concentrations, and potential style drift in the portfolio. If the diversification level is inconsistent with the fund's expected risk profile, it could explain the high short-term volatility.

B is incorrect. Investigating the personal investment strategies of the portfolio managers is irrelevant to risk monitoring as it does not provide direct information about the risk level or investment strategy of the retirement fund.

C is incorrect. While historic performance data can provide useful insights, relying solely on it for risk evaluation can lead to inaccurate risk assessments. This is because past performance does not guarantee future results, and it may not provide a complete view of the fund's current risk position.

D is incorrect. While understanding investor risk tolerance is a crucial part of investment management, it may not be directly useful for Allison in this context. She already has the expected risk profile of the fund and is aiming to assess the consistency of the fund's activities with those expectations. Consulting with investors might be more beneficial in setting risk expectations rather than confirming them.

Things to Remember

The RMU can use the following strategies to confirm whether the investment activities are in line with expectations:

- Evaluating whether the risk capital is spent in the expected themes for each portfolio (diversification).
- Evaluating whether the risk forecasting model is performing as predicted.
- Evaluating whether the forecasted tracking error is consistent with the target.

Q.2512 Vanguard Capital is a high-profile investment management firm, reputed for its disciplined approach and high return distributions. Recently, the firm added a new portfolio, the Global Growth Fund, to its investment offerings, targeting high-net-worth investors. As a part of their internal risk management protocols, Vanguard Capital regularly employs risk monitoring processes to ensure investment activities align with expectations. After a quarter of managing the Global Growth Fund, Steve, the Chief Risk Officer (CRO), has decided to conduct a thorough risk monitoring exercise. Based on the scenario, which of the following risk monitoring actions should Steve focus on to effectively confirm that the Global Growth Fund's activities are consistent with its stated investment objective and expectations?

- A. Assess the variance between the Global Growth Fund's actual returns and the market's overall returns.
- B. Evaluate the forecasted level of tracking error against the target for the Global Growth Fund.
- C. Check the consistency of the Global Growth Fund's investment performance with its previous funds.
- D. Investigate the professional qualifications and years of experience of the Global Growth Fund's portfolio manager.

The correct answer is **B**.

Risk monitoring processes often involve the evaluation of the forecasted level of tracking error against the target established by the mandate. In the context of the Global Growth Fund, the forecasted tracking error can provide insight into whether the potential risk inferred from the fund's positions is consistent with the risk level that was originally envisaged. This measure can provide an important signal as to whether the fund's investment activities are aligned with expectations.

A is incorrect. While the variance between the Global Growth Fund's actual returns and the market's overall returns could provide some insight into the fund's performance, it may not necessarily indicate whether the fund's investment activities align with its stated objectives and expectations. The fund could outperform or underperform the market due to various factors, not all of which may relate to its adherence to its investment strategy.

C is incorrect. Checking the consistency of the Global Growth Fund's investment performance with Vanguard Capital's previous funds might be informative in some contexts, but it may not provide a clear picture of whether the Global Growth Fund's activities are consistent with its own stated objectives and expectations. Each fund may have different investment strategies, risk tolerances, and investor bases.

D is incorrect. Investigating the professional qualifications and years of experience of the Global Growth Fund's portfolio manager may provide some level of confidence in the manager's ability, but it does not provide direct information about whether the Global Growth Fund's activities align with its stated objectives and expectations. The focus of risk monitoring should be on the actions and results of the fund, not the background of the manager.

Things to Remember

- Tracking error is a critical measure used in risk monitoring that represents how much a portfolio's returns deviate from a benchmark. This error can provide insights into a fund's risk level and its adherence to an investment strategy.
- A low tracking error indicates a portfolio's performance is closely aligned with the benchmark, signaling a more conservative investment approach. Conversely, a high

tracking error can denote a more active or aggressive investment strategy, as the portfolio often deviates from the benchmark's returns.

- Comparing the forecasted level of tracking error against the established target can confirm if the risk inferred from a fund's positions aligns with its expected risk level. This analysis can effectively reveal if a fund's activities match its stated investment objectives and expectations.
 - The analysis of tracking error forms a crucial component of risk monitoring activities and assists in maintaining a fund's transparency, integrity, and risk-return tradeoff.
-

Q.2514 Stellar Asset Management is a successful investment management firm with a long-standing reputation for consistent returns. Recently, the firm onboarded a new portfolio manager, Adam, to manage a portion of its flagship Global Equities Fund. Adam has an unconventional strategy that doesn't neatly fall into the traditional 'value' or 'growth' philosophies. Despite this, his past performance at other firms has been stellar. Adam is allowed to make significant allocations to innovative tech companies, which usually fall under the growth investing style. Six months after Adam's arrival, Michelle, the Chief Risk Officer (CRO), observes that the fund has outperformed the benchmark. However, it has also experienced higher volatility than before. She decides to conduct a thorough risk monitoring process to ensure the current investment activities align with the expectations for the fund's risk profile. Given the scenario, which of the following risk monitoring steps should Michelle prioritize to ensure the fund's investment activities are consistent with the fund's risk profile expectations?

- A. Assess Adam's personal investment philosophy and past performance at other firms.
- B. Monitor the marginal contribution to risk at the stock, industry, sector, and country levels.
- C. Survey the fund investors to understand their satisfaction with the current performance.
- D. Compare the fund's performance to other funds with a similar investment strategy.

The correct answer is **B**.

By monitoring the marginal contribution to risk at various levels, Michelle can better understand how each element contributes to the fund's overall risk profile. It will also provide insights on any concentrations of risk that might be outside the fund's risk expectations. In this case, it could help in identifying if the fund's recent volatility and outperformance are due to Adam's significant allocations to growth-oriented tech companies.

A is incorrect. While understanding Adam's personal investment philosophy and past performance can provide insights, it doesn't directly address the risk profile of the current investment activities within the fund.

C is incorrect. Surveying the fund investors might provide useful feedback on the perceived performance of the fund, but it doesn't directly contribute to the understanding of whether current investment activities align with the fund's risk profile expectations.

D is incorrect. Comparing the fund's performance to other funds with a similar investment strategy can offer useful benchmarks but doesn't specifically address whether the fund's investment activities are aligned with its own risk profile expectations. The other funds may have different risk tolerances and investment mandates.

Things to Remember

- The Marginal Contribution to Risk (MCR) is a risk measure that quantifies how much each investment or asset class contributes to the total risk of an investment portfolio. It's a useful measure for risk managers to understand which investments are adding most to the portfolio risk.
- By analyzing MCR at different levels such as individual stock, industry, sector, and country, risk managers can identify concentrations of risk and whether they align with the fund's risk expectations.
- An increase in the MCR of an investment suggests that it's contributing more to the portfolio's overall risk. If a particular investment's MCR exceeds the acceptable level, it may indicate a need for portfolio rebalancing or a revision of the investment strategy.
- Monitoring MCR is a critical step in the risk monitoring process, as it can help ensure that the portfolio's risk remains within the boundaries set by the investment mandate and aligns with the expected risk profile.

Q.2515 Transglobal Investments is a multinational investment firm that has built its reputation around a commitment to a value-based investing style. Recently, the firm hired Laura, a portfolio manager known for her knack in identifying undervalued stocks, to manage their Global Equity Fund. In the last quarterly review, the firm noticed that the Fund, while meeting the targeted tracking error, has seen a spike in the allocation of the portfolio to technology companies, which are traditionally regarded as growth investments rather than value investments. To understand this anomaly and ascertain that the portfolio management strategy is aligned with the firm's philosophy and expectations, the Risk Management Unit (RMU) is called upon. What would be the most effective step for the RMU to ensure that the investment activities of Laura are

consistent with the firm's value investing philosophy?

- A. Interview Laura to understand her rationale for the increased tech allocations.
- B. Analyze the performance of the fund over the last quarter to see if the returns have been impacted by the tech allocations.
- C. Investigate the risk decomposition to see if the increased allocation to tech stocks aligns with the firm's expectations.
- D. Review the client feedback about the performance of the fund to understand their perception about the tech allocations.

The correct answer is **C**.

Risk decomposition will help the RMU understand how different elements of the portfolio are contributing to the overall risk. In this case, if Laura's allocation to tech stocks is leading to a drift away from the firm's value investing philosophy (style drift), it could show up in the risk decomposition analysis.

A is incorrect. Interviewing Laura may provide some insight, but this step alone does not provide an objective or quantitative measure to determine if the portfolio's risk aligns with the firm's expectations.

B is incorrect. Analyzing the performance of the fund over the last quarter does not necessarily confirm that the portfolio's risk profile aligns with the firm's expectations. Good performance does not necessarily mean alignment with the firm's philosophy.

D is incorrect. While client feedback is essential for the firm, it does not necessarily align with the firm's risk expectations and philosophy. Client feedback often focuses on returns rather than risk or style consistency.

Things to Remember

- Risk decomposition is a method used in risk management to understand how different parts of an investment portfolio contribute to its total risk. This technique provides a detailed view of risk by breaking it down into its constituent parts.
- By investigating risk decomposition, risk managers can get insights into the sources of risk and return within a portfolio. This helps identify the sectors, industries, or stocks that are contributing most to the portfolio's risk.
- This process can uncover any style drift - a shift from one investment style to another, such as from value to growth. Style drift can potentially lead to a portfolio that does not align with the stated investment strategy or the investor's risk and return expectations.

- Through risk decomposition, risk managers can determine if any specific allocations, such as increased allocation to technology stocks, are aligning with the fund's expectations and investment philosophy.
 - Risk decomposition is an important tool in the risk monitoring process to ensure that portfolio's risk is consistent with the risk profile outlined in its investment mandate.
-

Q.2516 Omega Investments, a well-respected asset management firm, has a diverse portfolio with investments spread across equities, bonds, commodities, and alternative assets. As part of the risk management strategy, the firm ensures that it maintains an optimal level of liquidity to cater to both expected and unexpected cash flow needs. Recently, the firm has observed increased redemption requests from clients due to market volatility. In response, the Risk Management Unit (RMU) decides to review the firm's liquidity status. The RMU wants to measure the potential duration of illiquidity for certain assets, given the volatile market conditions. They decide to calculate the Liquidity Duration Statistic (LDS) for a comprehensive understanding of the portfolio's liquidity risk. What would be the most effective use of the Liquidity Duration Statistic (LDS) by the RMU to manage the liquidity risk of the firm's portfolio?

- A. Use the LDS to predict the price movements of the assets in the portfolio.
- B. Apply the LDS to estimate the potential duration it would take to liquidate a position without significantly affecting the market price.
- C. Utilize the LDS to evaluate the creditworthiness of the firm.
- D. Leverage the LDS to calculate the exact amount of cash needed to meet client redemptions.

The correct answer is **B**.

LDS is a statistic that helps estimate the potential time duration it would take to liquidate a position without causing substantial impact on the market price. Given the increased redemption requests and market volatility, LDS could provide valuable insight into the time it would take to meet these redemptions without affecting the portfolio's value significantly.

A is incorrect. The LDS is not designed to predict the price movements of the assets in the portfolio. It's primarily used to measure the liquidity of a position.

C is incorrect. The LDS doesn't evaluate the creditworthiness of the firm. Creditworthiness is generally assessed through measures such as credit ratings, financial ratios, etc.

D is incorrect. LDS doesn't calculate the exact amount of cash needed to meet client redemptions. It primarily provides an estimate of the time duration it might take to liquidate a position without causing a substantial impact on market price.

Things to Remember

- The Liquidity Duration Statistic (LDS) is a measure used to estimate the time it would take to liquidate an asset or position in the market without significantly impacting its price. This calculation takes into account both the size of the position and the liquidity of the market in which the asset is traded.
 - LDS can be particularly valuable during periods of market volatility, as it helps firms understand how quickly they could convert their assets into cash if necessary. This can be essential in times of high redemption requests, where maintaining portfolio value is critical.
 - LDS is a useful tool in managing liquidity risk. By estimating the potential duration of illiquidity, firms can better plan their cash flow needs and avoid costly or disruptive asset sales.
-

Q.2517 Unity Bank has an independent risk management unit. The RMU develops an inventory of risk data for use in evaluating portfolio managers and market environments. During an internal meeting, RMU members opined that the use of data must be restricted and should be used exclusively by the RMU. Which of the following statements is correct

- A. The members are correct.
- B. This risk data should be synthesized and routinely circulated to the appropriate decision-makers.
- C. The development of inventory risk data does not form part of the objectives of the RMU.
- D. This risk data should be synthesized and be shown to senior management during Board meetings.

The correct answer is **B**.

The risk data developed by the RMU should indeed be synthesized and routinely circulated to the appropriate decision-makers. This is because the primary role of the RMU is to create an inventory of risk data that can be used to evaluate portfolio managers and market environments. By circulating this data to the appropriate decision-makers, the RMU ensures that these individuals are equipped with the necessary information to make informed decisions. This process of synthesizing and circulating risk data is crucial in risk management as it allows for

the identification, assessment, and prioritization of risks. This, in turn, enables the organization to coordinate and economically apply resources to minimize, monitor, and control the probability or impact of unfortunate events. Therefore, the suggestion by the RMU members to restrict the use of the data to the RMU alone is not in line with the objectives of risk management.

Choice A is incorrect. The members of the RMU are not correct in suggesting that the use of risk data should be limited and exclusively used by the RMU. In fact, risk data should be shared with relevant decision-makers to ensure informed decision-making across all levels of the organization.

Choice C is incorrect. The development of inventory risk data does indeed form part of the objectives of an RMU. It's crucial for an RMU to create and maintain a comprehensive inventory of risk data as it aids in assessing portfolio managers and market conditions.

Choice D is incorrect. While it's important for senior management to have access to synthesized risk data, it shouldn't only be shown during Board meetings. Risk information needs to be routinely circulated among appropriate decision-makers so they can make timely decisions based on current risks.

Things to Remember

- Risk data is crucial for decision-making in financial institutions. It provides insights into potential risks and allows decision-makers to take proactive measures to mitigate these risks.
 - The RMU plays a vital role in risk management by creating an inventory of risk data. This data is used to assess portfolio managers and market conditions, thereby helping the organization to make informed decisions.
 - It is important for the RMU to routinely circulate the synthesized risk data to the appropriate decision-makers. This ensures that the decision-makers are equipped with the necessary information to make informed decisions.
 - The process of synthesizing and circulating risk data is a key component of risk management. It allows for the identification, assessment, and prioritization of risks, enabling the organization to coordinate and economically apply resources to minimize, monitor, and control the probability or impact of unfortunate events.
-

Q.2518 Omega Investments, a leading hedge fund, has a diverse portfolio of assets, including equities, bonds, and derivatives. Given the market volatility due to recent geopolitical events, the fund's Chief Risk Officer, Jennifer, decides to conduct a thorough liquidity risk assessment. She realizes that conventional liquidity measures may not fully capture the complexities of the fund's diverse asset classes. Remembering a novel measure she learned about during her Financial Risk Manager certification, the Liquidity Duration Statistic (LDS), Jennifer decides to explore its applicability to Omega's portfolio. In Jennifer's context, how might the Liquidity Duration Statistic (LDS) be useful in assessing liquidity risk for Omega Investments?

- A. LDS can measure the fund's ability to meet short-term obligations.
- B. LDS can estimate the time it would take to liquidate an asset or portfolio without significantly impacting its price.
- C. LDS can evaluate the amount of cash readily available in the fund.
- D. LDS can determine the number of liquid assets as a percentage of total assets in the fund.

The correct answer is **B**.

The Liquidity Duration Statistic (LDS) measures liquidity risk by estimating the time it would take to liquidate a given position without significantly affecting its market price. It's particularly useful for a fund like Omega Investments, which holds a diverse portfolio of assets with potentially different liquidity profiles.

A, C, and D are incorrect as per the explanation for B above.

Things to Remember

- Liquidity Duration Statistic (LDS) is a measure of liquidity risk that estimates the time it would take to liquidate a given position without significantly affecting its market price. This measure is particularly useful for diverse portfolios containing different types of assets, each with their own unique liquidity profiles.
- In volatile markets, the LDS can provide critical insights to risk officers like Jennifer, helping them understand how quickly they can convert assets into cash without causing a significant impact on the assets' market prices. This is an important part of liquidity risk management, especially for portfolios with complex or less liquid assets.

Q.2520 An organization has recently established an independent Risk Control Unit (RCU) and is in the process of defining its roles and responsibilities. Which of the following tasks would

correctly fall under the purview of the RCU?

- A. Selecting data analysis software and financial intelligence reports to guide trading activities.
- B. Determining VaR levels to align with risk targets outlined in the organization's risk strategy.
- C. Supervising analysts as they develop asset valuation models.
- D. Reviewing the efficacy of the risk measurement models being used by the organization.

The correct answer is **D**.

Reviewing the efficacy of risk measurement models is one of the main responsibilities of an independent Risk Control Unit (RCU). This involves examining and ensuring the accuracy and reliability of the models used to gauge risk within the firm.

A is incorrect. The selection of data analysis software and financial intelligence reports used to influence trading activities is a function generally performed by the trading or portfolio management teams, not the RCU.

B is incorrect. Although the RCU would monitor VaR levels to ensure they are within the acceptable range established by the firm's risk strategy, it is generally the responsibility of portfolio managers to decide on these risk-taking levels

C is incorrect. It's usually not the responsibility of the RCU to supervise analysts during the development of asset valuation models. The RCU's role leans more towards independent oversight and validation of models rather than their creation or implementation. This would typically be done by the portfolio management function.

Things to Remember

Beyond reviewing the efficacy of risk measurement models, other objectives of the RMU include:

- Ensuring accurate risk data handling: collection, monitoring, analysis, and distribution.
- Assisting in systematizing risk identification and management while promoting risk culture.
- Researching emerging risk topics, tracking risk trends, and promptly flagging unusual events.
- Facilitating risk-related discussions within the organization and helps implement findings.

- Boosting risk awareness, ensuring transactional alignment with guidelines and expectations.
 - Developing and employing risk measurement tools, using risk data for performance assessments.
 - Aiding management in understanding portfolio risks and performance sources, and flags significant deviations.
-

Q.2521 In a newly formed asset management firm, a team of risk managers is tasked with implementing a risk management process covering risk planning, budgeting, and monitoring. As they start to dissect the elements and rules contained within the risk plan, which of the following assertions would be accurate?

- A. Incorporating qualitative scenario analyses into a risk plan can aid in pinpointing elements that could potentially lead to the plan's failure.
- B. While a risk plan can establish volatility targets, it can't factor in how the company's main dependencies might influence these targets.
- C. Severe or outlier events should not be part of a company's risk plan but should be reflected in its strategic plan.
- D. The risk plan should include an estimation of return on equity, calculated using the return on risk capital for each allocation taken independently.

The correct answer is **A**.

A risk plan should absolutely include qualitative scenario analyses. These analyses can help in identifying factors that may cause elements of the risk plan to fail. Scenario analysis allows for the simulation of various outcomes under different conditions, providing insight into potential risks and how they might impact the organization.

B is incorrect. A risk plan can and should consider the effects of the organization's key dependencies on the established volatility goals. Any dependency that can affect the risk exposure of the organization should be factored in while setting risk metrics such as volatility.

C is incorrect. Extreme events, or "tail risks," should be incorporated into an organization's risk plan. Not considering such events could lead to substantial unexpected losses. While they might also be considered in the strategic plan, their place in the risk plan is crucial for a comprehensive approach to risk management.

D is incorrect. Yes, a risk plan should set an expected value for return on equity, but this should be done while considering the correlations among each activity's defined minimum acceptable

RORC level.

Things to Remember

- Qualitative scenario analyses play a crucial role in comprehensive risk planning. By simulating various outcomes under different conditions, they provide valuable insight into potential risks and how they might impact an organization. This process aids in identifying the elements of a risk plan that could potentially fail under specific conditions, thereby enhancing the robustness of the plan.
 - The planning process should factor in all relevant variables that can affect the organization's risk exposure, including the company's key dependencies. This ensures that risk metrics such as volatility targets reflect the true risk profile of the organization.
 - The inclusion of extreme events, or "tail risks," in the risk plan is essential. Ignoring such events could lead to substantial unexpected losses. These events represent low-probability, high-impact scenarios that can significantly impact an organization's financial health and operational stability.
 - While estimating the expected return on equity is a part of risk planning, this should be done considering the correlations among each activity's defined minimum acceptable Return on Risk Capital (RORC) level. This ensures a more holistic and accurate view of the potential returns and associated risks.
-

Q.2522 Camille is a risk manager at AlfaBeta Capital, a hedge fund specializing in a mix of equity and fixed income investments. In a recent portfolio review, Camille and her team identified a substantial position in a particular security, XYZ Corp, that they hold. To better understand the liquidity risk associated with this position, Camille decides to calculate the Liquidity Duration Statistic (LDS). The fund owns 10,000 shares of XYZ Corp, whose daily trading volume is typically around 100,000 shares. The firm has a policy of not exceeding more than 15% of the daily volume in any given security during liquidation to avoid a material market impact. Based on this information, Camille's calculations yield a value of 1 for the LDS of XYZ Corp. What does this value of LDS imply about the XYZ Corp position?

- A. AlfaBeta Capital would require 1 day to liquidate the XYZ Corp position without causing a significant market impact.
- B. The XYZ Corp position contributes 1% to the portfolio's overall liquidity risk.
- C. AlfaBeta Capital would experience a 1% market impact cost if the XYZ Corp position was liquidated immediately.
- D. AlfaBeta Capital could liquidate up to 1% of the XYZ Corp position daily without causing a significant market impact.

The correct answer is **A**.

The Liquidity Duration Statistic (LDS) represents the number of days required to liquidate a position in an orderly fashion, without causing a significant market impact. In this case, an LDS of 1 for the XYZ Corp position implies that AlfaBeta Capital would require 1 day to liquidate the position without causing a significant market impact.

B, C, and D are incorrect as per the above explanation.

Things to Remember

- Liquidity Duration Statistic (LDS) is a measure of liquidity risk that estimates the time it would take to liquidate a position without significantly impacting its market price. The lower the LDS, the quicker a position can be liquidated, and vice versa.

Q.2523 A portfolio manager is looking to reduce the holdings of four different securities in their portfolio due to liquidity risks. The following table provides information on each security:

Security	Market Value (CNY million)	Shares in Fund	Average Daily Trading Volume	Max Daily Volume for Liquidation (%)
P	100.00	400,000	600,000	25
Q	180.04	450,000	1,200,000	15
R	75.88	300,000	650,000	20
S	35.80	550,000	900,000	22

Which security will require the most number of days to liquidate fully, given the restrictions?

A. Security P

B. Security Q

C. Security R

D. Security S

The correct answer is **D**.

The liquidity duration statistic (LDS) gives us the time required to liquidate a position. It can be computed as:

$$\text{LDS} = \frac{\text{Shares in Fund}}{\text{Average Daily Trading Volume} \times \text{Max Daily Volume for Liquidation(\%)}}$$

Security	Days to Liquidate
P	$\text{LDS}_P = 400,000 / (600,000 \times 0.25) \approx 2.67 \text{ days}$
Q	$\text{LDS}_Q = 450,000 / (1,200,000 \times 0.15) \approx 2.50 \text{ days}$
R	$\text{LDS}_R = 300,000 / (650,000 \times 0.20) \approx 2.31 \text{ days}$
S	$\text{LDS}_S = 550,000 / (900,000 \times 0.22) \approx 2.78 \text{ days}$

Things to Remember

- Liquidity risk refers to the potential for an investor to be unable to buy or sell an asset without causing a significant change in its price.
- Average Daily Trading Volume is a key metric used to assess the liquidity of a security,

representing the average number of shares traded per day over a specific period.

- Max Daily Volume for Liquidation (%) indicates the maximum percentage of the average daily trading volume that can be traded in a day without significantly impacting the price.
 - Liquidity duration statistic (LDS) helps in determining the time required to fully liquidate a position based on trading volume and market constraints.
-

Q.2524 A risk manager at a large bank argues that the risk management unit (RMU) must monitor only the overall tracking error expectations. If the overall tracking error expectations are within the prescribed limits, there is no cause of concern. Which of the following statements is correct?

- A. Monitoring the overall tracking error expectations is sufficient.
- B. The overall tracking error analysis must be restricted to keep it within prescribed limits.
- C. Managers must identify how such tracking error is decomposed into its constituent's parts.
- D. The tracking error expectations can exceed the prescribed limits.

The correct answer is **C**.

The statement that managers must identify how such tracking error is decomposed into its constituent parts is accurate. This is because monitoring the overall tracking error expectations is not sufficient in risk management. It is crucial for managers to understand how these tracking errors are broken down into their individual components. This will enable the risk manager to assess whether risk is being incurred as per expectations, both in total and at the individual component level. By doing so, the risk manager can identify specific areas of concern and take appropriate measures to mitigate the risks associated with each component. This approach provides a more comprehensive understanding of the risk profile and allows for more effective risk management strategies.

Choice A is incorrect. While monitoring the overall tracking error expectations is important, it is not sufficient. It does not provide a complete picture of the risk profile as it fails to identify the sources of risk.

Choice B is incorrect. Restricting the overall tracking error analysis to keep it within prescribed limits may lead to overlooking potential risks. It's crucial for managers to understand how such tracking errors are decomposed into their constituent parts.

Choice D is incorrect. The statement that tracking error expectations can exceed prescribed limits contradicts basic risk management principles. Exceeding these limits could potentially expose the institution to unacceptable levels of risk.

Things to Remember

- Tracking error is a measure of how closely a portfolio follows the index to which it is benchmarked. The most common measure is the root-mean-square of the difference between the portfolio and index returns.
 - Many portfolios are managed to a benchmark, often an index. Some portfolios are expected to replicate, before trading and other costs, the returns of an index exactly (an index fund), while others are expected to 'actively manage' the portfolio by deviating slightly from the index in the hope of outperforming it.
 - Decomposing tracking error into its constituent parts allows managers to understand the sources of tracking error. This can be done using factor analysis, which breaks down the tracking error into contributions from various factors such as sector allocation, stock selection, currency exposure, etc.
 - Understanding the decomposition of tracking error can help managers to manage risk more effectively. For example, if a large part of the tracking error is due to sector allocation, the manager may decide to reduce the portfolio's exposure to that sector. Similarly, if a large part of the tracking error is due to stock selection, the manager may decide to review the stock selection process.
-

Q.2525 Bridgewater Asset Management recently set up a new division and hired Sarah as the Risk Manager. Sarah has been tasked to align the company's risk management process with the firm's overall investment strategy. She begins by reviewing the firm's Value at Risk (VaR), Tracking Error, and the three dimensions of risk management: risk planning, risk budgeting, and risk monitoring. After some consideration, Sarah proposes the following approach: "Let's define VaR as our threshold for the maximum dollar value of losses we are willing to bear for a specific level of confidence over a specific time. We can then use this VaR as a guide to allocate capital among different asset classes based on their risk and return preferences. Additionally, we can use tracking error as a guide for the amount of discretion that can be taken away from benchmark returns for active management. With these measures in place, we can effectively plan, budget, and monitor our risks." Which of the following accurately critiques Sarah's approach?

- A. Sarah's proposal is fundamentally flawed as VaR should never be used in risk budgeting.
- B. Sarah's approach is incorrect as tracking error cannot be used as a guide for active management discretion.

C. Sarah has correctly applied VaR and tracking error to risk planning, budgeting, and monitoring, but the frequency of risk monitoring needs to be established.

D. Sarah's approach is incorrect as the use of VaR and tracking error should be reversed in risk planning and risk budgeting.

The correct answer is **C**.

Sarah has correctly identified and applied the roles of VaR and tracking error within the context of risk planning, risk budgeting, and risk monitoring. VaR is used as the threshold for potential losses, guiding capital allocation among various asset classes according to their risk and return preferences. Meanwhile, tracking error serves as an indicator of the amount of active management discretion that can deviate from benchmark returns. However, Sarah's proposal does not mention the frequency of risk monitoring, which is a crucial aspect of an effective risk management process.

A is incorrect. VaR is indeed an appropriate tool for risk budgeting. It helps determine the threshold of acceptable losses and can guide the allocation of capital among various asset classes based on risk and return preferences.

B is incorrect. Tracking error can certainly be used as a guide for the amount of active management discretion. It helps determine how much a portfolio's performance can deviate from a benchmark, serving as a guide for active management.

D is incorrect. The uses of VaR and tracking error in Sarah's approach are appropriate. VaR should be used in risk budgeting and planning while tracking error is useful in determining the discretion in active management.

Things to Remember

- An effective risk management process consists of three related dimensions: risk planning, risk budgeting, and risk monitoring.
- From a portfolio management perspective, VaR can be determined for each asset class to guide capital allocation decisions based on risk and return preferences. This helps to achieve targeted levels of dollar VaR (risk planning).
- In contrast, tracking error can be used to determine the relative amount of discretion that the portfolio manager can take away from benchmark returns in attempts at active management (risk budgeting).
- Risk monitoring identifies any variations from the budget.

Q.2526 A risk manager argues that the risk management unit (RMU) must monitor the risk decomposition by monitoring the range of acceptable active weights at the stock, industry, sector, and country levels. Another manager adds that, in addition to the monitoring the acceptable active weights, the RMU must also monitor the range of acceptable marginal contributions to risk at the stock, industry, sector, and country levels. Which of the following statements is correct?

- A. Both managers are correct.
- B. Both managers are incorrect.
- C. Only the first manager is correct.
- D. Only the second manager is correct.

The correct answer is **A**.

Both managers are indeed correct in their suggestions. The role of a risk management unit (RMU) in an organization is to identify, assess, and prioritize risks. This involves monitoring various aspects of risk, including active weights and marginal contributions to risk. Active weights, which are portfolio holdings less benchmark holdings, need to be monitored at the stock, industry, sector, and country levels. This is because changes in these weights can significantly impact the overall risk profile of the portfolio. Similarly, the RMU must also monitor the range of acceptable marginal contributions to risk at these levels. Marginal contribution to risk is the amount of risk that each individual investment contributes to the overall portfolio risk. By monitoring these two aspects, the RMU can ensure that the portfolio's risk is managed effectively and efficiently.

B, C, and D are incorrect as per the explanation for A above.

Things to Remember

- Risk decomposition is a process that breaks down the total risk of a portfolio into its individual components. This allows risk managers to identify the sources of risk and to manage them more effectively.
- Active weights are a measure of how much a portfolio's holdings differ from the benchmark. They are important because they can significantly impact the portfolio's risk and return characteristics.
- Marginal contribution to risk is a measure of how much each individual investment contributes to the overall risk of the portfolio. It is important because it allows risk managers to identify the investments that are contributing the most to the portfolio's risk.
- Monitoring both active weights and marginal contributions to risk allows the risk management unit to manage the portfolio's risk more effectively. By keeping a close eye on these measures, the RMU can make adjustments as necessary to keep the portfolio's risk within acceptable levels.

Q.2528 Global Asset Management (GAM) has a reputation for a robust risk management process. It is known for its clear expectations and risk profiles that are outlined for their various asset classes, which allows them to remain consistent with their investment philosophy. GAM utilizes a popular risk-monitoring tool that uses a well-defined risk forecasting model. As a part of their risk monitoring process, they regularly check if the portfolio's actual tracking error is consistent with the targeted tracking error outlined in their risk plan. Recently, during a routine check, they found that the forecasted tracking error of their US Equity Fund was deviating from the target by a significant amount. They discovered that this was due to an over-investment in value-themed assets by one of their portfolio managers, even though GAM primarily focuses on growth-themed assets. However, the portfolio manager argues that while the style of the portfolio might have drifted, the overall risk level remains within the risk budget and therefore should not be a cause for concern. He suggests that this "style drift" provides an opportunity for higher returns without violating the risk limit set by GAM's client. Based on the vignette, which of the following best describes why the portfolio manager's argument might not be completely accurate from the perspective of effective risk management?

- A. GAM's tracking error was higher than its target, indicating an ineffective risk management process.
- B. GAM's client would not approve of a style drift even if it might lead to higher returns.
- C. A style drift might indicate that the portfolio manager is not adhering to GAM's investment philosophy, even if the overall risk level remains within the risk budget.
- D. The portfolio manager's decision to over-invest in value-themed assets implies that GAM's risk forecasting model is not behaving as predicted.

The correct answer is **C**.

A style drift can indicate that the portfolio manager is not investing according to GAM's stated investment philosophy. This means that even if the overall risk level stays within the set risk budget, the manager is not utilizing the risk budget in expected themes, which goes against the principles of effective risk management and GAM's philosophy.

A is incorrect. A higher tracking error does not necessarily indicate an ineffective risk management process. The tracking error deviation should prompt a question, and if it is material enough, it should prompt immediate corrective action.

B is incorrect. It's not about the client's approval of a style drift; it's about the manager's adherence to the firm's investment philosophy. The vignette does not provide information about the client's acceptance of the style drift.

D is incorrect because the portfolio manager's decision to over-invest in value-themed assets doesn't imply a flaw in the risk forecasting model. The issue is the manager's deviation from the firm's expected investment in them.

Things to Remember

- Style drift in portfolio management refers to a situation where the actual investments in a portfolio deviate from the stated investment style or philosophy.
 - Tracking error is a measure of how closely a portfolio follows the index it is benchmarked against. A higher tracking error indicates a larger deviation from the benchmark.
 - Risk budgeting involves setting limits on the amount of risk that can be taken in a portfolio, ensuring that the overall risk level remains within acceptable boundaries.
 - Effective risk management involves not only staying within risk limits but also adhering to the stated investment philosophy and style guidelines of the firm.
 - Investment philosophy outlines the core beliefs and principles that guide investment decisions and asset allocation strategies of a firm.
-

Q.2530 Northern Capital is a well-known asset management firm with a reputation for delivering solid returns for its clients. They pride themselves on their disciplined risk management process, with an emphasis on monitoring risks to ensure that investment activities align with their stated investment philosophies and strategies. Recently, they noticed that the forecasted tracking error for one of their funds, the Northern Growth Fund, deviated from their targeted tracking error level by 20 basis points. Upon investigating, they found that while the total risk levels were within the expected parameters, the fund was over-exposed to certain value themes, even though the fund is marketed as a growth fund. The risk managers are now trying to decide on the next course of action. Which of the following steps should Northern Capital take in response to this observation?

- A. Ignore the deviation since the total risk level is within expectations.
- B. Consider revising its targeted tracking error for the fund.
- C. Evaluate if its growth fund is experiencing a style drift and re-balance its portfolio if necessary.
- D. Change the investment strategy to include more value themes in the portfolio.

The correct answer is **C**.

The fact that the Growth Fund is over-exposed to certain value themes, even though the overall risk level remains within the expected range, suggests potential style drift. Style drift occurs

when a fund departs from its stated investment style or strategy. As the fund is marketed as a growth fund, the risk managers should investigate further and, if confirmed, re-balance the portfolio to realign it with the firm's stated growth philosophy.

A is incorrect. Ignoring the deviation just because the total risk level is within expectations is not advisable. The fund might be drifting away from its stated investment style, which is a cause for concern and needs to be addressed.

B is incorrect. Revising the targeted tracking error doesn't address the core issue at hand, which is the potential style drift. The focus should be on ensuring that the fund aligns with its stated investment philosophy.

D is incorrect. Changing the investment strategy to include more value themes would be inconsistent with the fund's stated growth strategy. Rather than altering the investment strategy to suit the drift, the firm should make sure the fund aligns with its original strategy.

Things to Remember

- **Tracking Error:** Tracking error measures the divergence of a portfolio's returns from the returns of a benchmark. It is a measure of how actively a portfolio is managed.
 - **Style Drift:** Style drift occurs when a fund deviates from its stated investment style or strategy, potentially leading to unintended risks or returns.
 - **Portfolio Re-balancing:** Re-balancing involves realigning the weights of assets in a portfolio to maintain the desired asset allocation and risk-return profile.
 - **Investment Philosophy:** An investment philosophy outlines the core beliefs and principles that guide an asset manager's investment decisions and strategies.
-

Q.2531 A fund manager is concerned about his portfolio's liquidity profile. He does not wish to exceed more than 15 percent of the daily volume in any given security holding. The daily volume of the security is 100,000 while the number of shares held by the manager is 10,000. Which of the following is closest to the liquidity duration of the security?

- A. 0.67
- B. 0.015
- C. 0.15
- D. 0.1

The correct answer is **A**.

$$\text{Liquidity duration} = \frac{Q_i}{(0.15 \times V_i)}$$

Q_i = Number of shares held in security i

V_i = Daily volume of security i

$$\text{Liquidity duration} = \frac{10,000}{(0.15 \times 100,000)} = 0.67$$

Q.2532 An estimate of liquidity duration for the portfolio taken as a whole can be derived by:

- A. Adding each security's liquidity duration.
- B. Weighting each security's liquidity duration by that security's weight in the portfolio.
- C. Computing the average of the securities liquidity duration in the portfolio.
- D. Weighting each security's liquidity duration by its market capitalization.

The correct answer is **B**.

The liquidity duration of a portfolio as a whole can be estimated by weighting each security's liquidity duration by that security's weight in the portfolio. This approach takes into account the fact that different securities may have different weights in the portfolio, and thus their individual liquidity durations will have different impacts on the overall liquidity duration of the portfolio. By weighting each security's liquidity duration by its weight in the portfolio, we can obtain a more accurate estimate of the portfolio's overall liquidity duration. This method is consistent with the principles of portfolio theory, which emphasize the importance of considering the weights of

individual securities in portfolio analysis.

Choice A is incorrect. Simply adding each security's liquidity duration would not provide an accurate estimate of the portfolio's overall liquidity duration. This is because it does not take into account the relative weights of each security in the portfolio, which can significantly impact the time it would take to liquidate the entire portfolio.

Choice C is incorrect. Computing the average of securities' liquidity durations also fails to consider their respective weights in the portfolio. An average might distort the true picture if there are securities with extreme values or if certain securities make up a larger portion of the portfolio.

Choice D is incorrect. Weighting each security's liquidity duration by its market capitalization may seem logical, but it doesn't accurately reflect a portfolio's overall liquidity duration. The weightage should be based on each security's proportion in our specific portfolio rather than its market capitalization.

Things to Remember

- Liquidity duration is a measure of the time it would take to liquidate a security or a portfolio without significantly impacting the market price. It is a critical measure in portfolio management as it helps in understanding the liquidity risk associated with the portfolio.
- The liquidity duration of a portfolio is not just a simple sum of the liquidity durations of individual securities. It is a weighted average where the weights are the proportions of individual securities in the portfolio. This is because different securities may have different weights in the portfolio and thus their individual liquidity durations will have different impacts on the overall liquidity duration of the portfolio.
- The liquidity duration of a portfolio can change over time. Factors that can affect the liquidity duration include changes in the market conditions, changes in the portfolio composition, and changes in the liquidity of individual securities. Therefore, it is important to regularly monitor and update the liquidity duration of a portfolio.
- Understanding the liquidity duration of a portfolio is important for risk management. It can help in making informed decisions about portfolio rebalancing, risk mitigation strategies, and liquidity management. For example, if the liquidity duration of a portfolio is high, it may indicate a high level of liquidity risk, and the portfolio manager may need to take steps to reduce this risk.

Q.2533 For fixed-income securities, the liquidity duration can be calculated by:

- A. Weighting each security by the security's coupon.
- B. Weighting each security by the security's residual tenure.

- C. Weighting each security by the security's weight in the portfolio.
- D. Way of discussions with portfolio managers.

The correct answer is **D**.

Liquidity duration refers to the estimated time required to liquidate a position in a security. For equity holdings, this is relatively straightforward to calculate as volume data are readily available. However, in the case of fixed-income securities, volume information is not as easily accessible. As a result, the calculation of liquidity duration for these securities often involves discussions with portfolio managers. These discussions can provide insights into the market conditions, the liquidity of the specific securities, and the potential time required to liquidate the positions. Therefore, the liquidity duration of fixed-income securities is often determined through discussions with portfolio managers, making choice D the correct answer.

Choice A is incorrect. Weighting each security by the security's coupon does not provide an accurate measure of liquidity duration. The coupon rate of a bond is related to its yield and interest payments, but it does not directly reflect the liquidity or trading volume of the security.

Choice B is incorrect. Weighting each security by its residual tenure also fails to accurately calculate liquidity duration. While residual tenure can impact a bond's price and yield, it does not necessarily correlate with the bond's liquidity or trading volume.

Choice C is incorrect. Weighting each security by its weight in the portfolio may seem logical, but this method also falls short in accurately calculating liquidity duration. The weight of a security in a portfolio reflects its relative importance within that portfolio, but it doesn't necessarily reflect the overall market liquidity or trading volume for that particular fixed-income instrument.

Things to Remember

- Liquidity duration is a measure of the time it would take to liquidate a position in a security. It is a critical aspect of risk management in fixed-income securities.
- Portfolio managers play a crucial role in determining the liquidity duration of fixed-income securities. Their insights into market conditions, the liquidity of specific securities, and the potential time required to liquidify the positions are invaluable.
- Unlike equity holdings, fixed-income securities do not have readily available volume data. This makes the calculation of liquidity duration more challenging and reliant on the expertise of portfolio managers.
- Understanding the liquidity duration of fixed-income securities can help in making informed investment decisions, managing risk, and optimizing portfolio performance.

Q.2534 In financial markets, the term 'liquidity duration' is often used. This term refers to a specific time period associated with a security. Which of the following best describes the concept

of liquidity duration?

- A. The number of days in which a position or a security can be purchased without having any material impact on the market.
- B. The number of days in which a position or a security can be liquidated without having any material impact on the position.
- C. The number of days in which a security can be liquidated with material impact on the market.
- D. The number of days in which a security can be purchased with material impact on the market.

The correct answer is **B**.

Liquidity duration is indeed the number of days in which a security or a position can be liquidated without having any material impact on the position.. This concept is crucial in financial markets as it provides an estimate of the time required to sell a particular security without causing a significant change in its price. This is particularly important for large investors or institutions that hold substantial quantities of a particular security. If they were to sell their holdings all at once, it could potentially cause a drastic drop in the security's price. Therefore, understanding the liquidity duration helps these entities to plan their selling strategies in a way that minimizes the impact on the market.

Choice A is incorrect. This definition is more aligned with market depth, which refers to the ability of the market to absorb large purchase orders without significantly impacting the price.

Choice C is incorrect. The concept of liquidity duration does not involve liquidating a security with material impact on the market. This would imply that selling off a security would cause significant changes in its price, which contradicts the idea of liquidity duration as it pertains to liquidating assets without causing substantial price fluctuations.

Choice D is incorrect. Similar to choice A, this option incorrectly associates liquidity duration with purchasing securities that have a significant impact on their market prices. This misinterpretation confuses liquidity duration with concepts related to buying pressure and its effect on asset prices.

Things to Remember

- Liquidity duration is a measure of market liquidity and is particularly relevant for large investors or institutions that hold substantial quantities of a particular security.
- It is a crucial concept in risk management as it helps in understanding the potential impact of a large sale on the market price of a security.
- Liquidity duration can vary greatly depending on the type of security and the overall market conditions. For instance, highly liquid securities like government bonds typically have a shorter liquidity duration compared to less liquid securities like corporate bonds or stocks of small

companies.

- Understanding the liquidity duration of a security can help investors and institutions in planning their investment strategies. For instance, if the liquidity duration of a security is long, it might be more suitable for long-term investors who do not need to liquidate their holdings frequently.
-

Q.2536 Global Finance is a renowned asset management firm with a strong risk management framework. Recently, during their risk monitoring process, they found that while their Asia-Pacific Equity Fund has an overall risk level within their expected limits, the fund is over-exposed to certain U.S. equity themes, deviating from the fund's stated objective of focusing on Asia-Pacific equities. This has raised concerns among the risk management team. Which of the following actions should Global Finance take based on this observation?

- A. Global Finance should re-evaluate their risk monitoring model as it seems to be producing inaccurate results.
- B. Global Finance should modify their fund's stated objective to include more U.S. equities.
- C. Global Finance should investigate the fund's investment activities for potential style drift and realign the portfolio if needed.
- D. Global Finance should ignore the deviation as the total risk level is within acceptable limits.

The correct answer is **C**.

The fact that the Asia-Pacific Equity Fund is over-exposed to U.S. equity themes suggests a potential style drift. Style drift occurs when a fund departs from its stated investment style or strategy. As the fund is marketed as an Asia-Pacific equity fund, the risk managers should investigate further and, if necessary, realign the portfolio to match the firm's stated investment philosophy.

A is incorrect. While it's important to ensure that the risk monitoring model is functioning correctly, the issue at hand seems to be related to a potential style drift, not an error in the risk monitoring model.

B is incorrect. Modifying the fund's stated objective to match the drift would be contrary to the fund's originally stated strategy and could mislead investors. The focus should be on ensuring that the fund's activities align with its stated strategy.

D is incorrect. Simply ignoring the deviation because the total risk level is within expectations wouldn't address the potential issue of style drift. Even if the total risk level is within limits, a deviation from the fund's stated strategy could lead to unexpected results and investor dissatisfaction.

Things to Remember

- **Style drift** is a term used in investment management that refers to a fund or portfolio straying from its stated investment strategy or focus.
 - Style drift can be indicative of changing risk-return profiles that might not be in line with investors' expectations. It can lead to unexpected outcomes and investor dissatisfaction.
 - It's essential to identify and address style drift promptly to ensure alignment with the fund's stated investment strategy. Regular risk monitoring and controls are crucial to detect and manage such situations.
 - Even if a fund's total risk level is within acceptable limits, a significant style drift could alter its risk-return characteristics and should not be ignored.
 - Realigning a drifting portfolio back to its original strategy is generally preferable to altering the fund's stated objective, which might mislead investors.
-

Q.2537 Phoenix Capital, a prominent asset management firm, has a successful Emerging Markets Equity Fund. Their Risk Management Unit (RMU) uses a proprietary model to estimate a forecasted tracking error. Recently, the RMU noticed that the forecasted tracking error for the fund was consistently lower than the target established by the firm's investment mandate. The RMU identified that this is due to the fund's portfolio manager having a more conservative approach and holding a larger proportion of assets in stable, less volatile securities. Which of the following actions is most appropriate for Phoenix Capital's RMU in response to this finding?

- A. Immediately instruct the fund's portfolio manager to invest in more volatile securities to increase the tracking error to the target level.
- B. Initiate a review of the proprietary risk forecasting model to ensure it is functioning correctly.
- C. Discuss the findings with the fund's portfolio manager and senior management to understand the reasons and evaluate the need for any portfolio adjustments.
- D. No action is required as a lower tracking error means less risk, which is beneficial for the investors.

The correct answer is **C**.

The lower forecasted tracking error compared to the target indicates that the fund is taking less risk than intended. While this might not necessarily be detrimental, it could result in returns that are inconsistent with investor expectations. The RMU should discuss this finding with the fund's portfolio manager and senior management to understand the reasoning behind the portfolio manager's conservative strategy and assess if any adjustments are necessary.

A is incorrect Forcing the portfolio manager to immediately invest in more volatile securities could result in unnecessary risk-taking, which is not aligned with prudent risk management principles.

B is incorrect There is no suggestion that the risk forecasting model is functioning incorrectly. The tracking error discrepancy is due to the conservative approach of the fund's portfolio manager, not a flaw in the model.

C is incorrect Although a lower tracking error means less risk, it may also mean that the fund is not meeting its stated objectives or investor expectations. Ignoring this discrepancy would not be a prudent approach to risk management.

Things to Remember

- **Forecasted tracking error** is a statistical measure that estimates the potential deviation of a portfolio's return from its benchmark. It's derived from the portfolio's holdings and forward-looking estimation techniques.
- An effective risk management process requires portfolio managers to maintain an appropriate level of risk in alignment with client expectations. Neither taking too much nor too little risk is desirable.
- Tracking error forecasts should be made for each individual portfolio and for the aggregate of all client-owned portfolios. These forecasts should then be compared with the tracking error budgets for reasonableness.
- Policy standards should determine the acceptable variance from the target. Unusually large variances should trigger inquiries, and material variances should prompt immediate corrective action.
- Such measures help to identify and manage unusual deviations across accounts, ensuring alignment with the client's risk-return objectives.

Q.2538 Chameleon Capital, an investment management firm, has a well-defined and time-tested conviction of focusing on high-quality growth stocks in the US market. According to the recent

risk reports, the firm is maintaining the forecasted tracking error consistent with the target established by the mandate. However, the risk decomposition shows that a significant proportion of risk is being taken on non-U.S. instruments, which is unusual for Chameleon Capital. As the Risk Manager of Chameleon Capital, how should you interpret this situation in the context of effective risk monitoring?

- A. The situation is not concerning since the firm is maintaining the target tracking error.
- B. The risk decomposition is irrelevant as long as the overall risk (tracking error) is within target.
- C. There might be a "style drift" in investment activities, indicating a departure from the firm's philosophy.
- D. This suggests that the risk forecasting model might not be behaving as predicted.

The correct answer is C.

Although the overall risk as measured by the tracking error is within the target, the change in risk decomposition indicates a potential "style drift". This is a concern because it might mean the firm is not investing in line with its stated philosophy, which can lead to inconsistent performance and unanticipated risk exposure.

Option A is incorrect because even though the tracking error is within the target, the change in risk decomposition should not be ignored as it might indicate a shift in investment strategy.

Option B is incorrect because the risk decomposition provides additional insights about where and how the risk is being taken. Ignoring this information can lead to a misunderstanding of the firm's risk profile.

Option D is incorrect. While a malfunctioning risk forecasting model can be a concern, the information provided does not necessarily suggest this. The issue here seems more related to the firm's investment activities rather than the performance of the risk forecasting model.

Things to Remember

- **Tracking error** is not only about the total expected deviation from a benchmark but also about the sources of that deviation. Managers should be able to decompose and explain the sources of tracking error.
- Risk management requires checking whether the risk taken aligns with the expectations, both at the total and at the constituent levels. If it doesn't, the investment strategy may be deviating from its stated philosophy, a situation often referred to as "style drift."

- For instance, a growth manager investing heavily in value themes may meet the overall tracking error target, but the risk style doesn't align with the stated investment approach.
 - Key elements of risk decomposition that managers should clarify and risk management units should monitor include the range of acceptable active weights and marginal contributions to risk, both at the levels of individual stocks, industries, sectors, and countries.
-

Q.2539 Allegro Investments, a renowned investment management firm, sticks to a rigorous risk monitoring process. Recently, their Risk Management Unit (RMU) found out that one portfolio manager is generating a forecasted level of tracking error, which aligns perfectly with the target established by the mandate. However, the same manager is found to be significantly over budget in specific sectoral themes with offsets in other areas. As the Chief Risk Officer at Allegro Investments, how should you assess this situation?

- A. The situation is not problematic since the overall forecasted tracking error is aligned with the target.
- B. The overspending in specific sectors is a matter of the manager's discretion and should not be intervened.
- C. The over-budget scenario in specific sectors indicates potential deviations from the established investment strategy.
- D. This situation suggests that the risk forecasting model is not functioning correctly.

The correct answer is **C**.

While the overall forecasted tracking error is within the target, being significantly over budget in certain sectors may indicate potential deviations from the expected risk decomposition, similar to a "style drift." Such deviation can hint at inconsistency with the manager's stated investment philosophy.

Option A is incorrect. While it's true that the overall tracking error is within the target, focusing only on the overall tracking error could miss important details about where and how risk is being taken.

Option B is incorrect because even though portfolio managers do have a degree of discretion, going significantly over budget in specific sectors could indicate that the manager is not following the established investment strategy.

Option D is incorrect because the information given does not necessarily suggest that the risk

forecasting model is malfunctioning. The problem seems more related to the manager's investment activities and how the risk is distributed rather than a problem with the risk forecasting model itself.

Things to Remember

- **Tracking error decomposition** involves breaking down the overall tracking error into its constituent parts to understand the sources of risk. It helps to ascertain whether the risk is being taken in accordance with the expectations at both the total and the constituent level.
 - If the risk decomposition does not align with the expectations, it could indicate that the manager is not investing in line with the stated philosophy, a situation often referred to as "style drift".
 - A style drift might occur when a manager, for instance, meets the overall tracking error target but concentrates the risk in themes that do not align with the investment style.
 - Risk managers should monitor the acceptable range of active weights and marginal contributions to risk at various levels, including stock, industry, sector, and country. These measures provide detailed insights into the portfolio's risk allocation.
-

Q.2540 Stronghold Investment Managers is renowned for its adherence to a strict risk management process, with established limits for its risk expenditure. Recently, during their risk monitoring process, the Risk Management Unit (RMU) noted that their flagship portfolio shows consistent tracking error forecasts from two different risk models, which are both comparable to the targeted risk level. What does this information imply about the risk process at Stronghold Investment Managers?

- A. The risk process is ineffective as different models should not produce similar tracking errors.
- B. The risk process is effective as the consistent tracking error suggests the portfolio's risk level is in line with the set target.
- C. The risk process is ineffective as the RMU should rely on one model to avoid discrepancies.
- D. The risk process is effective but could lead to over-optimization, given the alignment of two models' forecasts.

The correct answer is **B**.

A consistent tracking error forecast from two different risk models that aligns with the targeted risk level indicates an effective risk process. It suggests that the risk taken by the portfolio managers is in accordance with the client's expectations.

Option A is incorrect as similar tracking errors from different models indicate consistency and affirm the robustness of the risk management process.

Option C is incorrect because utilizing different models can provide a more comprehensive view of potential risk, as each model might consider different factors. If they show consistency, it's a positive indication.

Option D is incorrect because consistency in tracking error forecasts doesn't necessarily imply over-optimization. It simply suggests that the portfolio's risk level is being managed in line with set targets.

Things to Remember

- **Tracking error forecasts** estimate the potential risk in a portfolio. When such forecasts from different models align with the target risk level, it implies an effective risk management process.
- Consistency in tracking error forecasts across multiple models indicates the robustness of the risk process. It suggests the models are taking into account similar factors and reaching similar conclusions about potential risk.
- Using multiple models for risk forecasting can provide a comprehensive view of potential risks. Each model might consider different factors, and consistency across these models is a positive indication of effective risk management.
- Effective risk management means ensuring the risk taken aligns with the client's expectations. If tracking error forecasts consistently match the targeted risk level, it suggests that portfolio managers are taking risk in accordance with the client's expectations.

Q.2541 The excess returns of a fund are regressed against the excess returns of the benchmark. The slope coefficient against the excess returns of the benchmark is given by:

A. Alpha

- B. Beta
- C. The residual error
- D. The correlation coefficient

The correct answer is **B**.

In finance, Beta is a measure of the volatility, or systematic risk, of a security or a portfolio in comparison to the market as a whole. It is used in the capital asset pricing model (CAPM), which calculates the expected return of an asset based on its beta and expected market returns. The slope coefficient in a regression of a fund's excess returns against the excess returns of a benchmark is indeed the Beta. This is because Beta measures the sensitivity of the expected excess asset returns to the expected excess market returns. So, when we regress the excess returns of a fund (the dependent variable) on the excess returns of the benchmark (the independent variable), the slope coefficient we get is the Beta. It tells us how much, on average, the fund's excess returns will change when the benchmark's excess returns change by one unit.

Choice A is incorrect. Alpha refers to the excess return of an investment relative to the return of a benchmark index. It does not represent the slope coefficient in a regression analysis comparing a fund's excess returns to those of a benchmark.

Choice C is incorrect. The residual error in regression analysis represents the difference between observed and predicted values, not the slope coefficient that measures sensitivity of fund's returns with respect to benchmark's returns.

Choice D is incorrect. The correlation coefficient measures the strength and direction of a linear relationship between two variables, but it does not represent how much one variable changes for each unit change in another variable, which is what beta (the slope coefficient) represents.

Things to Remember

- Beta is a measure of a stock's risk in relation to the market or a benchmark. A beta of 1.0 indicates that a stock's price will move with the market. A beta less than 1.0 indicates the stock will be less volatile than the market, while a beta greater than 1.0 indicates the stock's price will be more volatile than the market.
- Beta is a key component for the Capital Asset Pricing Model (CAPM), which is used to calculate the cost of equity funding. The CAPM formula is: $\text{Cost of Equity} = \text{Risk-Free Rate of Return} + \text{Beta} * (\text{Market Rate of Return} - \text{Risk-Free Rate of Return})$.
- While beta offers useful information for stock investing, it has its limitations. For example, a beta based on historical data may not predict future risk. Also, beta assumes that stock returns are normally distributed, which may not be the case.
- Investors who believe in a passive investment strategy often buy index funds, which have a beta close to 1.0. On the other hand, investors who believe they can beat the market will look for

stocks with betas that are significantly higher or lower than 1.0.

Q.2542 The excess returns of a fund are regressed against the excess returns of the benchmark. The intercept represents:

- A. Alpha
- B. Beta
- C. The residual error
- D. The correlation coefficient

The correct answer is **A**.

Alpha is a measure of the active return on an investment. It is the performance of an investment against a market index or benchmark which is considered to represent the market's movement as a whole. The excess return of an investment relative to the return of a benchmark index is the investment's alpha. In the context of regression analysis, when the excess returns of a fund are regressed against the excess returns of a benchmark, the intercept of the regression line is known as Alpha. This is because the intercept represents the point where the expected excess return of the fund (when the excess return of the benchmark is zero) is equal to Alpha. Therefore, Alpha is the correct answer.

Choice B is incorrect. Beta, in the context of financial analysis, represents the sensitivity of a fund's returns to changes in market returns. It is not the intercept but rather the slope of the regression line.

Choice C is incorrect. The residual error refers to the difference between actual and predicted values in a regression analysis. It does not represent the intercept of a regression line.

Choice D is incorrect. The correlation coefficient measures how closely two variables move together; it does not represent an intercept term in a regression analysis.

Things to Remember

- Alpha is often used to measure the value that a portfolio manager adds or subtracts from a fund's return. A positive alpha of 1.0 means the fund has outperformed its benchmark index by 1%. Correspondingly, a similar negative alpha would indicate an underperformance of 1%.
- Alpha is a risk-adjusted measure. This means it takes the volatility (price risk) of a mutual fund and compares its risk-adjusted performance to a benchmark index. The excess return of the fund relative to the return of the benchmark index is a fund's alpha.
- Alpha is also a measure of the residual value. It represents the return of an investment when the market is at zero. In other words, alpha is the return on an investment that is not a result of

general market movements.

- Alpha can be used to directly measure the value added or subtracted by a fund's manager. There are limitations to using alpha to determine an investment's future prospects. Alpha does not account for the risk that the investment will lose money or for the opportunity cost of forgoing a risk-free investment.
-

Q.2544 A manager's excess returns are regressed against the excess returns of the manager's peer group. The output is used to determine:

- A. Whether the manager demonstrates skill over and above the peer group.
- B. Whether the manager's above-average returns are as a result of executing unauthorized trades.
- C. The absolute level of leverage employed by the manager as compared to the peer group.
- D. The absolute level of return generated by the manager.

The correct answer is **A**.

The primary purpose of regressing a manager's excess returns against the excess returns of their peer group is to determine whether the manager demonstrates skill over and above the peer group. This is done by examining the alpha term in the regression output. The alpha term represents the excess return that the manager is able to generate over and above the peer group, after adjusting for risk. If the alpha term is positive and statistically different from zero, it suggests that the manager has skill and is able to generate excess returns consistently. This is not a result of luck or random chance, but rather a reflection of the manager's ability to make effective investment decisions. Therefore, the regression analysis serves as a tool for evaluating the manager's performance relative to their peers.

Choice B is incorrect. Regression analysis in financial management is not typically used to determine if a manager's above-average returns are the result of executing unauthorized trades. Unauthorized trades would be identified through compliance and control mechanisms, not statistical analysis.

Choice C is incorrect. While regression analysis can provide insights into various aspects of a manager's performance, it does not directly measure the absolute level of leverage employed by the manager as compared to their peer group. Leverage levels would be determined through other financial metrics and ratios.

Choice D is incorrect. The primary purpose of regression analysis in this context is not to measure the absolute level of return generated by the manager but rather to compare their excess returns with those of their peers, which helps in determining whether they demonstrate skill over and above that group.

Things to Remember

- Regression analysis is a statistical tool used to understand the relationship between a dependent variable (in this case, the manager's excess returns) and one or more independent variables (in this case, the peer group's excess returns).
 - The beta coefficient in the regression analysis measures the sensitivity of the manager's excess returns to the excess returns of the peer group. A beta of 1 indicates that the manager's returns move in line with the peer group. A beta greater than 1 indicates that the manager's returns are more volatile than the peer group, while a beta less than 1 indicates that the manager's returns are less volatile.
 - Regression analysis also provides a measure of the goodness of fit of the model, known as the R-squared. This statistic measures the proportion of the variance in the manager's excess returns that can be explained by the peer group's excess returns. A high R-squared indicates a strong relationship between the manager's returns and the peer group's returns.
 - It's important to note that while regression analysis can provide valuable insights into a manager's performance, it is not without limitations. For example, it assumes a linear relationship between the variables, which may not always hold true. Additionally, it is based on historical data and may not accurately predict future performance.
-

Q.2545 Which of the following correctly defines the “survivorship bias.”

- A. A bias towards instances which made it past some selection process and overlooking others due to a lack of visibility.
- B. A bias towards instances with negative outcomes.
- C. A bias towards instances which have recently occurred.
- D. A bias towards instances which have positive outcomes.

The correct answer is **A**.

Survivorship bias refers to the logical error of focusing on the instances that 'survived' some process and inadvertently overlooking those that did not because of their lack of visibility. This bias can lead to false conclusions in several different ways. The survivors may be actual people, as in a medical study, or could be companies or research participants or students... or anything that must make it past some selection process to be included in the study or analysis. For example, if three of the five students with the best college grades went to the same high school, that can lead one to believe that the high school must offer an excellent education. This could be true, but the question overlooks the students from that same high school who didn't get into college at all. The overlooked group can potentially lead to a vastly different conclusion.

Choice B is incorrect. This choice describes a negativity bias, not survivorship bias. Negativity bias refers to the tendency to give more weight to negative experiences than positive ones, which is different from survivorship bias where the focus is on instances that have survived or made it past a certain selection process.

Choice C is incorrect. This choice describes recency bias, not survivorship bias. Recency bias refers to the tendency to weigh recent events more heavily than earlier events. Survivorship bias, on the other hand, does not consider when an event occurred but rather focuses on instances that have survived or passed a certain selection process.

Choice D is incorrect. This choice describes positivity or optimism bias, not survivorship bias. Positivity or optimism biases refer to the tendency of individuals to overestimate positive outcomes and underestimate negative ones in future situations while Survivorship Bias focuses on instances that have survived or passed through some sort of selection process.

Things to Remember

- Survivorship bias can significantly affect investment returns. For example, mutual fund companies often advertise the performance of their successful funds, while quietly closing or merging funds that have performed poorly. This can give the impression that the company's funds perform better on average than they actually do.
 - Survivorship bias can also lead to overly optimistic beliefs because failures are ignored, such as when companies that no longer exist are excluded from analyses of financial performance. It can also lead to the false belief that the successes in a group have some special property, rather than just coincidence.
 - Survivorship bias is a common logical error in the field of finance. For example, when a mutual fund company says "Our funds have beaten the market 3 years in a row", they are ignoring the fact that the poorly performing funds are closed and merged into other funds, thus 'disappearing' from the record.
-

Q.2549 A fund manager is concerned about his portfolio's liquidity profile. He does not wish to exceed more than 10 percent of the daily volume in any given security holding. The daily volume of the security is 100,000 while the number of shares held by the manager is 10,000. Which of the following is closest to the liquidity duration of the security?

- A. 1.0
- B. 0.1
- C. 0.01
- D. 0.001

The correct answer is **A**.

$$\text{Liquidity duration} = \frac{Q_i}{(0.10 \times V_i)}$$

Q_i = Number of shares held in security i

V_i = Daily volume of security i

$$\text{Liquidity duration} = \frac{10,000}{(0.10 \times 100,000)} = 1$$

Q.2553 A project has total revenues of \$20,000 and expenses of \$10,000. If the total risk-weighted assets in the project amount to \$30,000, then the return on risk capital (RORC) of the project is equal to:

- A. 0.67
- B. 0.33
- C. 0.5
- D. 1

The correct answer is **B**.

$$\begin{aligned}\text{RORC} &= \frac{\text{Net Income}}{\text{Total risk capital}} \\ &= \frac{\text{Revenues} - \text{Expenses}}{\text{Total risk capital}} \\ &= \frac{\$20,000 - \$10,000}{\$30,000} = 33.33\%\end{aligned}$$

Q.2554 The total assets of a company amount to \$30,000,000 while its liabilities aggregate to \$25,000,000. If the net income of the company for 2017 is \$3,000,000, then the return on equity (ROE) of the company for 2017 is closest to:

- A. 10%
- B. 60%
- C. 5.45%
- D. 12%

The correct answer is **B**.

$$\begin{aligned}\text{ROE} &= \frac{\text{Net Income}}{\text{Equity}} = \frac{\text{Net Income}}{\text{Assets} - \text{Liabilities}} \\ &= \frac{\$3,000,000}{(\$30,000,000 - \$25,000,000)} = 60\%\end{aligned}$$

Q.2751 Given the following information, calculate the liquidity duration in days for a security, assuming that we do not wish to exceed 15% of the daily volume in that security.

$$\frac{\text{Number of shares held}}{\text{Daily volume of security}} = \frac{2,000,000}{750,000}$$

- A. 17.8
- B. 3.1
- C. 2.5
- D. 15.4

The correct answer is **A**.

$$\begin{aligned} \text{LD} &= \frac{\text{Number of shares held}}{(0.15 \times \text{Daily Volume})} \\ &= \frac{2,000,000}{0.15 \times 750,000} \\ &= 17.8 \end{aligned}$$

Q.3030 Lisa is a portfolio manager at Goldman Sachs Asset Management. She is responsible for a portfolio of stocks and bonds, and her performance is evaluated annually. The company has an evaluation system based on performance measurement tools, which include benchmark comparison, risk assessment, and return attribution. As part of the evaluation process, the firm uses a "green zone" system to flag any deviations in tracking error or performance. What are the main objectives of performance measurement tools as applied to Lisa's portfolio management at Goldman Sachs Asset Management?

- A. The tools are used to forecast future returns and assess the potential market risks.
- B. The tools are used to assess whether Lisa consistently outperforms her peers and the benchmark on a risk-adjusted basis.
- C. The tools are used to identify the sectors or countries with the highest potential for investment.
- D. The tools are used to evaluate the individual performance of each security in Lisa's portfolio.

The correct answer is **B**.

Performance measurement tools aim to determine whether a portfolio manager, such as Lisa, consistently outperforms the benchmark and peers on a risk-adjusted basis. They also seek to ensure that the returns achieved align with the risks taken. The information obtained from these tools can help identify managers who can generate consistent excess risk-adjusted returns, providing insights that can be used to maximize the entity's long-term returns and profitability.

A is incorrect. While performance measurement tools do consider risk, their primary focus is not on forecasting future returns or assessing potential market risks. Instead, they assess the portfolio manager's actual results against relevant comparables such as benchmarks and peer groups.

C is incorrect. Performance measurement tools do not primarily focus on identifying sectors or countries with the highest potential for investment. They are used to evaluate the performance of the portfolio manager, not the potential of specific investments.

D is incorrect. While return attribution is a part of performance measurement and can attribute the source of returns to specific factors or securities, the overall objective of performance measurement tools is not to evaluate the individual performance of each security. Instead, they aim to assess the overall performance of the portfolio manager.

Things to Remember

- Performance measurement compares a manager's results to benchmarks and peer groups, aiming to assess consistent outperformance on a risk-adjusted basis.
- It helps determine if the returns achieved align with the risk taken and identifies managers generating consistent excess risk-adjusted returns.
- Performance is evaluated from a risk perspective (e.g., tracking error) and a returns perspective.
- Return attribution ensures returns result from intended risk-taking decisions, not luck.
- Sharpe and Information ratios are used as risk-adjusted return measures to compare performance and determine excess returns relative to risk.

Q.3168 Paul Collins is a chief investment officer at Welcome mutual funds. He is concerned about the liquidity situation in the current distressed market suffering from low volumes. He has a large holding of GetMax Pharma stocks with 200,000 shares. To lower the market impact cost, Paul has directed his portfolio manager that the desired maximum daily trading volume of GetMax should not exceed 10%. Since the daily market volume is currently 32,000 shares, the liquidity duration for GetMax Pharma shares is closest to:

- A. 32.0
- B. 1.6
- C. 6.25
- D. 62.5

The correct answer is **D**.

$$\text{Liquidity duration} = \frac{\text{number of shares of the security}}{(\text{desired maximum daily volume} \times \text{daily volume of the security.})}$$
$$\text{LD} = \frac{200,000}{(0.1 \times 32,000)} = 62.5 \text{ Days}$$

Reading 153: Portfolio Performance Evaluation

Q.2557 Consider the following data for a particular fund for the year 2017:

	Portfolio	Market
Average return	32%	25%
Beta	1.15	1
Standard deviation	40%	30%
Tracking error $\sigma(e)$	16%	0%
T-Bill rate	6%	

Determine Treynor's ratio of the portfolio and the market, respectively.

- A. 0.25 & 0.23
- B. 0.20 & 0.18
- C. 0.23 & 0.19
- D. 0.24 & 0.11

The correct answer is **C**.

$$\begin{aligned}\text{Treynor's ratio} &= \frac{(R_p - R_f)}{\beta_p} \\ \text{Treynor's ratio of portfolio} &= \frac{(32\% - 6\%)}{(1.15)} = 0.23 \\ \text{Treynor's ratio of market} &= \frac{(25\% - 6\%)}{(1.00)} = 0.19\end{aligned}$$

Q.2558 Consider the following data for a particular fund for the year 2017:

	Portfolio	Market
Average return	32%	25%
Beta	1.15	1
Standard deviation	40%	30%
Tracking error $\sigma(e)$	16%	0%
T-Bill rate	6%	

Calculate the value of alpha generated by the portfolio and the market, respectively.

- A. 4.00% & 2%
- B. 4.15% & 0%
- C. 3.15% & 2.15%
- D. 4.10% & 3.10%

The correct answer is **B**.

Alpha is the excess return generated by the portfolio when compared to the return expected using the Capital Asset Pricing Model (CAPM).

As per the CAPM, the expected return of the portfolio is computed by the following expression:

$$\begin{aligned}R_p &= R_f + \beta(R_m - R_f) \\R_p &= 6\% + 1.15 * (25\% - 6\%) \\R_p &= 6\% + 1.15 * 19\% = 27.85\%\end{aligned}$$

Expected return from the market:

$$R_m = 6\% + 1 * (25\% - 6\%) = 25\%$$

Therefore, the alpha (or the excess return):

$$\begin{aligned}\alpha &= R_{\text{average}} - R_{\text{CAPM}} \\ \alpha_{\text{portfolio}} &= 32\% - 27.85\% = 4.15\% \\ \alpha_{\text{market}} &= 25\% - 25\% = 0\end{aligned}$$

Q.2559 An investor purchases share of stock at $t = 0$ for \$300. A year later at $t = 1$, they purchase another share at \$320. At the end of year 2, both shares are sold for \$330 each. At the end of years 1 and 2, the stock paid a \$5.00 per share dividend. What is the time-weighted rate of return for this investment?

- A. 2.5%
- B. 10%
- C. 6.5%
- D. 5.5%

The correct answer is **C**.

Step 1

Break the evaluation period into two subperiods based on timing of cash flows:

Holding period 1:

beginning price = \$300.00

dividends paid = \$5.00

ending price = \$320.00

Holding period 2:

beginning price = \$640.00 (2 shares)

dividends paid = \$10.00 (\$5 per share)

ending price = \$660.00 (2 shares)

Step 2

Calculate the HPR for each holding period

$$\begin{aligned} \text{HPR}_1 &= \frac{(\$320 - \$300 + \$5)}{\$300} = 0.0833 \\ \text{HPR}_2 &= \frac{(\$660 - \$640 + \$10)}{\$640} = 0.046875 \end{aligned}$$

Step 3

Take the geometric mean of the annual returns to find the annualized timeweighted rate of return over the measurement period.

$$\begin{aligned} (1 + \text{time-weighted rate of return})^2 &= 1.0833 \times 1.046875 \\ \text{time-weighted rate of return} &= (1.0833 \times 1.046875)^{\frac{1}{2}} - 1 = 0.064931 \end{aligned}$$

Q.2562 Daisy Singh recently attended an investor awareness programme which sensitized the

general public to consider equity investment as an avenue for generating wealth. While she was excited to invest in stocks, Singh was a bit apprehensive about her expertise in stocks selection. While discussing the issue with her friend, she was informed about the mutual fund industry and how professionals in the mutual fund industry skillfully select stocks and bonds thus making an investment in mutual funds much easier for beginner investors like her.

As Singh starts looking for a mutual fund in which to invest her money, she notices that most of the funds had highlighted and advertised specific fund managers as the sole reason behind the fund's impressive performance. Though Singh was confident enough and was willing to invest money in mutual funds, she wondered what effect the change in fund manager would have on the returns generated by the mutual fund.

Daisy Singh is wary of management changes and is not willing to take any chances with individual managers. As a financial-savvy risk manager, what alternative would you suggest?

- A. She should invest in an index fund.
- B. She should invest in other assets but avoid mutual funds.
- C. She should invest in bonds and stocks, and select them herself.
- D. She must not invest in stocks herself.

The correct answer is **A**.

Index funds are a type of mutual fund with a portfolio constructed to match or track the components of a market index, such as the Standard & Poor's 500 Index (S&P 500). An index mutual fund is said to provide broad market exposure, low operating expenses, and low portfolio turnover. These funds adhere to specific standards (or rules) that stay in place no matter the state of the markets. Index funds are generally considered ideal core portfolio holdings for retirement accounts, such as individual retirement accounts (IRAs) and 401(k) accounts. Legendary investor Warren Buffett has recommended index funds as a haven for savings for the sunset years of life. Rather than picking out individual stocks for investment, the fund manager of an index fund replicates the performance of a specific index. Therefore, the role of the fund manager is limited and the fund's performance is not heavily dependent on their expertise. This makes index funds a suitable investment option for Daisy Singh, who is concerned about the potential impact of management changes on her investment returns.

Choice B is incorrect. Avoiding mutual funds entirely is not a recommended strategy. Mutual funds, managed by professional fund managers, offer diversification and can be a good investment avenue for novice investors like Daisy who may not have the expertise to manage their own portfolio.

Choice C is incorrect. While investing in bonds and stocks individually could potentially yield higher returns, it requires a deep understanding of the market and individual securities. As Daisy is a novice investor, she might lack the necessary knowledge to make informed decisions about which stocks or bonds to invest in.

Choice D is incorrect. The statement that Daisy must not invest in stocks herself does not

address her concern about changes in fund management impacting returns on mutual funds. Moreover, investing in stocks can be part of a balanced investment strategy if done wisely.

Things to Remember

- Index funds are passively managed funds that aim to replicate the performance of a specific market index.
 - Index funds have lower operating expenses compared to actively managed funds, making them cost-effective for investors.
 - Investing in index funds provides broad market exposure and diversification, reducing individual stock risk.
 - Warren Buffett and other prominent investors have recommended index funds as a suitable investment option for long-term savings.
 - Active fund managers aim to outperform the market through stock selection and market timing, which can lead to higher fees and potential underperformance compared to index funds.
-

Q.2563 A portfolio analyst achieved the following results:

	Portfolio A	Portfolio B
α	-1.0	1.75
β	0.5	0.6
Timing Coefficient	0.05	0

Which of the following is TRUE about the analyst's results?

- A. Portfolio A had an unsuccessful stock selection.
- B. Both portfolios have no evidence of an attempted timing.
- C. Portfolio B has evidence of successful timing.
- D. Portfolio B had an unsuccessful stock selection.

The correct answer is **A**.

Portfolio A had an unsuccessful stock selection. This is indicated by the negative alpha coefficient of -1.0. The alpha coefficient is a measure of the performance of an investment against a market index or other benchmark which the investment is expected to follow. A negative alpha indicates that the investment has underperformed compared to the benchmark. In this case, the negative alpha for Portfolio A suggests that the stock selection for this portfolio was unsuccessful, as it underperformed compared to its benchmark.

Choice B is incorrect. The statement that both portfolios have no evidence of an attempted timing is not accurate. Portfolio A has a timing coefficient of 0.05, which indicates some level of attempted timing, even though it may not be significantly successful.

Choice C is incorrect. There's no evidence to suggest that Portfolio B has successful timing as its timing coefficient is 0. A positive and significant timing coefficient would indicate successful market-timing ability.

Choice D is incorrect. The statement that portfolio B had an unsuccessful stock selection isn't correct because the alpha coefficient for portfolio B was positive (1.75), indicating a successful stock selection strategy.

Q.2564 A fund manager tries to generate excess returns by timing the market and by changing the composition of the fund between the market index and Treasury bills. The manager regresses the excess return of the portfolio on the excess return of the benchmark. The following result is obtained: Regression equation:

$$r_p - r_f = a_p + b_p(r_m - r_f) + c_p(r_m - r_f)^2 + e_p$$

Regression estimates:

Portfolio	Estimate
a_p	1.88
b_p	0.65
c_p	1.92
R^2	0.93

A positive c_p signifies that:

- A. The fund manager has successfully timed the market.
- B. The fund manager was not able to successfully time the market.
- C. The beta of the portfolio is constant.
- D. None of the above.

The correct answer is **A**.

The positive value of c_p in the regression equation indicates that the fund manager has successfully timed the market. This is because the term $c_p(r_m - r_f)^2$ in the regression equation is introduced to assess the impact of market timing on portfolio return. In this case, the positive c_p signifies that the fund manager's attempts to time the market have resulted in a positive impact on the portfolio's return. This means that the changes in the composition of the fund and the timing of these changes have been beneficial, leading to excess returns. Therefore, a positive c_p is an indication of successful market timing by the fund manager.

Choice B is incorrect. A positive c_p does not indicate that the fund manager was unsuccessful in timing the market. Rather, it suggests that the manager has been able to generate excess returns by adjusting the portfolio's composition between the market index and Treasury bills based on market conditions.

Choice C is incorrect. The beta of a portfolio being constant would imply that there is no change in risk relative to the market, regardless of changes in market conditions. However, a positive c_p indicates that there has been an adjustment in risk exposure based on changing market conditions, which contradicts with the concept of a constant beta.

Choice D is incorrect. As explained above, a positive c_p does suggest successful timing of the market by altering portfolio composition according to changing market conditions.

Q.2566 A fund manager wants to carry out a style analysis on the portfolio he himself manages for his clients. He identifies four style portfolios as indicated below:

Style portfolio
T-Bills
Small Cap
High P/E
Large Cap

The manager then regresses the fund's rate of return on the above style portfolios. The regression results are given below:

Style portfolio	Regression Coeff.
T-Bills	23
Small Cap	20
High P/E	28
Large Cap	29
α	23
R^2	94%

Which of the following statements is correct?

- A. 94% of variability in the fund's return can be attributed to security selection within the asset classes as well as the timing of the fund manager.
- B. 77% of variability in the fund's return can be attributed to security selection within the asset classes as well as the timing of the fund manager.
- C. 23% of variability in the fund's return can be attributed to security selection within the asset classes as well as the timing of the fund manager.
- D. None of the above.

The correct answer is **D**.

As observed from the regression output, R^2 is equal to 94%. This indicates that 94% variability in the fund's return can be explained by asset allocation to the portfolios indicated above. The four portfolio's viz. small cap, large cap, high P/E and T -Bills explain the 94% variance in the fund's return. **The remaining 6% of the fund's return variability can be attributed to security selection within the asset classes as well as the timing of the fund manager.**

Q.2752 Jensen's alpha will always produce the same rankings as the:

- A. Sharpe ratio
- B. Treynor measure
- C. M squared
- D. Information ratio

The correct answer is **B**.

Jensen's alpha will produce the same result as the Treynor measure, and the M squared and Sharpe ratio measures will produce the same conclusions regarding the existence of manager skill.

However, it is possible for the Sharpe and M squared to identify a manager as not skillful, although the ex-post alpha and the Treynor measure come to the opposite conclusion.

This outcome is most likely to occur in instances where the manager takes on a large amount of nonsystematic risk in the account relative to the account's systematic risk.

Things to Remember

- Jensen's alpha is a risk-adjusted performance metric that measures an investment manager's ability to outperform the market given the level of risk taken.
 - Sharpe ratio measures the risk-adjusted return of an investment or a portfolio.
 - Treynor measure, like Jensen's alpha, is a risk-adjusted performance measure that evaluates the excess return generated by an investment per unit of systematic risk.
 - M squared is a measure of the risk-adjusted return of an investment, taking into account a benchmark's performance.
 - Information ratio measures the active return of a portfolio relative to a benchmark, adjusted for the volatility of those returns.
-

Q.2754 You have been provided the following information about a portfolio:

Return of the portfolio	10%
Beta of the portfolio	1.1
Standard deviation of portfolio	6%
Return on the market	7%
Risk-free rate	3%

Calculate the Jensen's alpha for the portfolio.

- A. 3.2%
- B. 1.7%
- C. 2.6%
- D. 4.2%

The correct answer is **C**.

$$\begin{aligned}\text{Jensen's Alpha} &= R_p - [R_f + \beta(R_m - R_f)] \\ &= 10 - [3 + 1.1(7 - 3)] \\ &= 2.6\%\end{aligned}$$

Q.2757 A portfolio manager increases the number of stocks in his portfolio from ten to twenty, spread out across a range of industries. Assuming, there is no change in the excess return of the portfolio, what will be the most likely effect of this on the Sharpe and Treynor ratios of the portfolio?

- A. The Sharpe ratio will increase and the Treynor ratio will remain unaffected.
- B. The Sharpe ratio will decrease and the Treynor ratio will remain unaffected.
- C. The Sharpe ratio will remain unaffected and the Treynor ratio will increase.
- D. The Sharpe ratio will remain unaffected and the Treynor ratio will decrease.

The correct answer is **A**.

The Sharpe ratio is a measure of risk-adjusted return, calculated as the excess return of the portfolio over the risk-free rate, divided by the standard deviation of the portfolio's return. The standard deviation of the portfolio's return is a measure of the total risk of the portfolio, including both systematic and unsystematic risk. When a portfolio is diversified by adding more

stocks, the unsystematic risk, which is the risk unique to individual stocks, is reduced. This reduction in unsystematic risk leads to a decrease in the total risk of the portfolio, which in turn leads to an increase in the Sharpe ratio, assuming the excess return of the portfolio remains constant. On the other hand, the Treynor ratio is a measure of risk-adjusted return that considers only systematic risk, calculated as the excess return of the portfolio over the risk-free rate, divided by the portfolio's beta. Since diversification reduces only unsystematic risk and does not affect systematic risk, the Treynor ratio remains unaffected when the number of stocks in the portfolio is increased.

Choice B is incorrect. The Sharpe ratio measures the excess return per unit of total risk (standard deviation). By diversifying the portfolio, the manager reduces unsystematic risk, which in turn reduces total risk and increases the Sharpe ratio. Therefore, it's not correct to say that the Sharpe ratio will decrease while Treynor ratio remains unaffected.

Choice C is incorrect. The Treynor ratio measures excess return per unit of systematic risk (beta). Diversification does not affect systematic risk as it cannot be eliminated through diversification. Hence, increasing the number of stocks in a portfolio does not increase the Treynor Ratio.

Choice D is incorrect. As explained above, diversification does not impact systematic risk or beta; hence it doesn't affect Treynor Ratio either positively or negatively. Therefore, stating that increasing diversification would lead to a decrease in Treynor Ratio is incorrect.

Things to Remember

- Systematic risk is the risk that is inherent to the entire market or an entire market segment. It cannot be eliminated through diversification.
- Unsystematic risk, also known as specific risk, is the risk that is specific to a particular company or industry. It can be reduced through diversification.
- The Sharpe ratio is a measure of risk-adjusted return that considers total risk (both systematic and unsystematic).
- The Treynor ratio is a measure of risk-adjusted return that considers only systematic risk.
- Diversification helps in reducing unsystematic risk, leading to a decrease in total risk and an increase in the Sharpe ratio.

Q.3032 Bob White is a retail investor interested in a stock that is paying an annual dividend of \$3.00, selling for \$100 at present. His initial plan is to buy the stock, collect the \$3.00 dividend, and at the end of the year sell the stock in exchange for \$110. At the end of the first year, however, instead of selling his share, White purchases a second share and goes ahead to hold both until the end of the second year. During this time he receives a dividends of \$3.00 per share. At that point, he sells each of the shares at \$120. Compute the geometric average/time-weighted return on the investment.

- A. 11.05%
- B. 12.49%
- C. 13.25%
- D. 12.88%

The correct answer is **B**.

Step 1. Break the evaluation period into two periods based on timing of cashflows:

Holding period 1:

Beggining price: \$100

Dividend received: \$3.0

Ending Price: \$110

Holding period 2:

Beggining price: \$220 (2 shares)

Dividend received: \$6.0 (2 shares)

Ending Price: \$240 (2 shares)

Step 2. Calculate the holding period return, HPR, for each period:

$$\text{Period 1: HPR} = \left[\frac{(110+3-100)}{100} \right] = 0.13$$

$$\text{Period 2: HPR} = \left[\frac{(240+6-220)}{220} \right] = 0.12$$

Step 3. Calculate the total return for the entire period:

$$\text{Total return} = 1.13 \times 1.12 = 1.2656$$

Step 4. Compute the annualized time-weighted rate of return

$$(1 + \text{annualized time - weighted rate of return})^2 = 1.2656$$

$$\text{annualized time- weighted rate of return} = 1.2656^{0.5} - 1 = 0.1249 - 1 = 0.1249 \text{ or } 12.49\%$$

Q.3137 Suppose you purchase one share of the stock of AFC Bank at the beginning of year 1 for \$36. At the end of year 1, you receive a \$2 dividend, and buy one more share for \$30. At the end of year 2, you receive total dividends of \$4 (i.e., \$2 for each share), and sell the shares for \$36.45 each. The dollar-weighted return on your investment is:

- A. 16.93%
- B. 8.0%
- C. 12.35%
- D. 7.52%

The correct answer is **C**.

The dollar-weighted rate of return is defined as the internal rate of return (IRR) on a portfolio, taking into account all cash inflows and outflows

$$\begin{aligned} \$36 + \frac{\$30}{(1+r)} &= \frac{\$2}{(1+r)} + \frac{\$4}{(1+r)^2} + \frac{\$72.90}{(1+r)^2} \\ r &= 12.35\% \end{aligned}$$

.

Q.3140 The following statements compare a highly liquid asset against an (otherwise similar) illiquid asset. Which statement is most likely to be false?

- A. It is possible to trade a larger quantity of the liquid asset without significantly affecting the price.
- B. The liquid asset has a smaller bid-ask spread.
- C. The liquid asset has higher price volatility since it trades more often.
- D. The liquid asset has higher trading volume.

The correct answer is **C**.

The statement that a liquid asset has higher price volatility because it trades more often is incorrect. Liquidity and price volatility are two different aspects of an asset. Liquidity refers to the ease with which an asset can be bought or sold in the market without affecting its price. On the other hand, price volatility refers to the rate at which the price of an asset increases or decreases for a set of returns. It is a common misconception that a higher trading volume (which is a characteristic of a liquid asset) leads to higher price volatility. However, this is not necessarily true. While it is possible for a liquid asset to experience price volatility, it is not a direct result of its liquidity. Instead, price volatility is influenced by factors such as market sentiment, economic indicators, and company performance. Therefore, the statement in choice C is most likely to be false.

Choice A is incorrect. This statement is actually correct. In a liquid market, it is possible to trade larger quantities of an asset without significantly affecting the price due to the high number of buyers and sellers present.

Choice B is incorrect. This statement is also correct. Liquid assets typically have smaller bid-ask spreads because there are more participants in the market, leading to tighter spreads between buying and selling prices.

Choice D is incorrect. This statement too, is correct. Liquid assets generally have higher trading volumes as they are easier to buy and sell without causing significant price changes.

Things to Remember

- Liquidity refers to the ease of buying or selling an asset without affecting its price
- Price volatility is the rate at which the price of an asset changes
- Higher trading volume does not necessarily lead to higher price volatility
- Liquid assets have smaller bid-ask spreads due to more market participants
- Trading larger quantities of a liquid asset is possible without significant price impact

Q.3172 The Campbell family trust account had a value of \$4,700,000 on April 1st 2017 and \$6,400,000 on April 1st 2018. During the year, a contribution of \$800,000 was received but the trust manager did not note the date on which the contribution was received. Assuming that the fund values are given as end-of-day values, what would be the rate of return on the account if the contribution was received on April 1st 2017 versus if it were received on April 1, 2018?

- A. April 1, 2017: 19.14%; April 1st, 2018: 16.36%
- B. April 1, 2017: 19.14%; April 1st, 2018: 14.06%
- C. April 1, 2017: 14.06%; April 1st, 2018: 15.98%
- D. April 1, 2017: 16.36%; April 1st, 2018: 36.17%

The correct answer is **D**.

If the receipt was at the beginning of the period, then:

$$\left\{ \frac{[6,400,000 - (4,700,000 + 800,000)]}{[4,700,000 + 800,000]} \right\} = 0.1636 \text{ or } 16.36\%$$

If the receipt was at the end of the period, then the 800,000 contribution should not be taken into account when calculating the return earned over the past year as it already forms part of the 6,400,000:

$$\left\{ \frac{[(6,400,000) - 4,700,000]}{[4,700,000]} \right\} = 0.3617 \text{ or } 36.17\%$$

Q.3173 Rohit Sharma is a treasury manager at Zannata Industries, a large industrial unit producing equipment for farming. Rohit is analyzing a fund to park some of Zannata's excess cash. He has come across Kratnik Fund, a mutual fund with an average annual return of 12% over the last five years, a beta value of 1.35, and standard deviation of returns of 16.80%. During the same time period, the average annual central bank discount rate was 4.5% and the average annual return on the SENSEX 100 index was 18%. Using this information, the Sharpe ratio for the Kratnik Fund is closest to:

- A. 0.8
- B. 5.56
- C. 3.23
- D. 0.45

The correct answer is **D**.

$$\text{Sharpe Ratio} = \frac{(R_{\text{Kratnik}} - R_f)}{\sigma_{\text{Kratnik}}} = \frac{(12 - 4.50)}{16.80} = 0.45$$

Q.3174 David Hale is an Investment advisor preparing performance analyst of two large pension funds - Qintar Fund and Zombie Fund. The funds are being considered by the plan sponsor of Gallant Motor Companies Pension Fund as an addition to its portfolio. The minimum acceptable return for Gallant is 5.0%, which the sponsor has determined with the help of an independent actuary. The T-bill return over the last fiscal year was 4.5%. Over the same period, the return on the S&P 500 (which is used as the market index) was 10% with a standard deviation of 21% and a beta of 1.0. The most recent risk and return measures for both Qintar and Zombie are: **Qintar**: Return 16.5%; Standard deviation 38.1%; Beta: 0.8; and Downside deviation: 14.9% **Zombie**: Return 15.9%; Standard deviation: 35.6%; Beta: 1.25; and Downside deviation: 14.0% Given this information, the Sharpe ratio for the Zombie Fund is closest to:

- A. 0.15
- B. 0.62
- C. 2.31
- D. 0.32

The correct answer is **D**.

$$\text{Sharpe Ratio} : \frac{(r_p - r_f)}{\sigma_p}$$

The average portfolio excess return over sample period is divided by the returns' standard deviation over the said period, via the Share Ratio which is a measurement of the reward to volatility trade-off.

$$= \frac{(0.159 - 0.045)}{0.356} = 0.32$$

Q.3175 David Hale is an Investment advisor preparing performance analysis of two large pension funds – Qintar Fund and Zombie Fund. The funds are being considered by the plan sponsor of Gallant Motor Companies Pension Fund as an addition to its portfolio. The minimum acceptable return for Gallant is 5.0%, which the sponsor has determined with the help of an independent actuary. The T-bill return over the last fiscal year was 4.5%. Over the same period, the return on the S&P 500 (which is used as the market index) was 10% with a standard deviation of 21% and a beta of 1.0. The most recent risk and return measures for both Qintar and Zombie are: **Qintar**: Return 16.5%; Standard deviation 38.1%; Beta: 0.8; and Downside deviation: 14.9% **Zombie**: Return 15.9%; Standard deviation: 35.6%; Beta: 1.25; and Downside deviation: 14.0% Given this information, the M-squared measure for the Qintar fund is closest to:

- A. 8.10%
- B. 1.11%
- C. 6.70%
- D. 9.46%

The correct answer is **B**.

$$M^2 = \sigma_M(S_P - S_M)$$

Where:

$$\begin{aligned} \sigma_M &= 21\% \\ S_P &= \frac{r_P - r_f}{\sigma_P} = \frac{(16.5\% - 4.5\%)}{38.1\%} = 0.3150 \\ S_M &= \frac{r_M - r_f}{\sigma_M} = \frac{(10\% - 4.5\%)}{21\%} = 0.2619 \end{aligned}$$

$$M^2 = 21\%(0.3150 - 0.2619) = 0.01115 \approx 1.115\%$$

Alternatively,

P has a standard deviation of 38.1% versus a market standard deviation of 21%.

Therefore, the adjusted portfolio P* would be formed by mixing bills and portfolio P with weights $\frac{21}{38.1} = .55118$ in P and $1 - .55118 = .44882$ in bills. The return on this portfolio would be $(.44882 \times 4.5\%) + (.55118 \times 16.5\%) = 11.114\%$, which is 1.114% more than the market return (11.114% – 10%). Thus portfolio P has an M-squared measure of 1.114%.

*The slight difference is due to rounding error

Q.3176 David Hale is an Investment advisor preparing performance analyst of two large pension funds – Qintar Fund and Zombie Fund. The funds are being considered by the plan sponsor of Gallant Motor Companies Pension Fund as an addition to its portfolio. The minimum acceptable return for Gallant is 5.0%, which the sponsor has determined with the help of an independent actuary. The T-bill return over the last fiscal year was 4.5%. Over the same period, the return on the S&P 500 (which is used as the market index) was 10% with a standard deviation of 21% and a beta of 1.0. The most recent risk and return measures for both Qintar and Zombie are: **Qintar**: Return 16.5%; Standard deviation 38.1%; Beta: 0.8; and Downside deviation: 14.9% **Zombie**: Return 15.9%; Standard deviation: 35.6%; Beta: 1.25; and Downside deviation: 14.0% Given this information, Jensen's alpha for the Qintar fund is closest to:

- A. 0.315
- B. 0.15
- C. 0.076
- D. 0.046

The correct answer is C.

$$\text{Jensen's Alpha : } \alpha_P = r_P - [r_f + \beta_P (r_M - r_f)]$$

Given the beta of the portfolio and the average market return, this measure is the average return on the portfolio over and above the CAPM prediction.

$$= 0.165 - [(0.045 + 0.8(0.10 - 0.045))] = 0.076 \text{ or } 7.6\%$$

Q.3177 Jack Shiller is a portfolio manager at Quandra Investment. He has gathered the following information:

Thunder Growth Fund generated returns of 14% with a standard deviation of 25% and a Beta of 1.15. The index return over the same period is 12% with a standard deviation of 18% and a Beta of 1.00.

If the risk-free rate is currently 4%, which of the following represents the Sharpe ratio and the Treynor measure, respectively, for the Thunder Growth Fund?

- A. 0.56 and 0.12
- B. 0.70 and 0.19
- C. 0.08 and 0.02
- D. 0.40 and 0.09

The correct answer is **D**.

The Sharpe ratio is the difference between the Growth Fund's return and the risk-free rate divided by the Growth Fund's standard deviation = $\left[\frac{(0.14-0.04)}{0.25} = 0.40\right]$.

The Treynor measure is the difference between the Growth Fund's return and the risk-free rate divided by the Growth Fund's Beta = $\left[\frac{(0.14-0.04)}{1.15} = 0.09\right]$.

Things to Remember

- Sharpe Ratio: A measure of risk-adjusted return, calculated by subtracting the risk-free rate from the return of the investment and then dividing by the standard deviation of the investment's returns.
 - Treynor Measure: A measure of risk-adjusted return that relates the excess return of an investment to its systematic risk, calculated by subtracting the risk-free rate from the return of the investment and then dividing by the investment's Beta.
 - Beta: A measure of systematic risk that compares the volatility of a stock's returns to the volatility of the market as a whole.
 - Risk-Free Rate: The theoretical rate of return of an investment with zero risk, typically represented by the yield on government bonds.
 - Standard Deviation: A measure of the dispersion of returns around the average return of an investment, indicating the investment's volatility.
-

Q.3178 Samuel Badree is managing a hedge fund at Bumzee Investments. The hedge fund had a return of 13.3%, while its benchmark index, had a return of 8.9%. Over the same time period, the fund's volatility was 16.3%, while the index's volatility was 9.8%. Assuming that the fund's tracking error was 1.34%, and that the risk-free rate is 5.7%, what is the information ratio for this fund?

- A. 3.3
- B. 0.077
- C. 0.632
- D. 1.051

The correct answer is **A**.

$$\text{Information Ratio: } \frac{\alpha_p}{\text{TE}}$$

The portfolio's alpha is divided by the portfolio's nonsystematic risk (tracking error) to compute the information ratio. It is a measure of the abnormal return per unit of risk, which could be diversified away in principle by holding a market index portfolio.

$$\text{IR} = \frac{(0.133 - 0.089)}{0.0134} = 3.3$$

Q.3180 James Simpson, a balanced fund portfolio manager, has consistently outperformed the market in the past. Simpson wants to make his portfolio alphas benchmark-neutral. One of the positions in the balanced fund he manages is that of Tara Industries that has an alpha of 1.9%, an information coefficient of 0.23, and a beta coefficient of 1.53. If the benchmark has an alpha of 0.9%, the new modified alpha that would essentially reduce the position's beta to 1 is closest to:

- A. 0.523%
- B. 0.46%
- C. 3.277%
- D. 2.754%

The correct answer is **A**.

By making this alpha benchmark-neutral, a new modified alpha will be computed where the beta is reduced to one. To make the alpha benchmark-neutral, subtract the product of Tara Industries' beta and the benchmark alpha from the original alpha: $[1.90 - (1.53 \times 0.90)] = 0.523\%$

Things to Remember

- **Alpha:** Alpha is a measure of the active return on an investment, the performance of that investment compared to a suitable market index.
 - **Beta Coefficient:** Beta measures the volatility or systematic risk of a security or a portfolio in comparison to the market as a whole.
 - **Information Coefficient:** Information coefficient measures the correlation between the forecasted returns and the actual returns.
 - **Benchmark-Neutral Portfolio:** A portfolio that is constructed to have zero exposure to the systematic risk factors of a specific benchmark.
 - **Portfolio Management:** The art and science of making decisions about investment mix and policy, matching investments to objectives, asset allocation for individuals and institutions, and balancing risk against performance.
-

Q.3181 Graham Cook is an investment analyst at Gabriel Fund. He is assisting his portfolio manager in stock selection. He is currently considering two stocks for long-term investment. The first stock, Ketrick Fertilizer, has an expected return of 16% and a Beta of 1.6 with the market. The second stock, Sanbros Chemical, has an expected return of 13% and a Beta of 1.2 with the market. If the current risk-free rate is 6.7%, which stock should be included in the portfolio?

- A. Sanbros Chemicals because it has a lower Treynor measure.
- B. Sanbros Chemicals because it has a higher Treynor measure.
- C. Ketrick Fertilizer because it has a higher Treynor measure.
- D. Ketrick Fertilizer because it has a lower Treynor measure.

The correct answer is C.

The Treynor measure is the difference between the stock return and the risk-free rate divided by the stock's Beta:

$$\text{Ketrick Fertilizer's Treynor Ratio} = \left[\frac{(0.16 - 0.067)}{1.6} = 0.0582 \right]$$

$$\text{Sanbros Chemical's Treynor Ratio} = \left[\frac{(0.13 - 0.067)}{1.2} = 0.0525 \right]$$

Investors want more excess return per unit of systematic risk or Beta. Given only this information, this pension fund manager should choose Ketrick Fertilizer because it has a higher Treynor measure.

Q.3184 Zhong Ren is a performance analyst at JingShin Financial, a large Chinese investment bank. During the annual performance evaluation, Ren is analyzing his fund manager's performance. He has gathered the following information for his analysis:

- Manager's return: 7.6%
- Benchmark return: 6.2%
- Shanghai market index return: 8.8%

Based on the above information, how can the manager's performance be most accurately characterized?

- A. The manager earned an excess return from style but not from active management.

- B. The manager earned an excess return from active management but not from style.
- C. The manager earned an excess return from style and active management.
- D. The manager earned an excess return neither from active management nor from style.

The correct answer is **B**.

The manager earned an excess return from active management but not from style. Active management refers to a strategy where the manager makes specific investments with the goal of outperforming an investment benchmark index. In this case, the manager's return (7.6%) is higher than the benchmark return (6.2%), indicating that the manager has earned an excess return from active management. The excess return is calculated as the difference between the manager's return and the benchmark return, which is 1.4% ($7.6\% - 6.2\%$).

On the other hand, style return refers to the return attributable to the manager's investment style. It is calculated as the difference between the manager's return and the market return. In this case, the manager's return (7.6%) is less than the market return (8.8%), indicating that the manager did not earn a return from style. The style return is -1.2% ($7.6\% - 8.8\%$). Therefore, the manager earned an excess return from active management but not from style.

Choice A is incorrect. The manager did not earn an excess return from style. Style return is the difference between the manager's return and the market index, which in this case is -1.2% ($7.6\% - 8.8\%$). Since it's negative, it indicates that the manager did not outperform the market index through his investment style.

Choice C is incorrect. As explained above, there was no excess return from style as indicated by a negative value of -1.2%. Therefore, it's impossible for the manager to have earned an excess return from both active management and style.

Choice D is incorrect. The fund manager did earn an excess return from active management but not from style as explained above in choice A and B explanations respectively.

Things to Remember

- Active Management: Strategy where the manager makes specific investments to outperform an investment benchmark index.
- Style Return: Return attributable to the manager's investment style, calculated as the difference between the manager's return and the market return.
- Excess Return: The difference between the manager's return and the benchmark return, used to evaluate performance.
- Market Index: A hypothetical portfolio representing a segment of the financial market, used as a benchmark for evaluating investment performance.

- **Performance Evaluation:** Process of assessing the effectiveness of an investment manager in achieving investment objectives.
-

Q.3185 Johnny Nara is a research analyst at a large investment bank in the US. During an advisory relationship with his client, he performed an attribution analysis for the Sethi Investment Portfolio. He determined that the sector effect was 0.322%, the within-sector selection was -0.157%, and the allocation/selection effect was 0.061%. If the benchmark return was 8.441%, how much did the portfolio manager add value for Sethi, and what was the portfolio's return during the period?

- A. 0.226%, 8.667%
- B. 0.418%, 8.859%
- C. 0.226%, 8.215%
- D. 0.324%, 7.534%

The correct answer is **A**.

To calculate the added value for the Sethi Investment Portfolio by the portfolio manager and determine the portfolio's return for the period, you can use the information provided from the attribution analysis performed by Johnny Nara:

Attribution Analysis Details.

- **Sector Effect.** +0.322%
- **Within-Sector Selection.** -0.157%
- **Allocation/Selection Interaction.** +0.061%
- **Benchmark Return.** 8.441%

Calculation Steps.

1. **Sum the Attribution Effects.** Add together the sector effect, within-sector selection, and allocation/selection interaction to determine the total active return attributed to the portfolio manager's decisions:

$$\text{Total Active Return} = 0.322\% - 0.157\% + 0.061\% = 0.226\%$$

This represents the additional return generated by the portfolio manager's active management strategies over the benchmark.

2. **Calculate the Portfolio's Total Return.** Add the benchmark return to the total active return to find the portfolio's total return during the period:

$$\text{Portfolio's Total Return} = 8.441\% + 0.226\% = 8.667\%$$

This analysis indicates that despite some negative impacts from within-sector selections, the positive contributions from sector allocation and allocation/selection interactions resulted in an overall positive active return, enhancing the portfolio's performance over the benchmark.

Q.3186 Dale Charles is an equity fund analyst at Songyang Investments, a large mutual fund in South Korea. To thoroughly evaluate the results of a macro performance attribution analysis of a fund, Dale has gathered the following data on the fund:

Beginning value	\$100,000
Net contributions	\$100,000
Risk-free asset	\$103,000
Asset category	\$107,000
Benchmarks	\$104,000
Investment strategies	\$113,000
Allocation effects	\$116,000

Had the manager only engaged in a pure index approach, then instead of having a fund return of 16%, the return of the fund would have been closest to:

- A. 16%
- B. 7%
- C. 10%
- D. 13%

The correct answer is **B**.

$$\text{Return} = 7\% = \$107,000 - \$100,000 / \$100,000.$$

The Asset Category investment strategy assumes that the fund's beginning value and external cash flows are invested passively in a combination of the designated asset category benchmarks, with the specific allocation to each benchmark based on the fund sponsor's policy allocation to those asset categories. In essence, the asset category corresponds to a pure index approach. The dollar return would have been \$7,000 or 7% on the initial \$100,000.

Q.3187 Lilly Johana is a bond portfolio manager at Jackson Investment. She is in the process of decomposing the various sources of return to her bond portfolio that yielded a return of 10%. The actual treasury yield was 8%, which is 0.5% better than the expected yield of 7.5%. In addition, Lilly has ascertained that her portfolio benefited 0.50% due to sector allocation and 0.25% from allocation/selection interaction. Based on this information, how much of the portfolio's overall return is attributable to within-sector selection?

- A. 1.25%
- B. 1.00%
- C. 1.75%
- D. 2.50%

The correct answer is **A**.

Expected treasury yield	7.50%
+Unexpected treasury yield	0.50%
+Return from sector allocation	0.50%
+Return from allocation selection interaction	0.25%
+Return attributable to within-sector selection	1.25%(can be backed out given the other inform
Total return	= 10.0%

Q.3190 Jack Marconi is an equity strategist at Gandhara Investment and is evaluating the performance of four large-cap equity portfolios: Azgard, Lambda, Tricky, and Jackpot. As part of his analysis, Jack computes the Sharpe ratio and the Treynor measure for all four funds. Based on his finding, the ranks assigned to the four funds are as follows:

Fund	Treynor Measure Rank	Sharpe Ratio Rank
Azgard	1	4
Lambda	2	3
Tricky	3	2
Jackpot	4	1

The difference in rankings for Funds Azgard and Jackpot is most likely due to:

- A. Different benchmarks used to evaluate each fund's performance.
- B. A difference in risk premiums.
- C. Poor diversification in Azgard Fund relative to Jackpot Fund.
- D. None of the above.

The correct answer is **C**.

The Sharpe measure uses the standard deviation of returns as the measure of risk whereas the Treynor measure uses beta (systematic risk).

The Sharpe measure, therefore, evaluates the portfolio manager on the basis of both the rate of return performance and diversification.

For a completely diversified portfolio, one without any unsystematic risk, the two measures give identical rankings because the total variance of the completely diversified portfolio is its systematic risk.

On the other hand, a poorly diversified portfolio has a higher ranking on the basis of the Treynor measure but a much lower ranking on the basis of the Sharpe measure.

The difference in rank emerges from the difference in diversification.

Reading 154: Hedge Funds

Q.2571 Assuming that you are an analyst in a hedge fund consulting firm, you are tasked with studying the historical performance of hedge funds through a commercial database. You notice that the returns of the hedge funds listed in the database show a surprisingly high correlation with the average returns of funds-of-hedge funds (FoFs). Given your knowledge of hedge fund databases, how would you explain this phenomenon?

- A. The commercial database has managed to include all existing hedge funds, resulting in a comprehensive dataset.
- B. The high correlation is due to look-ahead bias in the commercial database.
- C. The high correlation is due to survivorship bias in the commercial database.
- D. The FoFs theoretically invest across all hedge funds, not just those reported in commercial databases.

The correct answer is **D**.

The structure of Funds-of-Hedge Funds (FoFs) is such that they invest in a broad array of hedge funds, not limited to those that report to commercial databases. Therefore, their returns should theoretically represent a wider hedge fund universe. The high correlation between the average returns of FoFs and those listed in commercial databases suggests that the funds represented in these databases provide a reasonable approximation of the overall hedge fund market's performance. Hence, despite the potential for selection bias in commercial databases - which arises from the voluntary nature of hedge funds' reporting - the observed high correlation implies that this bias might not be significantly distorting the performance representation in these databases.

A is incorrect. It is unlikely that the commercial database includes all existing hedge funds. The passage explicitly states that some hedge funds do not participate in commercial databases, thus contributing to selection bias.

B is incorrect. Look-ahead bias is not the reason for the high correlation. Look-ahead bias occurs when a study uses data that was not available during the period being analyzed. This bias does not necessarily lead to a high correlation between the returns of FoFs and the returns of hedge funds listed in commercial databases.

C is incorrect. Survivorship bias refers to the phenomenon where the performance of surviving funds is more likely to be included in the database than the performance of funds that have stopped operating. This can affect the aggregate performance of hedge funds in the database, but it does not directly explain the high correlation with the average returns of FoFs.

Things to Remember

- FOHFs (Funds of Hedge Funds) are investment vehicles that put money into a variety of individual hedge funds, offering investors comprehensive diversification in the hedge

fund industry at a cost.

- FOHFs are not restricted to investing only in hedge funds listed in commercial databases.
 - It is presumed that FOHFs invest in all available hedge funds, including those not listed in commercial databases, implying their average returns could be a more accurate representation of the overall hedge fund industry's average returns.
 - The significant correlation between the average returns of FOHFs and the average returns of hedge funds listed in commercial databases indicates that the latter is not a special subset but is representative of the whole hedge fund population.
-

Q.2573 Investment Strategies Corporation (ISC) is a newly founded hedge fund firm looking to invest in various strategies. As a risk manager, you are asked to conduct a review of potential strategies to determine which one would best fit the firm's risk profile and return expectations. The following strategies are under consideration: A) Equity Long-Short, B) Global Macro, C) Distressed Securities, and D) Convertible Arbitrage. ISC is looking for a strategy with substantial upside potential during both bull and bear markets, although they are willing to accept the associated risks such as illiquidity and default risk. They are not looking for a strategy that relies heavily on leverage or that requires high management skill to hedge the net market exposure. Which of the above strategies would you recommend to ISC?

- A. Equity Long-Short
- B. Global Macro
- C. Distressed Securities
- D. Convertible Arbitrage

The correct answer is **C**.

The distressed securities strategy might be the best fit for ISC based on their risk and return expectations. This strategy involves investing in the debt of firms that are near or currently going through bankruptcy. It has substantial upside potential during both bull and bear markets as recovery or restructuring can lead to significant returns. It is also known for its illiquidity and default risk, which ISC is willing to accept. It does not rely heavily on leverage, nor does it require the high management skill needed to hedge net market exposure like the Equity Long-Short strategy.

A is incorrect because the Equity Long-Short strategy requires high management skill to

manage net market exposure by balancing long and short positions. Also, it doesn't have the same potential for substantial returns in bear markets that distressed securities do.

B is incorrect because the Global Macro strategy heavily relies on leverage to take large positions based on macroeconomic trends. It doesn't fit with ISC's preferences, which mention a preference for a strategy that doesn't rely heavily on leverage.

D is incorrect because Convertible Arbitrage involves buying convertible securities and shorting the same company's stock, which relies heavily on leverage and requires extensive management skill. Additionally, the strategy does not possess the illiquidity and default risk inherent in distressed securities investing, which ISC has shown a willingness to accept.

Things to Remember

- Distressed Securities strategy involves investment in the debt of firms nearing or undergoing bankruptcy, providing considerable returns potential during both bull and bear markets due to restructuring or recovery.
- Illiquidity and default risk are inherent in distressed securities investments, as these securities can be hard to sell and carry a significant risk of the debtor defaulting.
- Unlike some other strategies, Distressed Securities does not rely heavily on leverage, thus minimizing the exposure to potential losses from borrowing.
- The success of a distressed securities strategy does not necessarily require high management skill in hedging net market exposure, which is often required in strategies like Equity Long-Short and Convertible Arbitrage.
- Although Distressed Securities can offer high returns, they are also associated with high risk; therefore, the firm should be willing to accept this risk-reward trade-off.

Q.2574 Quasar Hedge Fund (QHF) is a prominent player in the hedge fund market known for its innovative and sophisticated investment strategies. QHF has recently decided to explore opportunities arising from corporate events and seek asymmetric payoffs. The fund wants to leverage the price discrepancy between the current price of the target firm's shares and the final purchase price offered by the acquirer. Management at QHF is willing to take on risks associated with the potential cancellation or modification of deals. However, they are reluctant to get involved in strategies that are overly sensitive to interest rate fluctuations or require deep insight into distressed securities. Given their preferences, which of the following hedge fund strategies should QHF consider adopting?

A. Convertible arbitrage

- B. Merger arbitrage
- C. Fixed-income arbitrage
- D. Distressed securities

The correct answer is **B**.

B is correct. Merger Arbitrage aligns perfectly with QHF's preferences. This strategy is focused on announced corporate mergers and acquisitions, seeking to profit from the price discrepancy between the current and final purchase price of the target firm's shares. The fund is exposed to the risk of the deal's failure, but skilled managers can structure trades to limit losses and possibly even generate moderate returns when deals fail. Importantly, this strategy is not overly sensitive to interest rate fluctuations and doesn't necessitate a deep understanding of distressed securities.

A is incorrect because Convertible Arbitrage focuses on exploiting pricing inefficiencies in convertible securities and their related equity, which doesn't align with QHF's desire to profit from mergers and acquisitions. Furthermore, this strategy can be sensitive to interest rate fluctuations.

C is incorrect because Fixed-Income Arbitrage involves exploiting price differences in fixed-income securities, which makes this strategy very sensitive to interest rate fluctuations against QHF's criteria.

D is incorrect because Distressed Securities investing involves buying the securities of companies in or near bankruptcy. It requires a deep understanding of distressed securities, and it doesn't focus on pricing discrepancies as QHF prefers.

Things to Remember

- Risk (merger) Arbitrage event-driven hedge funds aim to capture the spreads in merger or acquisition transactions involving public companies after the transaction terms are announced.
- The spread is the difference between the transaction bid and the trading price. Typically, the target stock trades at a discount to the bid to account for the risk of the transaction failing to close.
- In a cash deal, the manager will usually purchase the stock of the target and tender it for the offer price at closing.
- In a fixed exchange ratio stock merger, one would go long the target stock and short the acquirer's stock according to the merger ratio to isolate the spread and hedge out

market risk.

- The principal risk in risk arbitrage is deal risk, which arises if the deal fails to close.
-

Q.2575 Joseph Mendoza, a seasoned financial analyst, is reflecting on the historical performance and portfolio construction of hedge funds versus equity indices. He has collected data from a series of studies conducted on 27 large hedge funds identified in 2000, and on two distinct portfolios of top 50 large hedge funds, formed based on the assets under management at year-end 2001 and 2010 respectively. Mendoza's main focus is on understanding the alpha performance, factor exposures, and the effectiveness of various hedge fund strategies compared to the S&P 500 index over different periods. Based on the data and research findings, which of the following statements is most accurate?

- A. Large hedge funds demonstrated persistent exposure to the U.S. equity market, and their alpha did not decline over time.
- B. The portfolios of the top 50 large hedge funds had a consistently significant and negative alpha.
- C. The portfolios of the top 50 large hedge funds, rebalanced yearly, demonstrated statistically significant alpha relative to the DJCSI and HFRI hedge fund indices.
- D. Hedge funds, when regressed against an 8-factor model, had significant exposure to stocks and bonds.

The correct answer is **C**.

The portfolios of the top 50 large hedge funds, both the one constructed based on the assets under management at year-end 2001 and the one based on year-end 2010 (the "foresight-assisted" portfolio), demonstrated statistically significant alpha relative to the DJCSI and HFRI hedge fund indices. This indicates that a strategy of investing in large hedge funds and rebalancing yearly could deliver superior performance compared to just investing in hedge fund indices. Additionally, the top 50 portfolios and the broad hedge fund indices, DJCSI and HFRI, outperformed the S&P 500 index during the 2002 to 2010 period.

A is incorrect because the research indicates that there was not a persistent directional exposure to the U.S. equity market for the large hedge funds. This means that the hedge funds' returns did not consistently move in the same direction as the U.S. equity market. Also, alpha, the measure of performance on a risk-adjusted basis, declined over time for these funds, contrary to what is stated in this option.

B is incorrect because, while the decline in alpha over time was noted for the hedge funds, there was no significant negative alpha observed for the portfolios of the top 50 large hedge funds. The decline in alpha was attributed to increased competition in the hedge fund industry, but it didn't reach a point where it became significantly negative.

D is incorrect because, when the returns of the hedge funds were regressed against an 8-factor model, it was found that the hedge funds had no significant exposure to stocks and bonds. This contradicts the statement made in this option.

Things to Remember

- Large hedge funds can potentially constitute a separate asset class with no significant exposure to stocks and bonds, based on regression analysis of their returns against an 8-factor model.
 - A strategy of investing equally in the top large hedge funds, rebalancing yearly, can yield significant alpha returns and demonstrates persistent exposure to emerging markets.
 - However, the alpha delivered by such strategies has been observed to decline over time, potentially due to increased competition in the hedge fund industry.
 - Portfolios of the top 50 large hedge funds have shown statistically significant alpha relative to the DJCSI and HFRI hedge fund indices, outperforming these indices and the equity market (S&P 500) during the 2002 to 2010 period.
 - Large hedge funds continue to attract institutional investors due to the alpha return they deliver relative to peers and their low exposure to the U.S. equity market.
-

Q.2576 One of the most common phenomena witnessed in the US Treasury market is that on-the-run T-bills have higher prices than off-the-run T-bills. Which of the following hedge fund strategies is most suitable to exploit this inefficiency?

- A. Global macro
- B. Distressed
- C. Managed futures
- D. Arbitrage

The correct answer is **D**.

Arbitrage is a financial strategy that aims to profit from price discrepancies in different markets or between different financial instruments. In the context of the US Treasury market, an

arbitrage strategy would involve exploiting the price difference between on-the-run and off-the-run T-bills. On-the-run T-bills are the most recently issued and therefore have the highest liquidity, which often results in them being priced higher than off-the-run T-bills, which are older issues with lower liquidity. An arbitrageur could potentially profit from this discrepancy by buying the cheaper off-the-run T-bills and selling the more expensive on-the-run T-bills, assuming that the price difference will eventually narrow. This strategy is commonly used by fixed income arbitrage hedge funds, which seek to generate profits by exploiting price anomalies between related fixed-income securities.

Choice A is incorrect. Global macro strategies involve making investment decisions based on the economic and political outlook of different countries. This strategy does not specifically target inefficiencies in the pricing of on-the-run and off-the-run T-bills.

Choice B is incorrect. Distressed strategies focus on investing in securities of companies that are experiencing financial distress or bankruptcy. This strategy has no direct relation to exploiting pricing discrepancies between on-the-run and off-the-run T-bills.

Choice C is incorrect. Managed futures involve trading futures contracts, which are standardized agreements to buy or sell an asset at a future date at a predetermined price. While this strategy may involve trading Treasury futures, it does not specifically aim to exploit the pricing discrepancy between on-the-run and off-the-run T-bills.

Things to Remember

- Arbitrage is a risk-free strategy in theory, as it involves simultaneous buying and selling of securities to take advantage of pricing discrepancies. However, in practice, it can involve risks such as execution risk and liquidity risk.
- Arbitrage opportunities are often short-lived, as the market quickly corrects the pricing discrepancies. Therefore, arbitrageurs need to act quickly to capitalize on these opportunities.
- Arbitrage requires sophisticated trading systems and algorithms to identify and exploit pricing discrepancies. Therefore, it is typically employed by hedge funds and other institutional investors, rather than individual investors.
- Fixed income arbitrage, which involves exploiting pricing discrepancies between related fixed-income securities, is one of the most common types of arbitrage. Other types include statistical arbitrage, merger arbitrage, and convertible arbitrage.

Q.2577 A company has issued bonds with an option to convert the bonds into equity after three years. Due to an error in the conversion factor, the convertible bonds appear undervalued when compared to the projected market price of the company's stock at the time of conversion. A hedge fund manager wants to exploit the trading opportunity arising from this underestimation. The most appropriate strategy is:

- A. Going short on the convertible security and long on the common stock of the company.
- B. Going short on the convertible security and short on the common stock of the

company.

C. Going long on the convertible security and short on the common stock of the company.

D. Going long on common stock and long on the convertible security.

The correct answer is **C**.

When a convertible bond is undervalued due to a mispriced conversion factor, it means that its price on the market is lower than its intrinsic value when compared to the projected conversion value in the company's stock. Therefore, the most direct way to exploit this is by buying the convertible bond, expecting its price to rise as the market recognizes and corrects the discrepancy. Simultaneously, to hedge against potential downside risk in the underlying stock, the trader can short the common stock. This way, if the stock price decreases, gains from the short stock position can offset potential losses from the long convertible position.

Choice A is incorrect. If the convertible bond is undervalued, it doesn't make sense to short it. Instead, one would want to buy or go long on it. Going long on the stock without a corresponding hedge also increases the risk.

Choice B is incorrect. Shorting both the convertible security and the stock doesn't take advantage of the bond's undervaluation. Instead, it suggests that both the bond and the stock are overpriced, which isn't the scenario presented.

Choice D is incorrect. While going long on the convertible makes sense in the context of its undervaluation, simultaneously going long on the stock doesn't provide the hedge that a typical arbitrage strategy seeks. If the stock price drops, both the convertible bond and the stock position could potentially suffer losses.

Things to Remember

- Convertible arbitrage is a market-neutral investment strategy often employed by hedge funds. It involves the simultaneous purchase of convertible securities and the short sale of the same issuer's common stock.
- The profitability of convertible arbitrage is primarily dependent on the price movements of the underlying stock. However, other factors such as interest rates, credit spreads, and company-specific news can also impact the strategy's performance.
- Convertible arbitrage can provide a hedge against market volatility. This is because the long position in the convertible bond tends to increase in value when the market rises, while the short position in the stock tends to gain when the market falls.
- Despite its potential benefits, convertible arbitrage is not without risks. These include credit risk (if the issuer defaults), liquidity risk (if the convertible bond or stock becomes difficult to trade), and execution risk (if the hedge fund manager is unable to establish or unwind the positions at favorable prices).

Q.2578 Richard Chang is a financial risk analyst at a global investment firm, Capital Dynamics. He is reviewing the performance of various hedge funds in which the company has significant investments. Richard is aware that principals (investors) and agents (hedge fund managers) have different risk-sharing preferences, leading to potential issues. One fund, Zeta Capital, has had a fantastic run in the past three years, consistently generating high returns. However, Richard notices that this performance was driven by high-risk strategies. He is concerned about the asymmetry in risk-sharing between his firm (as the principal) and Zeta Capital's managers (as the agents). Which of the following correctly describes the problem of risk-sharing asymmetry between principals and agents in the context of Richard's concerns?

- A. Hedge fund managers might underperform if they are not given enough freedom to take risks, leading to lower returns for principals.
- B. Hedge fund managers tend to avoid high-risk strategies due to the pressure from principals, leading to missed opportunities.
- C. Hedge fund managers, being compensated for performance and not bearing the full downside, might take excessive risks, which might not align with the risk appetite of the principals.
- D. Hedge fund managers always align their risk-taking decisions with the principals' expectations, leading to homogeneous investment outcomes.

The correct answer is **C**.

The risk-sharing asymmetry problem arises when agents (hedge fund managers) do not bear the full downside of their decisions, as they typically earn a performance fee for producing positive returns, but do not directly incur the losses. This asymmetry can incentivize them to take on excessive risk, potentially more than what the principal (investor) is comfortable with, leading to a misalignment in risk preferences.

A is incorrect because the problem of risk-sharing asymmetry is not related to hedge fund managers underperforming due to a lack of freedom to take risks. Instead, it is about the potential for them to take excessive risks, given they do not bear the full downside of their decisions.

B is incorrect because risk-sharing asymmetry does not arise from hedge fund managers avoiding high-risk strategies due to pressure from principals. In fact, the asymmetry can cause managers to embrace high-risk strategies because they do not fully share in the downside risk.

D is incorrect because the very nature of the risk-sharing asymmetry problem suggests that hedge fund managers' risk-taking decisions do not always align with the principals' expectations. The potential for disparate risk tolerances is a fundamental element of this issue.

Things to Remember

- The principal-agent problem arises from different risk preferences between the two parties, often due to their varied exposure to the consequences of the risk.

- Hedge fund managers, acting as agents, may take on more risk than principals (investors) desire because their compensation structure incentivizes performance but does not penalize them equally for losses.
 - This situation, known as risk-sharing asymmetry, can lead to misalignment of risk-taking behavior between principals and agents, potentially causing significant issues for investors.
 - Effective risk management and alignment of interests are crucial to mitigate this asymmetry and protect the principal's investment. This may include re-structuring compensation to incorporate downside risk or implementing strict risk control measures.
-

Q.2580 John McAdams, a pension fund manager, is considering diversifying the fund's portfolio by adding investments in hedge funds. However, John has serious concerns about the possible asymmetry of risk-sharing that typically comes with hedge funds. What strategy should John adopt to mitigate the potential risk asymmetry inherent in hedge fund investments?

- A. Distribute the funds across various hedge fund strategies to ensure diversification of risk.
- B. Choose a hedge fund manager who has an excellent reputation and manages portfolios for other large pension funds.
- C. Verify that the chosen hedge fund managers have a substantial portion of their own assets invested in their respective funds.
- D. Mandate daily position reports from the hedge fund to monitor closely for any possible risk asymmetry.

The correct answer is **C**.

It is advisable for John to ascertain that the hedge fund managers have a substantial portion of their personal wealth invested in the hedge fund they manage. This tactic aligns the fund managers' interests with the investors', mitigating the asymmetry in risk-sharing. If a significant part of the managers' wealth is at stake, they are more likely to manage the fund in a manner that delicately balances risks and returns.

A is incorrect because, although spreading investments across multiple hedge fund strategies may diversify the risk, it doesn't necessarily mitigate the risk asymmetry. This issue is primarily due to fund managers possibly not having a considerable stake or alignment of interests with the

investors. Hence, diversification may lessen portfolio risk but may not ensure interests alignment.

B is incorrect because choosing a hedge fund manager with an excellent reputation could be beneficial in reducing the risk of fraud and increasing trust, but it does not assure mitigation of risk asymmetry. Even renowned fund managers can potentially manage funds in a way that benefits themselves more than the investors.

D is incorrect because insisting on daily position reports could improve the transparency of the hedge fund's operations but does not necessarily alleviate risk asymmetry. Frequent updates on the fund's position don't guarantee that the fund manager's decisions are in line with the best interests of the investors.

Things to Remember

- Ensuring that hedge fund managers have skin in the game—i.e., substantial personal investments in the funds they manage—aligns their interests with those of the investors and can effectively mitigate risk asymmetry.
 - Managers having significant personal investments in the funds they manage often promote prudent risk-taking behaviors.
-

Q.2581 Samuel Ray, a risk analyst at Alpha Investments Bank, is evaluating the annual performance data of hedge funds and real estate funds for investment purposes. He takes note of two significant concerns:

- For the hedge fund dataset, any fund that discontinues its performance reporting is excluded from the database.
- In the case of real estate funds, assets are appraised only annually due to their illiquid nature and infrequent trading. What could be the potential implications of these data issues on Samuel's performance analysis?

A. The mean Sharpe ratio of hedge funds is underestimated, and the mean Sharpe ratio of real estate funds is overestimated.

B. The mean Sharpe ratio of both hedge funds and real estate funds is overestimated.

C. Both hedge funds and real estate funds show an overestimated average volatility.

D. Hedge funds present an overestimated average volatility, while real estate funds show an underestimated average volatility.

The correct answer is **B**.

The exclusion of poorly performing hedge funds from the database can lead to "survivorship bias," potentially resulting in an overestimated mean Sharpe ratio of the remaining hedge funds. For real estate funds, due to the infrequency of asset appraisal (known as "appraisal (smoothing) bias"), the returns may appear more stable than they actually are, thus possibly leading to an overestimated mean Sharpe ratio.

A is incorrect because the mean Sharpe ratio of hedge funds is more likely to be overestimated, not underestimated, due to survivorship bias.

C is incorrect because while survivorship bias could have an effect on the returns of hedge funds, it would most likely result in an underestimation of the true volatility across all funds. Similarly, for real estate funds, due to appraisal (smoothing) bias, the volatility is likely to be underestimated, not overestimated.

D is incorrect because survivorship bias would most likely cause an overestimation of hedge fund volatility, not an underestimation.

Things to Remember

- "Survivorship bias" refers to the skewing of results when discontinued or poorly performing funds are excluded from the analysis, leading to an overestimation of the performance and stability of the remaining fund pool. It is common in hedge fund data due to the tendency to only consider funds that have survived over the analysis period.
- "Appraisal (smoothing) bias" is seen when asset values are infrequently appraised, which is often the case for illiquid assets like real estate. This infrequent appraisal can give the impression of smooth, stable returns, potentially masking the true volatility and risk of the investment.
- Other potential biases to consider when analyzing fund performance include **Look-ahead Bias** (using information that wasn't available at the time of investment), **Time-period Bias** (choosing a favorable time period to improve the appearance of performance), and **Backfill (Instant History) Bias** (including a fund's historical returns in a database only after it has demonstrated success).

Q.2582 Alexa Thompson, a senior risk analyst at Beta Capital, is analyzing a collection of investment strategies that include both active and passive funds for potential inclusion in the firm's investment portfolio. She's particularly focused on two possible issues in her dataset:

- Alexa observes that the data of some of the active funds, which had poor performance and subsequently closed, are not included in the database.
- For some of the passive funds, the selection process involves including those funds that have outperformed the market benchmark in the recent past.

Considering these observations, which bias issues could Alexa potentially be facing while analyzing this dataset?

- A. Alexa is facing look-ahead bias in active funds and selection bias in passive funds.
- B. Alexa is facing selection bias in both active and passive funds and look-ahead bias in passive funds.
- C. Alexa is dealing with both look-ahead bias in both active and passive funds.
- D. Alexa is facing survivorship bias in active funds and selection bias in passive funds.

The correct answer is **B**.

B is correct. Alexa is indeed encountering selection bias in both active and passive funds, along with a look-ahead bias in passive funds. In the case of active funds, the exclusion of funds that closed due to poor performance represents a clear example of selection bias or survivorship bias. By omitting these underperforming funds, the dataset becomes skewed towards more successful active funds, potentially leading to an overestimation of their performance.

Similarly, in the context of passive funds, selecting only those funds that have recently outperformed the market introduces both selection bias and a look-ahead bias. The selection bias arises because only the outperforming funds are included, ignoring the underperforming ones that might have been removed or terminated. Simultaneously, there's a look-ahead bias because the decision to include these funds is based on future information (their recent outperformance) that would not have been available at the time of the actual investment.

A is incorrect because Alexa is facing selection bias in active funds, not look-ahead bias. Similarly, in passive funds, she is dealing with look-ahead bias, not selection bias.

C is incorrect because Alexa is not dealing with look-ahead bias in both fund types. She is encountering selection bias in both, and look-ahead bias in passive funds.

D is incorrect because Alexa is not encountering look-ahead bias in both fund types. She is dealing with selection bias in both, and look-ahead bias in passive funds.

Things to Remember

- "Selection bias" refers to situations where the data sample is not representative of the entire population. For instance, excluding poorly performing active funds from the analysis can overstate the overall performance, presenting a distorted view of reality.

- "Look-ahead bias" occurs when a strategy includes future information that wouldn't have been available at the time of decision-making. Selecting passive funds based on their recent outperformance may lead to overestimated returns as it involves using information not accessible at the investment time.
 - Other types of biases that might impact investment performance analysis include "Survivorship bias," "Backfill bias," and "Appraisal (smoothing) bias."
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Q.2583 Pamela Clark, a risk analyst at a prominent investment bank, is studying the performance of various hedge fund strategies. She is looking for strategies that are resilient during market-wide funding crises. The portfolio she manages consists of different hedge fund strategies, and she has noticed a convergence in terms of risk factors across these strategies during times of market stress. Based on historical events and strategy performance profiles, which of the following could potentially reduce the tail risk in Pamela's hedge fund portfolio during a market-wide funding crisis?

- A. Introducing leveraged positions to exploit market inefficiencies.
- B. Increasing allocation to merger arbitrage funds.
- C. Reducing overall portfolio leverage.
- D. Introducing managed futures into the portfolio.

The correct answer is **D**.

Introducing managed futures into the portfolio could potentially reduce tail risk during a market-wide funding crisis. Historical performance profiles indicate that managed futures, along with short sellers, are the only strategies that avoided losses during previous crises. Hence, managed futures could act as a hedge during market downturns and reduce the impact of credit-driven tail risk on the portfolio.

A is incorrect because leveraging, while can magnify gains, can also amplify losses. The use of leverage was one of the factors that caused significant losses during market-wide funding crises, as highlighted in the events leading to the 2007–2009 financial crisis.

B is incorrect because while merger arbitrage funds earned a positive return in certain stressful market periods, it is not a consistently resilient strategy during market-wide funding crises. The historical events suggest that only managed futures and short sellers consistently avoided losses.

C is incorrect because while reducing overall portfolio leverage can indeed lower the potential for large losses during a market-wide funding crisis, it primarily reduces general exposure rather

than specifically addressing tail risk. Tail risk refers to extreme outcomes that occur with low probability but have severe consequences. Although reducing leverage is a prudent step to mitigate risk, it does not provide a dynamic strategy that could protect or profit from extreme market movements.

Things to Remember

- "Managed futures" are investment strategies that involve going long or short in futures contracts in sectors like commodities, currencies, or bonds. They have demonstrated resilience during market-wide funding crises, making them a potential strategy to reduce tail risk.
 - "Tail risk" refers to the risk of extreme market movements that have a small probability but can lead to significant losses when they occur. Diversification, including strategies with non-correlated returns, can help manage tail risk.
-

Q.2585 Justin Boucher is planning to invest in a hedge fund. However, he is worried about the compensation structure of the fund. The compensation structure is designed in such a manner that the fund managers are entitled to receive 20% of the profits generated by the fund. Boucher feels the compensation structure would entice fund managers to take unreasonable and risky bets which may have a higher chance of going wrong. He recalls a recent article about hedge fund managers taking failed risky bets and then closing the fund to start a new hedge fund.

One of the ways of mitigating the risk of closure of the fund by unsuccessful fund managers is:

- A. To invest in a smaller hedge fund.
- B. To invest in a hedge fund with a small number of managers.
- C. To invest in a hedge fund with multiple managers.
- D. All of the above.

The correct answer is **C**.

Investing in a hedge fund with multiple managers can help mitigate the risk of fund closure by unsuccessful fund managers. This is because each manager would have a smaller stake in the total capital, reducing the impact of any one manager's poor performance on the overall fund. This strategy also promotes diversity in investment strategies, as different managers may have different areas of expertise and investment philosophies. This can lead to a more balanced and resilient portfolio. Furthermore, having multiple managers can increase accountability and reduce the likelihood of risky bets, as decisions are typically made collectively. Examples of large hedge funds that employ multiple managers include Bridgewater Associates, AQR Capital

Management, JPMorgan Asset Management, and Renaissance Technologies.

Choice A is incorrect. Investing in a smaller hedge fund does not necessarily mitigate the risk of fund closure by unsuccessful managers. The size of the hedge fund does not directly correlate with the risk-taking behavior or success rate of its managers. A smaller hedge fund could still have managers who take excessive and risky bets.

Choice B is incorrect. Investing in a hedge fund with a small number of managers also does not guarantee protection against the risk of closure due to unsuccessful bets. The number of managers is not an indicator of their propensity for risky behavior or their ability to manage risks effectively.

Choice D is incorrect. As explained above, neither investing in a smaller hedge fund nor investing in one with fewer managers can ensure mitigation against the risk posed by unsuccessful and excessively risky bets taken by fund managers.

Things to Remember

- Hedge fund compensation structures often include a performance fee, typically around 20% of profits, to incentivize fund managers to generate high returns.
- Risk-taking behavior in hedge funds can be influenced by the compensation structure, as managers may take excessive risks to chase higher returns and earn larger bonuses.
- Diversification within a hedge fund, including having multiple managers with different expertise and investment styles, can help reduce the impact of poor performance by any single manager.
- Investors should conduct thorough due diligence on hedge funds, including understanding the fund's investment strategies, risk management practices, and track record, before making investment decisions.
- Hedge fund closures can occur for various reasons, including poor performance, regulatory issues, or changes in market conditions, highlighting the importance of risk management and diversification.

Q.2586 During the early days of institutional investments in hedge funds, the favored route through which institutional investors invested in hedge funds was through:

- A. Funds of hedge funds.
- B. Direct investment in hedge funds.

C. Large investments in management funds.

D. ETFs

The correct answer is **A**.

During the early days of institutional investments in hedge funds, the most favored route for institutional investors was through funds of hedge funds (FOHFs). FOHFs are investment vehicles that invest in a portfolio of individual hedge funds. This method offered investors a one-stop diversification service to the hedge fund industry for a fee. The primary advantage of this approach was that it allowed investors to gain exposure to a wide range of hedge funds with a single investment, thereby spreading their risk across multiple funds. This diversification helped to mitigate the risk associated with investing in a single hedge fund, which could be subject to significant volatility. Furthermore, FOHFs often have lower minimum investment requirements than individual hedge funds, making them more accessible to a broader range of investors.

Choice B is incorrect. Direct investment in hedge funds was not the most preferred method for institutional investors during the early days. This is because direct investments require a high level of expertise and understanding of complex strategies that hedge funds employ, which many institutional investors did not possess at that time.

Choice C is incorrect. Large investments in management funds were also not the most preferred method for institutional investors to invest in hedge funds during the early days. Management funds often involve significant operational risks and lack transparency, making them less attractive to institutional investors who prioritize risk management and transparency.

Choice D is incorrect. ETFs were not the most preferred method for investing in hedge funds by institutional investors during their initial stages of investment. While ETFs offer liquidity and transparency, they do not provide access to the same level of potential returns as direct investments or fund-of-funds structures can offer.

Things to Remember

- Funds of hedge funds (FOHFs) are investment vehicles that invest in a portfolio of individual hedge funds, providing diversification and risk mitigation for investors.
- Direct investment in hedge funds requires a high level of expertise and understanding of complex strategies employed by hedge funds.
- Large investments in management funds can involve significant operational risks and lack transparency, making them less attractive to institutional investors.
- ETFs offer liquidity and transparency but may not provide the same level of potential returns as direct investments or fund-of-funds structures.
- Investors should consider factors such as fees, performance history, risk management

practices, and transparency when choosing an investment route in hedge funds.

Q.2587 After the collapse of Long-Term Capital Management (LTCM) and the dot-com bubble burst of 2001, the hedge fund industry witnessed:

- A. Tighter regulations
- B. Enhanced disclosure requirements
- C. A large arrival of institutional investors
- D. A decrease in assets under management

The correct answer is **C**.

The collapse of LTCM and the dot-com bubble burst in 2001 were significant events that shook the financial markets. However, these events led to a change in the hedge fund industry. Despite the turmoil, the hedge fund industry was able to generate higher returns with a lower standard deviation compared to the S&P 500. This performance attracted a large influx of institutional investors. Institutional investors, such as pension funds, endowments, and insurance companies, began to see the potential of hedge funds as an alternative investment that could provide diversification benefits and enhance overall portfolio performance. The arrival of these institutional investors brought a significant amount of capital into the hedge fund industry, leading to its growth and development.

Choice A is incorrect. While it's true that financial crises often lead to tighter regulations, this was not the primary shift observed in the hedge fund industry during this period. The main change was an influx of institutional investors, not a tightening of regulations.

Choice B is incorrect. Enhanced disclosure requirements were also not the primary shift observed in the hedge fund industry during this period. Although there may have been some changes in disclosure requirements, they were secondary to the large arrival of institutional investors.

Choice D is incorrect. There was actually an increase in assets under management due to the arrival of institutional investors, rather than a decrease as suggested by this option.

Things to Remember

- Long-Term Capital Management (LTCM) was a hedge fund that collapsed in 1998 due to significant losses in its investments and excessive leverage.
- The dot-com bubble burst in 2001 refers to the sharp decline in the value of technology stocks and internet companies after a period of rapid growth and speculation.

- Hedge funds are alternative investment vehicles that typically use a variety of strategies to generate returns for their investors.
 - Institutional investors, such as pension funds and endowments, are large entities that invest on behalf of a group of individuals or organizations.
 - Regulations in the financial industry are often put in place to protect investors, maintain market stability, and prevent systemic risks.
 - Enhanced disclosure requirements refer to the rules and regulations that require investment funds to provide more detailed information about their investments and operations to investors.
 - Assets under management (AUM) is a key metric used to measure the size and growth of investment funds, including hedge funds.
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Q.2588 A hedge fund manager manages multiple funds. In order to ensure diversification, the fund manager employs different hedge fund strategies. Such a diversification strategy fails:

- A. At times of stress.
- B. Due to excessive leverage.
- C. In case of large hedge funds.
- D. When short selling is allowed.

The correct answer is **A**.

Diversification is a risk management strategy that mixes a wide variety of investments within a portfolio. The rationale behind this technique is that a portfolio constructed of different kinds of investments will, on average, yield higher returns and pose a lower risk than any individual investment found within the portfolio. However, during times of stress, such as a financial crisis, the benefits of diversification can diminish. This is because during a crisis, correlations between asset classes can increase, causing the assets to move in the same direction. This convergence of risk factors can lead to significant losses, despite the diversification strategy. Therefore, while diversification can help to manage risk in normal market conditions, it may not provide the same level of protection in times of market stress.

Choice B is incorrect. While excessive leverage can increase the risk of a portfolio, it does not necessarily mean that diversification strategy will fail. Diversification is about spreading

investments across different asset classes and strategies to reduce risk. Even with high leverage, if the investments are well diversified, the overall risk of the portfolio can still be managed.

Choice C is incorrect. The size of hedge funds does not directly impact the effectiveness of a diversification strategy. Large hedge funds may have more resources to invest in a wider range of assets and strategies, which could potentially enhance their ability to diversify effectively.

Choice D is incorrect. Allowing short selling does not inherently cause a failure in diversification strategy. In fact, short selling can be part of an effective hedging strategy that contributes to portfolio diversification by providing potential profits when markets are falling.

Things to Remember

- Correlation between asset classes can increase during times of stress, diminishing the benefits of diversification.
 - Excessive leverage can increase the risk of a portfolio but does not necessarily mean that diversification will fail.
 - The size of hedge funds does not directly impact the effectiveness of a diversification strategy.
 - Short selling can be part of an effective hedging strategy that contributes to portfolio diversification.
-

Q.2589 An investor wants to replicate the hedge fund return without taking direct exposure to hedge funds. A fund manager informs the investor that various investment management firms now offer products which mimic the hedge fund index. The fund manager further asserts that these products typically have more liquidity and lower fees.

Which of the following options is most accurate?

- A. Hedge fund returns cannot be replicated without having direct exposure.
- B. Hedge fund returns can be replicated without having direct exposure.
- C. Such products have low liquidity.
- D. Such products have higher fees as compared to hedge funds.

The correct answer is **B**.

Hedge fund returns can indeed be replicated without having direct exposure. This is made possible through products offered by large investment management firms that mimic the

strategies of hedge funds. An example of such a product is the Liquid Alternative Beta (LAB) offered by UBS. These products are designed to replicate the hedge fund index, thereby providing similar returns to those of hedge funds. The advantage of these products is that they offer higher liquidity and lower fees compared to direct hedge fund investments. This is because they are not subject to the same lock-up periods and high management and performance fees that are typically associated with hedge funds. Therefore, they provide a more accessible and cost-effective way for investors to gain exposure to the returns of hedge funds without the need for direct investment.

Choice A is incorrect. Hedge fund returns can indeed be replicated without having direct exposure. This is achieved through the use of various financial instruments and strategies that mimic the performance of a hedge fund index.

Choice C is incorrect. Contrary to this statement, these products typically offer more liquidity than direct hedge fund investments. This is because they are often structured as mutual funds or exchange-traded funds (ETFs), which can be bought and sold on an exchange at any time during trading hours.

Choice D is incorrect. These products generally have lower fees compared to direct hedge fund investments, not higher. The high fees associated with hedge funds are often due to their complex strategies and the need for active management, while these replication products can achieve similar returns using less costly methods.

Things to Remember

- Direct exposure to hedge funds typically involves high fees, lock-up periods, and limited liquidity.
- Replicating hedge fund returns can be done through products like Liquid Alternative Beta (LAB) offered by investment management firms.
- Products that mimic hedge fund strategies often use financial instruments and derivatives to achieve similar returns.
- Investors can gain exposure to hedge fund returns through mutual funds or exchange-traded funds (ETFs) that replicate hedge fund indexes.
- Replication products offer higher liquidity and lower fees compared to direct hedge fund investments.

Q.2590 It has been observed that the tracking error of passive investable hedge fund indices and the tracker portfolio is consistently positive. One of the reasons for this is that:

- A. The indices offer better liquidity than the tracker portfolio.
- B. The tracker portfolio offers better liquidity than the indices.
- C. The high operational costs of the tracker portfolio.
- D. The low operational costs of indices.

The correct answer is **A**.

Liquidity refers to the ease with which an asset can be bought or sold in the market without significantly impacting its price. In this context, the investable indices are considered to offer better liquidity than the tracker portfolio.

Investable hedge fund indices are often constructed using liquid and widely traded assets or instruments. They are designed to be easily accessible and tradable for investors. On the other hand, the tracker portfolio may contain less liquid assets or strategies that are not as readily tradable.

When the tracker portfolio contains less liquid assets, it can be more challenging to buy or sell those assets in the market without impacting their prices. This can lead to higher transaction costs, slippage, or delays in executing trades, which can negatively impact the portfolio's performance compared to the more liquid investable indices.

As a result, the tracker portfolio may experience a positive tracking error, indicating that it underperforms the investable index due to liquidity-related issues.

Things to Remember

- Tracking error is a measure of how closely a portfolio follows the index it is benchmarked against.
- Liquidity risk is the risk that an asset cannot be sold on short notice without incurring a loss.
- Operational costs include expenses such as trading costs, management fees, and administrative expenses.
- Investable hedge fund indices are designed to be replicable and investable by investors.
- Tracker portfolios aim to replicate the performance of a specific index or benchmark.

Q.2760 Which of the following hedge fund strategies can be described as event-driven?

- A. Trend followers
- B. Equity long/short
- C. Global macro
- D. Distressed securities

The correct answer is **D**.

An event-driven strategy is one that seeks to take advantage of price inefficiencies that may occur before or after a corporate event. Distressed securities strategy is a type of event-driven strategy. This strategy involves investing in the securities of companies that are in financial distress or have filed for bankruptcy. The idea is that these securities may be undervalued because of the company's financial situation, and if the company can successfully restructure or improve its financial position, the value of the securities may increase. This strategy is risky because if the company's financial situation worsens, the value of the securities may decrease further. However, if the strategy is successful, it can provide significant returns.

Choice A is incorrect. Trend followers, also known as managed futures, are a type of hedge fund strategy that seeks to take advantage of upward and downward trends in the market. They do not typically focus on corporate events or pricing inefficiencies related to such events.

Choice B is incorrect. Equity long/short is a hedge fund strategy that involves taking long positions in stocks that are expected to increase in value and short positions in stocks that are expected to decrease in value. While this strategy may sometimes involve capitalizing on pricing inefficiencies, it does not specifically target corporate events like mergers or acquisitions.

Choice C is incorrect. Global macro strategies aim at profiting from large economic or political changes in various countries by implementing opportunistic investment strategies. Although these funds might invest based on anticipated global events, they do not specifically focus on individual corporate events which form the basis for event-driven strategies.

Things to Remember

- Event-driven strategies seek to take advantage of price inefficiencies before or after corporate events
- Distressed securities strategy involves investing in financially distressed companies or those in bankruptcy
- Trend followers (managed futures) focus on market trends, not corporate events
- Equity long/short strategy involves long and short positions based on stock value expectations, not corporate events

- Global macro strategies profit from economic or political changes in countries, not individual corporate events
-

Q.3033 Which of the following best describes the Dow Jones UBS Emerging Markets Hedge Fund Index, which is a subset of the Dow Jones UBS Hedge Fund Index? It helps to measure the aggregate performance of emerging markets funds which typically invests in:

- A. Equities and derivatives instruments of countries with emerging markets.
- B. Currencies of countries with emerging markets.
- C. Debt instruments of countries with emerging markets.
- D. All of the above.

The correct answer is **D**.

According to the DJCS, the emerging markets hedge fund index which is a subset of the Dow Jones UBS Hedge Fund Index whose task is to measure the aggregate performance of emerging markets funds. These funds invest in currencies, debt instruments, equities, and other instruments of countries with emerging markets, considered to be in a transitional phase between developing and developed status.

Things to Remember

- The Dow Jones UBS Hedge Fund Index is a widely recognized benchmark that tracks the performance of hedge funds globally.
 - Emerging markets refer to countries that are in the process of rapid industrialization and experiencing significant economic growth.
 - Hedge funds are alternative investment vehicles that use various strategies to generate returns for their investors.
 - Investing in emerging markets can provide diversification benefits but also comes with higher risks due to factors like political instability and currency fluctuations.
 - Equities, debt instruments, and derivatives are common types of investments that hedge funds may use to gain exposure to emerging markets.
-

Q.3034 Which of the following best defines a swap spread trade?

- A. A bet that the spread between the fixed side and the floating side will widen while staying within a reasonable range that can be evaluated from historical data.
- B. A bet that the fixed side of the spread will remain higher than the floating side of the spread while staying within a reasonable range that can be evaluated from historical data.
- C. A bet that the spread between the fixed side and the floating side will decrease while staying within a reasonable range that can be evaluated from historical data.
- D. All of the above.

The correct answer is **B**.

A swap spread trade is indeed a bet that the fixed side of the spread will remain higher than the floating side of the spread while staying within a reasonable range that can be evaluated from historical data. In a swap spread trade, the trader is essentially betting on the difference between the swap rate and the yield of a Treasury security of similar maturity. This difference is expected to remain higher than the difference between the London Interbank Offered Rate (LIBOR) and the repo rate. The trader uses historical data to evaluate a reasonable range for this difference. If the actual difference stays within this range, the trader stands to make a profit. If the difference moves outside this range, the trader may incur a loss.

Choice A is incorrect. A swap spread trade does not necessarily involve a bet that the spread between the fixed side and the floating side will widen. The direction of the spread (whether it widens or narrows) depends on various factors such as interest rate movements, credit risk, and liquidity conditions in the market.

Choice C is incorrect. Similar to Choice A, a swap spread trade does not necessarily involve a bet that the spread between the fixed side and the floating side will decrease. The direction of movement in swap spreads can be influenced by many factors which are not predictable with certainty.

Choice D is incorrect. As explained above, both Choices A and C are incorrect definitions of a swap spread trade, therefore Choice D which suggests all options are correct cannot be accurate either.

Things to Remember

- Swap Spread: The difference between the swap rate and the yield of a Treasury security of similar maturity.
- LIBOR: London Interbank Offered Rate, a benchmark interest rate at which major global banks lend to one another.

- Repo Rate: The rate at which the central bank lends money to commercial banks in the event of any shortfall of funds.
 - Factors influencing swap spreads: Interest rate movements, credit risk, and liquidity conditions in the market.
 - Historical data analysis: Traders use historical data to evaluate a reasonable range for the difference between swap rates and Treasury yields.
-

Q.5384 Paul Sampson is a seasoned hedge fund manager at Stellar Hedge Funds. Recently, he has been implementing strategies to increase alpha by taking on opportunities that differ from his benchmark, the S&P 500. As a result, he notices his fund's beta exposure is different from his benchmark. In response to this, Paul uses futures contracts to hedge all systematic risks other than exposure to the S&P 500, such that the portfolio's beta relative to the S&P 500 is 1.0. However, one of Paul's colleagues, Margaret, has expressed concerns about this strategy, suggesting that it may have unintended effects on the fund's risk and alpha. Given this situation, which of the following best describes the relationship between the risk and alpha in Stellar Hedge Funds?

- A. The fund's risk is increased as a result of Paul's efforts to independently manage alpha and beta.
- B. Paul's strategy to separate alpha from beta has no significant effect on the fund's overall risk level.
- C. The fund's alpha is potentially decreased due to Paul's strategy of minimizing undesired systematic risk.
- D. Paul's strategy of using futures to control beta exposure is unlikely to have any effect on the fund's alpha.

The correct answer is **C**.

Paul is implementing a strategy to increase the fund's alpha while keeping the fund's beta at 1.0 relative to the S&P 500. However, alpha is a risk-adjusted measure of return, and by reducing systematic risk (via futures contracts), Paul could inadvertently be reducing the potential alpha of the fund. In other words, if the risks that are being hedged out were in fact contributing to the fund's alpha, this approach could diminish the fund's alpha.

A is incorrect because Paul's strategy is specifically aimed at managing the fund's risk profile, not increasing its overall risk level. By hedging against systematic risk not associated with the S&P 500, he is attempting to control the fund's beta exposure, which should, in theory, manage or potentially reduce the overall risk of the fund, not increase it.

B is incorrect because Paul's strategy does indeed affect the fund's risk level, but it's aimed at managing it, not leaving it unchanged. The use of futures contracts to hedge against certain systematic risks is a clear action to control risk exposure.

D is incorrect because the use of futures to control beta exposure could indeed have an impact on the fund's alpha. If the hedged risks were contributing to the fund's alpha, this strategy could potentially reduce the alpha. So, it's incorrect to say it's "unlikely" to have any effect.

Things to Remember

- Alpha represents a risk-adjusted measure of an investment's excess return relative to a benchmark. It is usually seen as a measure of the value added by a fund manager's decisions.
 - Beta is a measure of systematic risk, or market risk, of an investment compared to the market as a whole. A beta of 1.0 suggests the investment's price will move with the market.
 - Active management strategies can aim to separate alpha from beta. However, such strategies may impact the alpha of the fund.
 - Use of derivatives like futures to control beta exposure can potentially impact the fund's alpha, especially if the hedged risks were contributing to alpha generation.
 - Although hedging strategies can manage and potentially reduce a portfolio's risk, they can also unintentionally influence other parameters like alpha.
-

Reading 155: Performing Due Diligence on Specific Managers and Funds

Q.2591 A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.

Which of the following statements is most accurate?

- A. Past performances often predict future performances; hence, the committee must rely on past performances as its best quantitative tool.
- B. The committee must not rely entirely only on past performances to decide on the strategy.
- C. The committee must examine at least the past 10 years of performances to select a strategy.
- D. The committee must examine performance data since the firm's inception to select a strategy.

The correct answer is **B**.

The committee must not rely entirely only on past performance to decide on the strategy to be adopted. It is often observed that an investor's previous success is not a guarantee of future performance.

Things to Remember

- Past performance is not always indicative of future results in the financial markets due to various factors such as changing market conditions, economic cycles, and regulatory changes.
- Investment committees should consider a combination of quantitative and qualitative factors in addition to past performance when making investment decisions.
- Relying solely on past performance can lead to biases and may not capture the full picture of an investment opportunity.
- Regular performance appraisals of fund managers are essential to ensure accountability and alignment with the firm's investment objectives.

- Long-term performance data can provide valuable insights, but shorter-term performance trends and risk metrics should also be considered.
-

Q.2592 A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.

Which of the following is the most appropriate statement with regards to the committee's role in deciding the investment strategy?

- A. Since the committee consists of multiple members, the investment decision is more robust than that which would be made by an individual.
- B. Members of the committee could sometimes not speak out during controversial scenarios, and the committee head may be able to direct the decisions for the whole team.
- C. Since the committee consists of multiple members, reaching a consensus may be difficult.
- D. Since the committee consists of multiple members, diverse opinions may cause confusion.

The correct answer is **B**.

In a committee setting, especially one where the head of the committee holds a significant position of power, there is a possibility that some members may not voice their opinions during controversial discussions. This could be due to fear of conflict, repercussions, or damaging their relationship with the committee head. In such situations, the head of the committee may end up directing the decisions for the entire team. This is particularly likely when the committee members have established high-level relationships with the committee head. Therefore, the decision-making process may not be as democratic or inclusive as it ideally should be, and the decisions made may not fully reflect the diverse perspectives and expertise of all committee members. This could potentially impact the robustness and effectiveness of the investment decisions made by the committee.

Choice A is incorrect. While it's true that having multiple members on a committee can lead to more robust decision-making due to the diversity of perspectives, this doesn't necessarily mean that the decisions will always be better than those made by an individual. The quality of decisions depends on the expertise and judgment of the individuals involved, not just their number.

Choice C is incorrect. Although reaching consensus in a group can sometimes be challenging,

this isn't necessarily a negative implication. In fact, it could lead to more thorough discussions and better decision-making as different viewpoints are considered.

Choice D is incorrect. Diversity of opinions within a committee does not inherently cause confusion; rather, it can foster innovation and creativity in investment strategies. It's up to the leadership of the committee to manage these differing opinions effectively.

Things to Remember

- **Group Thinking:** In a committee setting, there is a risk of group thinking where members may conform to the opinions of the majority or the committee head, leading to biased decision-making.
 - **Conflict Management:** Effective committees should have mechanisms in place to manage conflicts and encourage open communication among members to ensure all perspectives are considered.
 - **Decision-making Process:** Committees should have clear guidelines on how decisions are made, including the criteria used to evaluate investment opportunities and the role of each member in the decision-making process.
 - **Transparency and Accountability:** It is important for committees to operate transparently and hold members accountable for their decisions to maintain trust and credibility.
-

Q.2593 A large investment firm manages multiple funds. The investment committee decides on the investment strategy and takes the final decision on the investment opportunities available in the market. While making the investment decision, the committee relies heavily on the past performance of a similar strategy. The Chief Investment Officer (CIO) heads the investment committee with fund managers as members. The firm carries out a performance appraisal of its fund managers quarterly, and the CIO heads the appraisal committee.

With regards to the CIO heading the appraisal committee, which of the following is the most appropriate statement?

- A. The arrangement ensures that the interests of the fund managers, head of the committee, and the firm are aligned.
- B. The arrangement guarantees robust investment decisions by the committee.
- C. The arrangement may result in suboptimal investment decisions.

D. The arrangement may result in better discussions on investment opportunities.

The correct answer is **C**.

The arrangement may result in suboptimal investment decisions. This is because the dual role of the CIO, who heads both the investment committee and the appraisal committee, could potentially limit the open discussion and sharing of diverse views by the fund managers. Specifically, since the CIO also leads the investment committee, fund managers might feel intimidated or hesitant to express their opinions, especially if they contradict the CIO's views. This fear could stem from the concern that their performance appraisal might be negatively affected if they disagree with the CIO. As a result, this arrangement could lead to suboptimal investment decisions due to the lack of diverse opinions and robust discussions. It is crucial in investment decision-making to have a variety of perspectives to consider all potential risks and opportunities. Therefore, an environment that discourages open discussion could hinder the effectiveness of the decision-making process, leading to potentially suboptimal investment decisions.

Choice A is incorrect. While the arrangement may seem to align the interests of the fund managers, head of the committee, and the firm, it does not necessarily ensure this alignment. The CIO's dual role could potentially lead to conflicts of interest, as they might be biased towards strategies that they have approved or favor.

Choice B is incorrect. The arrangement does not guarantee robust investment decisions by the committee. Relying heavily on historical performance can lead to overconfidence and neglect of other important factors such as market conditions or changes in company fundamentals.

Choice D is incorrect. Although having a single person leading both committees might streamline discussions on investment opportunities, it doesn't necessarily result in better discussions or decisions. It could limit diversity of thought and increase risk of groupthink.

Things to Remember

- Conflicts of interest in investment decision-making
- Groupthink and its impact on investment decisions
- The importance of diverse perspectives in decision-making
- Risk of overreliance on past performance in investment strategies
- The role of the Chief Investment Officer (CIO) in investment committees

Q.2595 A group of investors is looking for an ideal investment management firm in which to channel their surplus funds. At a minimum, the investors feel that the ideal firm must have a robust control mechanism, excellent regulatory oversight, and a non-negotiable operational due-

diligence process. They are able to shortlist one firm with the following features:

- a. The firm has excellent regulatory oversight, without compliance-related issues for the last 10 years
- b. The firm managers have exhibited excellent control over the fund, and there haven't been any instance related to the breaching of concentration norms, or other caps as mandated by the management
- c. The firm has an excellent operational system but the returns of the fund are below the average returns posted by other funds in its category

Assume that you are an expert in risk and investment management. At a meeting called to determine the suitability of the firm whose characteristics have been listed above, what would your proposal be?

- A. To select the firm because it fulfills all the requirements set by the investors.
- B. To turn down the fund because its returns are below the average returns posted by other funds in its category.
- C. To turn down the fund on the basis that it doesn't have a clean regulatory track record of 20 or more years.
- D. None of the above.

The correct answer is **A**.

In this scenario, the investors have identified a firm that meets their key criteria of having excellent regulatory oversight, strong control mechanisms, and a rigorous operational due diligence process. However, the fund's returns are below the average returns posted by other funds in its category. Given the priorities and minimum requirements set by the investors, the investors should select the firm because it fulfills all the requirements set by the investors. While the fund's returns may be below average, the fact that the firm has a clean regulatory track record for the last 10 years, excellent control over the fund, and a robust operational system means that it meets the minimum criteria established by the investors. The investors' primary concerns appear to be regulatory compliance, control, and due diligence, which this firm satisfies. Furthermore, investment performance can vary over time, and investors may be willing to prioritize stability and risk management over short-term returns.

Option B is incorrect. It focuses solely on returns and disregards the other essential factors mentioned by the investors.

Option C is incorrect. The shortlisted firm has data observed over 10 years, not 20 years.

Things to Remember

- **Regulatory Oversight:** Regulatory oversight refers to the supervision and regulation of

financial firms by regulatory bodies to ensure compliance with laws and regulations.

- **Control Mechanisms:** Control mechanisms are processes and procedures put in place by a firm to ensure that operations are conducted efficiently, risks are managed effectively, and compliance requirements are met.
 - **Operational Due Diligence:** Operational due diligence is the process of evaluating a firm's operational infrastructure, processes, and controls to assess operational risks and ensure that the firm is well-managed.
 - **Investment Performance:** Investment performance refers to the returns generated by an investment over a specific period of time. It is an important factor for investors but should be considered in conjunction with other factors like risk management and compliance.
 - **Stability and Risk Management:** Investors often prioritize stability and risk management in their investment decisions, especially when considering long-term goals and risk tolerance.
-

Q.2596 Gong Fang attends an investor meeting conducted by a large investment management firm. In the meeting, the firm launches a new fund and follows it up with an excellent presentation on the past performances of the funds previously launched. After the presentation, the firm's CEO starts taking questions from prospective investors. In his case, Fang interacts with investor relations staff and gets his questions answered.

Fang feels satisfied with his interaction with the investor relation staff. This, combined with the CEO's readiness to answer any kind of query about the fund, convinces Fang to invest in the fund.

Suppose Fang approaches you in retrospect, just before making good on his decision to invest in the fund. What advice would you give him?

- A. That he should go ahead because his attendance of the investor meeting and interactions with the fund's staff constitute due diligence.
- B. That he must try to speak to the firm's employees of mixed seniority before investing in the fund.
- C. That he must try to dig into the past performances of the fund before investing in the fund.
- D. That he should turn down the offer.

The correct answer is **B**.

Investors must try to speak to people of mixed seniority to assess the performances of the fund. The goal must be to find out what is actually going on in the fund rather than to have a sales pitch by the investor relations staff. If the investors are unable to get access to the actual decision-makers, it's highly likely that their assessment of the performances of the fund will be incomplete.

Things to Remember

1. Due diligence is a critical step in the investment process. It involves a comprehensive appraisal of a business or a fund by a prospective buyer or investor to establish its assets and liabilities and evaluate its commercial potential.
2. Investors should interact with employees of mixed seniority within the firm before making an investment decision. This can provide a more comprehensive understanding of the fund's performance and potential risks.
3. Past performance is not always indicative of future results. Therefore, investors should not rely solely on past performance to make an investment decision.
4. Investors should conduct their own independent research and analysis before investing in a fund. This can help them make an informed investment decision and mitigate potential investment risks.

Q.2597 A fund manager states that: *"A firm's ownership structure impacts performance."* Which of the following is the most accurate statement?

- A. A firm's ownership structure helps in aligning the interests of the fund managers with those of investors.
- B. A firm's ownership does not affect its performance.
- C. The presence of large personal investments on the part of fund managers increases the firm's returns.
- D. The presence of significant personal investments on the part of fund managers decreases the firm's returns.

The correct answer is **A**.

A firm's ownership structure indeed plays a crucial role in aligning the interests of the fund managers with those of the investors. This is because when fund managers have a significant personal investment in the firm, they are more likely to make decisions that are in the best interest of the firm and its investors. This is due to the fact that their personal financial well-being is directly tied to the performance of the firm. Therefore, they are more likely to make decisions that will enhance the firm's performance, which in turn benefits the investors. This alignment of interests can lead to better decision-making, improved performance, and ultimately, higher returns for investors.

Choice B is incorrect because it contradicts the statement made by the fund manager. The structure of a firm's ownership can indeed have a significant impact on its performance. This is because the ownership structure determines who has control over the firm's decisions and resources. If the ownership is concentrated in the hands of a few individuals or entities, they may make decisions that are in their own best interest, rather than in the best interest of the firm or its investors. On the other hand, if the ownership is dispersed among many shareholders, the firm may be more accountable to its investors and more likely to make decisions that enhance its performance.

Choice C is incorrect because while the presence of large personal investments on the part of fund managers can potentially increase the firm's returns, it is not a guarantee. The performance of a firm is influenced by a multitude of factors, including the firm's strategic decisions, market conditions, and competitive environment. While a large personal investment by fund managers can align their interests with those of the investors and potentially lead to better decision-making, it does not automatically translate into higher returns. The fund managers' investment decisions, their ability to manage risk, and the overall market conditions will ultimately determine the firm's returns.

Choice D is incorrect because the presence of significant personal investments on the part of fund managers does not necessarily decrease the firm's returns. In fact, it can potentially increase the firm's returns by aligning the interests of the fund managers with those of the investors. When fund managers have a significant personal investment in the firm, they are more likely to make decisions that enhance the firm's performance, as their personal financial well-being is tied to the performance of the firm. However, it is important to note that the performance of a firm is influenced by a multitude of factors, and the presence of personal

investments by fund managers is just one of many factors that can impact the firm's returns.

Things to Remember

- Ownership structure impacts firm performance by aligning interests of fund managers with investors
 - Ownership structure determines control over decisions and resources
 - Ownership concentration can lead to decisions in the best interest of a few, not the firm or investors
 - Ownership dispersion can lead to more accountability and decisions enhancing performance
 - Personal investments by fund managers can align interests but not guarantee higher returns
 - Performance influenced by strategic decisions, market conditions, and competitive environment
 - Personal investments by fund managers can potentially increase firm returns by enhancing performance
 - Presence of personal investments by fund managers does not automatically decrease firm returns
-

Q.2598 An investment management firm is looking for a fund manager and a risk manager. The firm also has an employee referral programme through which present employees may refer probable candidates for various roles. In fact, the firm has a track record of hiring managers based on internal employee recommendations. An employee recruited via an internal recommendation is neither subject to background checks nor rigorous pre-employment interviews.

The firm hires a risk manager through the internal referral programme. The risk manager is put in a team-mandated with the management of a global equity portfolio. The managers report to the head of the global equity portfolio. The risk manager observes that the firm does not have a risk committee, but the CIO does receive daily reports on various risk metrics. He also observes that the model used to value exotic derivative products was last audited two years ago.

Which of the following is the most accurate statement?

- A. Hiring managers through an internal referral programmes reduces the risk of employee frauds.
- B. Hiring managers through an internal referral programme reduces the cost incurred when conducting background checks.
- C. Independent background checks must be carried out for each manager irrespective of the mode of hiring.
- D. Managers must not be hired through an internal referral programmes as it reduces the number of potential candidates.

The correct answer is **C**.

Independent background checks must be carried out for each manager irrespective of the mode of hiring. This is a crucial step in the hiring process, especially for positions that involve significant responsibility and risk, such as a risk manager. Background checks help to verify the credentials, experience, and track record of the candidate, ensuring that they are suitable for the role. In the context of the investment management firm, the risk manager is responsible for managing a global equity portfolio, a role that requires a high level of expertise and integrity. By bypassing background checks, the firm may be exposing itself to potential risks, such as hiring a manager who is not qualified or has a history of fraudulent activities. Therefore, it is essential that independent background checks are carried out for each manager, regardless of how they are hired.

Choice A is incorrect. Hiring managers through an internal referral programme does not necessarily reduce the risk of employee frauds. While it may increase the likelihood of hiring individuals who are known and trusted by current employees, it does not guarantee that these individuals will not engage in fraudulent activities. Without a thorough background check, there is still a risk of hiring individuals with a history of fraudulent behavior.

Choice B is incorrect. While hiring managers through an internal referral programme may reduce the cost incurred when conducting background checks, this practice can potentially lead to greater risks and costs in the long run if it results in hiring unqualified or dishonest individuals.

Choice D is incorrect. The statement that managers must not be hired through an internal referral programmes as it reduces the number of potential candidates is too absolute. Internal referrals can be one method among many for sourcing potential candidates, but should not replace other methods such as job postings or recruitment agencies which can provide a wider pool of candidates.

Things to Remember

Additional important concepts related to hiring practices and risk management in investment management firms include:

- Background checks are crucial for verifying the credentials, experience, and track record of candidates, especially for roles with significant responsibility and risk.

- Internal referral programs can be a valuable source of potential candidates, but should not replace other recruitment methods.
 - Risk management in investment firms involves monitoring and assessing various risk metrics, as well as ensuring compliance with regulations and best practices.
 - Audit processes for valuation models, especially for complex products like exotic derivatives, should be conducted regularly to ensure accuracy and reliability.
 - A risk committee can provide additional oversight and expertise in managing risks within an investment firm.
-

Q.2599 An investment management firm is looking for a fund manager and a risk manager. The firm also has an employee referral programme through which present employees may refer probable candidates for various roles. In fact, the firm has a track record of hiring managers based on internal employee recommendations. An employee recruited via an internal recommendation is neither subject to background checks nor rigorous pre-employment interviews.

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Which of the following is the most accurate statement?

- A. Risk managers reporting to portfolio managers results in better risk management.
- B. Risk managers reporting to portfolio managers increases the overall risk borne by a firm.
- C. Risk managers must report independently to the CIO.
- D. Risk managers reporting to the portfolio managers results in better risk mitigation.

The correct answer is **C**.

Risk managers must report independently to the CIO. In an investment management firm, it is crucial to maintain a clear separation between investment activities and risk management. This is to avoid any potential conflicts of interest that may arise if risk managers report to portfolio managers, who are primarily focused on generating positive returns. By reporting directly to the CIO, risk managers can provide an independent assessment of the risks associated with the

firm's investment activities. This structure ensures that risk management is not influenced by the investment decisions and can effectively measure and manage the risks associated with the firm's investment activities. Furthermore, the CIO, being a senior executive, has the authority to implement necessary risk management measures based on the risk manager's reports. This reporting structure also promotes transparency and accountability in the firm's risk management process.

Choice A is incorrect. While it might seem that risk managers reporting to portfolio managers could result in better risk management due to close communication, this is not necessarily the case. It can lead to conflicts of interest as portfolio managers may be more focused on returns rather than managing risks. Therefore, independent reporting structure for risk managers is preferred.

Choice B is incorrect. The statement that risk managers reporting to portfolio managers increases the overall risk borne by a firm is too broad and lacks context. While it's true that such a reporting structure could potentially lead to conflicts of interest and inadequate risk management, it doesn't automatically increase the overall firm's risk level without considering other factors like the firm's overall governance and control environment.

Choice D is incorrect. Similar to choice A, while closer communication between portfolio manager and risk manager might seem beneficial for better mitigation strategies, it does not guarantee improved outcomes due to potential conflicts of interest or bias in decision-making process from the portfolio manager side.

Things to Remember

- Internal employee referral programs can be a valuable source for hiring new talent within a firm.
- Separation of duties between investment activities and risk management is essential in maintaining a robust risk management framework.
- Regular auditing of valuation models for exotic derivative products is crucial to ensure accuracy and reliability of the valuation process.
- Risk committees play a vital role in overseeing and guiding the risk management practices within an organization.
- Transparency and accountability are key principles in effective risk management processes.

Q.2600 Which of the following statements is most accurate with regards to the audit of valuation models?

- I. Inputs and assumptions used in the model must be revisited periodically
- II. Inputs and assumptions used in the model must not be changed
- III. Inputs and assumptions used in the model must not to be subjected to an external audit

- A. I only
- B. II and III
- C. I and III
- D. All of the above

The correct answer is **A**.

The inputs and assumptions used in a valuation model must be revisited periodically. This is because the output of valuation models heavily depends on the inputs and assumptions used. To ensure that the output of valuation models are not out of sync, periodic stress testing must be carried out to ensure that the results of the model are robust. This process involves revisiting the inputs and assumptions used in the model to ensure they are still valid and relevant. This practice helps to maintain the accuracy and reliability of the model's output, thereby enhancing the credibility of the valuation process.

Choice B is incorrect. The statement that inputs and assumptions used in the model must not be changed is not a best practice in auditing valuation models. In fact, it's quite the opposite. Inputs and assumptions should be revisited periodically to ensure they are still relevant and accurate, reflecting current market conditions.

Choice C is incorrect. While it includes the correct statement that inputs and assumptions should be revisited periodically, it also incorrectly states that these elements must not be subjected to an external audit. External audits are a crucial part of ensuring accuracy and reliability in valuation models.

Choice D is incorrect. As explained above, both statements II and III are not considered best practices in auditing valuation models.

Things to Remember

- Valuation models are used to estimate the value of assets, liabilities, or businesses.
- Inputs and assumptions in valuation models can include factors such as discount rates, growth rates, and market data.
- Validation of inputs and assumptions is crucial to ensure the accuracy and reliability of the valuation model.
- External audits help provide an independent assessment of the valuation model's

accuracy and compliance with relevant standards.

- Regular review and updates of valuation models are necessary to reflect changes in market conditions and ensure the model's relevance.
-

Q.2601 An investor wants to invest in a certain hedge fund. While carrying out due diligence, the investor observes that the fact sheet of the fund promises a return that's way above the average return generated by the funds in its category. He also observes that the fund executes its deals through a broker-dealer. Moreover, the fact sheet indicates that the fund uses a model developed by its quant researchers to value its illiquid investments. These illiquid investments form 20% of the fund's total assets.

Which of the following is the **most** appropriate statement?

- A. Since the fund promises a return that's above the average return generated by similar funds, the investor must invest in the fund.
- B. The investor should not talk to current and prospective investors before investing in the fund as they may give misleading information.
- C. Claims of higher returns are generally an indication of misrepresentation by the fund and a probable case of fraud.
- D. The investor must invest in the fund in a staggered manner.

The correct answer is **C**.

Claims of higher returns are generally an indication of misrepresentation by the fund and a probable case of fraud. When a hedge fund promises returns that are significantly higher than the average returns of similar funds, it raises a red flag. This is because the returns of a fund are typically determined by the performance of the underlying assets and the skill of the fund manager. If a fund is promising unusually high returns, it could be an indication that the fund is misrepresenting the facts or figures. This could be a sign of fraud, where the fund is attempting to attract investors with the promise of high returns, only to misuse or misappropriate the investors' funds. Therefore, it is crucial for investors to conduct thorough due diligence before investing in a fund, especially when the fund is promising unusually high returns. This due diligence should include a review of the fund's investment strategy, the qualifications and track record of the fund manager, and the fund's historical performance. Investors should also consider seeking independent advice from a financial advisor or consultant.

Choice A is incorrect. Just because a fund promises a return that's above the average return generated by similar funds, it does not necessarily mean that the investor must invest in the fund. High returns often come with high risks, and the investor should consider their risk tolerance before making an investment decision.

Choice B is incorrect. It's generally beneficial for an investor to gather as much information as possible before investing in a fund. Talking to current and prospective investors can provide valuable insights about the fund's performance, management style, and potential risks.

Choice D is incorrect. Investing in a staggered manner might be a good strategy for some investors depending on their financial goals and risk tolerance, but it's not necessarily appropriate for all investors or all types of investments. The decision should be based on thorough analysis rather than just one piece of information about the fund.

Things to Remember

- High returns in hedge funds can be a red flag for potential misrepresentation or fraud.
 - Investors should conduct thorough due diligence before investing in any fund, especially when promised returns are unusually high.
 - Due diligence should include a review of the fund's investment strategy, the qualifications of the fund manager, and historical performance.
 - Investors should consider seeking independent advice from financial advisors or consultants.
 - Talking to current and prospective investors can provide valuable insights about a fund's performance and risks.
 - Investing in a staggered manner may be a suitable strategy based on individual financial goals and risk tolerance.
-

Q.2602 An investor wants to invest in a certain hedge fund. While carrying out due diligence, the investor observes that the fact sheet of the fund promises a return that's way above the average return generated by the funds in its category. He also observes that the fund executes its deals through a broker-dealer. Moreover, the fact sheet indicates that the fund uses a model developed by its quant researchers to value its illiquid investments. These illiquid investments form 20% of the fund's total assets.

Which of the following is the most appropriate statement?

- A. Trading via affiliate brokers results in better quotes.
- B. Trading via affiliate brokers is more costly than direct trading.
- C. Trading via affiliate brokers results in a lack of trading independence.

D. Trading via affiliate brokers may result in higher transaction costs.

The correct answer is C.

Trading via affiliate brokers results in a lack of trading independence. This is because when a fund trades through an affiliated broker, it may not be able to negotiate the best terms for its trades. The broker may have a conflict of interest, as they may prioritize their own profits over the best interests of the fund. This lack of independence can lead to suboptimal trading decisions, which can negatively impact the fund's performance and the returns for its investors. Therefore, it is generally recommended that funds trade through independent brokers to ensure that they are able to negotiate the best terms for their trades and to minimize the risk of conflicts of interest.

Choice A is incorrect. There is no guarantee that trading via affiliate brokers results in better quotes. The quality of quotes depends on various factors such as the broker's expertise, market conditions, and the specific assets being traded.

Choice B is incorrect. While it can be true that trading via affiliate brokers may be more costly than direct trading due to potential conflicts of interest and additional fees, this isn't always the case. The cost-effectiveness depends on a variety of factors including the broker's fee structure and negotiation skills of the fund manager.

Choice D is incorrect. Although it's possible that trading via affiliate brokers may result in higher transaction costs due to potential markups or additional fees, this isn't necessarily always true. The actual transaction costs depend on several factors such as the broker's pricing model and whether or not they offer volume discounts.

Things to Remember

- **Trading Independence:** It is crucial for funds to maintain trading independence to ensure that they can make decisions that are in the best interest of their investors without any conflicts of interest.
- **Broker-Dealer Relationships:** Funds need to carefully consider the relationships they have with broker-dealers to ensure that they are getting the best possible terms for their trades.
- **Illiquid Investments:** Valuing illiquid investments can be challenging, and funds need to have robust models and processes in place to accurately value these assets.
- **Due Diligence:** Investors should conduct thorough due diligence before investing in any fund to understand the fund's strategies, risks, and potential conflicts of interest.
- **Conflicts of Interest:** It is important to identify and mitigate conflicts of interest in the

investment process to ensure that decisions are made in the best interest of investors.

Q.2603 Which of the following is (are) accurate statements?

- I. Illiquid assets generate an illiquidity premium which makes the fund attractive
 - II. A high proportion of illiquid assets is a warning signal for investors
 - III. The model used for the valuation of illiquid assets must be independently audited
 - IV. The use of proprietary models which are not independently audited to value investments/assets is a warning signal
- A. Only I
 - B. Both I and II
 - C. Both III and IV
 - D. All of the above

The correct answer is **D**.

All the statements are accurate. Illiquid assets do generate an illiquidity premium, which can make the fund attractive to investors seeking higher returns. However, a high proportion of illiquid assets can be a warning signal for investors as it may indicate potential liquidity risk. This is because illiquid assets can be difficult to sell quickly without incurring a significant loss in value. The valuation of such assets is often not transparent, which can further increase the risk. Therefore, the model used for the valuation of these assets should be independently audited to ensure its accuracy and reliability. Furthermore, the use of proprietary models that are not independently audited can make the valuation process more opaque and increase the risk of fraud. Therefore, this is also a warning signal for investors.

Choice A is incorrect. While it's true that illiquid assets can generate an illiquidity premium, making the fund attractive, this isn't the only accurate statement among the options. Other statements also hold true in certain contexts.

Choice B is incorrect. Although both statements I and II are correct, they do not represent all the accurate statements in this context. Statements III and IV are also correct as per FRM guidelines which emphasize on independent auditing of models used for valuation of illiquid assets and consider use of unaudited proprietary models as a warning signal.

Choice C is incorrect. While it's true that both statements III and IV are accurate, they don't encompass all the correct answers here. Statements I and II are also valid under certain circumstances.

Things to Remember

- Illiquid assets can include investments like private equity, real estate, and certain types of bonds that are not easily traded on public markets.
 - Illiquidity premium is the additional return that investors expect to receive for investing in assets that are not easily tradable.
 - Liquidity risk is the risk that an investor may not be able to buy or sell an investment quickly enough at a fair price.
 - Transparency in valuation is crucial for investors to understand the true value of their investments, especially in the case of illiquid assets.
 - Independent auditing of valuation models helps ensure objectivity, accuracy, and reliability in the valuation process.
 - Proprietary models are developed by a specific firm for its own use and may not be as transparent or reliable as independently audited models.
 - Investors should be cautious when dealing with investments that rely on unaudited proprietary models for valuation.
-

Reading 156: Finding Bernie Madoff: Preventing Fraud by Investment Managers

Q.4848 John Murphy, FRM, works as the chief investment adviser at Trader Inc., an investment firm based in Colorado, USA. The following are some details about the firm.

- Stock broker: Smart Trades Brokerage Limited (partly owned by Trade Inc.)
- Paid external research provider: Smart Trades Brokerage Limited.
- Fee paid by firm after referrals: 8% (industry average is 10%)
- Compensation: All managers are required to invest at least 5% of their annual salary with the firm.

Which of the above creates an opportunity for fraud to be committed?

- A. The fact that the firm partly owns the brokerage firm.
- B. The fact that the firm relies on paid external research to make investment decisions.
- C. The fact that Trade Inc.'s referral fees differ from the industry average.
- D. The fact that managers have their money invested with the firm.

The correct answer is **A**.

The fact that Trader Inc. partly owns its official brokerage, Smart Trades Brokerage Limited, can potentially lead to fraudulent activities. This is because firms that are affiliated with broker/dealers tend to have significantly higher rates of fraud. The absence of an external oversight due to the use of an in-house brokerage creates a mechanism that can be exploited for committing fraud. This is a significant risk as it can lead to conflicts of interest, manipulation of trades, and other fraudulent activities that can harm the firm's clients and its reputation.

Choice B is incorrect. Relying on paid external research to make investment decisions is not inherently fraudulent. Many firms use this method to gain insights and perspectives that they may not have internally. It becomes a problem only if the research is biased or manipulated, which is not suggested in the question.

Choice C is incorrect. Having referral fees that differ from the industry average does not necessarily indicate fraudulent activities. The referral fees could be higher or lower based on various factors such as quality of service, client base, etc., and it's a business decision rather than an avenue for fraud.

Choice D is incorrect. Managers investing their money with the firm can actually be seen as a positive sign as it shows that they have confidence in their firm's investment strategies and operations. This aligns their interests with those of other investors and does not provide an

avenue for fraudulent activities unless there are undisclosed conflicts of interest.

Things to Remember

- Conflicts of interest in financial firms can lead to fraudulent activities and harm clients.
 - Regulatory oversight is crucial in preventing and detecting fraud in the financial industry.
 - Disclosure of ownership and affiliations with other entities is important for transparency and accountability.
 - Investment managers investing their own money can align their interests with those of clients.
 - External research providers should be monitored for bias and independence to ensure quality decision-making.
-

Q.4849 Bernie Madoff was sentenced to 150 years in prison with restitution of \$170 billion after being found guilty of running an elaborate Ponzi scheme and defrauding clients of billions of dollars. Why is a Ponzi scheme not sustainable for an unlimited period of time?

- A. Without an actual money flowing in, it is impossible to pay all the investors back with only money from new ones.
- B. Nonprofit organizations will lose all of their money and sue the managers for fraud.
- C. There is a lot of risk for the investors, but very little profit gain.
- D. The creator of the scheme eventually grows tired of maintaining it, but is unable to get out of it.

The correct answer is **A**.

A Ponzi scheme is a fraudulent investment operation where the operator, an individual or organization, pays returns to its investors from new capital paid to the operators by new investors, rather than from profit earned through legitimate sources. Operators of Ponzi schemes usually entice new investors by offering higher returns than other investments, in the form of short-term returns that are either abnormally high or unusually consistent. The scheme is named after Charles Ponzi, who became notorious for using the technique in the 1920s. The idea is to use money from new investors to pay off the old ones until no more new recruits can be found and the whole scheme collapses, with the newest investors losing everything.

Without an actual money flowing in, it is impossible to pay all the investors back with only money from new ones. This is because the scheme relies on the constant influx of new investments to pay returns to existing investors. When the number of new investors decreases, or when too many existing investors want to cash out their returns, the scheme collapses. The operator usually disappears with the remaining investment money, leaving the investors with significant losses.

Choice B is incorrect. While it's true that nonprofit organizations, like any other investors, could potentially lose their money in a Ponzi scheme and sue the managers for fraud, this is not the primary reason why a Ponzi scheme cannot be sustained indefinitely. The sustainability of a Ponzi scheme primarily depends on the continuous inflow of new investments to pay off older investors.

Choice C is incorrect. Although there is indeed high risk for investors in a Ponzi scheme with little profit gain, this does not directly contribute to the unsustainability of such schemes. The main issue lies in the structure of these schemes where returns are paid from funds contributed by new investors rather than from legitimate business activities or profits.

Choice D is incorrect. The personal feelings or intentions of the creator have no bearing on whether or not a Ponzi scheme can be sustained indefinitely. The fundamental flaw lies in its financial structure which relies on an ever-increasing flow of money from new investors to pay off promised returns to earlier ones.

Things to Remember

- Charles Ponzi and the origin of the Ponzi scheme
 - Key characteristics of a Ponzi scheme
 - Difference between Ponzi schemes and legitimate investment operations
 - Legal consequences and penalties for operating a Ponzi scheme
 - Red flags to look out for to identify potential Ponzi schemes
-

Q.4851 Which of the following investment firms are more likely to commit fraud?

- A. Firms whose clients include a high proportion of agents.
- B. Firms where a majority of the investors are large with sophisticated portfolios.
- C. Firms trading assets through an external broker as opposed to an affiliate brokerage.
- D. Firms that have custody of the client's cash and securities.

The correct answer is **A**.

Investment firms whose clients include a high proportion of agents, such as pension managers investing on behalf of employees, are significantly more likely to commit fraud. This is primarily because agents, unlike principals, do not bear the full cost of a fraud. Therefore, they can be easily swayed through gifts or kickbacks to overlook fraudulent activities. This creates an environment conducive to fraud, as the risk of detection and punishment is significantly reduced.

Choice B is incorrect. Firms where a majority of the investors are large with sophisticated portfolios are less likely to commit fraud. This is because these types of investors typically have more knowledge and resources to monitor their investments closely, making it harder for fraudulent activities to go unnoticed.

Choice C is incorrect. Firms trading assets through an external broker as opposed to an affiliate brokerage are not necessarily more susceptible to committing fraud. While using an affiliate brokerage could potentially create conflicts of interest, it does not inherently increase the likelihood of fraudulent activities. In fact, external brokers might provide additional oversight that can help prevent fraud.

Choice D is incorrect. Although firms that have custody of the client's cash and securities do have more opportunities for misappropriation, this does not automatically make them more susceptible to committing fraud. Many firms with custody arrangements have stringent controls in place specifically designed to prevent fraudulent activities.

Things to Remember

- Principal-agent relationship in finance
- Types of fraud in investment firms
- Conflict of interest in brokerage relationships
- Risks associated with custody of client assets
- Importance of regulatory oversight in preventing fraud

Q.4852 Which of the following best describes the role of disclosure in preventing investment fraud and ensuring transparency in financial markets

- A. Publication of investment strategies adopted by managers.
- B. Timely release of all information that could reasonably be expected to influence an investor's decisions.
- C. Timely release of past fraud record by an investment firm.
- D. Publication of fraud deterrents and disciplinary tools employed by investment firm.

The correct answer is **B**.

Disclosure refers to the timely release of all information that could reasonably be expected to influence an investor's decision. It encompasses both positive and negative news as well as operational elements that impact business at an investment firm. It contains information about the investment adviser's business, business practices, ownership, clients, and employees. It also entails information about the adviser's business practices, fees, conflicts of interest, disciplinary information, and affiliations.

A is incorrect. While the publication of investment strategies adopted by managers is a part of disclosure, it does not fully define the concept of disclosure.

C is incorrect. Timely release of past fraud records is an aspect of disclosure, but it does not encompass the entire definition of disclosure.

D is incorrect. Publication of fraud deterrents and disciplinary tools employed by an investment firm is a part of disclosure, but it does not fully define the concept of disclosure.

Things to Remember

- Disclosure is a key component of transparency in the financial markets, ensuring that investors have access to all relevant information to make informed decisions.
- Regulatory bodies like the SEC (Securities and Exchange Commission) in the U.S. mandate disclosure requirements for investment firms to protect investors and maintain market integrity.
- Disclosure can include financial statements, risk factors, investment strategies, conflicts of interest, and other material information that may impact investment decisions.
- Failure to disclose material information can lead to legal and regulatory consequences for investment firms and individuals involved.

- Investors rely on disclosure to assess the credibility and integrity of investment firms and make informed choices about where to allocate their capital.
-

Q.4853 John Henry, FRM, is carrying out a regression to assess whether a set of variables can help predict fraud at investment firms. He has gathered form ADV filings – both past and current – for different firms. He uses the data in three different models. Model 1 uses a cross section of only the current Form ADV filings. Model II uses both current and prior Form ADV filings. Model III uses all prior Form ADV filings except the current filings. Which of the models is more likely to perform better at predicting fraud?

- A. Model I.
- B. Model II.
- C. Model III.
- D. All of the models perform the same.

The correct answer is **B**.

Model II, which uses both current and prior Form ADV filings, is more likely to perform better at predicting fraud. This is because it incorporates a broader range of data, including both historical and current information. Research has shown that prediction models that rely on all prior Form ADV filings, including the most recent ones, perform better than models that only rely on a cross-section of ADV filings at a particular point in time. The panel model that uses data submitted in all prior years can predict 31.4% of frauds, compared to only 25.9% for the cross-sectional model. This suggests that the inclusion of historical data can significantly enhance the predictive power of the model, making Model II the most effective of the three.

Choice A is incorrect. Model I only uses the current Form ADV filings, which may not provide a comprehensive view of the firm's history and potential fraudulent activities. It lacks historical context which is crucial in identifying patterns or trends in fraudulent behavior.

Choice C is incorrect. While Model III incorporates past Form ADV filings, it excludes the current ones. This could potentially miss out on recent changes or anomalies that might indicate fraud.

Choice D is incorrect. The predictive accuracy of models can vary based on the data they incorporate and how they use it to identify potential frauds. In this case, Model II has a higher predictive accuracy as it uses both current and past Form ADV filings, providing a more holistic view of the firm's activities.

Things to Remember

- Model selection in regression analysis is crucial for the accuracy of predictions.

- Using historical data along with current data can improve the predictive power of a model.
 - Cross-sectional data may not capture the dynamics and changes over time in a dataset.
 - Including a broader range of data points can help in identifying patterns and trends in fraudulent behavior.
 - Panel data models that incorporate data from multiple time periods can provide more insights compared to cross-sectional models.
-

Q.4854 Which of the following best describes a Ponzi scheme?

- A. An investment firm forwards funds gathered from new investors to existing investors disguised as returns instead of investing the funds.
- B. An investment firm gets into positions that are riskier than investors can tolerate.
- C. An investment firm that reports losses even though it has made substantial profits.
- D. An investment firm buys securities other than those the investors have been promised as detailed in the prospectus.

The correct answer is **A**.

A Ponzi scheme is a fraudulent investing scam promising high rates of return with little risk to investors. The scheme leads victims to believe that profits are coming from legitimate business activity, and they remain unaware that other investors are the source of funds. A Ponzi scheme is run by a central operator, who uses the money from new, incoming investors to pay off the promised returns to older ones. This makes the operation seem profitable and legitimate, even though no actual profit is being made. Meanwhile, the person behind the scheme pockets the extra money or uses it to expand the scheme. To avoid having too many investors reclaim their profits, Ponzi schemes encourage them to stay in the game and earn even more money. The scheme continues to operate as long as new investors line up with their money, or until too many existing investors cash out. Therefore, choice A accurately describes a Ponzi scheme where an investment firm forwards funds gathered from new investors to existing investors disguised as returns instead of investing the funds.

Choice B is incorrect. While it's true that an investment firm taking on riskier positions than investors can tolerate is a form of misconduct, it does not accurately describe the operational mechanism of a Ponzi scheme. In a Ponzi scheme, the fraudster pays returns to earlier investors using funds obtained from newer investors rather than from legitimate investments.

Choice C is incorrect. An investment firm reporting losses despite making substantial profits may be engaging in fraudulent activity, but this does not characterize a Ponzi scheme. A Ponzi scheme involves paying returns to early investors with funds collected from new investors, not hiding profits.

Choice D is incorrect. Although buying securities other than those promised in the prospectus could be considered fraudulent behavior by an investment firm, this action does not define a Ponzi scheme. The defining characteristic of a Ponzi scheme involves using new investor funds to pay returns to earlier investors instead of investing those funds as promised.

Things to Remember

- Ponzi schemes are named after Charles Ponzi, who became infamous for running this type of fraudulent investment scheme in the early 20th century.
 - Ponzi schemes rely on a constant influx of new investors to continue operating and paying returns to earlier investors.
 - Investors should be cautious of investment opportunities that promise high returns with little to no risk, as they could potentially be Ponzi schemes.
 - Regulatory authorities like the SEC (Securities and Exchange Commission) actively work to identify and shut down Ponzi schemes to protect investors.
-

Q.4855 Which of the following is least likely a barrier to the implementation of fraud prediction methods?

- A. Assessing whether investors are compensated for the risk of fraud through performance or fees requires fund level data.
- B. Historical Form ADV files may not be available for scrutiny by the general public, denying investors an opportunity to assess the fraud record of firms.
- C. Information in Forms ADV may be provided in a format that is not suitable for statistical analysis.
- D. Form ADV data cannot be used to predict fraud.

The correct answer is **D**.

The assertion that Form ADV data cannot be used to predict fraud is incorrect and therefore does not present a barrier to the implementation of fraud prediction methodologies. Form ADVs are forms that investment advisors use to register with both the Securities and Exchange

Commission (SEC) and state securities authorities. The data submitted in these forms by investment managers can indeed be used to predict fraud. Research has shown that disclosures related to past regulatory violations, conflicts of interest, and monitoring can significantly predict fraud. Specifically, if an investor avoids the top 5% of firms with the highest ex ante fraud risk, as measured through their ADV filings, they can avoid 29% of frauds and over 40% of the dollar losses from fraud. Therefore, Form ADV data can be a valuable tool in fraud prediction.

Choice A is incorrect. Assessing whether investors are compensated for the risk of fraud through performance or fees does indeed require fund level data. This can be a significant obstacle as this data may not always be readily available or easily accessible, making it difficult to accurately assess the risk of fraud.

Choice B is incorrect. The unavailability of historical Form ADV files for public scrutiny can pose a significant challenge in implementing fraud prediction methodologies. Investors rely on these forms to assess the fraud record of firms and if they are not available, it hinders their ability to make informed decisions.

Choice C is incorrect. If information in Forms ADV is provided in a format that isn't suitable for statistical analysis, it becomes a major obstacle in implementing effective fraud prediction methodologies. Statistical analysis plays an integral role in predicting and identifying potential fraudulent activities.

Things to Remember

- **Form ADV:** Forms that investment advisors use to register with the SEC and state securities authorities, containing valuable data for fraud prediction.
- **Fraud Prediction:** Using data and statistical analysis to predict potential fraudulent activities in investment firms.
- **Investor Protection:** Assessing fraud risk helps investors make informed decisions and avoid potential losses.
- **Data Accessibility:** Availability of data, such as historical Form ADV files, is crucial for effective fraud prediction methods.

Q.4856 Which of the following best explains why firms whose clients include a high proportion of agents are more likely to commit fraud?

- A. Agents are more willing to give kickbacks to obtain preferential treatment from managers.
- B. Agents do not have the same level of financial knowledge as their principals and are less likely to monitor manager transactions.

C. Unlike principals, agents do not bear the full cost of fraud.

D. Agents' remuneration is not linked to investment performance.

The correct answer is **C**.

Firms with a high proportion of agents as clients are significantly more likely to commit fraud. This is primarily because agents, unlike principals, do not bear the full cost of a fraud. In a typical principal-agent relationship, the agent acts on behalf of the principal. However, if a fraudulent activity occurs, the principal bears the brunt of the financial loss, while the agent may not suffer to the same extent. This lack of direct financial consequence for the agent can lead to a higher propensity for fraudulent activities within firms that have a high proportion of agents as clients. This is because the agents may be more susceptible to unethical practices such as accepting gifts or kickbacks, as they do not bear the full cost of the potential fraud.

Choice A is incorrect. While it's true that agents might be willing to give kickbacks to obtain preferential treatment, this doesn't necessarily lead to an increased likelihood of fraud in firms with a high proportion of agent clients. The willingness to give kickbacks is more related to the individual ethics and integrity of the agents rather than a systemic issue within firms.

Choice B is incorrect. Although agents may not have the same level of financial knowledge as their principals, this does not directly contribute to an increased likelihood of fraud in firms with a high proportion of agent clients. In fact, less knowledgeable agents are more likely victims rather than perpetrators of fraudulent activities.

Choice D is incorrect. The structure or link between an agent's remuneration and investment performance does not inherently increase the likelihood for fraudulent activities in firms with a high proportion of agent clients. While it could potentially create conflicts of interest if poorly structured, it doesn't directly lead to higher instances of fraud.

Things to Remember

- **Principal-Agent Relationship:** In this relationship, the principal delegates decision-making authority to the agent, who acts on behalf of the principal. This relationship can lead to conflicts of interest and moral hazards.
- **Financial Knowledge Disparity:** Differences in financial knowledge between agents and principals can impact the monitoring and oversight of transactions, potentially leading to opportunities for fraud.
- **Cost of Fraud:** Understanding who bears the cost of fraud is crucial in assessing the incentives and motivations for fraudulent activities within an organization.
- **Remuneration Structure:** The way in which agents are compensated, especially in relation to their performance, can influence their behavior and ethical considerations

within the firm.

- **Ethics and Integrity:** The ethical standards and integrity of individuals, including agents, play a significant role in deterring fraudulent activities within organizations.
-

Q.4857 Which of the following best defines investment fraud?

- A. Investing funds in products other than those approved by the client.
- B. Getting into positions that are riskier than investors can tolerate.
- C. Reporting investment losses even though a firm has made substantial profits.
- D. Deceptive practices used by managers to get money from unsuspecting investors.

The correct answer is **D**.

Investment fraud is a term used to describe deceptive practices employed by managers or firms to extract money from unsuspecting investors. These practices can take many forms, but they all involve some level of deceit or misrepresentation. Managers or firms may use false or deeply misleading information to convince investors to make buy or sell decisions that are not in their best interest. The goal of these fraudulent activities is to extract money from investors, often resulting in significant financial losses for the victims. It's important to note that investment fraud is illegal and can lead to severe penalties, including fines and imprisonment.

Choice A is incorrect. While investing funds in products other than those approved by the client can be considered unethical, it does not necessarily constitute investment fraud. Investment fraud specifically involves deceptive practices used to mislead investors and extract money from them.

Choice B is incorrect. Taking on positions that are riskier than what investors can tolerate may be a breach of fiduciary duty or a failure to adhere to an investment policy statement, but it does not equate to investment fraud. Investment fraud involves intentional deception for personal gain or damage to another party.

Choice C is incorrect. Reporting losses when a firm has made substantial profits could be considered fraudulent activity as it involves misrepresentation of financial performance, but this alone doesn't define the broad term 'investment fraud'. The definition of investment fraud encompasses all deceptive practices used by managers with the intent of obtaining money from unsuspecting investors.

Things to Remember

- Investment fraud can take many forms, including Ponzi schemes, insider trading, and false advertising.

- Regulatory bodies like the SEC (Securities and Exchange Commission) play a crucial role in investigating and prosecuting investment fraud cases.
 - Investors should always conduct thorough due diligence and be cautious of promises of high returns with low risk, as these can be red flags for potential fraud.
 - Common warning signs of investment fraud include guaranteed returns, pressure to invest quickly, and complex strategies that are difficult to understand.
-